THE SAYLOR SERIES

WHAT IS MONEY?

MICHAEL SAYLOR & ROBERT BREEDLOVE

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The Saylor Series

By
Michael Saylor
and
Robert Breedlove

whatismoneypodcast.com

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Editor's Note

This book contains a lightly edited transcript of an audio podcast and reflects the rambling nature of a back-and-forth, in-person conversation, not a polished written essay. Bear that in mind as you read it.

For the purposes of organization and readability, episodes have now become chapters. If you want to watch or listen to the original conversation, please go to whatismoneypodcast.com.

Why a book? People have different learning styles and preferences. While some can track with and retain information by watching a video or listening to a podcast, many others prefer reading the text. Perhaps the greatest advantage to a book is the ability to highlight, annotate and return quickly to the most noteworthy sections. As I listened to *The Saylor Series* initially, multiple times I found myself wishing I could write down something that Saylor had said.

The depth of thought and the paradigm-shifting connections that Saylor provides across a wide range of topics is unmatched. I found it intellectually stimulating, exciting and optimistic, and I hope this book produces a similar experience for you.

Introduction

Hey everyone. Welcome to the "What is Money?" Show. I'm your host, Robert Breedlove, and our purpose in this show in general is the pursuit of truth. We're going to explore many topics in depth, and many of them will take us down to the proverbial Bitcoin rabbit hole, by pursuing what I call is the rabbit. And the rabbit is that question — that all-important question — "What is Money?" And this question is a seemingly inexhaustible generator of answers that have continuously reshaped my perspectives on the world. And I think they will for you as well.

This is an extended discussion with Michael Saylor. He's the CEO of MicroStrategy. Michael is the latest and arguably the greatest proponent of Bitcoin, and an ally of this space in its battle for truth and freedom in the world. And MicroStrategy is a NASDAQ-listed business intelligence firm, so Michael has very deep experience in the fields of technology, network architecture — things like this. And in fact, he was educated in the domain of scientific paradigm shifts and the impact of technology on civilization.

Ten years ago, Michael wrote a book called *The Mobile Wave* that depicted many of the impacts that he saw FAANG stocks would have on the world: Facebook, Apple, Amazon, Netflix, Google. He had laid out a case — an investment case, largely — for these companies, and their dominance in the global marketplace. And clearly over the past ten years—as we sit down in 2020 — those stocks have been stand-out performers and have become in many ways the new dominant monopolies in the world today.

Michael has a very deep understanding of these topics that I think predisposed him toward gaining a rapid understanding of Bitcoin. And as you'll see — or as you may have heard in other interviews — he really entered the Bitcoin space in 2020 and got very deep into the rabbit hole very quickly, in the wake of the COVID global lockdown situation. Michael's a very intelligent guy, very high energy, very hard working, and I think his acceleration into the

Bitcoin rabbit hole also demonstrates that a lot of this trail has been blazed for him.

A lot of Bitcoin maximalists have laid the foundation for others to gain a more rapid and clear understanding of the impact of Bitcoin. In the wake of that as you all probably know — you may have not heard — Michael's firm, Microstrategy, named Bitcoin as its primary treasury reserve asset. They initially invested $425 million into Bitcoin, and Michael personally disclosed that he holds about 17,000 Bitcoin himself, so he's got a lot of skin in the game to say the least. And I think you'll see why as we go through some of this.

In this discussion we're calling *The Saylor Series*, we're going to start from the first principles of energy, of anthropology, of technology, and really build a solid foundation for really gaining a deep understanding of Bitcoin's potential impact on the world. And Michael and I, to craft this series, we iterated on its discussion framework, and we finally arrived at his overarching thesis — which he was kind enough to lay out in very sophisticated form — and he goes very deep on the topics we've laid out here, which starts very early, like the Stone Age, and we go all the way into modernity. So this is a long narrative arc that's super fascinating, very interesting stuff, and clearly, it takes us some time to build up to Bitcoin, but the journey itself is purposeful and it's well worth it.

The early chapters will include a lot of Michael talking — so a lot of him speaking solo about his bedrock thesis on energy, anthropology, technology, things like this. And then as we build into modernity and to Bitcoin, it will become much more of a dialogue conversation as we go back and forth about Bitcoin and things of that nature. So I realize this is really long form content, but I assure you, I can promise you, you're gonna find it deeply meaningful. I myself found the feeling of chills at times, there were various epiphanies I had going through this. But this was just dynamite content. And I think it's a great view into the mind of Michael Saylor, and it makes a very powerful case for Bitcoin and how much it's going to reshape the world.

So I promise you that you'll find — despite the time it may take you — you're gonna find this extremely intellectually satisfying, perhaps even philosophically satisfying. We go really deep on a lot of topics. So I hope you enjoy it. And I firmly believe the insights that come out of this will actually reshape your worldview. So if this is the kind of content you're interested in, and you're really interested in going deep and getting to truth, I think you're in the right place today. So with that, let's jump into Chapter 1 of *The Saylor Series* here on the "What is Money?" Show.

Chapter 1 | The Rise of Man Through the Stone and Iron Ages

"Money is the highest form of energy that human beings can channel. Bitcoin is channeling human ingenuity into making it better, and every commodity is channeling human energy into making it worse."

Robert Breedlove: Michael Saylor, thank you for joining me.

Michael Saylor: Well happy to be here, Robert. Thanks for inviting me.

Robert Breedlove: For a man that runs a company named MicroStrategy, you may have just executed the most brilliant macroeconomic strategy there has ever been. How does it feel?

Michael Saylor: It's been a busy quarter I would say. Really busy. It's been a busy year. January 1st of this year [2020], the year started out one way, and then it became something altogether different in March, and it became something altogether different again by June. And now we're in September. I look back on it, and certainly there's a lot of things I didn't expect. I joke with people, If I'd gotten what I'd wanted, I wouldn't have gotten what I needed. I wouldn't have been nearly as successful if at any point in time I'd got what I wanted. I'm sure this is not what I wanted when I started the year, and for a while I thought it wasn't terribly a good thing, but now as we move toward the end of the year, I see the silver lining here and I'm glad these things happened, which is fascinating.

Robert Breedlove: The world in a lot of ways got a wake-up call this year, right? On a lot of different levels. And for you particularly it was the melting ice you were sitting on, that maybe started to melt a little bit faster.

Michael Saylor: You know, there are quotes from Lenin's era. There's Trotsky's quote: "You may not be interested in war, but war is interested in you." And this year we launched one war on COVID, and another war on currency. And so we were caught up in kind of two wars, in two dimensions. And then there's Lenin's quote: "There are decades when nothing happens, and there are weeks when decades happen." And this was that year in both of those ways.

I'm grateful that our company is an enterprise software company — our value proposition is we ship software to large enterprises to help them think better. And the value proposition is intact, even improved, by all of the changes this year. And our cost structure and our operational systems — the way we operate — were dramatically impacted, and we had to adjust. But I would say, this was the year that digital transformation went from being a bromide or a dalliance, to being something that you really had to internalize. This was the year that digital transformation really did transform you at the core of your being. I mean, it transformed my ideas about: money, sales, marketing and services, what product offering we should deliver to the market, the marketplace in the future in general. And it's been thrown around, it's been a buzzword for a decade — maybe for two or three decades. But this is the year when you got it viscerally in your bones, if you had been dragging your feet the least amount.

Robert Breedlove: Great points there. As if the world wasn't changing quickly enough — as we've progressed further into this digital age — it's as if COVID was just a massive accelerant on the entire process. So not only are things transforming much more quickly now, moving to digital much more quickly, but are likely to change even more quickly, exponentially so, into the future. With that, the theme of this conversation today is deep conversations. And I know you're a deep thinker, and I really appreciate the media work that you're doing, and the voice that you brought to the Bitcoin community. I'd like to jump in, kind of like a first principles look at history, and what got us to today? What got us to this digital age that's changing so quickly? And where do we see it going? And I *know you've thought deeply about this, and maybe we could start just at the beginning, so to speak, with the story of technology?*

Michael Saylor: Okay. Well, I mean the phrase that runs through my head is, "There has never been such a thing as a fair fight." Humans have been struggling for millions of years, right? In order to rise, first to become the apex predator in nature, but if you look at our struggle against nature, there's never been such a thing as a fair fight. I remember seeing eagles fly along a mountainside that'll hone in on a goat, and it focuses on a baby goat, not the parent goat — catches it from behind, grabs its foot, and drags it off a cliff, and then backs off and waits while the goat goes bang, bang, bang, and hits rocks every fifty feet and is smashed to death five-hundred feet below, where the eagle circles down, lands on the goat, eats it — leisurely. Nothing fair about it. You know, you feel sorry for the goat, and then you realize, this is not a human being. This is nature. It's not fair.

Then you see lions — and it's not like one lion chases down one gazelle — it's eight lions chase sixty-seven gazelles into a channel with three other lions waiting. And one gazelle is forced to take the right side because it gets crowded out by the other sixty gazelle, and that one—bam. — it's dead for no reason other than that it just happened to be on the right side of the herd, and there's nothing fair about it. Nature's not fair, and when you think about the plight of Man the amazing thing is we actually evolved to be the apex creature on this planet.

A single individual, on their own, has almost no chance. There's that scene in Jurassic Park, and there's the bully, and he just is mean to the little dinosaur creature, some kind of small raptor, and he's like the size of a little dog and he kicks it around, and he's a bully and a sadist. And there's this part where he gets trapped in the park —he's walking and he sees that little creature, and it nips at its legs and he kicks it, and then he turns around and he sees its got a friend and there's two of 'em. And he looks around again and there's four of 'em. And then they jump on him and he knocks them off and there's sixteen of 'em. And then they all jump on him and he fights them off and he gets up and he runs and now there's thirty-two of them.

The human being—the modern American that lives in their world of shopping centers and cars and air-conditioned houses and locked doors and 911 and policemen they can call and feeling safety — they look at nature through a zoo. And they look through the bars and that's nature. Or it's in paintings and it's all just so romantic. Right? They don't have this view of nature. The view when there's sixty-four of those things — and the horrifying realization that that guy is as sure as dead. He's a dead man walking — he's gonna die. There's not a damn thing he can do, it doesn't matter if he has a bazooka, it doesn't matter if he has a machine gun. It doesn't matter — he's going to die. At some point, in the next 48 hours, he's going to fall asleep and they're going to eat him. And that's the human condition.

So when you think about that 3 million years ago the first question is, How did we even make it here? It's pretty obvious that, in that circumstance, if you're alone — you're dead. You're gonna have to have someone guard your back. And my heart goes out to that Adam and Eve wherever they were — you needed two, three four — you needed a tribe. You needed someone to watch, because when you fall asleep, something is going to eat you.

Watch a pack of wolves hunt. The one that kills you? It isn't attacking from the front. You're not going to get to fight it off. It's gonna be an asymmetric attack from the rear, while you're asleep. And so, the importance of human beings using their brains and thinking is incredibly important. And you start to think about — how do we survive in a hostile universe? We have to figure out how we can get harder, smarter, faster, and stronger. And that takes us to the beginning of Man.

If I look back at Stone Age technology and you ask, "How do we even emerge from this incredible, terrifying scrum?" There are just key technologies that you decide you kind of like in a hurry. One of them is fire. One of them is missiles. One of them is hydraulics. There's a lot more that we could talk about, but if we start with fire: fire is like the prime energy network of the human race. It all started for us with channeling energy. Fire's a chain reaction where we're releasing a latent energy in matter — we're converting matter into energy.

Robert Breedlove: Which is like the stored sunlight that we're releasing, right?

Michael Saylor: Yeah, stored sunlight—you're that human being, and you want to rise above the tigers and the packs of wolves and the other creatures and the snakes in the jungle. How are you gonna do that? You're going to have to tap into and channel energy.

And that's why Prometheus has such an incredible mythic place. Prometheus is to Satoshi as fire is to Bitcoin. Bitcoin's a fire. It's a fire in cyberspace. And most people don't realize it, but it has its antecedents, right? And fire came along first, and when you think about what it means — and most people don't necessarily think very hard about it, you always had it.

If you're an individual, what can you do with fire? Well, you can start by starting a fire so you don't freeze to death. That's pretty useful. The fire will scare away the animals. So I start the fire, I can sleep around it, I cannot freeze. I can also put it around my camp and maybe like a snake that would've slithered in and eaten me will go away. I can scare away insects and smoke away insects with it — that's useful. I can hunt with it. I can start a fire, and I can drive the prey away from the fire. And if I'm smart, I drive the prey from fire off the cliff. I wait for them to trip, I go to the bottom of the cliff, I find one that broke its neck.

You ever get in a fight with a horse? Or a fight with a hippopotamus? Or an elephant? It's not gonna end well. This idea of heroic hand-to-hand combat is a great idea in the movies. It's an awful idea in reality. And if you went back a million years, you would find that your great-great-great-great-great-grandparent thought you were pretty frickin' stupid to fight hand-on-hand with anything or anyone.

So I hunt with it — I cook with it. There's a lot of biologists and the paleotheorists that make a very compelling argument that human anatomy actually evolved because we mastered fire. And when you're cooking something, you're pre-digesting it. And if you pre-digest something, not only do you increase the scope of the foods that you can consume, you also accelerate and you also increase the efficiency with which you can convert that food into calories by a factor of 10:1 or 20:1. And if you can actually metabolize the food 10x more efficiently, your digestive tract shortens, and the energy that your body expends in order to digest food can be redirected — probably to your brain. Right?

Animals that don't cook food have small brains. A human being that can cook food can have a very short digestive tract, can eat anything—we're omnivores — we can go anywhere, we can metabolize calories that are very efficient. It only takes us 10 or 15 minutes a day to get all the calories we need. There are animals that have to graze all day to get the calories they need. So fire is critical for that.

It's critical for seeing. You channel your fire and you can light up a cave, a camp, a tent, you can light up anywhere. And with seeing comes communicating. You ever travel through the ancient world you see they have all these watchtowers. The Romans built watchtowers. You put a fire in the tower, you can see it for miles and miles away. You create a signal system. A certain presumptive arrogance or ignorance among modern men — we think kind of everything worth doing was done in the past 2,000 years or 3,000 years. I kind of figure 100,000 years ago, people were doing all of this stuff. We might not have the writings of it, but they were pretty smart.

So I'm gonna use the fire for all of those things and eventually for communicating, but once I figure that out, I can use it for hardening, right? I can cook things, I can harden the tip of a spear, I can use it to work metals — eventually we used it to work metals, and that ushered us from the Stone Age to the Bronze Age to the Iron Age. Fire is intrinsic to all sorts of manufacturing processes. And of course — I give you 1,000 acres of forest, Robert — how are you gonna clear it?

Robert Breedlove: Sounds like fire would be the easiest way.

Michael Saylor: Tractor, in 100,000 B.C.? Fire. You're gonna burn it. I see these discussions: "Well paleo-man, they're all hunter-gatherers and they're all just like walking around chasing after things that are running away from them." I doubt it. Like if I dropped you into 100,000 B.C., I don't think you would've — solo — chase after a bunch of stuff with four legs. I think you would start by finding a canyon and start a fire on one end and dig a trench on the other end and let a mastodon trip on it and break its neck.

Robert Breedlove: It's sort of life's way to take the most energy-efficient strategy, right? That's why the eagle drags the goat off a cliff and lets gravity do its work. Instead of trying to fight it out. And it's interesting that you bring up fire and it's almost as if we were using it to energize our strategies in the world. I think as you put it earlier, channeling energy through our intellect. I think the one piece that maybe we didn't hit on as much is the intellect itself developed through trade and interaction. That's how we are more than the sum of our parts is through our cooperation. And that's at the kernel of all economics, right? We have these ideas, we swap them, they become better over time, and we get to energize better and better strategies.

Michael Saylor: [You know] the phrase, "You're playing with fire — be careful you might get burned"? And what makes human beings unique is, as far as I can see, they're the only animal that plays with fire. And from the point we started to play with fire, we started to evolve at a very rapid rate. Genetically, intellectually, sociologically we evolved.

And we talk about the fire of truth and the fire of faith. It's like the keeper of the flame. The keeper of the flame really means something when you have a city, a village, a civilization, a tribe, and you've got a fire and it goes out, you might very well die. You don't want that fire to go out, at all. Bitcoin and the Bitcoin blockchain is a fire. We don't want it to go out either. We talk about feeding the fire of Bitcoin and we talk about feeding the fire of faith and simply being the keeper of the flame. It was an old idea thousands and thousands of years ago. I suspect it's the difference between life and death for humanity for a million years.

You've started the fire—it's all good. But now once you go back 100,000 years you'd be running around in 42 degree temperature while it's raining on you. Or what happens when it goes to 20 degrees? If it's cold and you're wet and the fire goes out, you're going to die. It's not an academic thing — it's a serious thing. So human beings harnessed fire and it made all the difference, and then along comes the next set of thoughts.

If you can harness fire, maybe you can develop a brain, and maybe you'll live long enough to use it. The next observation is: you ever wrestle around with a lion or a tiger or a bear? Pick any animal that you wish to kill — you ever wrestle with a dog that weighs 80 lbs? Not easy. Would you like to fight with one? How do you feel about fighting with ten? How do you feel about trying to run one of these things down?

I read about in Runner's World, runners want to tell you about how humans were always made to run, because ancient mankind chased its prey and it could run 20–30 miles a day, and we just run them down until they get tired. Okay well that's one idea, and maybe we did. But you ever try to catch something that's running away from you while you're hungry around dinner time? I don't really want to run for 20 hours straight until I tire it to death — I have a better idea, which is hit it with a missile. And by the way I literally mean missiles. I mean a primitive sling or an arrow, and I think they found arrowheads that go back 100,000 years. They're old. Most people think of the slingshot and they think about the kid's slingshot with the rubber band that kids play with.

Robert Breedlove: This is more like David and Goliath's sling, right?

Michael Saylor: Yeah. If you study Roman history and you go back 1,000 years before Rome, they had slingers. I mean the Balearic Islands like Ibiza they were very famous for slingers. And if you read about them what they'll say in the ancient texts is, the natives of the Balearic Islands were raised from age 3. Before they could speak they were raised to operate a sling. The sling is about 6–8 feet long, it's made of animal fiber, and you know, you don't throw a baseball — and you've seen highlights, right? — if I increase the lever at the end, if your arm was 12 feet long, you'd generate some serious leverage. A whip action. So those slings give the average person the equivalent of a 10-foot long arm, and they practice with those for years, from age 3.

You can imagine after 15 years of practicing, you get pretty good. And they weren't slinging little light stones or the shit that you pick up on the seashore. They're actually forming lead bullets. Everybody thinks, Oh yeah. Bullets are from guns. Well, they're not. I mean, people invented bullets thousands of years before guns. Guns were just the latest idea putting bullets together with gunpowder. The lead bullets probably came along 10,000 B.C. And maybe more, maybe 100,000 B.C.

This is a straightforward idea — if your life depended upon it, you would figure it out. And the figuring out is, you get yourself a very condensed bullet, you put it in the sling. You ever see a good pitcher? A good pitcher can place the ball 90 feet away plus or minus 4 inches? Can a good pitcher hit you in the head if you're standing at the plate? Now imagine someone that's pitching a 1–1.5 inch lead bullet from 50 yards away that can hit you on the head every time. Because the Romans said they could.

So now think about — and this is how bad it is, right? We'll talk about this in a bit, but to make the point, these guys could stand 100–200 meters off, and from 200 meters off they could actually hit an animal on the head, or another human being in the head. I mean it didn't matter if they hit you on the head — if they hit you on the torso they're gonna rupture your ribs and you may have organ failure. There's even stories — Livy, when he writes about the Second Punic War, he writes about Roman slingers, and they sling so many of these things that they pretty much break all the bones of the Gauls beneath their armor. They're wearing leather armor — their ribs are broken. And if they're not wearing leather armor and they get hit, just like getting hit with a bullet in your helmet it may still give you a concussion.

It was never a fair fight. Try taking your 8-foot-long spear and having a fair fight with a wolf or a pack of wolves? No. A bear? No. Humanity wouldn't be here if we hunted or defended ourselves using spears or using — these close quarter swords and clubs — they're all very romantic and they film well in Hollywood movies, in gladiatorial combat because you've got the two adversaries that are in the same frame, but if you go back a million years, the adversaries were never in the same frame. If you made it this far and you were a human being, you mastered the art of death from above. I mean, killed from a distance and nobody knew that you were there. Not a modern invention. Not only would you stand back 50 meters or 100 meters, you would stand up — and by the way, you would be up at the top of the hill where you have gravity working for you — that gives you more range. You would be back. If you were really smart, Robert, if I told you there's a bunch of whatever creature on the plain and any one of them can eat or trample you, wouldn't you like to stand up 20 feet on a cliff that they cannot run up? Stand up 20 feet above them, wait for them to come by, blast them either with a sling or use a bow and arrow, and if you miss — what happens if you miss?

Robert Breedlove: Load up and try again, right?

Michael Saylor: How many chances do you get? Til you run out of bullets. Now what happens if you walk down on the plain with your beautiful spear and sword with all of your buddies standing next to you and fight it out?

Robert Breedlove: Risk of ruin. You can never take that on. I think this is very interesting and it also highlights another difference that we have from animals. I know that humans are one of the few animals that rely on visual acuity as their primary sense. I think it's humans and predatory birds. And that also comes down to our dexterity, right? Our ability to handle and manipulate bow and *arrows, slings, these types of missile weapons. It just highlights the difference with us and everyone else. And those two things too are both intimately related with speech and thought and other tool-making, so I think that's very interesting.*

Michael Saylor: You think of the idea of missiles — I need my eyes, my brain, I need to set up the kill zone. Oh, by the way, I left off one other observation: it's 500,000 years ago — you want to kill something, you're downwind from it, the sun is to your back, you're above it, and you have a missile. And hopefully you have a chance. It's like, you really want to live, you're gonna go find that spot. You're gonna say, "At this point in the day, the sun is gonna be to my back, the prevailing winds are going to be blowing in my face, I'm going to be 20 feet up, oh there's a path up here but guess what, I'm gonna block that path because I don't want the bear running up to eat me once I start [shooting]. Then I've got a hundred missiles, and again, it's not a fair fight." There's only two types of human beings: there's the type that figured that out and that's your grandparents, and there's the type that were a bit sloppy about one of those things and they didn't make it. They're gone.

Robert Breedlove: This all calls to mind Sun Tzu, author of The Art of War. I'm gonna paraphrase here but he said, "Terrain is the most important aspect of any battle." It's almost like the smart general only goes into battle essentially knowing that he's won based on these preparations that you're describing, right? The sun at your back, wind at your back, high on the hill, undercover, plenty of missiles.

Michael Saylor: Technologies that are dominating today, they're dominating because they're able to deliver force faster, harder, stronger, smarter. If you're going to dominate, how do you deliver force harder, faster, stronger, smarter? And I could think of a hundred examples in history and they all — you tend to see those things, so if it's got the characteristic that it can be made harder, smarter, stronger, faster, there's something compelling about it.

That's why digital gold is thousands of times better than gold. Because you've got all those dimensions to work on. That's why the natural creature — gold is a rock, a bear is a bear, a mastodon is a mastodon. They're not getting harder, faster, stronger, smarter, they're just doing what they do. Human beings are. But only because of innovation.

So missiles are just a tool but they're illustrative. Fire is an energy network and an energy source. It's a battery. An energy source and you can deliver it in a certain way.

And then that takes us to hydraulics, which is power from water, and water is a network. And we talk about elemental forces: fire and water. Well, you ever look at the ocean, and what the ocean does — wave action is incredible energy. But another source of energy is buoyancy. You every try to pick up a 2,000 lb weight and carry it on your back, up a hill? Or just across. Put a 2,000 lb weight on a carriage, put it on the back of a donkey, drag it onto skids — problem, right? And this particular case can't be solved with fire, we can't easily burn it.

On the other hand, if you needed to move 2,000 lbs you put it on a barge, you put it in the water, and the water pushes back 2,000 lbs, and I can push it with one hand. And the mastery of hydraulics is fascinating. I went to MIT. MIT's mascot is a beaver. And we have rings that have the beaver on 'em, and we talk about, "Why have you got a beaver?"

The answer is, "the beaver is nature's engineer." And the beaver is this near-sighted, short, waddling creature. It shows up, looks around, sees the water flowing. It can just be what, bobcat bait or whatever, their dinner, or it can do something about it. And what the beaver does it just pretty unbelievable. The beaver starts chopping down trees. First the beaver figures out where to chop down the trees, then the beaver chops down the trees, then the beaver turns the tree into a dam, then the beaver channels the river into the dam, creates a pond, floods the pond. It's like it's reading the terrain.

After it's got the pond it creates a lodge in the middle of the pond with an entrance underwater, and then it creates its life in that lodge. That pond-water — water is elemental to life, and that pond creates a vibrant ecosystem and in that ecosystem lots of things grow. And lots of creatures benefit, I mean the ecological diversity improves, and it's good for all the plant life, the wildlife — people lament the loss if the beaver screws up forests, and the beaver is just doing its thing and if there's a storm and the dam gets messed up, the beaver swims out in the middle of the thunderstorm or hurricane or whatever it is and fixes the dam. It's a very industrious creature. And you just sit back and you're in awe. You think, Huh? How did a creature figure that out? And then what does that mean to humanity? Of course, it means a lot to humanity.

I've been all over the world, and I've been to the desert, I've been to Riyadh, I've been to UAE, I've been to Singapore, I've been to Miami Beach. People think oil is money. They think oil is power. And let me tell you: oil is not really power, it's not wealth. Water is wealth. Water is the key to life. If I gave you $10 billion and as much land as you wanted in the desert, you can't create life. The cost for you to create a pond in the desert, the cost for you to actually create a park with oak trees — if I gave you $10 billion and you lived in the desert, Robert, what would you do with the money? What's the first thing you would do?

Robert Breedlove: Buy Bitcoin and exit the desert?

Michael Saylor: Spoken like a progressive. So let's parse that: what if you didn't know about Bitcoin and I gave you the $10 billion and you lived in the desert?

Robert Breedlove: I guess you'd be looking for trading partners with water.

Michael Saylor: You'd exit the desert, right? So what do they do? You buy yourself a jet, you buy yourself a villa in the south of France. You buy yourself a yacht that floats in the Med. And then you figure out how to live your life, because the cost to grow a palm tree in the desert is $20,000 a year. You want 100 palm trees — it's

$2 million a year to have 100 palm trees. You want 4 acres of grass in the desert? You have $100 million a year? For 10 acres of grass? By the way, you still can't have it. Even if you spent $100 Million a year to put 10 acres of grass in the desert, the sandstorm comes. And it wipes you out.

Water is elemental to life. We underestimate how important it is. Until you start paying to create it. Yeah, you can stay alive in the desert, but a person making $50,000 a year that lives in a city that has parks and rainfall and a nice temperature lives better than a billionaire in the desert. It's just that powerful.

Now coming back to hydraulics. Hydraulics will generate power, right? I can harness running water running down a hill and create a turbine, then I can create a mill with that — that's interesting. Again, back to the hunter-gatherer thing. If you dropped me 100,000 years ago and you said, "Okay, well Mike, use your brain. Go hunt and gather." I'd be like, screw that.

What would I do? I would go find a stream with a little bit of elevation, maybe a mountain stream that had fresh water — you can drink it. And I'd find a big enough one that had fish in it, and then I would find a point where I could divert the stream to create a pond — I mean if the beaver can do it, I can probably do it. I would do some digging, I would divert the stream, I would create a lock, and at that point in the year when the salmon or whatever are running I might just flip that lock and I would actually divert the stream into my pond and create myself — maybe it's a 100-foot wide pond, maybe it's a 20-foot wide pond, maybe it's a 300-foot wide pond. Maybe I would waste a lot of the water — I don't care. And I would let 500 of those little fishies get trapped in my pond.

I'm not chasing after them with the stick like in Blue Lagoon. There's no such thing as a fair fight. I'm not fishing with the hook. My idea of fishing is with dynamite. I'm gonna blow up everything in the lagoon, and I'm gonna walk and pick up the fish. But in the absence of dynamite, I'm just gonna divert the water. What's the flow rate of water? How many fish swim by you? I want them all. Put them in the pond, then what do you think I'm gonna do? I'm gonna go pull out one a day. And I'm gonna let the other fishies swim around, and if the winter comes and the pond freezes over, that's okay. I'm gonna chip a little hole in the pond and I'm gonna walk out every day and I'm gonna reach in and grab my fish and I'm gonna eat my fish and I'm not chasing after stuff.

When you chase after stuff you twist your ankle and if you break your ankle you're dying. Right? If you chase after stuff and then a wolf pack catches you from behind, or you piss off an angry mastodon. So hydraulic power, it's the water, it's gonna bring you something to drink, it's gonna bring you something to eat. Maybe if I'm worried about the little creepy crawly creatures or whatever, I'm going to dig a trench around where I live, and they're gonna have to cross the water to get to me. Maybe I use water — a moat. If I live on the seashore, I'm gonna find a natural tidal basin. And in that tidal basin, I'm gonna let creatures crawl in. I went to Maine once. You ever watch lobstermen?

Robert Breedlove: I've seen it on TV, never in person though.

Michael Saylor: So if you don't know anything about lobstermen, you'd think, "Oh, well these are guys out hunting lobsters with the trap." When you go watch the lobstermen operating, you realize, they're not hunting lobster, they're not catching lobster — they're farming lobster. Big difference.

They'll create a trap, they'll put some kind of herring or something in the trap that the lobster wants to eat, they'll drop it, they'll put 10 of those cages down, and then they wait. Lobsters are lazy, lobsters crawl into the cage, they grab the food, they get stuck in the cage. They pull the trap up, they find a big lobster, they keep that one. They find little lobsters, they throw them back in because they need them to keep growing. They're creating agriculture to feed the lobster, the lobsters living in happy lobster hotels their entire life — it's not so bad, Robert.

If I said to you, I'm gonna give you free room and board to age 70 and then I'm gonna eat you. Well, not so compelling. But if I said to you, Robert, I'm gonna give you free room and board until you're 750 years old, and then your life is gonna end, or, you can make it on your own and you'll suffer a horrific death being eaten up by a barracuda at age 35, you might think it's not so bad living in your lobster hotel 10 times longer than you live naturally. It's not like these lobsters would've made it very far. They're liking it. They're domesticated lobsters.

Robert Breedlove: Nature tends to pursue the most energy-efficient strategy available to it, right? You're the eagle dragging the baby goat off a cliff or the lobster enjoying the lobster hotel, or you're the man diverting the stream to capture a bunch of salmon — you have a tendency to want to do the least but achieve the most, right? That's kind of the nature of productivity itself.

Michael Saylor: I'm channeling energy.

Robert Breedlove: Yeah. And not wasting any of the process. Channeling it as efficiently and as usefully as possible.

Michael Saylor: The pyramids got built 2,000 years before Cleopatra and Caesar. How'd they haul it up there? And some of the most fascinating videos I've seen on YouTube are those that show how they built hydraulic elevators to move the 2-ton or 4-ton stone up by floating it up the channel to the side of the pyramid. And I totally believe that's how they did it. They actually used hydraulics to construct the pyramids, it makes so much sense.

Robert Breedlove: I haven't seen that actually. I've only seen the rolling them along the logs. How are the hydraulics constructed?

Michael Saylor: You have a tube of water. If I put something in the bottom of the tube that's lighter than water with a float attached to it — I'd put a rock with a float, maybe you'd take animal skins and you blow them up with air, it will float up the tube and pop out the back. So what you do, is have the tube be able to hold the integrity. I can't do it for 1,000 feet, but I can do it for 20 feet. It's like the way a lock works in a canal, like I'm going into a lock, I close the gate, I flood the lock, it lifts the barge, I open the other lock, and I got out. So imagine a series of locks that I use to actually life 100,000 tons of stone using water. Very interesting. I think that that's how it was done, in my opinion.

But water can be used for farming, for fishing, it can be used for security. It can be used for sanitation. In fact, without understanding water and the dynamics of water, there is no cradle of civilization in the Aegean or anywhere.

I'll give you another interesting vignette. I go to Santorini, and Santorini's built out on this caldera looking down —

Robert Breedlove: It's a white city, right?

Michael Saylor: Yeah, it's a beautiful city. Well in the 20th century you can take the elevator up from the port and the city is 500 feet above from the port. And then on the way back I see donkey rides and you can take a donkey up to Santorini from the port or take a donkey down. I was in my fitness craze and I'm like, "Well, I'm not riding a donkey down, but I think I'll just walk down." I mean I think I can just walk — I don't need to take the sissy elevator. So I start walking down these steps and the steps aren't terribly difficult. What do you think I see as I'm walking down the donkey steps?

Robert Breedlove: Donkey doo?

Michael Saylor: How much of it do you think I see?

Robert Breedlove: Probably more than you wanted to.

Michael Saylor: A river. A river of donkey excrement. Mind you, this is like 3–4 tourist donkeys taking down the occasional tourist. A river, of donkey excrement. And you can hardly avoid it, you're hopping this way and that way, and then my brain starts working, and I start thinking, "Hmm? What happens if there's 100x as many donkeys and what happens if they're walking through a city? And what happens if it doesn't — how to you clear this stuff?" It's not like they had hydraulic hoses and they could just clean — by the way, they're not cleaning it in the 21st century in Greece. It's a river of donkey crap. I mean, they haven't figured out how to clear it 3,000 years later.

So let's go back 3,000 years and let's do the thought experiment: what's it like to live in the city using animal power to move stuff around?

Robert Breedlove: It'd be the place to live.

Michael Saylor: Awful. But also unsanitary. I mean, fly-invested typhoid-infested, germ-infested. And when it gets dry and this stuff desiccates and it blows through the air, you're going to be breathing it, smelling it, it's just going to be awful. So, you want to drink it and eat it? I mean, this is not just a matter of creature comfort — you're gonna die. You can't actually bring together a bunch of human beings unless you work out the sanitation problem.

And it was then and there that it dawned on me very viscerally, there's a reason why all the streets in Mykonos are so narrow that you can't get a horse or a cart through them. They're walking cities. There's a reason that the equestrian class were the Roman knights. The knights were the equestrian class, and what it meant in ancient Rome was the top 1%, or the top 0.1%. The nobles of Rome were the equestrian class. What it meant to be rich and powerful was: you had the right to bring a horse into the city. Nobody else could. The problem was not that they couldn't afford the horses. The problem is if they allowed anybody to bring a horse into the city, it would be so unsanitary as to render the city uninhabitable.

So most ancient cities, if you wanted them to work, you would have to have them human-powered, and now you've got this dilemma of how do you move goods around? Tell me, how do I move things around cleanly? I need a clean energy source that is not going to foul my sanitary system, that it's not gonna actually kill me. And of course it's a boat.

What do you want? You want 25 cities with ports on an inland sea with at least a season 6–9 months a year where you can cross from one point to another point without being bashed and killed on the rocks. So you need a fairly mild sea, but you have to have water, because if you have the same 25 cities on land and you're gonna use horse or animal power in order to move goods and services back and forth, it's just so dirty, so unsanitary, that your civilization's probably not gonna get off the ground.

Robert Breedlove: So the Mediterranean was ideal for this, it sounds like.

Michael Saylor: The Mediterranean is the perfect ocean, because you can oftentimes navigate without leaving the sight of land. It's hard to get too lost. There's a lot of ports.

Robert Breedlove: Very placid, relatively.

Michael Saylor: And there's a lot of stops. If you look at all of these empires: the Phoenician empire, the Roman empire, the Venetian empire, the British empire, if you actually tour all the great ports in the Mediterranean — all the really good ones — the story goes something like this. Maybe you go to Bonifacio in Corsica. Well in 1,000 A.D. this was the Phoenician port. And then the Greek empire came along and it was an Athenian port, 500 A.D. And then the Carthaginians kicked them out and it was a Carthaginian port. And then the Romans kicked them out and there was a Roman port. And then after the Romans stole them the Venetians took this over it was a Venetian port. And then eventually it became a British port, you know? That's the story of Malta, that's the story of Corfu, that's the story of lots of different ports in the Mediterranean.

And the reason why is, if you want to dominate the Mediterranean, you need to have a port within one or two days' sail that you can hide in whenever the mistrals blow. And if you control that network of ports when the weather goes bad, you go into the port and your ship doesn't get sunk. And if you don't and you're like a week away from a port that's friendly and the weather gets bad, you get dashed against the rocks and you just die. And that's the end of it. So these are all nautical networks and they're all based upon terrain.

The Mediterranean was a good crucible for the beginning of a civilization. And when you put together the incredible power of hydraulic transportation and then you consider the consequences of not having it, you realize: you can't really develop the economic density. We haven't touched much on agriculture. We could but the general theme is the same: when I drop you and you find a fruit tree, you're not gonna go, "Oh duh. There's a fruit tree in this clearing. I'm gonna walk 18 miles to the place where I can find a different fruit bush. And then I'm gonna walk 5 miles back to the place where the fish are." You're gonna actually pick up a fruit tree and plant like 100 fruit trees next to your fishing pond. You're not stupid. Like, paleolithic man — there's every reason to believe they were smarter, stronger, and tougher than we were.

Robert Breedlove: I think it's a great point that all of these inventions were leading towards increasing economic or energetic density, and that's what actually provided the bedrock on which to build a civilization.

I was trying to get a quick overview and feel free to jump in if I'm missing anything, but it started out with: We can't handle an animal one-on-one. It's kind of our wits that make us who we are. Through our wits, we're able to communicate and coordinate with one another. One thing we didn't get into is the Yuval Harari Sapiens thesis where he says that man came to dominate the world because we can tell and believe stories. Like we're actually able to abstract and represent reality in symbols, and that's what gives us the ability to make tools, and so on and so forth.

So with those strategies that are often cooperative, we've energized them with fire as our base — I guess we're harnessing the energy of the world and ancient sunlight through using fire — and then the other interesting thing about fire is that it actually accelerated our own evolution, right? Our cognitive development was increased because we're able to liberate more calories from food and whatnot. It also gave us the ability to make harder, stronger, *better tools in terms of metalworking. I think that's a great point too, is that mankind actually changes his own course of evolution through the conscious decisions we make, like the tools we make in turn make us. Which I think is a really interesting point to touch on later as well with money.*

Then we had missiles, so we could actually take advantage of our visual acuity, which is something unique to people — and our dexterity — and actually hunt animals at a distance, and hunt them on a terrain that's advantageous to us. Then we had to tap into water, because we are water first of all — humans are 70% water — we have to consume a lot of water very frequently. I think that's the quickest way to die. It's like oxygen first — we have to have that most frequently — water second —

Michael Saylor: 3 minutes without air, 3 days without water, 3 months without food.

Robert Breedlove: Exactly. And I've read too that people going without water actually cry tears of blood — it's actually one of the symptoms — it makes your eyes bleed. So not only is water clearly this life-giving substance that we have to have access to fresh water and be able to implement into our agricultural systems and whatnot, but it's also a tool for overcoming gravity, so we can actually construct larger-scale structures and conduct commerce at scale. So I think that's a great first principles view on what makes us unique.

Michael Saylor: And we gotta do all of that to get to the Iron Age. And we come back to this issue being harder and stronger. We harness that fire and we start to work metals and we move into bronze and we move into iron, and I think the Roman Empire is a great model for the way that human beings interact with technology and the way that they interact with a competitive world and become both antifragile and get harder, smarter, faster, and stronger.

This same thing was going on in other parts of the world, but I'll focus upon Romans for a bit. You go read Livy's history of Rome and he writes about how the Roman republic had 700 good years. 700

years before even it went to empire. And we started with this idea of Roman politics. You've heard the phrase, "Beware the Ides of March," in Julius Caesar, and people think of it as, Oh, well that's when someone's gonna kill Caesar, but it's really referring to the fact that for 700 years, the Romans got together on March 15th and had an election, every year.

The Romans were the most organized of all of the civilizations we can find in the ancient world and that's how they grew to dominate — they were just organized. One of their forms of organization is — and this is a thing of beauty — they're running a process where every year, March 15th, they have their election. They appoint two consuls; they appoint all of their officers. The consuls then, they conduct about 2 weeks' worth of religious ceremonies. They all worship, they appease the gods, they're getting psyched up. They're reminding themselves that they're unique — they're celebrating.

Simultaneously they raise an army. They train the army. We go from March 15th all the way through to May 1st — 6 weeks. In those 6 weeks they get organized, celebrate, get excited, wait for a good omen, and they're really getting ready, and then the campaigning season starts May 1st. Everybody that knows anything about Europe and the Mediterranean knows that the weather gets good on May 1st. The problem before May 1st? It rains, there's storms, if you set out to sea or you set out across terrain before that time period, if the cold doesn't get you, the storm's gonna get you or your ship's gonna sink or something. Ultimately, in the history of all of these wars, more people die from natural causes than they die from bullets from the enemy. The number one danger is: nature's gonna kill you. So they were all basically doing these summer campaigns.

May 1st, they started to campaign. That goes through June, July, August, September — all good months. If they're still fighting something, maybe around October they wrap it up. They go into winter quarters by November. November, December, January, maybe they're having November, but certainly December, January, February — that's winter. They're not doing anything because the elements are a much bigger threat than the enemy is. And if you know anything about the Med, you can't navigate the Med in the winter.

Even in the modern day, no one would want to go yachting in the Med in the winter — it's just not comfortable, you get storms, weather is very uncertain. So all this time, they're rest, they're recuperating, they're regrouping, they're politicking—you can imagine. They're discussing with each other who's most suited. That guy's long in the tooth, that guy's lost his step, this is the up-and-comer — you support me, I'll support you. They're working through that consensus back in Rome, and they're remembering what it's like to be a Roman.

And then along comes March, and then they decide who's gonna do what — you're gonna be a tribune, you're gonna be a consul, you're gonna be a governor — let's put in place an administration. And they're always rotating and they go and they do it again. And if they send off the best and the brightest and the guy takes an arrow in the back, well next year there's another guy.

Scipio Africanus, one of the most famous Roman generals of all time, he rose to power in his early 20s after all of his uncles and his father died in the Second Punic War. His entire family is getting wiped out, but there's always another Roman. Always another one. From a very early age.

And so the political system had a certain elegance to it because it was tied to the calendar — it was tied to nature. It was a natural cycle, and it took into account the need of human beings to celebrate each others' successes. I go campaign, I come back, I get a triumph. It took into account their need to have a common faith. The faith is critical. If we're not all Romans and we don't all believe the same things, why are we gonna die for each other? Faith mattered. But the weather mattered. And you know, people don't realize they did it every year on March 15th, because if I told you the weather's gonna get good on May 1st and you need an army, when would you start?

They're pretty smart — 700 years of it — that's the Roman way. And then they also took into account human motivation, which is: everybody's got an ego. Everybody needs their turn. Nobody can hog all the power, so even if you were the greatest general this year, you've gotta give it up to someone else next year. And as long as they kept turning — and if I'm the second-most powerful family, you're from the most powerful family, maybe I'll support you for consul with the understanding that it's my turn next year. And then maybe my family will fight and die for you. Because we have a chance at glory next year. But at the point where you take over and you tell me, well, you think you're gonna keep the job for the next 62 years — at that point, the fabric of the civilization starts to break down, because that equity and that citizenship and that sense that we're protecting the Roman way of life, starts to degrade to, we're just protecting someone's dynasty — screw them.

Robert Breedlove: So the dynamism of the hierarchy keeps it revivified and fresh, and they're harmonized with nature. That's very interesting.

Michael Saylor: Antifragile. Romans are antifragile. They're always going off to fight, always, always, always. It's just a history of war after war after war after war. You know, a typical CEO, you could be in your job 10 years, 20 years. I'm 55, I've been in my job for quite a while. But it's not uncommon for someone in modern day America to be doing a job and become CEO somewhere between the age 40 and age 65. Not uncommon. In fact, the captain of a yacht will oftentimes be 40 and they'll stay as captain until they're 65. You might do the same job for 20–25 years.

I once took a tour of the US military, and I was treated like a senator. It was an orientation tour, and they would take to an army base, a navy base, an air force base, a marine base, Camp Lejeune, Fort Hood, etc. And one of the things they did is they took us onto an aircraft carrier, the John Stennis. And so I landed on an aircraft carrier and then I got to tour the carrier, and then I got to meet the captain. The average age of the soldiers on the aircraft carrier is 19. Average age. There's 5,000 people on an aircraft carrier — 19. The officers are in their 20s, some in their early 30s.

Do you know? The oldest man or woman on the carrier, Robert? The oldest? The old man is 41. So I start talking to ’em. And the #2 is 38. If you’re the #2 Head of Operations, it’s an 18-month gig, and if you’re the captain, it might be 36 months. And so I start talking — and this is a nuclear-powered aircraft carrier. These guys can start a war. It’s like 1/12 of the firepower of the US Navy. It could take down all but like 3 countries in a heartbeat. Maybe you could take down any country in a heartbeat. It’s a pretty important job, Robert, wouldn’t you say?

Robert Breedlove: Absolutely.

Michael Saylor: So I said to him, “Tell me the path to get here.” And he goes, “Well I went to the naval academy and I did this for a few years, and every 1–3 years I move to a different command, and I finally made it as an XO, #2 officer, 2 years ago, and I got promoted 6 months ago and I’ll have this command for 24 months or something.” And I said, “Well let me get this: so there’s only like 12 of you, right? So if you’re 1 of the top-12 most talented officers in the entire navy.” Like how many people are in the navy? Hundreds of thousands of people in the military. There’s just 1 of the 12 most important jobs in the US military, bar none. I’m like, “So, in like, 12 more months you’re leaving?” He said, “Yeah, I’m leaving.” I’m like, “Why wouldn’t they want you to do this job for 20 years? Like, you could start World War III. Why would you take the risk of changing and putting someone else on the job? What if they screw it up, you know?”

By the way, Robert, we don’t do that in any other part of our economy. We don’t actually put 40-year olds—with term limits of 36 months — in charge of cities, states, countries, companies. We don’t even put ’em in charge of yachts. If you had a 100-foot pleasure craft, you wouldn’t do it. I’d be like, I’d find one captain — I’m keeping the guy for 20 years. I’m not changing him. You wouldn’t actually do that with a person that cooks your food, or mows your lawn. So, why do you think this guy’s gotta go after 36 months? Any guesses? Because the answer’s gonna blow your mind.

Robert Breedlove: What jumps to mind is if he were to get paid off or corrupted or something? But I really don't know.

Michael Saylor: That's not a bad idea. That's like the forest ranger principle. We rotate the forest rangers in order to keep anybody from bribing or corrupting the forest rangers so they don't misuse public national park resources. Brilliant idea, right? A better-off forestry service and a great anti-corruption technique. But that's not why.

I'm standing on the deck of this aircraft carrier talking to this guy whom a lot of people would think like he's just a junior executive, maybe we're ready to give him a whatever. And I said, "So tell me again? Why you've gotta leave this job even though you're the best guy in the navy to do it? You're obviously hyper-talented?"

He goes, "Well, Michael, there's a lot of really, really good people coming out of the academy every year. And everybody needs their turn." Everybody needs their turn. You're talking about a 21-year old lieutenant coming out of the naval academy saying, These people signed up to commit their life and their career and potentially sacrifice their life to be a navy officer. And at the pinnacle is their hope that they can be the captain and have their own command, and at the pinnacle of that is captain of an aircraft carrier. And if you want people to love and fight and die and cherish your institution, you've gotta get out of the way and give 'em their chance. Everybody needs their turn. And you start thinking, Maybe we overestimate ourselves—right? This is why, again, these decentralized organisms like Bitcoin are superior to a company, because as Charles de Gaulle said, "Graveyards are full of tombstones of indispensable men."

Robert Breedlove: 100%. This makes me think too of…as a kid, there was this notion that anyone—I grew up in Tennessee—anyone could be the American president. Now that's maybe kind of silly and not actually the case, but that notion seemed to give people, at least kids, this motivation to really want to be patriotic and part and parcel for their country. So it seems like something about the *possibility of achieving the highest level within an organism or an organization sort of gives people maximum motivation or something like that.*

Michael Saylor: There's a certain pride of being a naval officer. When the head of a carrier looks at you — when you've started your career — and says, "You know, one day you'll have your turn at this. You and I are comrades. You're as good as me. You're the future of the navy." That's what will cause people to lay down and die for you.

Robert Breedlove: That's inspiration.

Michael Saylor: That respect. And that's what the Romans had in those 700 years — the height of the republic. It's "You're a Roman first. This year, maybe you're under the command of your family's #1 adversary, but next year that'll be your command." And that's the Roman way.

There's a certain submission to nature and the organism is greater than any one individual and any one family. It's continually refreshing itself. We have to have a constant flow of new talent, new leadership. Someone drops the baton, someone else picks up the baton. That's what made the Romans great. They suffered no kings among them.

You look at the Second Punic War and then maybe it was the Second Macedonian War? Eventually the Romans went to fight against Philip of Macedon. He was a king. And he had an awful son, and his one son got fighting with his other son, and convinced the father to murder the second son, and then the father realized that he'd made a mistake because his first son lied to him. And the father was a nutcase crazy guy, and the son was kind of crazy, and it corrupted the entire society, and the Roman consul was just the most talented general and he knew that — the way it worked was, his officers were from every other competing family in Rome. If that general was lazy, drunk, cowardly, or stupid, it got reported back by the officer corps to Rome and to the Senate. And so they were slightly gossipy, but the point is, when you know that everybody's watching you and that you can be replaced and will be replaced next year and your future is uncertain, it brings out a higher degree of professionalism, and that's the competition in the market. You're a miner, and they cut off your electricity, well your mining rig stops and the mining shifts somewhere else.

Robert Breedlove: Absolutely. That's the reason entrepreneurs in the free market are accountable to the preferences of their customers, right? Because they constantly face the existential threat of customers going elsewhere. Whereas the opposite would be true in a monopoly. The monopolist doesn't have to give two shits about his customer's preferences, because they have no other choice. So I think it's really interesting.

Michael Saylor: So the Roman political system, it bred harder, stronger, faster, smarter individuals. No apologies about it, from age 3, this is the way it is. And Rome comes first, and everybody else's interests are subjugated. And when you look at that, they started with that system — a good system. — I mean people forget about it. 700 years as a republic. Find me another republic that lasted 700 years.

Then you've got the Roman army, and that tells a different story. The Roman army takes us back to this issue of there was never such thing as a fair fight. The Romans weren't fighting fair. They would've laughed at you. The Roman approach to this was to take my illustration of the slinger on the cliff and take it to a whole new level. The Romans manufactured ballistas, catapults, every sword and shield to be the same length, every soldier took the same step the same length, everything was the same.

You could be an 8-foot tall goliath, and the Roman 5' 10" tall normal dude is gonna beat the crap out of you because you're not gonna get within 12 feet of him because you're gonna take a spear in the gut from the 12 guys standing to his left and his right as you charge.

In all of these time periods, all of these wars, and you read Livy and he describes them very in depth, over and over again: hundreds and hundreds and hundreds of battles, and they always consisted of the Romans maneuvered to get the high ground, the Romans maneuvered to get the enemy out on the plain, the Romans unleashed the artillery onto the enemy while they stood and obliterated 10% of them. Then the Romans unleashed some more artillery.

There's one story where the Roman army cornered the Gauls. The Gauls are on a mountaintop, on a hillside, and the Romans are below — they just surround them, they stop, they start to pummel them, and they rain down hell from above — bullets, boulders, napalm, and Livy writes, before a Roman even took a step toward the enemy line, half of the Gauls were dead and 80% of them had been maimed or incapacitated from the bullets. And then the Romans start to move up and do something. It's like, there's none of this "Let's just charge into battle and fight it out with our sword," right? Never happened that way. It was always going to be: find a way to get an advantage.

And by the way, the technology is very critical. The Romans had a military industrial complex. There's a way to do it, you're gonna do it in a certain way. They had an entire body language, and entire system of how you're going to act.

If you want something which is eye-opening, there's this story of the Roman navy from the First Punic War. The Carthaginians dominate the Mediterranean, the Romans are a land power. The Romans don't know anything about naval power — but it tells you a lot about the Roman psyche and intellect. The Romans are getting beat up by the Carthaginians because the Carthaginians control the ports and they have the fleet. The Romans have no fleet. One day, a storm kicks up, and the storm drives a Carthaginian naval ship into a Roman port — it's blown into the port by a bad mistral. The Romans capture it. They take it apart to try to figure out how the Carthaginians make their ships. You can't make this up. This is the most amazing thing in history.

They find out that the Carthaginians make their ships from a kit from reusable standardized parts, and not only are all the parts standardized, the Carthaginians have labeled each part with the instructions with where it fits and the number of the part. The Romans deconstruct the entire thing, steal the entire blueprint. 90 days later they made 150 ships. You think these guys were screwing around? They're not screwing around. It's like, everything's, "Oh you know, I'm gonna take my time to figure this stuff out." No.

War has a way. War has a way of quickening your activity. Bam. I'm losing? I find a ship, that's the DNA, that's the formula of the ship — 150 ships. To make a long story short, the Romans win the First Punic War and they vanquish the Carthaginians and they become the naval power. Of course, it's not that the Romans invented everything, it's just that the Romans stole every good idea from every civilization from the Greeks, from the Carthaginians, from the whatever that they crushed. And because they lasted, we're able to read their histories.

It kind of blows your minds that in 500 years — by the way, you think the Carthaginians invented that? Maybe they stole it from the Phoenicians. 500 B.C., if you want to win wars, you don't just make ships. And you don't just train hard. And you don't just make wagons, right? Roman roads. The Romans had standardized parts, a standardized gauge for a wagon wheel — every Roman wagon rolling on the road is carving ruts in the road. That gauge has to be standardized, you can't just make any wagon, you have to make it exactly the same.

For all those people that recoil at standardization, well that Roman wagon gauge eventually became the standard width of a railroad track in Europe. And then the standard width of a railroad track everywhere. So if you want to know how wide a Roman chariot was, or a war chariot or any chariot, just go stand on a railroad anywhere on Earth — the Roman's gave that to you. And the reason why they did it that way is because, if you build wagons with different gauges, they fall in the rut, they snap the axle, and that's death.

So it's like Henry Ford said, "You can have it any color as long as it's black." No. You can't have it any way you want it. It's not that every civilization figured this out. It's just that every civilization which insisted upon doing it a different way with different bells and whistles got crushed to death.

There's an analogy with this in the Bitcoin world too. When you come up with a different feature it's like, "It would just be 10% better, you know, if you made your wagon 10% wider: it would hold 20% more and you would need 10% less and your transaction costs would be less."

Robert Breedlove: It points towards path dependence too. The fact that a technology already took a certain path, it kind of has an inertia that's carried the width of the gauge of the wagon wheel to the width of the modern rail. I thought it was interesting too you pointed to the civilizations that won out, the Romans over the Carthaginians, are the ones that studied their history. So they're actually gleaning insights from civilizations that had come before them, which again hearkens back to that Yuval Harari concept of our ability to abstract our learnings into symbols like language or whatnot, and then pass them from generation to generation such that the most successful strategies take advantage of the collective learnings up until that point. Versus trying to just do something from scratch on your own. Like we all stand on the shoulders of giants, so to speak.

Michael Saylor: Yeah, those roads were the logistics network of the Roman empire, and if you can move goods and services, if you can move armies faster inside your borders than your enemies can move inside their borders, then you're gonna win. Well, you've got a major, major advantage. If everybody lays down a railroad track that's a certain width and you come up with an idea for a car that's got a different width — who are you gonna sell it to?

People talk about protocols being important, right? Well TCP/IP wasn't the first protocol. Roman roads probably weren't the first protocol either, but the point is, protocols matter, and it's arrogant — by the way, I'm sure that the Egyptians had protocols to build those pyramids. Standard sizes, widths, weights and measures. Those protocols matter a heck of a lot. And if you don't have them, it's impossible for people to cooperate. So we've talked about money a lot as being essential for civilization to cooperate and allowing us to specialize, but all these other logistics protocols or military protocols are in their own way equally important.

I'll make one last point on Roman engineering on aqueducts. The Romans understood the importance of hydraulics, and they took it to a new level. They actually created aqueducts that would bring water from up to 70 miles away to a given city. There are a lot of coastal cities on the Med that are uninhabitable — the natural economic density is really a function of the amount of water per year, so if the amount of water per year is based on rainwater, maybe you can have 500 people live in the city. And if you bring the aqueduct it goes to 5,000 or 50,000. And so the economic density requires the hydraulic flow for sanitation and just to keep everybody alive.

So engineering the roads, the aqueducts, it's the rails upon which the entire Iron Age civilization was built. And Romans — if anything — they're engineers. And they elevated engineering, above all. What is engineering? I am an engineer. I think engineering is an incredible honorable, ethical life-affirming profession. The basic credo of the engineer is, I look at nature, and I look at the circumstances that I'm surrounded by, and I use my intellect and every material and technique at hand in order to construct a better world for everyone and everything that I love. That's the credo. I'm not going to be a victim of circumstance. I'm going to actually change my circumstances with my intellect and that might mean build a bridge. It might mean build an aqueduct, build a road, build a ship — whatever it is, just like the beaver builds the dam, the engineer builds the world. Look at any city where you take the bridge down, and try to figure out how life changes. And it's pretty consequential.

I'll leave you one more vignette on Rome and then we'll move on I think to the Dark Ages. I have a holding company, it's called Alcantara. And Alcantara is based upon something I saw in Alcantara, Spain. It's a Roman bridge — it stood for 2,000 years. And if you go underneath that bridge you'll find a Roman inscription in Latin, where the Roman engineer whose name is Gaius Julius Lacer said, "This bridge will stand for all time." They took their engineering seriously.

Robert Breedlove: I recall from Taleb's writing that the architects of the Roman aqueducts — to give the architects skin in the game, so to speak — he would be required himself or even with his family at times, to stand beneath the aqueduct as the scaffolding was removed. So he knew it was his life and possibly his family's life on the line, should his architectural abilities be incompetent. Again, it's a protocol. It's a protocol and an incentive or a disincentive to malperformance for that architect, to take his profession very seriously. Another thing that came to mind, Rome as being incipient to all these civilizational technologies and protocols we use — what ultimately led to their downfall was their monetary protocol being compromised. It was the debasement of the coin, I think it started with Nero, and eventually led to the forking into the East Roman empire —

Michael Saylor: And their political protocols compromised. A series of civil wars — Caesar's being the most famous, but a series of civil wars where the political protocols broke down, even before the monetary protocols broke down. You can see they're all related and at some point the integrity of the society broke down, and when they lost their integrity across all of these areas, the collapse of the political system begat the collapse of the military system. The religion — how do you maintain your patriotism in Rome when one Roman army is fighting another Roman army?

Robert Breedlove: Right. Was there an inflection point that you recall, that led to all these protocols being compromised?

Michael Saylor: Around 50 B.C. it all started going bad. There's a whole series of wars in Julius Caesar's youth, and the rise of a series of strong men and the weakening of the Roman republic. Taleb I think makes great points in his books about how it's the death penalty in Babylon if you screwed with weights and measures. Or if you're a builder and your house collapses, your own family dies. There are definitely skin in the game points, and they just remind you that in a society that respects natural law, nature is not going to pity you and she's not listening for excuses.

I know this is not quite relevant but I can't help but state it: The richest man in China a year or two years ago was out on summer vacation in the South of France. The guy's worth $20–$30 billion. And he decided to take a selfie or get a photo, and he stood up on a rock wall at some ancient ruin in the South of France, and while they were taking the photo, he slipped and fell off the side of the wall and plunged 50 feet to his death. Again, somewhat emblematic of the point — it doesn't matter if you have an army of lawyers and $1 billion of clout — in that last 2 seconds of his life, he was punished by violating the law of gravity with a death sentence.

You know, gravity doesn't care who you are. Nature doesn't care. The richest man in China, out of a billion people, and he was sentenced to immediate death — no appeal, in a split second, for being careless. And when society forms all of these appeals and excuses and they let everybody off the hook, it's like, well, if you make a market in accepting excuses and lawsuits, you're gonna get a lot of lawsuits and a lot of excuses.

Robert Breedlove: And then Too Big to Fail institutions were interrupting the evolutionary impulse. We're not learning at a business and civilizational level when we preserve institutions artificially.

Michael Saylor: I'm very persuaded by all of the points made by Taleb and also by the Paleotheorists about the importance of pain in life. Pain is a natural feature, and you can learn a lot of things by pain, right? You try to pick up a chair the wrong way, you do something in the wrong fashion, the pain is a feedback and it's information. And when you try to cut off the pain flow through anesthetics or steroid shots or cortisone —

Robert Breedlove: Or QE.

Michael Saylor: Or QE, or an appeal, or a lawyer, or a bribe, or whatever it is you avoid paying the price or the consequences for your misstep. Try to suspend gravity? Well good luck with that. If you could've suspended gravity for $1 billion in that one second, how much other screwy stuff would've happened everywhere else in the world while gravity was suspended for that one guy? The damage would've been maybe a million times worse. Gravity is key.

Robert Breedlove: There's a good reason you can't compromise nature's protocols, and we should mirror that through natural law.

Michael Saylor: That would be the healthy approach. That's the Paleotheory. That's the theory of antifragility. That's the theory of Austrian economics and capitalism properly understood, and Darwinian evolution and natural equilibrium.

Key Insights from Chapter 1

- Three primal technologies helped us build everything: fire, missiles, hydraulics.

- All energy is essentially solar.

- There's never been a fair fight. The way humans out-compete in nature is by engaging in predetermined unfair fights. We ensure we have an asymmetric advantage to preserve ourselves and obtain the most energy for the lowest energy expenditure.

- The Romans became dominant due to their self-organization. They realized the value of an established protocol. There's a connection between protocol standardization and economic output and prosperity.

- The basic credo of an engineer: “I look at nature, and I look at the circumstances that I’m surrounded by, and I use my intellect and every material and technique at hand in order to construct a better world for everyone and everything that I love. I’m not going to be a victim of circumstance. I’m going to change my circumstances with my intellect.”

- Societies that deeply respect natural law tend to succeed and out-compete others.

Chapter 2 | The Rise of Man through the Dark and Steel Ages

"We need to optimize society for free exchange and experimentation. That's how we create wealth in the world, solve problems and increase life expectancy."

Michael Saylor: We have the rise of Man through the Stone Age, and it took quite a long time. But we got here — that's impressive. We have the Roman rise, and we saw a sophisticated society mastering energy networks, logistics networks, advanced tools, political processes, in order to dominate their sphere of influence. By the age of the Antonines, around Trajan, Hadrian, Marcus Aurelius — this is like 100 years, it's a golden age of Rome — it's like 90 A.D. to 180 A.D., the average life expectancy of a Roman is 72.

And so you've got baths, you've got writing, you've got civilization, you've got sanitation, you've got aqueducts, you've got roads — they've got stuff pretty well organized in sophisticated machines. And then of course we take a hard left-turn into the Dark Ages and stuff just starts to break down. And it's a reminder that nothing is certain — if you miss a key insight, you could waste 1,000 years. And stuff could've happened a different way — it didn't.

For example, let's take the printing press: the Chinese developed printing presses way back—2,000 years ago. They never really thought to commercialize them, and the Chinese alphabet's a pictographic alphabet, so you would've needed 25,000 different type pieces in order to have movable type in a printing press, and so they had the wrong language to develop printing presses.

The Romans had the Roman alphabet which is an ideal language because you can actually print anything with just like 26 or 50 pieces of movable type, and they had all the knowledge but for some reason — anybody could've figured it out, Robert. —All you gotta do is walk in your boots through the mud and step on a nice floor and look down, and there's the idea for a printing press. Tracks in the mud. Right? They had paint, they had dyes, and you know, they're just oh so close, and then they don't hit it, and we wait until whatever, 1453 or something for Gutenberg to work this out, and we're gonna suffer for 1,000 years.

It's amazing to me how certain things can be just right in front of a civilization and they just won't grab 'em. Like if you go to St. Peter's square to the Vatican and you look at that column — it's now St. Peter's column but it used to be Trajan's column. And Trajan's column was put up to celebrate the triumph of Trajan and his wars. And it's a bas-relief, a marble relief of the Dacian wars, and it wraps, spirals up the column, has a little staircase in the middle of it, a work of art, and architecture, and at the top, eventually the Roman Catholics put the statue of St. Peter, I guess. And in the ancient Roman times, that had Trajan's statue on it. And Trajan is standing there in his Imperial robes, and he's holding his hand up in the air and do you know what he's holding in his hand?

Robert Breedlove: I don't know.

Michael Saylor: He's holding the world. And it's round. He's holding the globe in his hand, and this is 100 A.D.. Fast-forward 1,400 years and people think the world is flat. It's like the Romans knew the world was not flat. And if they hadn't ripped his statue off the column 1,000 years earlier or whenever they ripped it off and buried it somewhere, they would've known the world is round, and it's so ironic that you could learn and then forget such a basic thing.

Robert Breedlove: So there is the possibility of significant civilizational regression if we ignore those learnings that we've accumulated over time.

Michael Saylor: Yeah you can go backwards and you think, "Well, we've got it and if we do this we're gonna leap forward into a new progressive era, but if we kill it at a pivotal point, maybe people forget it ever existed — letting the fire go out."

I'll tell you another one that blows my mind. If you go to the Museum of the Native American Indian in Washington D.C., and you walk around, you see there's an entire civilization, I mean they're pretty much hunter-gatherers, and if you want to understand how hunter-gatherers lived, I mean, the American-Indian, the Native Americans, the way they were when Europeans showed up in America is probably your best proxy for what the Paleolithic man was, I guess. But amongst all the artifacts they gathered, you walk around, you eventually find a pottery wheel. And the pottery wheel, it's a wheel in stone, and it's got a beautiful piece of pottery, and the sign says, glowingly, Native Americans had very sophisticated pottery and they knew how to mold clay and they used this pottery wheel to do it. And I look at the wheel, and I think, for 5,000 years, nobody in the entire continent thought to take the wheel and turn it this way and roll it? Like, they knew wheel, they knew wheel was useful — I give you the wheel, Robert. But they never invented the wheelbarrow. They never invented the wagon. There's no rolling stock anywhere in America.

Robert Breedlove: Incredible.

Michael Saylor: And think about the consequences of that insight. How is it possible that not a single person in the entire continent ever thought they might want to turn the wheel this way up? But it's such a profound idea. And yet you can go 1,000 years and not get that idea.

Robert Breedlove: Do you think there's any modern examples that jump to mind that something that's maybe glaring at us in the modern age that we're ignoring? Or maybe something that maybe we've forgotten?

Michael Saylor: The blockchain and Bitcoin is a wheel. I can use this stuff if I turn it this way, you know? They're not complicated ideas, you know? Pretty good privacy, private key encryption, hashing — no one of them on its own, I mean they've all been sitting there and someone says, "What if I just do that with that and I start this fire. Rub two sticks together, or maybe we just never think to rub two sticks together."

Robert Breedlove: This calls to mind your example of the Roman hierarchy as well that because everyone knew they were being watched, it maximized their accountability and therefore their performance and competence. The Bitcoin network's sort of the same. All the nodes and letters looking at one another's activity to make sure everything's above board all the time. It's instantly auditing itself constantly in real time.

Michael Saylor: The Romans were healthy as long as they kept tension and dynamism and this incredible competition in their ecosystem, and when they lost it, they lost their edge. And then when you look at the classic Guns, Germs, and Steel-type narratives like, the Europeans, they got hardened and tougher because they were always fighting with each other, and living with your animals made you tougher, and having people come through from Asia made you tougher. And as soon as you try to insulate yourself from those stressors—

Robert Breedlove: Taleb calls the stressors — they are the information. So you're actually cutting yourself off from the information flow that's driving your adaptation.

Michael Saylor: So one way or the other we meandered through that, but it strikes me that it didn't have to last as long as it did. Life expectancy plunges down to 30 years, from 72 years, and a world without technology, if you don't have aqueducts and sanitation and rules like "don't bring your horse into the middle of the village because he's gonna crap all over everything and you're going to get sick." In the absence of all those orderly rules, then you have just a degradation of the human condition.

And we started to crawl out with the Renaissance, and what's interesting there I think is if you look at all of the great cities in the world, all the great cities grew as nexus, as the central node of an empire. So Rome was the nexus of an empire. Carthage was the nexus of an empire. When they lose their empire, they collapse.

The only way you generate enough money to make a great city, you have to scrape a tax off of all the energy, all the commercial value, in a large place. A fisherman casts a wide net to capture the fish — that's your empire. The Roman schtick was, "Pay us 10% when it comes into Rome and 10% when it goes into the next port and we'll take 20% or maybe 30% out of the value added or whatever we're gonna take."

If you go into Venice and you look around the grand canal, you see all these palazzos — they're all just warehouses. I bring a ship into Venice, I off-load my cargo, and then it goes out the back and it gets barged up and down the canal, and it gets transshipped to a new ship. In that world, each ship ran this route: Venice to Alexandria was one trade route, Venice to Rome was another trade route, Venice to Istanbul was another trade route.

You're at the nexus, you're running these shipping networks, and of course you've gotta bribe the guy in Istanbul, or maybe your son marries his daughter, and that's how you get to come in and out of Istanbul without getting murdered or without getting your stuff stolen. And eventually all the families in Venice intermarry, like I know you, you're my second-cousin, that's how we don't cheat each other. And there's no way—you can't solve the traveling salesman problem—there's no way you can take a ship from Istanbul to Alexandria to Venice to Rome to Ibiza or wherever I would go — Barcelona — because I would basically get overtaxed or extorted in each port unless I actually was friendly.

So the way that works is, you have a hub and spoke system, and there's always one central city, and there's always one set of families or companies, and they intermarry and they trust each other, and they just agree, I'm gonna buy wheat for a dime in Alexandria, I'm going to bring it to Venice and I'm going to sell it to you for 50 cents, you're going to take it to Barcelona and sell it for $1, and you're gonna pay a nickel in tax to the guy on the other end. You're gonna get your 45 cents, I'm gonna get my 45 cents, those guys get paid $1, those guys get paid a nickel. We can play with, What's the mark up, right? It might be I buy it for a quarter, sell it for 0.50 and you sell it for 0.75 to the dudes that sell it for $1.

But invariably, it's the people at the center of the network that are actually getting 10, 20, 30, 50% of all the commerce. Which is all the energy. And what's the definition of a smuggler? Or a pirate? The definition of a smuggler is someone who doesn't want to give me half their stuff. So, how do I stop that? I have to have a navy that goes to kill them.

So you have the Carthaginian navy stopping smuggling so they can tax half the stuff. And that's why the Romans can't — Carthago delenda est, right? — Cato the Elder said in every speech he gave in the Senate for 20 years, Carthago delenda est — Carthage must be destroyed. Why? Because two people can't shake down the same guy of half his stuff — there's nothing left.

If I take half your stuff as a tax, I can't — it's like the tax wars. The Carthaginians are taxing the shippers and the Romans want the money. Therefore, the Romans must defeat the Carthaginians. Now the Romans tax you, then they fall, then the Venetians rise. Now the Venetian navy controls the Med.

If you ever go to Venice, there's this great Renaissance painting, The Battle of Lepanto. And the Battle of Lepanto is when the king of Spain allies with the Pope from Rome, the head of the Roman Catholic Church, and with the Doge of Venice. And those three navies fight the Turks, the invading Muslims from Istanbul — and they beat them. And it's the triumph of Christendom.

But it gets you thinking: why were they all killing each other in the first place over the Med? And you realize it's because they're fighting over control of the mercantile network. And then the next interesting observation is, you've heard the word Roman Catholic Church, right? You ever hear the phrase, Venetian Catholic?

Robert Breedlove: I don't think so.

Michael Saylor: There was a time when the Venetians terminated the Catholic Church in Venice. The Catholic Church in the entire

Venetian empire didn't terminate with the Pope in Rome. It terminated with the Doge in Venice. And so that means all tithes, all due, goes to Venice and stops.

You can't have an empire, unless the person at the top of the empire is also in control of the religion, because if you don't control of the religion, then someone else takes half your money, you see? It's all about energy flow. And the energy is flowing — by the way, the Pontifex Maximus originally refers to the Roman consul, to the head of Rome.

Augustus Caesar was the Pontifex Maximus. So the Romans had eventually the high priest. So the Roman consuls were the high priest of Rome for 700 years. If you're elected the general, you're also the priest. You run all the religious ceremonies. They didn't separate those two.

So the church and the mercantile empire and the power to tax migrates from Rome to Venice. When did Venice start to decline? When they lost control of the church. At the end of the day, they start sinking, because you can't afford to maintain those buildings.

You can't afford 100-story buildings in Manhattan unless you're the center of a financial empire. I have to basically buy your bonds for 68 cents on the dollar and sell them for 82 cents on the dollar and cut the difference.

You know there are a lot of securities on Wall Street where there's only two market makers: there's the one bank and the other bank and they trade with each other, and if you're buying, you're buying at the top of the spread and you're selling at the bottom of the spread. And the spread is 2%, Robert. So if I turn over $1 Billion worth of bonds, I'm paying $20 million in commissions, and there's a monopoly there, and the $20 million is just flowing into what? It's flowing into the building. My buddy works for that bank and I work for this bank and we drink together and we just kind of joke about the 200 basis point spread.

Robert Breedlove: Interesting. So it's these groups sort of competing to be the head gatekeeper in a way. But to preserve that gatekeeping, it's intimately connected to the church — control of *religion, I guess you might say. So does this somehow connect the actual connections between money and religion? Even today we have "In God We Trust" on the US dollar bills?*

Michael Saylor: It takes us to the Reformation. By the way, if you go to Amsterdam, Amsterdam is a city of canals. If you've ever seen a distribution center, the trucks come in one side, they goe through a very complicated set of conveyors, and they goe out the other side. These trucks come from the manufacturers, those trucks go to the stores or the locales, and the distribution center is a maze of conveyors. Amsterdam's that, but it's that for barges, before we had machines. That's what Venice is. That's what they did in every mercantile center.

And when you get to Martin Luther's time, you realize one of the key drivers is there's no way that we can rise or elevate our civilization if we have to send all of our money to Rome. You see this struggle throughout medieval history — William the Conqueror had that struggle, you see the struggle of the Northern European German nobles, and then of course it punctuates itself with Henry VIII, who eventually forms the Anglican church so he can be the Pontifex Maximus, and if the church terminates with the King of England, they don't have to ship any money down to Italy, nor do they have to ask permission to change their alliances and get married and do what they will. It's useful to have God on your side.

And so that drives a lot of stuff throughout the Renaissance and you can say that Northern Europe broke off from the Roman Catholic Church for religious reasons, or you could say the Northern European powers that be created the religion for political and economic reasons, to break off from the Roman Catholic Church.

But either way, it's kind of a triumph of history that everybody's forgotten that there used to be lots of — why do they call it Roman Catholic if there weren't other Catholics? How many different branches of the Catholic church do you think there were? Like, a thousand. They just kind of coalesced over time.

Here's the general principle: everywhere on Earth where you see a big city, it was the center of an empire. Paris, London, Hong Kong, New York, Venice, Rome. Everywhere. And by the way, everywhere where you see a city that's fallen upon hard times, that's been destroyed — its empire lapsed. Carthage, Troy, right? You could name them on and on and on. Venice—had an empire, lost an empire. And that's because you can't physically create this kind of economic density.

I mean, you go to Paris and you look at the Cathedral of Notre Dame and you look at how much human effort went into creating Notre Dame — there are people that are selling postcards and bottled water in the shadow of Notre Dame Cathedral making money off of tourists, where the guys inherited the concession from his father's father's father's father, and if you go back 10 generations, the concession was 700 years old. They are living off of the vestiges of an empire long past, and now the question is: what are the empires of the future? Where do they form?

And that takes us really to the Steel Age, the 19th century, the robber barons and the like, and you can see — with shipping networks, those canals gave way to free ports and eventually gave way to container ships, and container ships totally remade everything and they shifted power to Singapore and Hong Kong and companies like Maersk.

Ultimately it's a low-energy, componentized way to move things around. The most efficient to move anything on Earth is modern containers. By the way, the biggest rage and technology in service today is containerized software via Kubernetes and Docker. Which is the same principle as: put all your stuff in the container ship, and then the container goes onto a ship, they get standard loading facilities into a port, they've got standard train cars and standard trucks — everything's standardized. And the cost and the transparency of that was cut maybe by an order of magnitude with that innovation.

Robert Breedlove: Right. Standardization once again. So I do have a question about these empires that we've discussed historically: ostensibly the purpose of an empire like this is to basically preserve the walls of the city, protect the peace, honor *private property rights within that dominion, enforce contract law, ensure that there's a non-violent means for dispute resolution such that commerce can be conducted fluidly. And you mentioned the Empires of the future — this is something I think about a lot, is that we've always needed this monopolist on violence to sort of honor private property rights within their jurisdiction. But another really interesting aspect of Bitcoin is it's kind of the first private property right that is agnostic of government completely. You don't need government to enforce your right to hold private keys. It's like holding physical gold or any other bearer instrument. So I wonder how much that plays into the relevance of an empire in the historical sense going into the future?*

Michael Saylor: Yeah, a traditional empire is producing security. The #1 export in the United States is security. Like literally, I can live in Miami Beach and I don't worry about someone shooting me across the intracoastal waterway. And if I was in certain other parts of the world I'd be surrounded by 100 bodyguards — so, security.

They have that saying about Genghis Khan: "When the Mongols controlled all of Asia, a virgin could've ridden from one end of the empire to the other with a pot of gold on her head and not be molested." The Mongols weren't screwing around either. You intercept their mail, if anything gets stolen, anybody gets hurt, they show up with an army and they murder everybody for 100 miles in every direction, trying to make the point: don't eff with the system.

Now most of these empires generally provide this kind of security for their citizens, not always for the non-citizens or the aliens or the slaves or the whatever — the underclass. But they do manage to establish them.

Bitcoin is security, and Bitcoin's #1 value proposition is security. It's security of energy. If energy is translated to money and money is translated into Bitcoin and you store it on the Bitcoin network, you're securitizing your assets in a cloud of — behind a wall of — encrypted energy. And in that regard, it provides an important right and empowerment to the individual.

It hasn’t quite addressed the physical security element. When do we actually come up with something that will surround your person with a field of repulsive energy. Your own force field. Then you will have accomplished the same thing in the physical domain that Bitcoin does in the virtual domain. But if I secure your life force — money is energy, is life force, money is power — if I secure your power, that can be converted into physical security either by allowing you to travel away from the threat.

We have the example — Nazi Germany in the 30's — all the Jews had their money locked up in Germany and so the way the system worked is they operated as bankers and they allowed people to launder their money out of Germany and they would take a hair cut of 10, 20, 30% initially, then 50%, then 70%, then 90%, until pretty soon it was like a 90% hair cut to get your value out, and so consequently, people didn’t want to leave.

And so if you don’t have your monetary power or your assets secured virtually, then your physical security is always at risk because you can’t leave nor can you protect yourself, right? You can’t pay for anything to protect yourself, and you can’t get out, or you can’t pay to get out.

You would think that probably one of the most useful or one of the most compelling use cases of Bitcoin would be if I’m a refugee trying to flee a war zone. Because it’s either that or gold, and the problem with gold is there’s a lot of people with guns — they’re gonna take it from you. And at least with Bitcoin, you could pay the guy with the gun half the money when you started, and the other half when you got there, and the worst he could do is blow your head off, but he’s not getting the money. Whereas gold, he just blows your head off — takes the gold.

So yeah, I think it’s second order beneficial to physical security in a lot of different ways and it makes the world better, but it’s first order beneficial to economic security.

If we think about the Steel Age, these rail networks are another network to deliver energy faster, stronger, smarter, harder. Of course, at the nexus of every transportation hub, there’s an economic center, so rail heads — if a great city is in a port, it’s at the center of a railroad juncture. And a lot of times at the center of the rail juncture you can tax all the trains.

So if goods come from Spain into Paris and they're going to Germany, the French get to take a tax right there in Paris. The rail heads became a nexus. The other fascinating thing about the railroads is they became really instrumental to logistics, movement of armies, and they drove economic and political power.

One interesting example is Winston Churchill. When he was like 25, he wrote a book called *The River War*, before he was famous for anything in fact. He was quite the adventurer — he went off to fight in the war in the Sudan under Kitchener. And the entire book is about the British working to win a war in the Sudan, and they had to take Khartoum — it's like going 1,000 miles up the Nile, maybe 1,500 miles up the Nile. — across desert, and why they would even want to do it is an interesting question, but here's the bottom line: the entire outcome of the war comes down to the question, "Can the British build a railroad to Khartoum?" And if the British succeed in building a railroad, they win. And if they can't build a railroad, they can't provision the army and they can't move the heavy equipment and they lose the war.

The entire war — it's called the River War — but it ought to be called the Railroad War. It's just about building a railroad across the desert, and at the end of the day, as soon as the railroad arrived, a bunch of guys with explosives and Gatling guns showed up, they devastated everybody and it was not a fair fight at all. Like Gatling guns versus guys with spears and musket loaders — it's kind of very unpleasant.

And if you read it through the modern view, it doesn't necessarily leave you a good taste in your mouth. But it's a reminder that a lot of times, the difference between winning and losing, and living and dying, is via railroads.

I think there's a similar story in the American Civil War, if you look at the way railroads functioned. Even the conquest of the continent, the Union Pacific Railroad. Once the railroad crossed the continent — it's Manifest Destiny — you think the United States was gonna dominate without that railroad? What is it, 1,000x more expensive to move stuff over land than over water? And Google is very good at saying, "We don't do anything that's not gonna be 10x or 100x better" — I mean it's a Silicon Valley trope. Don't bother to do it unless you're gonna break through this 100x. Well, all these things were 100x better, but a railroad we take for granted.

I give you 5 tons of stuff — carry it from New York to California. Count the amount of energy it's gonna take you. Now put it on a railroad car — try again. You think that's not faster? And stronger? 1,000x faster? 1,000x stronger? Something like that. Orders of magnitude.

And that takes us to John D. Rockefeller. Before Standard Oil comes along, people are actually hunting whales — they're getting in wooden ships, chasing around in the Indian Ocean to kill a whale, boil down its blubber and make kerosene and burn a lamp. Not a very efficient way to gather energy.

So then along comes oil, and oil is 1,000x more efficient way to get energy, and what Standard Oil was, was it was first an energy producer, but it was also an energy storage device. It's a battery. Because the best way to store energy is put it in a tank.

It was also an energy network. Standard Oil, they bought up all the — they didn't actually buy the fields, they bought the refineries. They refined the oil, they stored the oil, they bought up all the tanker cars, they bought up all the tanker ships, they locked down all of the networks, and they basically had an energy network. They actually had guys driving around with carriages to deliver their energy to every single retail store. They had retail distribution. They would even give away the furnaces to sell the energy.

They did a Jeff Bezos thing. You know — Amazon Prime. People think Jeff Bezos invented it? He didn't invent it. He read the biography of John D. Rockefeller. Rockefeller invented it all. He did it all. He gave away the razors to sell the razor blades, he's the first guy to realize that you have to form a cartel or you have to form some kind of understanding of scarcity. If there's no scarcity, there's so much volatility that the market is chaotically destructive.

So you had the world's first serious energy network there, and it's such a powerful network that 100 years after Rockefeller's dead, those companies are still around, worth $1 trillion. They're still instrumental.

Robert Breedlove: Yeah. What does that mean? You said that Rockefeller realized that he had to institute a cartel in order to impose scarcity on the market, such to offset volatility? Can you elaborate on that?

Michael Saylor: It comes in booms and busts. Like people would — you talk about scarcity. The problem with using a commodity as money is when the price goes up, people produce too much of it. Well so he — it was a very inefficient industry with massive volatility, and so he consolidated it to drop the volatility so that they could standardize every component along the way. So that he could drive to a lower energy level, a more efficient energy system.

Ultimately, if you want to look at the human condition — you ever get on one of those rowing machines and you row as hard as you can for an hour, and a kilowatt-hour is hard to come by. You start to think — if I rowed all day long and I was an Olympic-level rower, I think the sum total of my effort is like 29 cents? Like, if you calculate the value of all your human effort in modern energy cost? A quarter? A nickel — some people couldn't even do a nickel worth of work, right? And you think about where we got to, and we got through to that by channeling this energy, and it must be 1,000x, 10,000x more energy.

In fact, just a general theme you see everywhere where there's an explosion of innovation and an explosion of vitality, somebody tapped into a 1,000x more energy or figured out how to deliver the energy. Oil was originally all about kerosene and lighting, and then heating, and then of course the automobile shot it up by a factor of 10 and that was the killer app.

But if you go on to some other industrial-era robber baron networks — here's an interesting one: Kraft, Hershey's, and Post Cereal. They're technology companies. People don't think of them as that. Before they came along, there was no breakfast. People didn't eat breakfast. Kraft and Post figured out how to box corn flakes, and what they did is they stabilized food energy and put it into a container that didn't bleed the energy, that didn't leak energy.

If I give you a bottle full of tomato sauce, and I just make it in my kitchen, and you put it in your refrigerator, it will spoil over some period of time. How much? Well, there's bacteria in it. The origin of branding — the Kraft brand, the Hershey's brand, or whatever brand — the origin of branding is I make it in a clean room, hermetically sealed. Everybody has to clean up, scrub down, sterilize, get the bacteria off, I have machines to load the can, to seal the can, to seal the bottle. I stamp it with my brand.

The #1 value proposition wasn't that it's good ketchup. Or it's good tomatoes or good soup or good whatever. The #1 value proposition is it's not gonna kill you. You ever actually make some leftovers and leave it in your refrigerator for 2 months and eat it? Here's a better one: why don't you make something in your kitchen, and then leave it in your closet for 2 months without a refrigerator, because they didn't have refrigerators. Frozen food came along later.

And Marjorie Merriweather Post became one of the richest women of the century, A) because her father gave her Post Cereal, and they were able to stabilize starch in a box at room temperature, and then B) because she bought the first frozen food company, and she realized the ability to actually freeze food was gonna be a game-changer. And before that, no one had ever frozen food before. You think they're not technologists? It's a food energy company.

Robert Breedlove: Energy storage technologies.

Michael Saylor: I mean, you need energy in food form, in nutritional form, to not die. If I could store it — now it's like, it's very interesting here, right? Can I take electricity from Detroit and deliver it to you in San Francisco today? No. Can I deliver electricity from Detroit to Grand Rapids, Michigan? Yes. When it gets to you, how long can you store it in your battery? It bleeds 2% a month. You'll probably lose it all in the year. Can I ship food from Detroit to Grand Rapids? Yeah. When it gets there, how long can you store it in your cellar? A day? A week? If the answer's 2 weeks, there's no national business there. There's no national brand. The answer needs to be 3 months or 6 months — now there's a national brand. Now it matters.

So these guys that were launching these businesses, they were really launching clean room manufacturing plants that captured energy or something of value. They were stores of value, Robert. A store of value that would not decay or bleed due to bacterial infestation or spoilage. And because of that, the brand became important, just like the Standard brand was important — the kerosene's not gonna explode — it's clean, right? Is the gasoline clean? Is the ketchup clean?

You go to Hershey, Pennsylvania, they got a factory — it's a work of art. It's more complicated than most computer programs. They wrote a computer program in steel. Like, don't eff that up, right? There's no version 2 coming — write your program in an analog computer welded in steel that takes up a football field — that's what they did. And in one end goes milk and eggs and out the other ends come boxes of chocolate bars — 50,000 an hour. And it's not just they come out perfectly uniform, it's they come out without bacteria in them and you can put them on the shelf and they won't make your kids sick. And therein is the rise of all of those CPG companies.

Robert Breedlove: I love the perspective you have on all investment being an investment in technology. Because that is what we are doing, right? We are making tools, protocols, technologies, that basically improve our productivity, and that's how we advance is these layers of innovation on top of one another, such that basically every business is a technology business.

I think that's a very unique insight I have never heard before. And I think the other thing is, the interesting connection between this hyper-sanitation — so really make sure there's no bacteria or anything that could cause harm to the user of energy — which would be the eater of the food, how the hyper-sanitation's connected to the preservation of the food, or the energy store. And that seems to somehow kind of mirror Bitcoin in a way. That it's got a hyper-sanitary ledger, right? There's never any error in the blockchain whatsoever, such that it preserves its sanctity maximally *across time. Like it's the best preservation technology of value because there's no detritus in it.*

Michael Saylor: It's the best branded asset in the history of the world. Maybe it's the first time someone came up with a way to cryptographically brand a commodity. Which is an interesting idea.

Now with regard to businesses, I would say that all of the great businesses, the growth companies, were all technology companies—in their time. And eventually, almost by definition, they stop growing when they're no longer cutting-edge technology companies, right? There are other businesses that are more like rent-seeking businesses, or they're concessions. The guy that sells bottled water in the shadow of the Notre Dame Cathedral, that's a business — it relies upon political largesse, maybe there's a real estate business, right? I own that real estate, and then I sell you water or lemonade on that real estate.

There are those kind of businesses. But the growth companies — Standard Oil was a growth company, U.S. Steel, Boeing, IBM — in that period when they're growing, they're a technology company. And that means that all growth stocks are technology companies by definition. You can buy another stock that's not a growth company or a non-tech company, but it's probably not a growth stock. In order to grow a non-technology company, you could then make the assertion that you're gonna need to do financial engineering, like roll-ups — like I'm gonna buy every McDonald's or every restaurant across the country with debt, maybe that's a way to grow it.

Or you need a concession from a regulator. It's now illegal for you — I'm the only person that can operate airports in the United States. You can grow that way. I need a concession, a political concession, and I suppose there's a place for innovative marketing, but I'm not — if there's not a compelling technology breakthrough, I'm just not a big fan of the marketing thing, you know? Except for, if the marketing breakthrough comes about due to technology, for example, I created a company that got famous because I'm famous on Twitter and YouTube — that's not a bad idea. I'm the best marketer on the new media — maybe that can work. But at the end of the day, those are not gonna be $1 trillion.

You know, in their day, if the dollar's debased by a factor of 100, at least, well John D. Rockefeller was worth $300-$400 million in his money. Multiply it by 100, you know? Multiply it by 200. He got to be worth $1 billion I think. So he probably was worth $200–250 billion in our money, and that was in a day where you didn't have access to all the deflationary tech services that are effectively free. So relatively speaking they had extraordinary power, but they were all technology capitalists.

It's important at this point for us to just look at the impact steel and aluminum through this entire era. Carnegie created an empire based on steel, and Andrew Mellon's empire was substantially based on aluminum. Steel is an elemental force for the civil engineering industry, and aluminum became that elemental force for aviation.

Without steel, really there is no modern city. You build a building in wood, it's 2 stories. Build a building in bricks or masonry, it's 5 stories max. In order to create New York or London or any great city, you need steel and you need of course, an elevator. Straightforward things, but of the two of them, steel is the harder development. The elevator you can probably figure out — it's a counter-weight on a pulley. Whereas steel is iron laced with carbon, and it's really hard.

How hard? Think about how complicated it is in order to refine steel and shape steel when it's molten, and it melts through just about anything you might put it in, or on. If you read any books on steel like American Steel I think by Richard Preston about Nucor, they talk about steel refinery blow-outs. If you actually have an accident in a steel refinery and if the molten steel falls on the concrete, there's water vapor in the concrete. So molten steel superheats the water vapor, and what happens when water vapor gets hot, or water gets hot? It expands. Explodes.

Molten steel on concrete turns the entire refinery into a bomb, and it blows up, and it kills everybody for 100 meters in every direction. So technology, or not technology? Harder technology — everybody thinks they're in a technology business today. Nobody deals with technology that's as dangerous and as tricky as what Carnegie and those early steel refineries were dealing with.

Or the DuPonts handling nitroglycerin. Like, what do you think happens when you mishandle nitroglycerin? When you mishandle crypto, you lose some money. When you mishandle nitroglycerin, everything gets blown up — again, for half a mile in every direction. And aluminum, again, not so easy either. So these are really difficult technologies, but really elemental, because the difference between steel and no steel is, do you build a 100-story skyscraper?

I guess you could say, you might do something with iron, but iron's just got problems. Steel is the perfect material for civil engineering. It's cheap, it's got extraordinary tensile strength, if you paint it or protect it from corrosion, it will last forever. Literally forever, Robert. In fact, if you build a steel ship and you punch a hole in it, you can weld the hold with another piece of steel, and the weld will be stronger than the original cold-rolled steel. It's that strong.

So in the world of strong — this is the strongest of strong materials. It's strong, it's cheap, Carnegie figured it out — they built every bridge with it. No steel — no bridges. No bridges — no Manhattan. Think about what happens if you blow up the bridges in Manhattan. Everybody would starve to death. You can't even get the food in fast enough. And of course, you can't hold the skyscrapers. So all modern civil engineering is based on steel.

So then along comes aviation, and they try to build a plane with steel — what happens? You ever see a steel airplane?

Robert Breedlove: No.

Michael Saylor: It's the perfect material. It's cheap, it's indestructible—why not build an airplane with steel?

Robert Breedlove: Too damn heavy.

Michael Saylor: Just one little nuance. Just one little problem. But the fact that it's better than aluminum in every way, shape —

aluminum's 20x more expensive than steel, you know? And it flexes, and it's difficult to work — it doesn't matter. Steel doesn't fly. Iron doesn't fly. Wood flies. Canvas flies. Fabric flies. But you try to find a metal which is stable, that's going to be a structurally sound metal for aviation, and aluminum's the one. No aluminum — no aviation. Nothing. Nothing. You're talking about balloons, right? Or maybe you've got the right flyer. But it's an elemental force, and on that element, you make that breakthrough, then you have a $1 trillion industry based on that breakthrough — how do you work it, how do you create it, how do you use it.

Robert Breedlove: It's so interesting that these raw material breakthroughs then have so many follow-on consequences. Like first and second order consequences, and to the point that you were saying: no steel — no city. No aluminum — no aircraft. And then we have to think about how much commerce is actually conducted through the city and through the aircraft. I mean it's foundational to global civilization as we know it.

Michael Saylor: And they all kind of come down to networks that move energy around. Standard Oil was an energy network. The railroads are energy networks. By the way, no railroad — no tanker car with energy on it, right? The railroad is the energy network moving the oil around. The airplanes — another energy network — moving high frequency cargo around, and information around. And each one of them build on another one. And then the food networks. The result of all of them is, there are large corporations and huge opportunities for wealth creation, if you get to the node of the network.

One last point on the Steel Age before we move on that's worth noting is: average life expectancy at 1900 is 50 in America. It was 70 under the Romans, it was 30 in the Dark Ages, it might have been 32–33 in the Revolutionary War in the US. We crawl back up to 50, and then by 1950, it's 70. And so probably the most rapid expansion in the quality of life in thousands of years is in this 50 years from 1900–1950, because of the conquest of infectious diseases. And that's all really a function of discovering the science of microbes and sterilization — understanding that you need to be sterile. And then the second is antibiotics.

Those two things together were extraordinary. Antibiotics alone and Penicillin is responsible for the defeat of tuberculosis, and tuberculosis killed a billion people. The White Plague. When you caught tuberculosis, it was a death sentence. I think it killed Chopin — it killed all sorts of people. A billion people — more than any war. And of course, in this particular case, if you look at history books on the 20th century, they give it like 2 paragraphs. If you were weighting the text based on the significance of what happened, something like half of all the history of the 20th century ought to be just about Penicillin, antibiotics, and sterilization — half. And everything else can be the other half. But in fact it's not even 0.01%.

The only event with a measurable impact on mortality rate in the 20th century is the flu epidemic around 1920 where you could see a blip. You can't see an impact on the average life expectancy of any other event — including all the wars — World War I doesn't register, World War II doesn't register. It's kind of like, the impact of technology, of modern medicine and antibiotics and networks and cheap energy and sterilization and sanitation so dwarfs every political thing that took place in the century that what you really got is, I think, just a small blip in 1920. And that was like, different estimates: 100 million people died, people say as much as 10–20% of the population died in that 1–2 year time frame, and they're still debating what that is, but it's the only event you can see on the chart, and all of the activities of all the politicians and all the ideologies and everything we fought over, turned out to be not as relevant as defeating tuberculosis.

Robert Breedlove: And it was derived from a Mycelium? Is that correct? It was left in a sink overnight accidentally, something to that effect?

Michael Saylor: Yeah. And it's an accident. We get it from a fungus and it's accidental. And powerful. I dunno. I guess your takeaway from that is what people do? There's stuff, and don't try to channel people in any particular direction too much, because nature's a bit more complicated.

Robert Breedlove: I think Taleb makes the point that the vast majority of our innovation occurs through trial and error. This tinkering impulse that people naturally have, versus this image of the inventor alone in a room laboring for 20 years straight and all of a sudden he has a breakthrough. It's more like people working and tinkering all over the world and communicating that lead to these breakthroughs.

Michael Saylor: Yeah, half of science is accidental. And half the stuff gets discovered but the issue is no one decides to commercialize it or they don't engineer it into the solution. There's William Gibson's phrase, "The future is already with us, it's just not evenly distributed."

Robert Breedlove: Exactly.

Key Insights from Chapter 2

- Money is a network of trust, and that's why the state has always fought to control and monopolize it, because it gives them essentially unlimited power and wealth to be able to confiscate that from the entirety of their citizens.

- We collapsed fixed costs by standardization.

- The #1 export of an empire is security. That's what the US is today — we export security. Bitcoin fits into that picture because its #1 value proposition is security of your time and energy in a medium that cannot be compromised. For the first time in history, we have a place to secure our life force, our wealth, that is independent of any political happenings in the world. It’s an apolitical money, which is very essential to its value proposition.

- Kraft, Hersheys, and Post foods were in the business of selling stabilized energy. Food is inherently unstable, especially before the invention of the refrigerator, and these innovators figured out how to stabilize food energy at room temperature.

- The freezer and refrigerator enabled us to suck the entropy out of food and store food energy over extremely long periods of time. By contrast, fiat currency is a high-entropy storage device. It leaks a lot. It suffers from spoilage over time, whereas Bitcoin could be considered the deep freeze, the absolute zero of storing monetary energy. It sucks all the entropy out of it, and you know that you’ll own a guaranteed fraction of the money supply for all time.

- Bitcoin is the best branded asset in history, because Bitcoin does do exactly what it says it will do. Nothing less, nothing more. And there’s no element of human corruption that can change that. Bitcoin is this leaderless institution that just runs the rules of the protocol, and it doesn’t change, it doesn’t bend.

- One of the most important discoveries in the history of man, penicillin, came from an accident. So we need to optimize society for free exchange and experimentation — that’s how we create the wealth in the world, solve problems and increase life expectancy.

Chapter 3 | Technology Themes through History — Harder, Smarter, Faster, Stronger

"Unbreakable is better, antifragile is best."

Michael Saylor: I remember the name of the Jewish bankers: it's the Warburgs. If you want a great vignette on why it's important to have capital that's mobile, study or just read any history of the Warburgs and what their family went through, and what all the Jewish families went through in the 30's in Germany.

What the bankers had to do to move capital out of the country, and you get a very poignant punctuated appreciation for the need of having secure capital that you control. So there's some, I think, general themes of technology throughout history, if you look across all of these. The general theme is: human beings marching toward being harder, smarter, faster, and stronger. And everyone that's succeeded — every empire, individual, company — they all found a way to be harder, smarter, faster, stronger. George R.R. Martin, he wrote a bunch of books — he's known for Game of Thrones, but he also wrote a bunch of books on superheroes, or at least he called them wildcards. And it's about a bunch of superheroes or super-people that have genetic mutations, and they get it from some virus, and it's a universe full of normal people, and then jokers — people who have been hideously deformed by this virus, and then aces — people that have been given superpower.

And he takes a very modern view of it, and he starts to analyze superpowers: a guy figures out he can shoot laser beams out of his hands, another woman can be invisible, or some dude can be a thin man. And he goes, "There is a lot of these superheroes, but after a while we quickly realized that the most powerful ones were the ones with defensive superpowers."

Defensive attributes overcome all the offensive attributes in a battle. So for example, the guy that's indestructible with no other superpower, can pick up a machine gun and kill everybody. And you can shoot him a million times, but he's indestructible. Whereas the guy with the laser beams jumping out of his hands, he's just as mortal as you and me. And you hit him with one rock—he goes down and he's dead. So it's the impossible-to-kill ones that end up winning all the battles between superheroes in the smackdown, and the ones with the sexy things, or sexy, really cool superpowers — they're useless. Because the guy that's got none of those — he just gets a bazooka and he just fires. Or he's like, I'm indestructible so I'm gonna set off an atomic bomb in the city and kill everybody but myself because I'm indestructible.

It's pretty easy to win the war if you're indestructible. And it causes you to come to a conclusion about history and who wins in history: unbreakable is better, antifragile is best. So the superhero that's good is the one that you hit him and you can't hurt him. But the superheroes that are most frightening are the ones where you shoot him with a laser beam and now they can shoot your laser beams back at you. And you mind-read them and they absorb your mind-reading and they can mind-read you. And you fly at the speed at the sound and now they can fly at the speed of sound. Like that character, what is it —

Robert Breedlove: I think it's Rogue in X-Men?

Michael Saylor: Yeah, or Sylar in Heroes. Maybe Rogue. It's the one that absorbs your power and plays it back at you. You know, pretty soon it's like, I am indestructible and I have every other power and you just better get out of the way. And that's antifragile, right? I'm looking for the fight. I'm looking for people because I'm just gonna absorb their thing.

You see that theme through history: the really successful governments, empires, organizations, they're continually fighting, they're continually absorbing — the Romans absorbed the Carthaginian's navy in 90 days. It's easier said than done. It wasn't like some Roman politician said, “These are Carthaginian ships — they’re inferior to us. We will invent our own ships.” It’s like, “Not too proud to copy, not too proud to absorb it.”

This kind of principle can explain the success the United States in the wars of the last century or more. Why did we win World War I, World War II? Why is America America? Well, there’s two narratives. One is that we’re more patriotic and we’re righteous and we’re better. The other one is: the Europeans were fighting in Europe and we were fighting in Europe. So we were fighting where they were soft. When you’re dropping bombs in the middle of suburban Germany, you’re attacking the soft underbelly of the citizens in Germany, or the soft underbelly of the citizens in France, or the Battle of Britain.

This is the citizen population being ravaged by a war. And that destroyed their economies, right? Destroyed their culture. Same happened in Russia, Italy. It’s happened in China, and the Chinese fought a war in China’s soil. The Japanese fought sort of a portion of a war on their own soil with the atomic warheads. On the other and, the US hadn’t fought a war on US soil since 1865. Okay? So who’s hard?

Back to this missile idea, right? If you can throw a missile 3,000 miles and your enemy can throw a missile back 300 miles back at you — it’s pretty simple: I have a rifle, you have a pistol. I stand back 1,000 feet — your pistol’s good for 200 feet. I take 100 shots, eventually 1 hits — you’re a perfect shot, it doesn’t matter. Longer arms.

So this idea: are you unbreakable? Or back to terrain: what is the terrain? The geography determines a lot of things. If you take a modern ship: take a modern yacht built in the year 2020, with every piece of modern equipment on it. Now try to sail it to Afghanistan. You probably won’t get there. Something will probably break. Try to sail it to Rome without stopping in a port along the way. You probably won’t make it. Robert, nobody does. They all stop and hide somewhere. It’s really hard to move stuff over the ocean thousands of miles. It’s really hard.

Now, why don't you try to sail an armada of 1,000 ships 3,000 miles in order to invade America? Germans couldn't sail 13 miles across the Straits of Dover. The Chinese couldn't cross from Japan to China during the kamikaze, and that's not very far. It's really hard. There's only one guy that ever pulled it off—William the Conqueror in 1066. Nobody else managed to cross 13 miles. That's your hydraulic moat. It's really hard. It explains the culture of the U.K. It explains the culture of the United States. It explains a lot of other culture. It explains the balance of power, I think.

If you just say, "Here's someone that you couldn't kill who could kill you." I had the option to fight on your turf, but you would never have the option to fight on my turf. Now, that in and of itself is the reason that security is so incredibly important to the Bitcoin network. What is important is that it not break. If I'm indestructible, I don't have to be the fastest, the shiniest, the whatever anything. I just have to be indestructible. If I'm indestructible — the indestructible man goes and buys a machine gun and he's the most powerful superhero in the room, right?

The indestructible blockchain goes and buys a digital app, or another crypto network, an off-chain app, or a Lightning, or a Binance, or a Coinbase, or a Square Cash or whatever, and it's just fine. It's the most powerful network in the room. Because the thing that made it the winner was its indestructibility, its immortality. Not the fact that it could shoot laser beams from its fingers, or it was beautiful, or it could fly with its pretty wings, or it was invisible.

All of these other things? Privacy, transactions — all these other things, they're interesting, but they pale compared to the characteristic of immortality. Being indestructible in the here and now — pretty useful superhero power. Being indestructible and never aging — double useful superpower. And as for the rest, if you're an open software protocol, someone's gonna code an extension to your protocol to give you every other cool power that anybody wants. I can find a way to construct every other superpower by hook or crook, and that makes me antifragile and now I'm like the frightening impossible-to-stop superhero.

Robert Breedlove: That's a great analogy. Calls to mind that old saying that, "If you wait long enough by the river, you can see the bodies of your enemy float by." It's like, because Bitcoin is unchangeable and it's just a rock-solid, absolute scarce quantity of 21 million, and that is the primary property in money that matters — I love the analogy that you gave too of, when someone asks you about the EPS of your company, they only care about the fully diluted cap table, right? Who gives a shit if you've got 18 outstanding — whatever. Just tell me the fully diluted number. And that's what Bitcoin is, right? It's like, you know that it doesn't matter if there's only—3 million have been lost, or 18 million and a half mined so far and 3 and a half to go — all that really matters is that fully diluted figure of 21 million. And that the scarcity cannot be compromised, and then that it has enough antifragility on top of that to absorb other market-proven features — that's all you want in money.

Michael Saylor: Yeah. The mining rate, it's like a secondary particular effect that happens in the first few years of its evolution, but exponentially it becomes irrelevant. It's a second order driver. The first order driver is: is the hardness infinitely hard? And then adoption and utility and inflation in the fiat reference frame that it's operating in.

A couple more points that are worth making on technology and human history. Humans triumph throughout history by channeling energy. All of these things come down to: how did I channel energy? So, navy power. The British Navy ruled the sea, the Phoenicians, the Venetians, the Greeks, the Romans. The Americans today control the seas via their aircraft carriers and via naval power. And naval power is just bringing overwhelming stronger force to bear that's faster, that's harder, than any other force anywhere on Earth. I park an aircraft carrier somewhere and death rains from above. So we've seen all empires are driven by a naval power.

In time, we invented airplanes. And then air power became the thing. And air power is again all about death from above: the most powerful navy idea is airplanes flying off an aircraft carrier. First it was the battleship: show up in your harbor — I blast everything. Then it became air power. And if I could get above you — it is impossible for a modern nation state to function if it does not control its own air space. You saw that in the First Gulf War, the Second Gulf War. The United States, especially the United States air force, could pretty much bomb any country back to the Stone Age in a matter of a week, once you lose your air supremacy, once you lose control of your air space. There's no hope for you. You can take out every port, every power reactor, every generator, every serious building, they're all gone. Death from above. There's nothing you can do. You're at the bottom of the gravity well.

And with technology today, someone can sit out 6 miles up — you can't shoot down a plane moving at high speeds 6 miles up. You have to fight the gravity well to get there. They're just working opposite: it's an energy thing, right? So I'm moving from a lower energy state—I'm on the ground moving enemies around, to a higher energy state — I have a boat, I'm moving heavy masses around, to a higher energy state — I'm above you dropping munitions on you.

Then after that, it became nuclear power. You harness nuclear power and that ended World War II pretty definitively, in a hurry. Then we go to space satellite power. And then you go up the gravity well some more, and that manifests itself primarily with GPS. If you look at the impact of having a Global Positioning Satellite, that's what allows me to drop a laser-guided bomb or a drone anywhere on Earth and guide it day or night to anything. And that's a massive source of American supremacy — controlling those GPS satellites. And if you lose them — big problem, too.

One of my favorite books is *The Moon Is a Harsh Mistress*. And in it, Robert Heinlein, he writes about a bunch of Loonies—basically Libertarians that live on the moon that believe in making their own way, and their motto is, "There ain't no such thing as a free lunch." Abbreviated to TANSTAAFL, and he introduced that idea, and he introduced that idea. And they take it to the extreme, which is like, you're air's not even free on the moon—you pay for everything. You pay for water, you pay for food, you pay for air. Everybody pulls their own weight. No one waits for a bail-out. There's no bail-out coming from anybody.

And they're not big fans of authority. They don't like being told what to do. And the plot of the book winds on the premise that they're being oppressed by Earthlings who are making them work as slaves and their oppressors are taking 90% of their economic production (basically their future) and shipping it back to Earth. They're gradually suffocating and starving them death. It's just onerous. Taxation without representation. But onerous.

So the Loonies decide to fight back with the help of an A.I. computer. The first A.I. computer I can remember. The name of the computer is Mike, by the way. And their technique is they realize that they've got a catapult, and they start tossing rocks down the gravity well. They just take a big rock, put it in their catapult, and it doesn't take that much force to toss it off the moon because they're low-G, they're tossing the rock out of a 1/8th G or whatever, and it heads in a trajectory toward the Earth, and of course starts picking up speed. By the time it lands, it's like an atomic warhead. They just start tossing 'em at random places until the Federation Empire that's oppressing them decides that maybe it's had enough.

And of course the lesson is: it's humans channeling energy, but they're channeling gravitational energy — and a lot of it. And there's a lot more energy than the stuff that we have. All through human history — from a million years ago to today — it's really the story of intelligent people looking around for where the energy is coming from. And how do I channel it in a network to achieve something harder, smarter, faster, stronger?

Robert Breedlove: That's such an interesting point. Back to the terrain of the battle — they're holding the ultimate high ground being on the moon. Another thing this calls to mind is, I don't know if you've read the book Sovereign Individual?

Michael Saylor: I don't know if I've read it all — I've heard of it.

Robert Breedlove: Written in the late 90's, it's been relatively prescient in a number of domains. They predicted the rise of what they called narrowcasting, as opposed to broadcasting, which we would call social media today. They predicted the rise of anonymous digital cash which would be disruptive to the nation state, which we would think Bitcoin is. And it also predicted the digitization of securities, so other property rights going away from a government function to be provided via software.

And one of the things that was interesting there is that they compared digital space to the high seas, basically saying the reason why whoever controls the high seas controls the world is because that's the most frictionless environment. You can move things with the least energy on the seas, once you can conquer them. So whoever controls that domain sort of controls the world. But also — because it's frictionless — it's very difficult to project dominion over the high seas. So we don't have law, really — we have international waters everywhere in the world. And they compare that and analogize it to digital space, where it's a frictionless space that whoever is dominant there will be dominant in the 21st century, but it will also be very difficult to project dominion. So it will be unregulated, largely. They're using that to kind of make the case for digital cash, and the fact that people would be able to escape oppressive regimes. Once you get your capital in the digital space, you're no longer under the thumb of the government of the local oppressive monopoly. So I just think it's interesting how it just always comes down to the terrain, right?

Michael Saylor: Yeah. And I agree with you on that. I agree that the digital domain has weakened all the geographic terrain constraints. It's clear that information is flowing across borders and even when people try to stop it, they're defeated by VPNs and anonymous browsing and other techniques. And we see places where information's illegal — it's leaking. Where it's illegal to move capital — it's leaking. Where it's illegal to move information — it's leaking. Where one person's sedition is another person's news, or truth. That's something I wrote about in my book *The Mobile Wave*.

You might be interested to know, I think in 1998, we created a product called Narrowcaster.

Robert Breedlove: Oh really. I didn't know that.

Michael Saylor: Yeah. And we made a lot of money off it, and it did just that. It actually sent out personalized information. So I was always a big fan of that, so I think it's interesting the way that's evolving. Probably the key thing there is: technology is empowering the individual and giving them more sovereignty. And it's harder and harder in those ways for the government to control what information you get, and then how you manage your assets. But it's a very twisty, complicated subject, because the government's getting more power over you in some areas, they get less power over you in other areas, and it's spinning fast.

Robert Breedlove: And you could argue that that trajectory that's taking us closer to a reflection of natural law, because natural law would say that sovereignty inheres within the individual. Our rights are not given to us from government, they actually inhere naturally within us. So it seems to be that — and it's complicated because we've never seen a world like that, right? Again, we've always needed government to sort of keep the peace and enforce contract law and satisfy property rights, but now in the digital age there's at least the possibility of some of those functions falling to software.

Michael Saylor: Yeah it's more and more things dematerializing to software and they move into cyberspace. It seems like it gets harder for any one government to assert control over the things that take place virtually. So there's a lot of virtual empowerment going on, but there's a lot of other things going on too. I guess I would end this section with just a couple of dynamics that I think I've seen over and over again with technology.

When I was at MIT I studied the history of science and I read about the structure of scientific revolutions and I studied all different types of technology diffusions. And there's certain things that pop out. A lot of times you'll see an S-curve where it'll start slow and it'll accelerate until it reaches diminishing returns, and then all of a sudden it's not an interesting technology anymore. You see that a lot. But one of the lessons of history is: it's very difficult to figure out when the curve starts, and it's hard to stop it once it starts, but it's a lot easier to invest in it and to forecast it once it starts. But before it starts, people will say, "Well, next year this will commercialize—or in 2, 5, 10 years." You could be off by 2 years, you could be off by 100 years.

You remember Google Glass? How big that was gonna be? And how that was a lead balloon, and how many years is it after Google Glass and it's still not gonna happen next year either. Technology fails until it succeeds — that's another trend and another theme. So you see over and over again people pick at technologists and they'll say, "Oh it'll never work. It'll never work. It'll never work." In the year 1902, before the Wright brothers flew that next year, if you'd asked the most learned people on Earth, they would've given you 100 reasons why airplanes will fly, and they would give you 100 examples of how they didn't fly, and they would talk you to death.

And the thing about these paradigm shifts is, oftentimes it's not the learned individuals that make the breakthrough. Aircraft weren't designed by professors of aeronautics and astronautics and engineering and fluid dynamics and mathematicians — they all failed miserably. The Wright brothers were bicycle mechanics. They had a bicycle shop — tinkerers. They're the ones that ushered in the age. And they did it against all conventional wisdom.

On the other hand, Leonardo da Vinci tried to develop his own flying contraption. Every brilliant person for hundreds of years tried it — failed. All failed. And then some dudes in Dayton, Ohio that have a bicycle shop succeed. It happens like that.

The story of John Harrison inventing, discovering a method to determine longitude on the ocean — it's articulated in the book, *Longitude*, it's an amazing story. Everybody knew how to find latitude, it was easy, it was just like, against the North Star. But longitude's very hard because the Earth's spinning all the time and it required massively complicated calculations.

They gave it to Newtonian professors at Cambridge and Oxford — the smartest mathematicians and astronomers in the world — no one could figure it out. Finally, they posted a £10,000 prize and John Harrison—who was not a mathematician, he's a clockmaker — figured it out. And the way he figured it out is he made a clock that was accurate, and they put one clock on the ship, and they set it to the Greenwich Observatory, the royal observatory, which is in Greenwich, England. Hence the term, Greenwich Mean Time. Greenwich Mean Time comes because you sail down the Thames past the royal observatory in Greenwich, you set your clock on that, and then you have another clock, and then every day at noon — if you're in the Caribbean you look up and then you set that clock — and then you just subtract the time on the one from the time from the other, multiply it by 15 degrees and you've got your longitude.

It was a tinkerer's solution to a problem everybody else wanted to solve with star charts and tables and complex math. But of course, ships' captains in the middle of the Caribbean or in a storm, they can't do complex math. They just wanna stay alive. So you see lots of examples in history where people try to do something and they either accidentally discover it like Penicillin, or a tinkerer discovers it, and not the educated — and then all the professors crap all over it, and they all swear it'll never work. And the tendency is either to reject it — people don't like change — or it doesn't get embraced until everybody dies, or until there's a war.

War has a way of opening your mind because when someone drops a bomb on your head and you're burning or screaming, then the pain causes you to take a more open-minded attitude. So a lot of stuff happened after World War II and happens in various wars, and you can trace it through human history. That drives innovation faster. Generational change will drive it faster. People keep trying it, and then just because it's failed 1,000 times doesn't mean it won't succeed, but when it does succeed, then it's probably unstoppable for a while. And hence, yeah you just get to this phrase, which is kind of punctuation for modern technology: Arthur C. Clark writes, "Any sufficiently advanced technology is indistinguishable from magic."

And it was a great quote, it's a quote that I put on the back of my prospectus when my company became public in '98. I believed it then — we have thousands and thousands of examples of magic things in our lives today. Literally magic things, that are just technical things. It's always been that way, and to a primitive man, a rifle looks magical. Or an airplane. And now, we're starting to deal with some software technologies and digital technologies that don't just look magical to a primitive man, they're gonna look magical to a modern man. Because they just really are getting pretty magical.

Key Insights from Chapter 3

- Defensive attributes tend to be even more important than offensive attributes. If you are optimized for survivability, that tends to make you a dominant competitor over time.

- Bitcoin has dominated the digital space for monetary networks. It's impossible to compete with it as money because it's already perfected all the properties of money, and it's done so in this digital terrain that all other competitors would be competing for: it's a very low friction domain, so you can move things at the speed of light in the digital space. And you can defend things with cryptography at very low cost. It gives us a way to store the vital life energy that is money behind a wall that cannot be breached.

- Like the moon in the sci-fi novel, digital space is the ultimate high ground. It enables you to safeguard billions of dollars of capital, which is life force, by memorizing a 12-word seed phrase. You can move it around the world at the speed of light, permissionless, at virtually zero cost, unlike the fiat system.

- We don't need nation states to honor the property rights of money anymore. That entire function of trust and security has fallen to distributed software. Bitcoin is a permissionless medium for storing your wealth in a way to no one can compromise. You’ve converted property rights to information.

- It’s the tinkerers and not the theoreticians that bring about breakthrough.

Chapter 4 | Bitcoin: The First Digital Monetary Energy Network

"Being a cynic and being a pessimist about technology is generally a losing trade, because you're assuming that human beings won't invent anything new and have no capability to do it differently."

Robert Breedlove: In this chapter we're going to be talking Bitcoin theory. It's a field you wouldn't have thought you would've heard about even 5 years ago.

Michael Saylor: Well, when I think of Bitcoin, I think: this is the first digital monetary system in the history of the world. We've tried others, they just didn't work. This is the first one that's perfected, that's functioning. It's the first one to cross $100 billion in market cap and now it's about $200 billion in market cap. $200 billion means $200 billion of monetary energy, and if I look at all of the great digital networks — Apple, Google, Facebook — when they cross $100 billion of monetary energy, then that's a legitimizing step.

Generally, when they get there, 95% or more of the investing community doesn't believe in them. Sometimes 99% doesn't believe in them. But they're too big to fail — they're fires that have been unleashed into the society, and they're burning, and the effect is exothermal. What we have in each of these networks is: we have the collapse, a dematerialization of some product or service or virtue or some ineffable quality be it friendship or mobile devices or information. It's collapsing into a lower energy state. And as it collapses into a lower energy state, huge amounts of energy in the form of profit, cash flow, and value get given off.

Apple can ship a better camera to a billion people overnight for a nickel. Facebook can improve the way that you communicate to your loved ones overnight for a nickel. And Google can package the

Library of Alexandria in the palm of your hand and ship it to a billion people overnight for a nickel. And when you have these massive dematerializations of value and they get on a network with a network effect, it's almost like — well you see a crystallizing structure where you've got an amorphous substance and as it crystallizes we go from steam to water to ice. Collapses, gives off energy. And what Bitcoin is, is it's that first digital monetary network, digital monetary system. It's collapsing into a much more efficient form, it's giving off energy. And that just brings us back to that entire subject of how important energy is to the human race.

Robert Breedlove: Let me just ask you a question there: there's a chart with phase transitions of say, water, of going from ice to water to steam as its temperature increases, and it shows increases in temperature, and when it actually goes into the phase transition it flatlines. So it's like all that energy is being reallocated to I guess changing the molecular structure for the next state. Then the temperature starts to increase again as it goes into water, and then it flatlines again before going into steam. So I guess what you're getting at is that energy becomes transmuted into the next state before it can start to give off energy in the form of profit, productivity — it's giving back economic substance to its users. And I think that's sort of the analogy you're drawing there?

Michael Saylor: Yeah it's a wild thing when all the monetary energy leaps from gold to Bitcoin. Or when it leaps from fiat to Bitcoin. There's this phase transition, and we see it throughout all areas of science, but right now this is just the first time in human history that we see this creation of pure digital monetary network. And I want to replace monetary network with energy network, because monetary energy is energy. And money is energy. In fact, money is the highest form of energy.

So if we ask the question, What is money? Money is the highest form of energy that human beings can channel. So if I look back through time, human beings as a species prosper by channeling energy. And when we mastered fire, we channeled chemical energy.

And when we mastered missiles we channel kinetic energy. And when we master water and hydraulics, we're actually channeling gravitational energy.

The idea of an aqueduct is, well, I'm using gravity to move water 70 miles. Or I'm running a water wheel, or I'm floating a 2-ton block in the water and the gravity's pushing down on the water and the water is pushing up. And when I dam a stream and I generate hydro energy, well that's gravity being converted into energy. But if I dam the stream to divert a bunch of fish into my pond, I'm still using gravity.

Now, I can channel gravity by dumping a bunch of rocks on your head, but it's not nearly so easy to create a river of rocks as it is to just tap into a river of water. And so the mastery of fire and water is the mastery of chemical energy, gravitational energy, eventually thermal energy, and the modern era morphed into the mastery of electrical energy and atomic energy.

And of course there's conservation of energy, and when we look at all of these energy networks — 100 guys with bows and arrows are an energy network. I'm moving kinetic energy from this side of the battlefield to that side of the battlefield. And a civilization at the mouth of a river with cities up and down the river is sitting on an energy network, just like the Aegean, and the Greek civilization was sitting at the middle of an energy network and they were using gravitational energy. And wind energy — another form of energy between sails and gravity — I'm taking advantage of these energies.

So the theme is: humans prosper by channeling energy. Now, what's the most efficient energy network in the history of the world? Well it's about to be Bitcoin. Because the challenge of humanity is: how do I store energy and transmit energy across time and space and domain? And by domain I mean perhaps governmental domain — like how do I move my energy from New York to Tokyo? And this becomes an interesting question.

Let's take a typical power grid: well I generate power, I channel chemical energy into electrical energy, I lose like 35% of the energy in the coal or in the fossil fuel. When it gets onto the grid, I move it over a high voltage line, and I can move it up to about 500 miles and I lose 2% of the energy. Now it has to get stepped down to 240 volts or lower voltage even, to get into your house. As the voltage steps down, I lose more energy — it's about a 4% loss. If I had pure energy at the power plant, I'm gonna lose 6% of the energy to put it into your house 250 miles away. I can't send 2,000 miles away — I can't send it 10,000 miles away — energy will not move from New York to Tokyo. But I can do New York to Schenectady.

Now when it gets into your house, you have to use it immediately. You can't store it. So let's say I wanted to store it — I need a battery. Well the absence of a battery prevents a mechanism. The mobile wave is a function of lithium ion batteries in the palm of your hand — no lithium ion battery, no smart phone. Now we're working with modern batteries. Tesla: it's all about the battery, right? And Elon Musk has really driven battery technology.

So let's say I put a battery in your house. And you pull energy into your house. Well, you've lost 6%. Now a typical battery — a good one — is gonna lose 2% per month. Okay, that means you're gonna lose 24% of your energy a year. Well what does that sound like? It sounds like 24% inflation a year. It sounds like hyperinflation — it could get worse, right? Hyperinflation is 100% inflation a year. Let's say that I have a battery which loses 20% of my power a year, well my half-life on my energy that I pulled off the plant is 3.5 years in 10 years. I'm down to 12.5% of my energy.

So the entire civilization is based upon electric power grids and networks, and yet it's not that good. I mean you really can't store that much power. Anybody that ever put their computer, they charged it and left the computer for a month or 2 months and you whipped it open, it's like — it's drained.

Robert Breedlove: It's dead, yeah.

Michael Saylor: Okay so now, let's say I want to take $100 million. By the way, I could take $100 million of money, and I can buy $100 million of electricity in New York, and I can distribute it to 10 million people in New York — as long as they use it today. So if they don't use it today, it starts to bleed out. And so this is the loss on the network.

Robert Breedlove: And in a monetary sense, we would say that energy really lacks durability, right? And I think this is important too to tie this back to money, that gold itself — to your point — was an energy network. Whatever productivity couldn't be allocated towards something more economic, we would go and mine gold, such that gold became this claim on savings of humanity—which is what money is. And those savings themselves are the result of all our collective energy utilization up until that point. We've been transitioning energy into capital, and then gold, or money, becomes the network that commands that capital, and then what I think is interesting, too, is that the scarcity of the gold actually reflects the scarcity of the energy. So it maps onto it in a way.

Michael Saylor: A brilliant insight. And now let's play a thought experiment. Let's take our $100 million worth of monetary energy and let's put it in a power network that runs on copper, and then let's put it into a gold network. If I put it into a copper energy network, I have 24% bleed rate per year by the time it gets to the battery. I lose 6%, and I can't get it more than 500 miles. Okay, so it's a very short term, short duration, here and now energy network. Let's put it into a gold network: I put $100 million into gold. Now I can move that $100 million of gold 100 miles — would I lose 6%? Probably not 6%. I could probably move $100 million of gold 100 miles for $10,000-$100,000 depending on how much security I need. So we're talking about 10 basis points, instead of 600 basis points of loss. 10 basis points.

So gold is a more efficient way to move large amounts of energy for short distances. What if I want to move $100 million of gold 10,000 miles? Well, that's about 3,000 lbs of gold, 1.5 tons. So I put it on a global express, it costs $10,000 an hour, I put some dudes with guns on it, I fly 16–18 hours around the world — that's about $180,000 plus another $70,000. $250,000. If I have to fly the plane back — let's just assume I don't. It's $250,000, so now we're up to like 25 basis points. 0.25% is the cost to move it around the world once. Okay?

Now what if I want to deliver $100 million of gold 100 years into the future? What if I wanted to deliver $100 million of energy 100 years into the future on copper and batteries? Well my half-life is at 24% at 2% bleed a month. My half-life is 3.5 years. It's gone completely. And everybody with any common sense knows if you put your laptop charged in your attic for 100 years, it will not be charged in 100 years. You cannot store electricity on a copper network or a Lithium ion battery, it's no good. I put it in gold, put it in a vault.

So let's say I put it in a vault in J.P. Morgan in 1900 in the United States of America. And the United States is the most successful country in the 20th century: we win every war, and J.P. Morgan remains as a bank and the vault in New York remains. In that case, assuming a 2% mining rate, assuming a stock to flow of 50 and miners mine 2% more gold a year, the half-life of a gold battery is 35 years. I go from $100 million to $50 million in 35 years, to $25 million in 70 years. So about $12.5 million in 100 years.

So I've depleted my gold battery 87% if the United States wins every war, and if J.P. Morgan isn't a corrupt institution and doesn't fail, and if no one drops a nuclear bomb on New York City. If those things don't happen then I will get 12% of my money back.

Robert Breedlove: So in addition to betting on gold, which is governed by natural law, you're also assuming this counterparty risk in the form of US Government, in the form of J.P. Morgan. You have to bet on stability in the geopolitical landscape as well. Because the gold has to be secured, and it has to be secured by institutions.

Michael Saylor: What if you put a $100 million worth of gold into a bank in Frankfurt in 1900? What if you put it in the largest bank in Japan in Tokyo in 1900? Name a city and a bank you could have put it in in 1900 that will still be there in the year 2000? It's a short list. London, Zürich, New York. You would've failed in Paris, Berlin, anywhere in Eastern Europe, you would've failed in Moscow, you would've lost it all in Beijing, you would've lost it all in Tokyo, you would've lost it all anywhere south of the Rio Grande, you would've lost it everywhere in Africa —

Robert Breedlove: And in the US, possibly with Executive Order 6102 would have impacted you in the US?

Michael Saylor: And you would have lost it in the US.

Robert Breedlove: Yeah.

Michael Saylor: Okay. Isn't it ironic? So can't we reduce it down to maybe Switzerland — maybe? It's an interesting exercise for the reader, but the counterparty risk on gold is at the municipal level, the state level, the federal level, and the corporate level. There's a phrase: "Over a long enough timeline, mortality rate is 100%."

Robert Breedlove: Yeah. In the long run we're all dead. I think Keynes may have said that.

Michael Saylor: He said it, Robert Heinlein wrote a book called *Lifeline* and he said, "Over a long enough timeline, the mortality rate is 100%." And so in this particular case, back to our gold network. We have a gold network and we want to move money, so I put $100 million in gold and I want to move it around — well it's 25 basis points every time I want to move it around. If I want to move it once a quarter, it's 1% bleed a year if I move it once a quarter. If I mine it it's 2% bleed a year. That gets me to 3% bleed a year. Divide that into 70 and every 22 years — that's the half-life of gold.

Energy on a gold network has a half-life of 22 years at best. When you throw in the counterparty risk and the need to move it around — let's just assume that we move it around so you don't lose it. You're just down to now the issue of technology and commodity risk. And this is an important point: gold is the king of commodities. Gold is the greatest of all human commodities. But my first job at DuPont was I built computer simulations of commodities and specialty chemical networks, and let me tell you what people in that business think: they think commodity is a dirty word.

The first thing I learned is: commodity is awful. Nobody wants to be in a commodity business. And here's the reason commodities are awful: if I actually create a factory that creates a commodity, say, gasoline — the only thing it can do is create gasoline. If I invest $10 billion into gasoline refinery, my fixed cost are $10 billion. The ideal, rational price for me to make a profit? Probably at $4 a gallon? But my variable cost is $1.50 a gallon, because I've got all these billions of dollars in the factory.

What happens is, when I create commodity refineries, when the price below the profitable point, I will still keep running the factory, because I've got a variable margin. I'm generating cash flow even though I'm driving the price down for everybody else in the business. So it's possible in the commodity business for every single producer to be losing money and for them to all be acting irrationally while pumping out the commodity below cost. If you're a gold miner, what can you do other than mine gold? Once I've gone and I've invested $100 billion in mining gold, if the price of gold is cut in half, but my variable cost is $400 an ounce and it's $800 an ounce, I'm mining gold and selling it at $800 an ounce. I'm selling it at $700 an ounce. I'm selling it at $600. When it gets to $500, I'm selling it because the market is not rational — I can't transmute my $100 billion of gold mining capital into Google stock.

Robert Breedlove: Right, the switching costs are too high. It's just not possible.

Michael Saylor: And maybe I've been captured by the government. Maybe a certain government wants to mine gold. Maybe I can't legally stop even if I wanted to stop. So when you have a commodity business where people have specialized capital and they make those investments, what happens over time everywhere in every industry in every commodity in the history of the world is: the producers overproduce the commodity because, in the phrase of Hotel California, you can check in anytime you want, but you can't ever leave. It's a one-way route. You go in, you can't get out.

So that takes us to this real issue of gold, or the real risk of gold, which is: gold prices goes up by a factor of 10, capital gets attracted into commodity production. It's a feedback loop: people produce more, gold price goes down, people keep producing to try to recover their cash flows, lots of intelligent people become desperate. When men are desperate, they invent new techniques to produce more gold. They keep producing because they don't have a choice. They will produce down until the variable cost equals the price, then they will keep producing below variable cost, because it's possible that they're in a situation where they can lay off — for example: a government that takes ownership of a gold mine will produce below variable cost in order to maintain jobs.

Robert Breedlove: They'll subsidize it, yeah.

Michael Saylor: Where has this happened? Automobiles and airlines. This is why you don't ever want to be a budget airline because a government will operate airline flights at a variable cost loss in order to avoid shutting down the airline, right? If you're Singapore and you turn off the airlines because they're not profitable, you turned off your bridge to the world. The politicians will run the airline below variable cost. If I want to keep jobs — and how many countries want to project jobs — if I want to project jobs, I will produce something and sell it below the variable cost. We do that with anything that is politically charged when a government decides: education, healthcare, transportation, automobiles, local manufacturing, security, defense — defense is a great example of something that we produce whether we like it or not at a cost that's higher than potentially the value and use of it. And we can't stop. It creates this industrial complex.

Robert Breedlove: A good point there that I think reinforces your earlier point too: you said that gold was the greatest commodity in history. I think the point there is that it is the greatest commodity in *history because it commands human time or commands savings. It commands the collective output of capital that humanity has ever created. So in that way it's kind of like the smartest form of energy, because human beings — our ingenuity, our time, our ability to see the world — we are the greatest form of economic energy in the world, and gold is the instrument that commands that energy.*

Michael Saylor: For thousands of years, it was the best commodity that we could produce to store our energy in. Partly it was hard, but it's not the hardest thing to produce. I think there are other commodities that are harder to produce. But it was the best combination of being hard and then being durable and being non-toxic—right? There are toxic things that kill us. There are things that aren't durable that are unstable. I'm sure we figured out how to produce gold before we figured out how to produce Uranium or Polonium. So there's other stuff, but gold —

Robert Breedlove: The other important piece too is that gold is indestructible. Such that every ounce we ever mined is pretty much still part of the extant supply. I think it's two Olympic-sized swimming pools of gold that we've ever produced.

Michael Saylor: You want stability, right? It takes us back to why Marjorie Merriweather Post was the richest woman in the world. Because Post cereal was starch—it was stable in a box for a year. Stability at room temperature. Like, Coca-Cola is stable at room temperature in a can. And gold is stable energy.

So yeah, it's energy but nonetheless it's a commodity, and you know what they call a business when it's been totally wrecked? They call it commoditized. And so every businessperson forever has always strived to avoid being commoditized. That's the origin of branding, right? We branded sugar water, we branded Gucci bags, we branded everything. We patent things, we brand things, because we want to avoid the inevitable result. The result is: as soon as something is commoditized and open to the public and anybody can produce it, its value goes to the variable cost of production and then it goes below.

For example, Apple Computer — worth $2 trillion today — Google, worth more than a trillion, Facebook. These are valuable networks. But are they the most valuable networks? No. They're not the most valuable networks to humanity — they're the most valuable non-commoditized networks. Because if I go to New York City and I pull the plug on Google, it's inconvenient. But if I go to New York City and I pull the plug on the power company, it's deadly. If I cut off your power and your water or even turn off the bridge — people die. If I turn off Google, Facebook, and Apple — nobody in New York City's going to die, right? And so people forget this.

And this is the danger of putting all your wealth, of storing your wealth in Apple stock. The world thinks Apple, Google, Facebook — Big Tech — they're a store of value. And post-pandemic, everybody surged into the NASDAQ five, and the NASDAQ, because of Big Tech equity — this is a store of value. I'll be safe here. We'll you'll be safe there for a year, or two years, but you know, General Electric and General Motors and Standard Oil were once, they were the most important networks on Earth, and they changed humanity a lot more than Google, Apple, and Facebook did.

You want to change your life? Try to go a week without electricity and see if there aren't riots, murders, mayhem. So people don't ask the question, "Why is it that I get my electricity for nothing? Why do I get my water for nothing?" Because you try to go 3 days without water, 3 days without electricity, you see what that's like. And the answer is: because those two things got declared as public utilities — they are so important that nobody is allowed to have an (unregulated) monopoly on them.

As soon as Standard Oil became so instrumental that it changed the Western world, politicians got interested in Standard Oil. And if your power company said, "We just decided to jack the cost of electricity by a factor of 10," would you pay it? Sure you'd pay it, right? Would you complain? Who would you complain to? A politician, right?

So we've got these networks, they're really important. But eventually if they're important enough, they become commodity networks. And so that's an interesting characteristic. The reason that gold doesn't work over time is — we have two examples. It doesn't work over time because people produce 2–3% more of it every year, and over 100 years that means you lose 90% of your energy. It doesn't work over time because there's counterparty risks, and the Polish bank, through no fault of their own, got overrun by the Nazis in World War II, and Beijing bank got overrun by first one regime then another regime then a third regime. So that's another reason it doesn't work.

And a third reason it doesn't work is if the people become threatened by the network — so for example 1933, Franklin Delanor Roosevelt found gold to be inconvenient — if the people become threatened, they complain to the politicians. The politicians might go ahead and take action, and in this case, the reason they were able to take action is because all the gold was sitting in the same place.

So if the gold's sitting in a vault and we know where it is, and it's under the control of institutions, the institutions are under control of government and therefore that heightens the counterparty risk because of the centralized nature of gold. So the best case for a gold network is you're gonna lose 90% of your energy over 100 years. But the likely case is you're gonna lose 95%-98% of your energy over 100 years. And if we look at Nicholas Taleb's range of outcomes, if you take the 100 biggest cities in the world and you put your gold in the best bank in any of the hundred cities in the world, it looks like in 95–96 of them or maybe 99 of them, you lost all your money. You lost all your gold.

Robert Breedlove: So that highlights too one of the shortcomings of gold— that the economies of scale lead to its centralization. Because it is so heavy and hard to transact — compared to Bitcoin, that's non-corporeal, it can be transmitted at the speed of light — because gold is so heavy to settle, that that leads to its centralization in bank vaults, and that becomes the ultimate honey pot for politicians and governments, frankly. And the other attack *vector we didn't discuss is that temptation is always given into. As soon as things get dicey, governments immediately monopolize that gold energy network, which is the most important in the world.*

Michael Saylor: Let's say—it's not fast enough. Every good technology is smarter, faster, stronger. Every technology. Smarter, faster, stronger. So coming to digital gold versus gold: physical gold is not fast enough. How fast is it? If I wanted to move $100 million of gold, as we talked about, it's gonna cost me $250,000. So that's an impedance. But how long is that gonna take you? A week? A month? Somewhere between — you wanna move 3,000 lbs of gold from New York? I've got a month I'm guessing, if you want to get all the protocols set up.

So you're talking about a quarter-million dollars and a month to move the gold. Assuming I needed another custodian and I used Bitcoin, and I wanted to move $100 million to Bitcoin — and this is where Bitcoin critics are just utterly wrong and missing the point. — they all think, "Oh, well it takes 30 minutes and $5 to move Bitcoin." And they're comparing it to a new crypto network that has no value on it. And they're saying, "I can find a way to move it in 5 minutes for a nickel" — but that's not the point.

The appropriate comparison is to gold. How long would it take to move $100 million of gold? Because there's $250 trillion in assets in the alt-assets. And there's only $25 billion of real assets in the altcoins or the alt-cryptos. So how long does it take me to move the $250 trillion around? And when you think about that you realize that Bitcoin would move it in 30 minutes instead of 30 days. Okay? That's 1,440x as fast. A thousand times as fast a minute versus 1,440 minutes in a day. And so it's a thousand times as fast, but then it's $5 versus $250,000. So that's 50,000x cheaper.

So now we've got people saying, "Oh well it's very energy inefficient" — blah blah blah. Well, it's not, really. It's inefficient in the way that it's inefficient to create an electro high speed transit system, a mass-transit system. It was expensive to build the mass-transit system, it got really cheap to move onto the rails.

If you look at the history of railroads, the biggest thing in the 19th century was railroads. It was pretty expensive to create the railroads. It became really cheap to move on the rails. So we've created crypto rails in order to make it 50,000x cheaper to move. But it's not just that it's 50,000x cheaper, it's that it's 1,000x faster and 50,000x cheaper, and then when you start to multiply 1,000 x 50,000, and you realize — it starts to be 50,000,000x faster, and then you start to add that third dimension, which is maybe a computer that thinks about this while you're sleeping 1,000 times, and you realize eventually you get to 50,000,000,000x faster.

And now we've got to a new scientific metaphor: superconducting networks. There's impedance going through an electric power network and you're losing power, so the solution is that I need to get this superconducting. I gotta cool the network down to close to near-zero, and it's expensive. And the point is: Yeah, it's expensive to get to near-zero, and then the impedance disappears in the network and the friction goes away. And what could you do if the friction went completely away?

Robert Breedlove: Right. You're in outer space. With the smallest amount of energy you can move something billions of miles. And I love that analogy too that you're getting to a lower energy state, and that eliminates the frictions to conduct them. So you achieve superconductivity and in a way that's what Bitcoin is. It's a monetary medium completely free of the noise of unexpected inflation, so you're actually conveying pure price signal. Even gold we didn't quite have that.

Michael Saylor: And I love your analogy because I am an aerospace engineer. You could think about when you encrypt monetary energy on the Bitcoin network, it's like achieving escape velocity out of the gravity well. What we've done — we paid a price to get out of the gravity well. Throw a baseball on a baseball field, it goes a couple of hundred feet. Get out of the gravity well, throw the baseball, it'll go around the Earth forever.

So how much more distance do you get out of the baseball if you pay the price of getting out of the gravity well? It's not like 10x better, it's not 100x better, it's not 1,000,000x better, it goes to infinity, and it never stops. And that's the breakthrough that people don't get, it's like, what can I do if I had vacuum, and I was rid of friction? And yeah, there's a price to pay. And that's your phase change and your state change, and that's why I would say: Bitcoin is the most efficient system for channeling energy through time and space in the history of mankind. We've never figured out how to channel energy with no impedance and store energy with no power loss.

But let's come back to the outer space analogy: take your flashlight and shine it in your basement. Take your flashlight and shine it on your baseball field. Now get into outer space and take your flashlight and shine it, or flip it the other way: the Hubble telescope. How much better are the photos you get from the Hubble telescope than the photos you get from a telescope that has to shoot through the atmosphere?

Robert Breedlove: Right. It's totally free of distortion.

Michael Saylor: It's a billion times better. It's like you just can't really imagine the world when you're trapped in an energy well.

Robert Breedlove: This analogy holds too for the institutional counterparty risk: it's almost as if once you escape the gravity well, you're also free of institutional counterparty risk. I don't need to worry about the stability of the United States and J.P. Morgan to transmit Bitcoin 100 years into the future, you only need to be concerned about the stability of the energy network, which is maintained by the collectively self-interest of the world, in theory.

Michael Saylor: Yeah. It's something that's just altogether unique and we've just never had it before. If you now conceptualize that and you go through your thought experiment, you realize: we need a monetary system, and our monetary system—the three in front of our face are, let’s take it—fiat is a monetary system, gold is a monetary system, Bitcoin is a monetary system.

If I put my $100 million of monetary energy — I have energy. I take energy, I sell it on the grid, you give me money. I take my money, I put it into the US dollar bank — it’s in fiat. I wait 100 years, and it’s 7–8% asset inflation rate, I have a half-life of 10 years, so I get cut in half 10 times. Okay? $100 Million…$50 million…$25 million…$12 million — it’s gonna get painful.…$6 million…$3 million…I only cut in half 5 times.…$1.5 million…that’s 6 times, now it gets really painful.…70 basis points…35 basis points…17 basis points…8 basis points.

By the way, 8 basis points if you don’t get hyperinflation. 8 basis points if your nation wins every war. So we’re not even — what to say is: you’re losing 99% of your energy. 99% of your energy is being charitable.

Now let’s generate $100 million worth of electricity by burning coal or nuclear power or peddling on my flywheel or rowing on my row machine — however you got to it — windmills. Let’s sell it to the grid, take the $100 million, and let’s to go J.P. Morgan and let’s buy $100 million worth of gold and let’s have them custodian for me. Put it in their vault — I guess I could just take it with me, it’s 3,000 lbs. It’s $2,000 an ounce, or whatever the number is. Okay so 1.5 tons — no. I gotta put it in a vault, right?

So I put it in a vault and I pay for custody fees and going in there’s a fee, and then I’m paying, whatever, 20, 30, 40 basis points a year to keep it, and then the miners are out there doing their mining thing and it’s probably 200 basis points worth of additional gold. So that’s 250 basis points a year. And if I watch it and assuming it’s just a dead rock and it’s heavy, I’m not moving it — I’m doing nothing with it, I’m just staring at it — then I’ve just gotta add on the counterparty risk and then the fracking risk, or the technology risk.

The fracking risk is — academics always opine about shortages. Academics have been saying there’s an oil shortage, and energy shortage coming. They’ve been saying it since the Club of Rome in 1973 or something. They said it in the ’50s, ’60s, ’70s, ’80s, and

’90s, and the world always predicted that we were gonna run out of oil or energy in about 10 years. I studied this at MIT — I created computer simulations about it.

There’s an entire school of thought — system dynamics — that studies these things. And the flaw in the reasoning is: it’s a linear interpretation of the world instead of a closed feedback or a non-linear interpretation of the world. The linear interpretation of the world is: we’ve got 10 years worth of oil because that’s what our main proven reserves are. The closed loop interpretation of the world is: when we actually get to 5 years worth of reserves, people start looking for more reserves. And so, no company wants to carry on their balance sheet more than about 10 years worth. And then they just keep finding more, but they don’t publish it.

We’re not factoring in human will to live, human ingenuity. People have a tendency that when you tell them they’re about to run out of something, they reprioritize, they think a little bit harder, and they come up with an innovative solution. So that’s what happened with fracking. We’re gonna run out of oil, we had a crisis, and eventually the price of oil went high enough.

The people sat down and said, You know? If we invent a new chemistry, and if we raised some capital, we can go ahead and implement fracking, and we doubled the amount of oil. And we did it fast. We had 5 million barrels a day for like 40 years and that was conventional wisdom and everybody thought, That’s it. And then we went the next year to 6 million, the next year to 7 million, the next year to 8 million, the next year to 9 million, the next year to 10 million. And I watched it happen, and I watched all of the big investment bankers — J.P. Morgan and the like — they went and they raised billions of dollars from investors and they invested in these fracking companies, in Chesapeake Energy and all these others popped up, and we were awash with oil.

And the next thing you read is that we have too much oil. Because it’s a commodity. Because if you put $100 billion into anything, you invent something new. This is why being a cynic and being a pessimist about technology is generally a losing trade, because you're assuming that human beings won't invent anything new and have no capability to do it different.

Robert Breedlove: And this has its roots in what I would say is kind of the Malthusian fallacy. Where he said, "We're gonna run out of food, there's gonna be mass starvation." And it just fails to take into account the non-linearities associated with innovation. When we get our back against the wall so to speak, we get smarter, we figure out new ways of extracting new resources or growing food. And it's impossible to project that. You don't know when those breakthroughs are coming but when they do come, it releases a lot of energy. It releases a lot of productivity.

Michael Saylor: Yeah, Malthus is the iconic example of just being utterly wrong over and over and over again. And if you study the history of science, the history of science is very simple: the non-scientist and non-believers that will tell you why it's impossible, and then the creative, innovative scientist who thought, I'm gonna ignore that—and just go try it.

And 99% of the population generally will just tell you why it's impossible and be cynical and be critical and they're fearful. And the 1% will say, I think I'll just go try it. And of course the 1% is generally right. I mean, they're wrong until they're right. The technology fails until it succeeds. But if they just keep trying, the likelihood that you'll invent electricity or airplanes or antibiotics or better techniques for agriculture or mobile phones or YouTube—the likelihood is high.

It is highly likely that someone in the future will come up with a way to extract all the energy that you're ever gonna need from some element the size of a sugar cube. We don't have it yet, and I can probably find a million conventional thinkers that'll tell us why it's impossible. The same guys that told John Harrison he couldn't discover longitude with a clock, and the same guys that told the Wright brothers they couldn't fly. It would be those same guys—and they'll be right until they're wrong. And in this particular case, that's a good thing.

That's a good thing if what you want is abundance. But that's why it's a crippling intellectual mistake to run a monetary standard on a commodity that can be produced by man. Ultimately, you have to run a monetary system on math, right? As you pointed out before — mathematical money — because 2 + 2 = 4, and as long as 2 + 2 = 4, human ingenuity is not the enemy. And this is a basic sociological principle, really, which is: do you want to design a system assuming that people are stupid and will not evolve and cannot defeat it? Or do you want to design a system assuming people are smart and channel the energy of human ingenuity into making the system work better?

And this is why gold is defective, and why a decentralized crypto network of which Bitcoin is the most successful in the history of the world — that's why it is effective, because Bitcoin is channeling human ingenuity into making it better. And every commodity is channeling human ingenuity into making it worse, as a money.

So if we come back to this idea, Bitcoin is the ultimate energy network. We're gonna bleed 99.5% of our energy on a fiat network. We're going to bleed 95% or more of our energy on a gold network. Once you calculate the fully diluted Bitcoin count — 21,999,999.98, as I heard from Andreas the other day, that number just slightly less than 21 million — once you've done that, then you just realize that it's a lossless monetary energy system through time. Through space, it has a slight loss in the form of transaction fees, but it's a good thing, and the transaction fees on the Bitcoin network are like a little bit of impedance and/or a little bit of gravity, a little bit of friction. And the goddess of wisdom that created the universe gave us a little bit of friction — it's a good thing. No gravity, no friction — your life gets really really complicated.

Robert Breedlove: No resistance, no growth.

Michael Saylor: And so there's nothing wrong with just a slight bit of friction. That's why the idea that I've gotta drive transaction fees to zero is a silly idea. It's like, No. What we want to do is drive inflation or the loss of energy over time to zero, and then we want there to be a slight loss of energy when we reorganize all of the monetary energy. When I send $100 million from New York to Tokyo, I don't mind spending $5. I probably won't mind spending $50. When Bitcoin has $250 trillion of energy into it, there's no reason why you can't pay $25 billion in transaction fees.

People forget — again — this is the problem of the crypto community: they're fixated upon a prototypical coin network that's a lab experiment, and they're comparing it to Bitcoin, instead of comparing Bitcoin to actual monetary or asset networks in the real world. So for example here's a real asset network: it's called real estate. I have $100 trillion of real estate. You have a house. Let's say you have a $1 million house. You wanna hold — this is a good example — let's assume that real estate is my energy network. If you wanna actually carry a $1 million house 100 years — in Florida, there's a 2% real estate tax. You would pay $20,000 a year every year for 100 years, assuming that the house was capped and not reappraised, and so you're in essence going to lose the house in 50 years. Right? Under the best of circumstances.

You're gonna lose all your wealth in about 20 years if you store your wealth or your monetary energy in a real estate network in Florida. So if you go to any other real estate jurisdiction, they've all got different tax rates over time, but this is why you can't really store energy in property, because the tax rate generally will bleed you out in somewhere between 20-100 years.

Now that's the inflation rate, or the energy loss rate over time. What about over space? What if you wanted to transfer a $1 million house? What if I want to buy it from you? So you want to exchange — heat exchange, energy exchange — you want to exchange the energy in the house? Well it's a 6% transaction fee, and at the point that I said, Robert I want to buy your house, I'll give you $1 million bucks for it. How many days 'till closing? 30? Probably 30. Okay, so you just paid $60,000 and waited a month in order to do your transaction. Now compare that to Bitcoin again? 30 minutes? $6 bucks? 30 days, $60,000.

This is why we don't use property as an energy network. By the way, some people do. You can ask people point blank, How are you gonna actually give your money to your granddaughter? Well I'm gonna buy property. But where? California? Florida? Where? It's the same counterparty risk. It's worse than gold. You can move 3,000 lbs of gold in 30 days for $150,000-$200,000. You can't move $100 million worth of property in 30 days to another country.

Robert Breedlove: And you're also taking the counterparty risk again. The other thing with property is that it's non-fungible, so the liquidity of the market is much smaller than say gold or Bitcoin. And you run the risk of that area having some natural disaster or some other event that makes it uninhabitable or unappealing.

Michael Saylor: It's illiquid right? It could take you 3 years to sell the house.

Robert Breedlove: That's right.

Michael Saylor: It'll take you 3 years to find a counterparty. It'll take you 30 days to do the transaction. Now we just come back to this fee, right? What are the transaction fees on the Visa network? What are the transaction fees across any monetary network? It's pretty routine to pay 1%, 2%, 3% to move something around. If Bitcoin gets to be $100 trillion and there's 1% transaction fees, it's going to be — well pick any number and multiply it by 1%. It's $1 trillion a year in transaction fees. Nothing wrong with that.

What's the entire size of the — you know what the spreads are in the bond industry? Like, I used to buy and sell convertible debt. There were 200 basis point spreads. You could buy at 98% of par value, you could sell at 96% — the banks got in between. So all of the financial system is built on taking a spread. That's why New York City has tall buildings. We've talked about this before. Wherever there's a node in a network — a rail head, Venice, Paris, London, New York — wherever there's a node in a central network where there's exchange, there's a transaction fee. And if you're lucky it's only 1% or 2%.

When you're unlucky — I mean there's a reason people refer to free ports. Free port meant that when you pull your ship into our port we weren't gonna steal it all. You know what the great breakthrough is in Singapore? Here's the breakthrough: we're gonna have a port in the middle of the Pacific where when the ship comes into our port, we don't take all of their cargo. Or we don't take 10% of their cargo. That's your idea? Yeah. That's my idea. We're gonna let them stop here and not seize 10% of their stuff. Wow. That's a brilliant idea.

By the way, that's such a unique concept that Singapore is Singapore. It is the greatest port in the Pacific — because it's so rare that a country agrees not to take 3% of what you have when you stop. You can't even come into the United States without filling out a customs form where they charge you a 10% duty on whatever you have in your possession. The point is: it's very common to take 10% of what you have when you come and when you leave. That's why those cities are cities. That's why those empires are empires.

So when someone sits around and they whine about $5 in transaction fees is too great, they're whining because it's more expensive than their laboratory experiment on their scientific workbench that no one's using. Yeah, I can conceptualize hypothetically in my perfect world a perfect system where it was better. But the real world is $100 trillion worth of real estate and $250 trillion worth of bonds and stocks and gold and silver and other property. And that stuff's moving around with transaction fees which are high enough to have paid for all the buildings in London, Paris, New York, San Francisco, Beijing, Tokyo, Venice, and Rome.

It's not a new idea to charge transaction fees. It's not a problem. And the beauty of Bitcoin is, as more miners come on, they create a very competitive industry, and if a miner charges too much transaction fees, someone else is gonna drive the cost of transaction fees down, and if the revenue from transaction fees falls below the variable cost of running the mining rigs, people are gonna take mining rigs out of production eventually — unless a government wants to subsidize them. Unless a government's gonna be subsidizing the crypto rails which create the 21st economy, and that's a reason why mining is a bit riskier as a business than owning Bitcoin. Because you're getting into a commodity business where you may get driven down to the variable cost of the electricity, or below the variable cost of the electricity if someone else wants to get into the business and they can. It's good for Bitcoin, it's good for everything built above the chain. Caveat emptor if you wanna get into the commodity — or the business of encrypted energy.

Robert Breedlove: Right. I think that's a great point too that you bring up the transaction fees on the Bitcoin network are set at fair market value. It is a freely competitive industry such that all of the transaction fees are a consensual exchange, and the value paid in those transaction fees goes — with very little loss — directly to supporting the security of the network. Whereas ostensibly these government fees—they're non-consensual, they're conducted under a monopolized area. A lot of that value being extracted — 10% in, 10% out — is not going to securing the property rights that you're bringing in and out. It's a very small piece of that. Most of it's going into the political coffers.

Michael Saylor: And politicians, they don't even hide that. They'll say, "We just decided to tax this in order to pay for something unrelated."

Key Insights from Chapter 4

- Money is the highest form of energy that human beings can channel.

- Bitcoin is the most efficient system for channeling energy through time and space in the history of mankind.

- Bitcoin is a lossless energy network.

- The best case for a gold network is you’re gonna lose 90% of your energy over 100 years. But the likely case is you’re gonna lose 95-98% of your energy over 100 years.

- It’s a crippling intellectual mistake to run a monetary standard on a commodity that can be produced by man.

Chapter 5 | Channeling Monetary Energy Across Time and Space

> *"A crucible of virtuous innovation requires a vacuum insulation layer."*

Michael Saylor: We're talking about Bitcoin as the first digital monetary network. I'm positing this thought experiment: I have a $100 million block of energy. I can generate $100 million worth of energy through chemical processes, kinetic, gravitational, electrical, atomic power. Let's just assume that I did some amount of work in order to generate that energy. I converted that energy into money. Money is the highest form of energy.

Now I've got $100 million block of money. You can do a lot with $100 million. It equates to about 3,000 lbs worth of gold, it converts to $100 million of currency, and we could do the calculation in Bitcoin — a little bit less than 10,000 Bitcoin at this point in time. Now, I want to do two things with it: I want to channel it through time, and I want to channel it through space.

The beauty of Bitcoin channeling through time is that there really is no loss over time, there's no inflation. You've only got the 21 million Bitcoin, so as a ratio to 21 million it's infinitely hard. And so that's a big check, and if we contrast it to gold and fiat: you're gonna lose 99.5% of your energy in fiat. You're gonna lose 98% of your energy in gold. You're gonna keep all your energy in Bitcoin.

So clearly, channeling energy through time is important and it's critical. Channeling it through space really refers not just to moving it around the world, but it means moving it across domains at well from counterparty to counterparty. We can see the inefficiencies in fiat counterparty to counterparty: you're locked into 9-to-5 banking hours, you can't do big transactions on a Saturday, there are certain jurisdictions where you can't do transactions at all, you're always working through a counterparty which is its own risk.

Of course, if we go to gold, fiat is faster, but it's still got complexities. Gold is slower — it's gonna take a month to physically deliver gold, and so: super slow, super expensive. Neither fiat nor gold make nearly so much sense for channeling energy through time and space. I think that the best metaphor when I think about this is it's like I want to cross from London to New York City and I need to cross the Atlantic — and that's the journey. And if I'm gonna do it with fiat currency it's like a big rubber raft with a leak in it, and I'm gonna be continually blowing up that raft every day to keep it from sinking on me. And I'm going to get battered this way and that way, and it's a bit of a venture.

On the other hand with gold, it's like getting in a wooden ship. And it's gonna rot over the course of many many years, but maybe for the first 2, 3, 4, 5, years — compared to the rubber raft — I feel pretty good. But it's a heavy ship. Wood is heavier than rubber, so it's kind of harder to get going and it's harder to maneuver it but it's a bit more solid. But ultimately it's organic and it's going to decay.

And then if I put it into a crypto container — a Bitcoin — it's like a steel vessel. I have a steel hull — a steel hull container ship really, is the best metaphor — 1,000 feet long, 60 feet wide, moves 30 knots. The thing that's special about steel is: steel is indestructible as long as you maintain it. The maintenance means: you have to keep it from corroding. It can be attacked by a corrosion, and the way you maintain it is you paint it. You sand it down every 5–10 years, you repaint it. If you maintain it, it will last longer than you will last. In fact it will last essentially forever — hundreds of years. If you punch a hole in a steel hull and then you weld it, the weld is stronger than the original steel. So it is the indestructible, extremely strong, repairable element.

And it doesn't take a rocket scientist to figure out that if you're going to actually engage in commerce all around the world, you want a bunch of steel container ships. You do not want wooden ships, and you certainly don't want inflatables. And that's the distinction and that's why Bitcoin is really such an extraordinary invention.

We talked about getting out of the gravity well — the other metaphor is: I have $100 million worth of energy. If I can get it into a vacuum and vacuum seal it—it's like when we want to keep food forever, we vacuum seal the food—we want to keep bacteria from attacking it. We want to keep natural parasites and pathogens from attacking our food energy. And so the way that you protect your food energy and stabilize it forever is you vacuum seal it. And the way you protect your monetary energy from parasites and from decay and from rotting is you vacuum seal it. And that is the genius of Bitcoin.

When I say that the destiny of money is to be encrypted, it's like vacuum sealing your food. You're taking monetary energy in fiat that can decay, debase, diffuse, and you're encrypting it into a crypto-container such that no one else can touch it, no one else can screw with it. And when you think about all these miners, these miners are power plants, they're plugged into analog energy, but they're modulating the analog, physical, pure energy through the SHA-256 protocol to make it a wall of encrypted energy, if not a cloud of encrypted energy and the rails of encrypted energy. If you want to pass through that, if you want to attack that, you have to go through that wall of SHA-256 encrypted energy. And all of our monetary energy is protected and floating in the cloud behind that wall. And it's a fairly unique wall. And that's the majesty of Bitcoin.

Robert Breedlove: Right. Already the most powerful computing network in the world. We've never had something as cryptographically secure as the Bitcoin network. In its early stages it's already the most powerful computing network humanity has ever seen, with still a lot of room to grow.

Michael Saylor: Yeah. And I guess that takes us to our next interesting question, which is: How big can the Bitcoin network get? And how is it going to absorb more power? I'm measuring: money is power —by the way, the cliche, Money is Power, it's not a cliche. — it's a deep reality. Money is the highest form of power, the superset of all power. Kinetic energy?—I can take bows and arrows and guns, club you to death and take your money — take your value. With a war, I can convert kinetic energy into power. Atomic energy into power. Chemical energy I can burn and create power. Gravitational energy — dam a river and create power. And ultimately that power becomes money.

But I can also do the opposite — I can take a money and I can convert it back into the other things. But money is the amalgam of all powers that mankind has managed to collect. And so the question really is: How much money, how much power can fall into Bitcoin? And there's a lot of debate in the community. For example the most famous model is the S2F model. And a lot of Bitcoiners talk about stock to flow and it's like, well, as Bitcoin gets harder, its price is going up. Well, I mean the power in the network is directly proportional to the price because it's equal to 21 million times the price. That's the power in the network. You could almost think of it as a voltage or something too — the higher the voltage, the more energy passes through the electrical network.

So I see the price as the voltage, and I think that its stock to flow is an interesting transcendental model — a particular model — as we're moving through a short period, early history of Bitcoin, where the rate of block rewards is falling dramatically. But I think that as you look toward the future of the chain as it asymptotically goes to 21 million and as the block rewards go to zero and the entire network flips over to transaction fees, then you start to think, Well, just because it's hard, that's not gonna explain why it's valuable.

It's hard to synthesize Polonium or Uranium. By the way, it's really hard to create steel. Really hard. Like steel power plants, if the steel overflows, it hits the concrete, it blows up — kills everybody. It's hard to deal with nitroglycerin. There's a lot of hard things in the world. And there's some things that are nearly impossible — I will probably never walk on the sun. Impossible. But, the impossibility of something doesn't make it valuable. What makes it valuable is some other dynamics.

I have a simple model, and I tweeted this model the other day: I think the success of Bitcoin and the network power ultimately is a function of the adoption, the utility, the productivity, and the inflation. Those four. And they're big broad concepts that are vectors, they're not like one number — they're ideas.

The adoption idea is this: a monetary network is gonna be as big as the amount of asset holders that adopt it as their primary treasury reserve asset. So for example, if a million people with $100 each decide to HODL their $100 and they put $100 million into Bitcoin, the Bitcoin network's worth $100 million. If a company $100 million decides to HODL their $100 million in Bitcoin, the lowbrow or the historic colloquial term is HODL — Hold On for Dear Life or just HODL or save, whatever — and the highbrow term would be, adopt as a treasury reserve asset.

Robert Breedlove: Same thing. Yep.

Michael Saylor: If I'm filing an SEC statement I would say, We're adopting the treasury reserve asset. And if I was just a Bitcoin maximalist, I loved it and I got it — I'm HODLing. And that's why I love the HODLers. That's why they're my people — I love them. They're HODLing. And they figured it out before I figured it out. And I give them credit: they figured it out and that's genius. But I'm not too proud to learn.

Some other — you gotta believe, if I looked a million years ago and some dude comes up with fire, I'm like, I'm not the guy that's gonna say, I'm not doing that — that wasn't my idea. Could I borrow some of that fire, Sir? And thank you. Thank you, for bringing life to me.

So, adoption: the number of individuals and the number of entities that adopt Bitcoin as a treasury reserve asset — not measured in numbers, measured in monetary energy in whatever US dollar equivalents, but they're converting their fiat (Euros, Yen, Pesos, Dollars) — the total US dollar amount of fiat that is converted into Bitcoin for the purposes of holding as a treasury reserve asset. You see that the traders, the speculators — they're short term interesting, but over the long term they don't really have any value to the network, right? Someone that's going to buy $1 million of Bitcoin today and sell it tomorrow? They're not going to drive network energy, because they're going to rob the network of energy as fast as they fed the network. They are mercenaries.

The Roman Empire was built on citizen soldiers willing to fight and die to protect what they believed in. At the point that you're hiring mercenaries to fight and not die as long as you pay them — your empire's over. So, the HODLers — people that adopt Bitcoin as a treasury reserve asset — they're citizens of Bitcoin. They're citizens of the network. And if you decide it's 20% of your treasury reserves — like, it's a matter of faith.

How committed are you? If you really committed like, say, Microstrategy — I had $500 Million worth of money I didn't need. I had two choices: I can buy the stock back or buy Bitcoin with it. Well, I bought as much stock back as the shareholders wanted to sell to us, because that was the appropriate, respectful thing to do — to put a tender offer. And then whatever's left, we convert. The treasury is our reserve asset, which is our reserve energy. It is our life force. Treasury is life force. It's not unlike fat. Fat is nature's organic battery. You carry 30 lbs of fat, that means you'll probably be able to live an extra 60–90 days without food when the going gets tough. And so it's an organic battery. A treasury is an organic battery. Or it's a monetary battery for a family, an individual, a company, or a country. You have no treasury? You're gonna die very quickly when the crisis hits. You're gonna be insolvent.

Robert Breedlove: In some ways, through an economics lens, I often describe this as a cash balance. Whatever you're holding it in is basically an insurance policy on the uncertainty of the future. Because this cash balance or this treasury reserve gives you pure optionality in the marketplace, you're best able to contend with unforeseen developments.

Another thing, just to jump back to the power and money relationship, I think it's very interesting to look at power through the physics definition. Physics defines power as work over time. That's what gold was: it was reflective of our collective work over time, and it was a claim on that work over time in the form of capital. And the other thing—you were drawing the analogy to the vacuum *sealing to protect it from parasites — I think we could all agree that work is kind of the opposite of theft, right? You don't create any value by theft, you just steal the value of others' work. And that's essentially what inflation is. So, money being power, there's a big incentive to store it in a medium that is as much protected from theft as possible. And I think that's getting to the core value proposition of what Bitcoin represents.*

Michael Saylor: Yeah, that is definitely the proposition, right? It's an encrypted energy crucible in which we store the precious energy of life. And we're insulating it via an encrypted vacuum layer from all the things that would bleed off or steal that heat or steal the energy, be it a political exchange or just physical — there are a lot of ways to lose the energy of life. And Bitcoin solves that problem.

Now how do we get energy into this system? There are a lot of metaphors for heat exchange, and if you look at closed systems, a closed system in thermodynamics is: mass cannot leave or enter — only heat. So let's assume that 21 million Bitcoin is the mass—it cannot leave or enter. Heat can come. If I'm buying Bitcoin at a price higher than the rolling 3-year average, or the rolling 200-week average — I'm heating it up. And if I'm selling it at a price lower than the rolling average for whatever time frame you care about — let's say that 4 years is your time reference frame — if I'm selling it at a price lower than the 200-week moving average, I'm cooling it down.

So people see Bitcoin trading day-to-day, and they're like, Oh it's lower than yesterday. That's not what I see. I see it as: If it's trading at $10,000 and the 200-week rolling average was $7,000, I see it heating up. I see its capacitor gathering energy. And that's a very important way to think of it. Now we come back to this issue of adoption: you can adopt Bitcoin with varying degrees of commitment. It's like joining a religion, right? So I adopt Bitcoin as my treasury reserve asset, and then do I commit 50% of my reserves to it? Or 1%? Or 99%? Or 100%? What is the true adoption rate? How much do you really believe it? Is it a hedge? Is it a 1% hedge à la Paul Tudor Jones? Or is it a 100% commitment à la

Michael Saylor? Right? I said, No hedge. No speculation. 100% commitment. Just like your body is 100% committed to storing excess energy in fat. 100% commitment. There's no other thing.

I like your analogy of an insurance policy, except that I would say: an all-powerful insurance policy with no caveats. Because the problem with most insurance policies is I have an insurance policy on my restaurant, but it doesn't cover pandemics. And so I'm bankrupt. Have you noticed how many insurance policies are not paying off in the pandemic? Because there's a carve-out: "Well, we only insure against a — only if this happens and that happens, and if you didn't do this and if nobody said that and if the government didn't do this, then in that case I will give you some money."

Robert Breedlove: Right.

Michael Saylor: If you were driving your car and you were drunk, I don't pay the policy. If you were sober I pay the policy, unless someone says that you were erratic and I don't pay the policy. And so that's the problem with buying an insurance policy. If I have $100 million of energy and I buy a $1 million insurance policy that pays off $100 million—if it pays off. Well, that's cheap. And I get to take the $99 Million and buy something else with it, but I've assumed risk — I've taken on counterparty risk in order to get the insurance policy. You could say that $100 million of Bitcoin in your treasury is an insurance with no counterparty risk.

Robert Breedlove: This I think brings up a very deep point. I actually tweeted about this today. There's a quote from Carl Schmitt that says, "He who gets to decide the exception is sovereign." So there's rules they're all abiding by — or protocols — but whoever gets to make exceptions to those rules or protocols really has the power to act as they see fit in the world because they get to bend the rules. And that's what is so deeply profound about Bitcoin is that it is a set of rules that we cannot break. It is an exception-proof money supply. And it's unlike anything we've ever had. And by eliminating that attack vector such that any group could be *sovereign over the money, it effectively makes everyone individually maximally sovereign. And that's one of the great breakthroughs of Bitcoin.*

Michael Saylor: And that's why it should be a treasury reserve asset. If you look at the corruption and the decimation, the destruction, in our society, and the devastation in our economy, there are two things that everyone fears: individual bankruptcy and corporate insolvency. Becoming insolvent as an individual or as an entity.

How many times have you heard the story, "That was a really good company, but they took on too much debt and they couldn't make their interest and they were broken up and they closed the factories and they moved away and now we're eat manufactured slop from a machine because our favorite vendor got destroyed."

There are a lot of companies that get destroyed. How does it happen? Well, they have treasuries in fiat, the fiat's inflating, so the CFO thinks, "Well, I cannot afford to invest this. It's not politically acceptable to invest in stocks or equity that keeps up with inflation. If I invest in bonds, I can't keep up with inflation, and so I start buying my stock back, and I borrow money to buy my stock back. So if I have $100 million in treasury and I borrow $100 million and then I buy back $150 million worth of stock, I get the stock up, but now my treasury is -$50 million in net treasury instead of +$100 million, right?"

So I went from having a positive position—it's like you have enough fat to live for 30 days without food, to a negative position — you have no fat, you're in debt, and there's a banker that clicks on a Feed You button every day, and if the banker decides to pull the plug on your credit line, you're instantly eviscerated.

And so a company in debt with no liquid assets has no energy. It's in energy debt. So what that means is, Maybe you generate $100 million in cash flow a year and you've got $500 mMillion or $1 billion in debt and you need $90 million as EBITDA to cover the covenants. That means you have a $10 million cushion. That means you have a $2 million a quarter cushion. That means that in any given quarter, if you were short $3 Million on a $500 Million number, that $3 Million kicks you into default on your debt and renders you potentially — it renders your capital worthless. It can drive your equity to zero. And it can render you insolvent, you lose your sovereignty and the creditor can take control of your company — you lose everything.

It's the same as: I go onto an exchange and I'm trading at 50:1 or 100:1 leverage. The idea that I'm going to pledge 1 Bitcoin and get control of 100 Bitcoin, and if it goes down by 100 bucks I'm going to be utterly wiped out is a pretty silly idea. I would not recommend it to anybody other than an addicted gambler. If your goal is, I'm okay losing all of this—and that's just like going to Vegas or going to Bitcoinville — if that's what you want to do, then call it a gambling habit. But there's nothing rational about it.

The leverage could be thought of as every entity — be it a government or a corporation — is draining their life energy, pouring it out, when you flip from having a positive treasury to a negative treasury. I have drained my entity of my life force, and I'm living at the pleasure of my creditors or my counterparties, wherever and whoever they may be. I've lost my sovereignty.

Robert Breedlove: Absolutely.

Michael Saylor: Individuals do it too, right? Corporations do it and individuals do it.

Robert Breedlove: Absolutely, and I think you're plucking the thread of how the incentive schema related to fiat currency fragilizes the economy in a systemic fashion. To your point, there's no adequate store of value, so people at the individual and corporate level are forced into the use of leverage. The assumption of debt. Because the real debt load is actually eroding via inflation over time, so there's an incentive to take on debt. But this is a very short-sighted strategy, because now you are hyper-exposed to the inherent volatility of the future. So once there's any form of shock—COVID or any other form of economic shock — you can go to zero *immediately. You get wiped out. Versus holding cash — makes you antifragile. You can absorb these shocks. You can actually capitalize on these shocks, such that if things get priced lower in the market, you can go and acquire things at a discount.*

Michael Saylor: And in a simple world, if you had cash that was non-inflationary, if the politicians adopted Austrian economics and said we're gonna print cash with zero inflation, then it's simple for the citizens of the society to save in cash, and you have deflation. And the cash will actually be worth more over time. And that's what Jeff Booth points out.

But every technologist knows this. If there was no inflation, the cash would buy more over time, and my entire simple monetary strategy would be: save cash and over time I'll improve. But in a currency war, the political system declares war on the currency and makes the cash toxic. And if it's 2% toxic, that's like injecting a mild drug into your veins. When it becomes 10% toxic, it's like a poison in your veins. When it becomes 20% toxic, this is like saying I'm gonna put a healthy on chemotherapy, and I will pump toxic chemicals through a healthy person's bloodstream, because currency is like blood — it's carrying the energy of life, the oxygen. Oxygen is energy. Your blood carries energy to keep you alive — currency is carrying energy.

If a sovereign nation is injecting massive inflation into its own currency, it's injecting a toxicity into its own circulatory system, and it's somewhere between either — one metaphor is chemotherapy — another metaphor would be Type 2 Diabetes. I'm injecting so much sugar into your system that your insulin response is failing and you're becoming insulin-resistant, and your body becomes diabetic and I give you metabolic disease because I am just pumping the liquidity too hard into the currency, which is the blood of life.

Robert Breedlove: Which is possibly where we are today, right? The stimulative responses by the central banks don't seem to be having the same effects as they used to.

Michael Saylor: Yeah, if I pump enough sugar into your body, you could become diabetic first, and then the pancreas fails and the liver fails, and you have organ failure, cancer, and death. That's what happens if you overdose on pure sugar. I guess the equivalent would be hyperinflation. If I inject enough money in the system that the currency loses all of its energy carrying—like oxygen carrying capability? — energy carrying capability. I just lose it all. At that point the organ failure is, how how do I buy electricity? How do I buy food? How do I get on a bus? Like all the transit systems, the food systems — how do I pay soldiers and policemen and firemen? That's organ death of the society. That happens at hyper-hyperinflation.

Robert Breedlove: That's a great point. You said it loses its energy carrying capacity. And it's also — there's a connection here to information, because the price signals become totally disrupted. The money means nothing in that instance. So capital is not being allocated efficiently whatsoever, because prices are completely — you completely lose this economic nerve signal that we call the price signal. So it's losing its energy carrying capacity, its information carrying capacity, and you end up with cash in the streets like we have in Venezuela today.

Michael Saylor: The phrase "toxic shock" comes to mind. Toxic shock. The toxic shock of the system. But let's move back to our happy subject. How does the Bitcoin energy network grow? Well, adoption. So a million people adopt it as their 90% treasury reserve asset and they have $1 million each, and now you've got $1 million times a thousand is a billion, a million times a million is a trillion, right? So a million millions gets you to a trillion in the network, they put the money in the network and the network is the syndication of all of the treasury energy of the people that choose to adopt it as their primary treasury reserve.

So when a million people put a million dollars in it's a trillion dollar network. When a thousand companies put a billion dollars in, it's another trillion dollar network, so now you're up to $2 trillion.

When a hundred companies put a billion in or $100 billion in — you go the next level. So as individuals, families, private companies, public companies, government agencies, small governments, municipalities, states, small countries, mid-size countries, big countries, all the non-profit organizations — as they adopt this as their treasury reserve asset, they don't need to adopt it as their currency, medium of exchange. They don't need to adopt it as their unit of measure.

As it becomes their store of value, their treasury reserve asset, as they adopt it, the network syndicates all their energy, and the decision of Apple to buy $100 billion worth of it would accrue to the benefit of a million HODLers that bought $100,000 each or $1 million each or $50,000 or whatever they bought. The beauty of it is that everybody is pro rata, *pari-passu* benefiting. And it's in everybody's interest to bring everybody else in, right?

And it's a network effect, that as someone comes in, someone else comes in, the price moves up, the people that were in it get a benefit, and now we get to this next dynamic — with me pausing to say: without adoption, without people believing that this should be their treasury reserve asset, without that, you have nothing. So it's not enough to say it's just hard. People have to love Bitcoin more than they love gold, silver, Apple stock, Amazon, Facebook — whatever.

By the way, we didn't touch on it much, but if we consider Bitcoin as an energy network versus Facebook or Apple as an energy network, the issue there is: I gotta look out 100 years and say, "will I be able to put $100 million in Apple stock, hold it for 100 years — and what's my exposure?" And of course your exposure is: income tax on the company, sales tax on their product, tariff exchanges, regulatory interaction, income tax on their employees, all sorts of other taxes you can come up with, plus many other regulatory actions including and likely being the ultimate regulation of these things as public utilities, because if they weren't regulated as public utilities, the richest guy in every city would be the guy that owns the electrical power plant. And that guy would be richer than Jeff Bezos or Bill Gates or anything.

The only reason that these guys are rich is because they're in a new novel unregulated area that was not deemed to be important. And the problem is, as soon as it is deemed to be important enough that you can't live without it, then it becomes a God-given right to have access to your YouTube or to your iPhone or to your whatever, and now the entire value proposition differs.

I mean we live in a society right now where people think equity is the perfect store of value. What they don't realize is: it's possible for the equity to go to zero. For example, a nationalized power station has zero equity. It still works. When stuff gets nationalized, be it education, power, electricity, technology — the equity goes to zero. You can have something with an equity zero and the debt has value. You can have something where the equity has no value, the debt has no value, but the underlying vendors are getting paid. It can shift back and forth — it's a political thing.

So with regard to Bitcoin and the value of the network, it comes down to people making the commitment to adopt it as their treasury reserve asset at all levels. It's just as good for us if Norway adopts it for their treasury reserve as it is the Rockefeller Foundation or the Hughes Institute or Harvard or Stanford or MIT or for a private company or a public company or an agency of the government. Maybe the county or a city that you live in or the fire department or a union or a pension fund. And then there's all the investors, right? Hedge funds, pensions funds, insurance companies. There are a lot of entities that have monetary energy. They either need it to operate or they stored it up in trust for their shareholders or for future generations, etc. So as that energy flows, the network strengthens.

Now that's the first order dynamic. It's simple network effect. We're syndicating our power. No different than — let's say that there are a hundred of us in a football field, and you organize 50 people to be on your tug-of-war team, and I can only get 5 on mine, and the rest are all singletons — your team wins. If there's a lesson of history, the lesson of history is: the most organized team always wins. Usually the Romans kicked everybody's ass for nearly a thousand years. Because they were the most organized. When they put their petty differences aside for 700 years, they beat everybody else. And then when they started fighting with each other there's the decay and eventually when they were disorganized, that disorganization causes a deterioration in power. So it's the most organized that wins. It doesn't matter if you're in 8th grade football or whether you're in high school or in college. Whatever it is. Organization is always critical.

Robert Breedlove: And I think you're pointing to another strength of the Bitcoin network here. And also debunking what I would call the alternative narrative that money needs to be adopted as a medium of exchange to proliferate as a network. You do have to HODL or adopt it as a treasury reserve asset—in an economic sense — to create reservation demand for that asset. You're taking that asset off the market. You're reducing its supply, thereby increasing its price. And that's what creates the bootstrapping effect, this monetization process. And that again hearkens back to this evolutionary path where you have it as a store of value first. After it's created enough value, it can be used as a medium of exchange, because those early HODLers had more of an incentive to use it as a medium of exchange. And then finally when its been widely accepted enough, it's being used as a unit of account. To your point about organization, HODLers are all perfectly aligned to 21 million. It's just the energy efficient strategy is to just HODL. It's very simple, hard to disrupt, hard to disorganize because there's not a lot of activity on the part of the HODLer, so it seems like an indomitable strategy in the market for money.

Michael Saylor: That's why, if you understood Bitcoin, you would never say something so silly as, I know when to buy it and I know when to sell it and I just trade. The people that are trading in it don't really understand it, because if they understood it they would just buy it and HODL it. And sometimes people think they're accomplishing something, but they're accomplishing not that much.

If a hundred entities buy $1 billion each, then the value of the network's gonna go up by more than $100 billion, and if they just buy it and park it in a cyber-vault and don't touch it for 100 years, it doesn't matter. It's going up. The fact that they poured their energy into the container is actually what caused the energy network to grow. The moving in and out is actually wasting and dissipating energy. You're just dissipating energy to come and go, and ultimately the success of the thing — I hear these people, they talk about store of value, medium of exchange.

Well, Bitcoin is the high frequency store of value and a low frequency medium of exchange. That's what it needs to be and technically that's what it is. And if you understand that, once you get it, you realize you shouldn't fight that.

So for example: I buy $1 billion of Bitcoin. Every second, it keeps anybody from stealing my Bitcoin. Every second, it's storing the value. No government, no parasite, no thief, no hacker is taking my billion dollars every second for the next million years. It's working. In the same way that I put that energy into a vacuum package and every second it's staying vacuum sealed, and the whole point was to live forever. And so if I gave you a little crypto-field that made you live forever and never age, you would say, high frequency longevity device — it works pretty well. And you wouldn't have a problem — there's nothing wrong with living forever. It's immortal energy. So that's high frequency.

The problem is, people don't recognize that every second of the day they're being attacked. You are being attacked. Every second of the day there's bacteria and there's viruses trying to kill you. If I said, I'm gonna spin up a field that stops them from killing you, it's gonna work a million years, and you're not gonna notice it — you would think that's a pretty good trick. But that is in fact what happens when you actually store value.

Now, low frequency medium of exchange: in order for Bitcoin to have its antifragile properties and utility, we have to be able to move it on occasion. Maybe once a year I move it from one cyber crypto bank to another one, or one cyber vault to another one, or once a decade. Or maybe when I'm gonna die I need to transfer it to somebody or split it between my daughter and my son. Or maybe not, maybe I just have one key and I give it to my one child. They give it to their one child, they give it to their one child — then it never gets transferred.

Maybe once a year I have to take 5% of my Bitcoin out of my vault, convert it to fiat, and then break that into 100,000 little parts and put it on Apple Pay and use it to pay for Ubers and Domino's Pizza and credit card bills. Okay. Once a year I take a chunk out of my piggy bank, my crypto bank, and I put it into fiat, and I do whatever I'm gonna do with it — maybe. And maybe—if the world works the way you'd think it might work—I just put $100 million into the vault, I never take it out, I just borrow against it. I just borrow $3 million a year, $1 million a year — tax free.

I don't recognize income, I don't generate a capital gain tax, I don't generate an operating income tax. If I borrow money in cash and if my Bitcoin is going up 20% a year — that's the real yield—and I can borrow money at 5%, then my effective arbitrage is 15%. So it never makes sense to sell it, ever. In fact, if you look at people that use real estate as a store of value, the way it works is, my family buys a block in Manhattan for $10 million in 1900. It goes up 8% a year—it doubles every 10 years. It's worth $20 million, then $40 million, then $80 million, and then $160 million, and by the time you get out 80 years, you got a billion dollars worth of real estate in Manhattan.

You're not selling it. You haven't done one transaction in a hundred years. All you've done is pledged it as collateral against the loan and you've borrowed $42 million against it and you're paying 3% interest. And that $42 million — that's not income. You didn't pay 42% tax on $42 million — that's not income. That's not a capital gain. It's not a long term capital gain. It's not a short term capital gain. It's a liability. You've generated $42 million of liabilities against a non-taxed asset. The only way it's getting taxed is you're just suffering from real estate taxes, right? But, if you're in a jurisdiction where inflation is high, real estate tax is low, interest is low, then your secret to living well forever tax-free is just borrow against your stationary assets.

And so that's the news of this week: Donald Trump has $400 million in debt and paid no taxes for a decade. But he's not unique.

You could substitute every real estate magnate, a family that had generational wealth in real estate, they all did it. All of 'em.

How do rich people live well and pay no taxes? Not by selling stuff. Not by transacting stuff. The way that they actually live is they just park an asset on the balance sheet, and they never ever ever ever trade it. They just finance it or borrow against it — and once in a blue moon, once in a decade, somebody wants to pay me triple what it's worth — and maybe I do it, but oftentimes, the Warren Buffett school of thought is, the taxes kill you. And so the ideal holding period for an asset is forever. He says it's forever like I'm committed to it — yeah. But it's the taxes that murder you, and so the ideal holding period is forever because then you can pledge it and borrow against it.

How do you get rich? You buy an asset, it goes up, you borrow against it to buy another asset, and it goes up, you borrow against it, you buy another asset, and it goes up, and pretty soon you've generated all these assets that are highly appreciated with massive built-in capital gains that you're never ever gonna recognize. Obviously that can change in different types of jurisdictions where politicians decide they're just gonna tax you on unrealized capital gains, but —

Robert Breedlove: Let me ask you: How do you balance that with not becoming fragilized by leverage? Is it just kind of a threshold that you would never borrow more than say 10% of the value of the collateral — something to that effect? Is that how you protect yourself?

Michael Saylor: The fragility comes from two primary metrics: collateral coverage and the mark to market frequency of the loan — the duration of the loan. How frequently is the collateral marked to market? So for example, if you borrow money to buy a house and it's a 30-year mortgage and if the bank says to you, they're never gonna mark it down. Right? There's some loans like a mortgage loan where they'll never get marked down. They might get marked up — you can go to the bank and you can petition them to revalue your loan but nobody ever went to the bank and said, Mark my house to the market after the real estate market crashed.

So if you buy a $1 million house with an $800,000 loan, and you got $800,000 in debt and it's never marked to market then there's not that much fragility as long as you can make the $25,000 a year interest payment, assuming 3%, right? So that's not that risky — I mean a little bit risky, you can't make $25,000 and you lose it — but on the margin, if I didn't have the million bucks or I didn't have the $800,000 and somebody was willing to give me the $800,000 house for free, and all I did was pay $25,000 a year — I think that's a pretty good trade off. The worst that happens is they take the house back and I go do it again somewhere else. So that's not very fragile.

How does it get fragile? It gets fragile when you buy a $1 million worth of stock on a $800,000 loan and you've got loan-to-value of 80%. And now, the stock trades down 20%. And the stock trades down 20% and the banker marks the collateral every day. And if you're in a swap with the bank, they will mark it to market every day. Every single day, they calculate the value of collateral, and if you're under you have to wire them the money the next day. And if you're over, they in theory have to wire you the money. And you have to mark it to market every day — that's risky.

So if you were gonna buy $100 stock, you probably don't want to borrow more than 50%, but if you borrow 50% you damn well better think that the volatility's not gonna be 50%. You can't afford to. So probably in that case, if you thought the volatility was that it could lose half its value, you'd want to borrow no more than 25%. And if you wanted to hold the stock even if it loses 80% of the value you can't borrow more than 20% loan-to-value, right? So the big risk, the hyper-risk is one of these Bitcoin exchanges that go 100:1 in leverage and I mark to market every hour. What if I mark to market every minute, Robert? I mean that the BitMex liquidation. I basically bought 100 Bitcoin, I pledge 1 Bitcoin, it gets marked to market every minute, and if the price goes down $100 I get wiped out and there's a crash. And it happens in 3 seconds. So that's risky.

If you want to take the opposite point of view, I have 100 Bitcoin and I'm gonna borrow the equivalent of 1 Bitcoin or 10 Bitcoin in value, and as long as it doesn't go more than 90% down, I'm good. But if you wanted to save for one century, you would say, I'm gonna borrow against the collateral but I want you to agree that you won't ever mark it to market — that's what real estate loans are.

Or a bank: I'm gonna loan you money against your artwork or against your boat or against some other interesting collectible, and every year or every 5 years our appraiser is gonna reappraise it. So the issue is the frequency of the appraisal, combined with the volatility of the asset combined with the political regime you're in combined with the loan-to-value. If you're in a political regime where it's unacceptable to let real estate values go down, then you can reasonably expect that it's not likely your house is gonna be worth 20% of what you bought it for because the politicians won't let that happen. But they might not protect your Picasso painting.

So if I borrowed money against a Picasso painting and the banker said we're gonna mark it to market every month, that's not as good as once a decade. So when you're thinking about risk, you're thinking about, How liquid is the collateral? And how frequent is the mark to market? And then how much is the loan-to-value? And that's how you get to a question of risk and what you should do and not do.

I'm not saying you have to do it, by the way. You could just sell the Bitcoin highly appreciated for cash, pay the tax, and take zero risk. And look, if the interest rate was 18% and you thought that the economy was gonna grow at 3–4%, then you would sell and you wouldn't take on the debt. So it's a function of interest rates as well, and productivity.

Let's go back to this issue of our power equation: adoption, utility, productivity, inflation. We talked about adoption, and my point there was you're just syndicating — if the world was static, in a static world where there's $1 trillion of assets, if you get $100 billion on the network then that's better than $10 billion. And $500 billion is better than $100 billion. And in a static world it's just all about recruiting and getting people to join the network. But the world is not strictly speaking static. The next thing is dynamic, and technology is what makes it dynamic.

And that's where utility comes in. So if I can take Bitcoin and I can buy it from Square Cash, it's got more utility. If I can send it as $22 — if I can send it from Square Cash — if Square will convert my Bitcoin into dollars and send it in a split second, it's got more utility. If Apple Computer builds it into Apple Pay, and I can link a small wallet with 1% of my Bitcoin into Apple Pay and I can zap that around on my iPhone, it's got more utility.

If Kraken creates a crypto bank and they offered me 4% interest and they'll offer it to me with institutional low-risk counterparty and they represent to me that they've got $100 billion of insurance and I can give them my crypto wallet and I get 4% interest on it or 6% and I trust them, the utility just went up again, and Bitcoin becomes more valuable.

If I get to the point where I can manage my crypto keys using my retina scanner face ID and give speech instructions and I can say, "Send Robert $37 of my crypto in cash" — and if it always works and I just did it and it's more secure than a hardware key and I don't have to remember my 12 word seed key or whatever and it never fails — utility went up. Right?

If I can say, Robert — maybe I've got a girlfriend, Lisa— I could say, "Send Lisa flowers on her birthday every year for the next decade" — click. And it's jacked into my crypto—utility went up. There's a lot of ways utility can go up. Buying stuff, selling stuff, etc. It's all a function of technology. If the Lightning Network works. In the ideal world, back to my example, I have X money — $10 million — I have $100,000 in my checking account. I say, "Move $100,000 in checking, leave the rest in the bank yielding 7% interest" — tie my checking account into Apple Pay, link that to my Uber account, my sister's Uber account, my Domino's account, and use it to pay off Netflix, Google, this, that, and the other thing, and pledge it as trusted collateral on some dating network to show that I'm a real person, and then use it to automatically pay all my fees on my domain registrations every year when they become due. There I just set it. I just did it — that's utility. That's what we're looking for.

Robert Breedlove: It's connecting it to productivity, right? It's enabling you to accomplish greater results with the same or less efforts. So the utility is a reflection of your productivity.

Michael Saylor: And that's a great segue to element 3 of the network power, and that's productivity. Well, we have a million HODLers and they put $1 million each into the network and now we have $1 trillion in the network. Adopting it as your primary treasury reserve assets means sweeping your excess cash flows into Bitcoin, so those million people put $1 million each, but how much money do they make each year, and how much do they save each year? So let's say they make $100,000 a year, and they save $10,000 a year. So $10,000 times a million — $10 billion. So in that case $10 billion in fiat gets converted into Bitcoin every year. That's the productivity.

If they all get a raise — next year it'll be $11 billion. The next year it'll be $12 billion. If they all start their own business, the near year it'll be $30 billion. If the economy's growing and they're inventing cool stuff and one of them becomes the next Michael Dell or Bill Gates or whatever, that person's gonna put in $100 billion or $50 billion. If you get lucky and one of your HODLers is Jeff Bezos he's gonna put in $37 billion. So the productivity of the individuals is going to sweep cash flows into the network.

The same is true with the productivity of the corporations. So Microstrategy, we put $500 million into a network, we make $50 million a year. If we generate $50 million a year after tax, we sweep that into the treasury, right? And do that 10 years in a row, our initial $500 million is going to become $500 million more of discounted cash flow.

So now we're just back to some basic finance theory: What's the value of the stock? It's equal to the balance sheet cash value plus he discounted the value of the cash flows over time. So, the treasury that gets put in is the initial slug, and the discounted value of the cash flows of all the people in the network is the next value proposition.

Of course, this kind of dovetails nicely with economic theory, because if the overall worldwide economy is flat and not growing, then the cash flows are not gonna grow. If the overall economy tanks and starts to deteriorate, the cash flows will deteriorate. If I destroy the economy, cash flows are going to zero, no value will accrete. And if I invent an atomic overthruster that gives you infinite energy in a sugar cube, presumably productivity is gonna go through the roof and cash flows are gonna go through the roof.

So ultimately we're getting a bunch of people to join the network and then all of our fates are intertwined with all of our productivities. We've created a cyber economy. Just like Warren Buffett said, "Never bet against the United States."

The United States was a 20th century physical economy and every business in it was working to the benefit of every other business in a competitive capitalist Darwinian ecosystem. Well now we're actually creating a cyber economy where you can be in a relationship with anybody else to the extent that, Robert, you adopt as a HODLer and I adopt as a corporate treasurer. If I'm successful, you benefit. If you're successful, I benefit. Obviously, if we can get Apple Computer to put $100 billion in and then sweep $50 billion a year, we all benefit. And when a country does it, they benefit.

And of course, if the people that join the network are more responsible, if they actually are productive and they save more than they spend, or they earn more in revenue than they spend in cost, then the network grows. And if they spend more on revenue, flip it the other way — a million HODLers all of a sudden save their money and then they start going crazy and partying and quit their jobs and buy Lambos and blow it all on champagne and gambling, they start drawing down their balances in the Bitcoin network and they sell their Bitcoin for fiat. And if they're selling it for fiat they're draining it out of the network. So the network is going to accrete with virtuous economic behavior and debase and dilute with vices.

Robert Breedlove: I love the example that you use of the United States being this Darwinian economic ecosystem of value creation. Because it was indeed a place in the world with the lowest impediments to free trade that actually led to America creating the most wealth and the most capital. And in many ways I think Bitcoin *positively embodies a lot of the founding principles of American free market capitalism. You have inviolable property rights in Bitcoin, which in the American ecosystem actually marginally disrespected through Quantitative Easing and fiat currency printing. There's a rule of law here in the US, so we have non-violent dispute resolution and enforcement of contract law, and clearly the Bitcoin network is the most adept network at reaching global consensus we've ever created. And then kind of from the first view, honest money or hard money would be something else that a real capitalist system puts out. So it's almost as if America as an experiment was the closest thing to pure capitalism we had prior to Bitcoin. Because again the nation state always gives in to that temptation to violate the money supply and thereby violate the private property rights of the citizens.*

Michael Saylor: Now Robert, you jogged my memory, or jogged my thoughts. I've described Bitcoin as a swarm of cyber hornets behind a wall of encrypted energy. Well, the United States is a swarm of military assets: the navy, army, and air force, behind a wall of water. The insulation is the Pacific Ocean, the Atlantic Ocean. 3,000 miles of water — you have to go through the army, the navy, and the air force, and that protected a bunch of capitalists, a bunch of entrepreneurs. In service of the goddess of wisdom, in this case in service of the American Way.

The American businesses pursuing the American Dream are unhindered by interruption because they're behind a wall of water. You know if you go to Poland, they're good people too. I've been there. I have a lot of Polish employees — they're brilliant. Between Russia and Germany. Again, Trotsky's point, "You may not be interested in war, but war is interested in you." It's kind of hard to go about your business when people are rolling over you with tanks this way and that way. And the equivalent of that in a monetary system or an energy system is with someone stealing your energy. Okay?

How do I keep my coffee warm? I put it in a thermos. I need to insulate my energy so that no one steals it. If I take your Starbucks coffee and I put it in a cooler of ice and I drop it in the ice your coffee's getting cold. And so the wall of encrypted energy is the insulator. The wall of water is the insulator. The insulator is the insulator. The vacuum is the insulator. If you want a crucible of virtuous innovation, you need that vessel that serves as the insulator against all the forces that would attack it from without or within from fear with this process. And that is Bitcoin, right? We're creating that.

And having said all that: adoption, utility, productivity — those three things create that monetary chemical reaction of sorts. And the last piece is inflation. But inflation is almost just the way that we translate the energy — it's a translation coefficient of the energy into a frame of reference of the dimensionality or the domain where I'm spending it.

I mean, I'm gonna have to translate my monetary energy into rubles or pesos or dollars or euros or yen if I cross into one of those domains, because it's their world, not our world. So if the dollar didn't inflate, then would Bitcoin go up? If the United States had a perfect monetary policy — no inflation — could Bitcoin succeed? Yeah. It could succeed.

If people adopt it — because it's got technical advantages, right? If they adopt it as a reserve asset, I mean their choice is that versus stock versus bonds versus property — you still have the issue of: How do I commute energy through time and space? So I would still adopt it.

Technology will still get better. I mean if there was no inflation you would still like the iPhone. You would still like Zooming to me if there was no inflation. Technology would get better, and we'd have productivity. We'd be inventing stuff: fusion, better materials, etc. And when we created stuff like Zooming and we put together Zoom with YouTube, we would talk, and someone would put it up on YouTube and 100,000 people would see it and 3 people would have an idea and something would happen that wouldn't have happened otherwise.

You don't need inflation for Bitcoin to be successful anymore than you need it for Google or Apple or Amazon to be successful — it just happens that in an inflationary environment it accelerates — 2% inflation will grow it 2% faster, 10% inflation will grow it 10% faster — in theory if a bond is pure energy and if bonds are inflating at

20%, then that means that Bitcoin will have a 20% real yield in that currency where you see that energy inflation. And it'll be different relative to the frame of reference of every single domain or every country depending upon how they choose to manage their currency.

They could in theory — I could peg the dollar, like in Singapore or the UAE I peg to the dollar. I could peg to gold. If I peg to gold, it'll be a 3–4% differential. If I peg to the dollar it could be a 10–15% differential. If I peg to the peso it could be a 32% differential. So all of the value of Bitcoin relative to the people in the ecosystems in the domain will vary. And of course, that's why if I'm in Lebanon or Argentina, this is even more insanely valuable to me than if I'm in Switzerland.

Robert Breedlove: Absolutely. I would just maybe add that 2% inflation would increase adoption say by 2%, but as you increase the inflation rate, I think it would actually be non-linear, because if you take the extreme example of hyperinflation, everyone would pile out of their currency into the dollar or into Bitcoin — something that was more reliable, such that, as you increase from say 2% to 10%, people's inflation expectations actually increase, which gives them further incentive to move into Bitcoin or something alternative.

Michael Saylor: It's multivariate, non-linear, and one dynamic is: Hyperinflation panics people into it, high inflation pushes them into it, low inflation encourages them into it, but we're marking the value of the Bitcoin to all the assets — the tangible assets, all the products and services and assets in the domain which is being inflated — and that's also having a frame of reference impact. So the frame of reference changes literally when it's $1 million for a cup of coffee — Bitcoin's gonna be worth $1 billion, right?

Robert Breedlove: Yeah. That is the Number Go Up technology.

Michael Saylor: So it becomes powerful in that regard. So each of these four effects — I haven't written them out as a formula or equation. If I was writing it I would put:

*Price of BTC = f(adoption)*f(utility)*f(inflation)*f(productivity)*

Because in fact, they're all vectors that are all the time dynamically varying across many dimensions. Each of these is a general idea and they all convolve with one another in order to drive network power. But when you take them all into effect, then you just realize Bitcoin is this energy network. It's gonna gather energy and as people perceive it, they will adopt it, as people adopt it they'll want to integrate it with more utility, as they do that it's natural we can expect human productivity will increase, we can expect technology to advance. If we only have 0.1% adoption, it's like having all of the gas in 1% of the chamber and I take away the barrier. You could expect it will — it's not getting less. The genie is out of the bottle. The genie is going to expand. And then betting on some governments to inflate is not a highly risky bet.

Robert Breedlove: That's a good bet.

Michael Saylor: Probably you will get that. And that is Bitcoin network power dynamic. That is the dynamic there, and everybody that's marketing Bitcoin, they're contributing to it. Everyone working on Bitcoin technology is contributing to it. Everyone simply HODLing Bitcoin is contributing to it. Everybody that hates it, or everybody attacking every other asset or every time another asset fails or another currency weakens it contributes to it. And then just the relentless passage of time contributes to it.

Robert Breedlove: I think this is a brilliant and unique way to look at the network effects of Bitcoin. And I also find it interesting that at the center of this vortex is the highest expression of truth we've ever had. Bitcoin is literally this system of converting energy into indisputable truth about who owns what UTXOs, and the 21 million again is kind of like the third certainty in life. We've had death and taxes, and now in the socioeconomic sphere at least we have this *number 21 million we know that can't be violated. And that's what's spurring all these effects.*

Michael Saylor: Bitcoin is a cyber economy based upon the principles of truth, respecting the laws of thermodynamics, respecting Newton's laws. If you're gonna worship the goddess of energy, you better respect the laws of conservation of energy. And it is that conservative monetary energy system. The first conservative monetary energy system that we've ever invented—anybody can choose to be a member: any individual, any family, any company, any government, can choose to become a member of this closed energy system.

And conservative energy is truth. It starts with this simple principle: energy can be neither created nor destroyed. You could lose control of it. You have it — you could lose control of it and it can dissipate and you can lose it. And so that doesn't mean you can be lazy or sloppy. You have to channel the energy. But Bitcoin is the best system in the history of the world for controlling, storing, and channeling energy, and that's why it's destined to be successful.

Robert Breedlove: I love — I've never heard it put like that—that conservative energy is truth. That ties it back to how we started this conversation of the eagle dragging the goat off the cliff side: it's employing the least energy necessary to accomplish the greatest result. And that's what you want to bet on. That is the winning strategy on which you want to bet. And it points towards the kernel of all economics, which is: scarcity gives things market value. Scarcity is the driver of market value. Things that are hard to obtain and have utility are what give them value in the marketplace.

Michael Saylor: The ultimate scarce asset in the universe is energy. You can't create more of it. If I give this much to you, you can't wiggle your fingers and make it twice as much. If you lose it, it's gone. And you can play all these games and thermodynamics — it's a great field because everybody thinks — they're all looking for that perpetual motion machine. The laws of thermodynamics, we used to paraphrase them at MIT in a snarky way, we'd say, "You can't win, you can't break even, and you can't get out of the game."

The laws of thermodynamics. It's like, from a layman's point of view: You can't cheat. There is no cheating in thermodynamics. It might look like you got something for nothing — even Maxwell's demon. He posed, "Maybe I could actually fight or reverse entropy and get order from disorder by dividing a chamber and I have a demon and there's a little door and molecules are bouncing around, and what if my demon opened the door when the molecule bounced from the right to the left and closed it before the molecule bounced from the left to the right? I could over time with randomness get all of the bouncy molecules on one side of the chamber and I could reduce entropy." And they couldn't figure that out. They called it Maxwell's demon.

Well, what's wrong with that argument? Doesn't that break the laws of thermodynamics? And the answer came along 100 years later when some IBM computer scientist pointed out that information is building up in the head of the demon, and the information is in and of itself creating entropy, and so no you're not cheating. Once you actually account for all the information, disorder, and energy in the system, it did respect the laws of thermodynamics. You can't cheat time. You can't cheat space. There is ultimately conservation.

And Isaac Newton — all of And Isaac Newton — all of Newton's laws: conservation of mass, conservation of energy, $F = ma$, it's the basis of physics, the basis of mechanics, the basis of every machine we've built that works, it's the basis of all of our heat exchange. Scientists and engineers don't have a high opinion of economists. And one of the reasons why is that it hasn't been important for economists to understand closed systems, isolated systems, servomechanisms, conservation of mass and energy.

$E = mc ^ 2$ matters. What it means is: If there's mass, it becomes exponentially expensive to move it around. E is the energy. You wanna move stuff fast? You need to take the mass to zero. And that's how you move stuff fast. So economists, maybe what they were doing didn't matter before Bitcoin. You could say, maybe

Bitcoin is the first time that technology crashed into economics. You have energy, you have technology, you have math crashing into economics and now you couldn’t really be a competent economist without appreciating closed systems, energy efficiency, math, conservation of everything.

What it means is: If there’s mass, it becomes exponentially expensive to move it around. E is the energy. You wanna move stuff fast? You need to take the mass to zero. And that’s how you move stuff fast. So economists, maybe what they were doing didn’t matter before Bitcoin. You could say, maybe Bitcoin is the first time that technology crashed into economics. You have energy, you have technology, you have math crashing into economics and now you couldn’t really be a competent economist without appreciating closed systems, energy efficiency, math, conservation of everything.

Robert Breedlove: Yeah. Another thing this calls to mind and maybe a unique way to look at Bitcoin: We can’t break the laws of thermodynamics. Those are the rules of the game within which we are operating in physical reality. And Bitcoin in a way maps onto that system very nicely, because it gives us an economic system in which we cannot break the laws. It perfectly respects and aligns itself with the laws of thermodynamics in the economic sphere. And another thing this made me think of was — this was a framework I got from you on inflation — was that, CPI is low, but everything you want is inflating rapidly. You could probably plot that on a spectrum—as, the things that are more energy intensive to create are inflating more rapidly, and the things that are created easily are tweaked and controlled and dumped into that CPI bucket.

Michael Saylor: Because the laws of thermodynamics apply even if you don’t wish they did. You can’t get out of the game.

Robert Breedlove: No free lunch in the universe.

Key Insights from Chapter 5

- It can be very expensive and cumbersome to move fiat currency across different jurisdictional domains—which inhibits fiat's original purpose, which was to increase the portability of money.

- Bitcoin can absorb superior competitive features from the marketplace. It can actually repair or improve itself in a way that makes it even stronger than its original form.

- Think of the SHA-256 algorithm as an encrypted vacuum sealing of the monetary energy stored in the Bitcoin network. It gives it this super-high resistance to impurities or uncertainty or nefarious actors (like central banks). It gets the uncertainty out of your money.

- Bitcoin is even more valuable than a standard insurance policy in that it has no counterparty risk. It's money that is maximally resistant to political risks at all scales. If you're holding cash in a form that can't be confiscated, inflated, stopped, then no matter what eventuality you encounter or what circumstances you encounter in dealing with the uncertainties of life, you have a pool of pure capital optionality with Bitcoin. And that cannot be said for any other asset. There is no other money that can provide you that degree of assurance.

- Sovereignty is simply the authority to act as one sees fit, the ability to conduct an action consistent with your purpose or aims. Bitcoin is the only money in history that maximizes our ability to do that.

- Territory is the most decisive factor in determining the outcome of any battle (Sun Tzu). And as Bitcoiners, the moral, intellectual, and philosophical high ground that we occupy is virtually unassailable.

- Bitcoin is a high-frequency store of value and a low-frequency medium of exchange. Every second you're holding Bitcoin, it's performing its function. Anyone that's saying Bitcoin doesn't have utility, they don't understand inflation. They don't understand store of value.

- Bitcoin is extensible, which means its protocol is adaptable. You can add other features to it, you can build businesses on top of it, you can connect APIs to it. There's a great degree of programmability that Bitcoin enables that something like gold just does not.

- Bitcoin is the first monetary network that maps perfectly onto thermodynamics.

Chapter 6 | Digital Gold: Harder, Smarter, Stronger, and Faster

> *"How about a world where one rich person makes loans to 10,000 middle class people every week? That's inconceivable with gold. That's not possible with stocks or bonds."*

Michael Saylor: We talked about Bitcoin as money, but sometimes money has a lot of elements — as store of value, medium of exchange, unit of measure — and it can be simplifying and clarifying if we just focus in on Bitcoin as an asset.

So Bitcoin is digital gold. And if we use that metaphor some people look at it in a very constrained sense and some people look at it more broadly, but when I say Bitcoin is digital gold, it's harder, smarter, stronger, and faster than traditional gold. There's a lot of depth behind it, and I don't know that it's fully appreciated.

Let's just start with harder: We talked a bit about how Bitcoin is harder because of its 21 million cap, and its stock to flow is exponentially going to infinity. And that's the easy part of harder because you're exactly comparing it to gold. And if gold has a stock to flow of 50 and Bitcoin has a stock to flow of infinity then it's harder. But there's another element of harder that we don't talk about much, and it's really Nicholas Taleb's Antifragile harder. Bitcoin is an antifragile element. I've used the metaphor of cyber hornets, but I think people think it's a cute metaphor, but I'm not really thinking it's a cute metaphor. What I'm meaning is, it's literally a swarm of cyber hornets that keep getting more powerful that you can't kill that are going to get harder and stronger and faster and they're going to eat you if you try to stop them — and that's really hard. People don't think about it like that. But I'll give you an example, or another metaphor from history: The Great Wall of China.

So the Great Wall of China was a defense, and it's a material defense — it's like a bunch of stones stacked up and it's very expensive to create it, and it was meant to keep the Mongols out. And it's a fragile defense, because it's literally a static stone structure and it didn't keep the Mongols out. Eventually they found a weak point in one of the gates, cracked through, and then they slaughtered all the Chinese, defeated the empire — took over. Now how is that similar to what Alexander the Great's father said — Philip II of Macedon — he's attributed this quote: "No citadel is impenetrable as long as it has a road wide enough for me to fit a donkey up it with a pot of gold on its back."

Robert Breedlove: Interesting.

Michael Saylor: And what he meant was: It doesn't matter how strong your defenses are if I can bribe the gatekeeper. The third example we have in modern history is the Maginot Line, where the French built this impenetrable defense against the Germans and the Germans simply went around it through Belgium — by cheating. They hacked it. They just ducked it.

If you think about the lessons of military history, one of the lessons is: The best defense is a good offense. And another way to look at it is that if you want a good defense, you need an active defense — a moving defense. You're a boxer in a ring and someone wants to hit you — your best defense is that you move quicker than them and you hit faster and harder than them while they're getting ready to hit you — and so they can't.

Robert Breedlove: Right, it has to be adaptive. The defense has to respond to the nature of the offense continually.

Michael Saylor: Continually. Continuous adaptation, absorption of your enemy's capabilities and reaction and you play them back at them. And so when we think about gold — I've affectionately deemed gold the dumb rock — it's a rock. It's not intelligent. It's not a life force. A coral reef is a life form. All forms of life — they're living. They're adaptive. If you kill the weak element, the ones you didn't kill get stronger.

Bitcoin is composed of a number of elements: there's software — the software is running on hardware. The hardware is running in facilities that are plugged into the firmament of some domain. There are people that are running that software, that are operating those facilities, that are building that hardware. And those people are also acting in the political domain and the economic domain, right? There are people in the Bitcoin ecosystem that are improving the Square wallet, they're improving the Binance exchange. So they're improving the back end server exchanges, they're improving the front end client software, they're improving the hardware wallets, they're improving the Bitcoin Core software, they're improving the ASICs that the miners are running, there are people negotiating with governments everywhere on Earth to get better electricity, there are people that are negotiating with politicians to get better regulatory treatment, there are people that are marketing Bitcoin and there are educators — yourself, on YouTube. There are people on Twitter, there are communicators, there are analysts that are creating analytical functions that are being used to drive and channel the network. They're all moving. They're all evolving.

There are strong exchanges getting stronger. There are weak exchanges getting weaker, right? We see that all around us. There are countries where Bitcoin mining is flourishing, there are countries where Bitcoin mining is under attack. Gold is a dumb rock — it is sitting on the floor. Gold is not going to move itself. Gold is not going to feel pain. It's not gonna feel pity, it's not gonna feel inspiration — it's just going to lie there and wait. And Bitcoin is a different thing. So the antifragility of Bitcoin comes from the fact that everybody in the ecosystem feels the same pain in the same way. If Bitcoin's price goes down, every Bitcoin HODLer feels it. If Bitcoin is hacked and goes to zero, every Bitcoiner loses their life's energy. We are a hive creature all integrated with one another—and when there is that pain it spreads very quickly through the entire ecosystem. The information flows rapidly.

So if we consider threats to Bitcoin — a lot of people, they always whine — I hear this whine, “Well what about the quantum computer? What about the quantum computer?” And I wanna say, yeah. And what if an asteroid hits the Earth and kills us all? Or, what about a pocket thermonuclear pencil that blows up the entire universe and what if your enemy gets it before you did? I mean there’s a lot of silly notions of that — What if somebody had the impossibly powerful weapon and they’re evil and they decide to use it on you?

You could look at it that way, but I think a more constructive way to look at it is: If someone comes up with a computer that runs twice as fast as the current computers. The most profitable use of the computer is Bitcoin mining. If someone comes up with something which mines, that generates SHA-256 crypto hashes faster than the current generation of Bitcoin mining equipment, the most lucrative thing you can do with it is plug it into the network and start to contribute more hash rate. And so, who do you think is gonna notice first when that thing happens? If someone comes up with a better — Well what happens when it gets so powerful that it can break SHA-256? Well, dudes, we’re just gonna go to SHA-512. Or we’re gonna go to the next thing — 1024. And anybody that ever studied computer science knows: We start with 16, go to 32, go to 64, go to 128, 256, 512, 1024, and so on and so forth — and the numbers get pretty fricking big. And then when they get done with that they flip to the next protocol, the next protocol.

Who do you think is gonna figure that out? The people that have $500 billion of risk to figure it out? Or someone that’s got no money at all but thinks they might just want to solve the problem just so they can crack the world? And it’s like, yeah, maybe. But it’s just as likely that some dude’s gonna come up with a revolver that shoots like photon bullets and they’re going to crash the United States government because they’re the first dude with a photon bullet revolver.

It’s an interesting idea — I’m a lot more persuaded that the Pentagon with their hundreds of billions of dollars is gonna come up with a photon bullet gun or the laser rifle or the whatever. So the entire network is channeling innovation. You come up with better hardware? It's gonna go into the miners. You come up with better client software? It's gonna go into the wallets. You come up with a better software, or a way to write software? Don't you think people in the Bitcoin Core community are gonna use that way to write better software? And if it's too dangerous, the community is gonna slow it down — otherwise, if there's a political threat, everybody that's adopted Bitcoin as a treasury reserve asset has a vested interest in dealing with the political threat.

And if they can't deal with it — if you are a creature with two arms and I wrap a tourniquet around one of your arms and I choke off the blood flow, the arm dies — use the other arm. When you're an octopus, you've got eight. When you're a hydra, you've got a million. So the entire creature's just going to move to a place where there is oxygen where it can live. And if it can fight, it will fight. And if it can't fight, it will morph itself into a different domain, and unless you figure out a way to kill it everywhere at the same time — utterly, down to the last node — you're probably not gonna kill it.

And it seems to me that nobody's really got a vested interest in doing that and if they did it's just not clear. Pick one government — one government attacks it, it becomes more useful to every other government. And they all can't agree on anything, ever. And even if they could agree on anything, ever, they still can't do it, right?

Robert Breedlove: Yeah. I love the quote from Philip II of Macedon about the road being wide enough for a donkey with a pot of gold. That relates to me to the famous quote from Charlie Munger: "Show me the incentive and I'll show you the outcome." And I think that is one of the key aspects of Bitcoin is that, even if you have this breakthrough — quantum computer, whatever it is — you're still incentivized to contribute that innovation to the network. That is the most lucrative, most energy-efficient strategy possible, even in that breakthrough environment. And to your point about cryptography — it's always been a cat and mouse game. It's always been kind of the defense, someone trying to crack it, and then the defense adapts. To your point it's a dynamic defense, and that's why *it's ongoing. That's why it's so relevant. And when you look at it through that lens, the most brilliant aspect of Bitcoin is that incentive schema, frankly — you're always incentivized to contribute to its longevity. Which seems to make it antifragile, to your point.*

Michael Saylor: The one thing in the Bitcoin ethos is they hate gatekeepers. So gatekeeper is a very powerful metaphor: The reason that the Great Wall of China fell was because there were gates and there were gatekeepers. And the prosperity of 100 million people could be destroyed by corrupting the one on the gate. As soon as the one gatekeeper opens the gate, the 100,000 person army comes through.

And herein you see the problem with hiding behind a wall, or hiding behind a wall with a gate in it. A much better idea would be: I have my own 100,000 person army and when that army comes I stand against them. And that's what the Romans knew. The Romans never sat and cowered behind a gate, they knew — and anybody in a war knows — you have to go and leave the citadel and you have to actively fight the army that would kill you, because if they manage to surround you and isolate you, no amount of defense is going to hold them back. They will find a way through, and that's the story of history.

Gatekeepers are your weakness. There's no gatekeepers in Bitcoin, and there's no gatekeepers in a decentralized proof of work network that's energy-intensive. That's one of the problems I think of like proof of stake networks or the — every time someone tries to come up with a "more efficient way to do it", well if you're going to secure a $1 trillion network with $1 billion of stuff staked, the problem with that is all I need is another $1 billion to bribe that person. Or half a billion. Half a billion bribes half that network and then I topple the entire trillion and I get it all, right?

So when you have anything where I'm playing games where I'm trying to use a small amount of energy — and money is energy, right? — a small amount of money, a small amount of energy. If you're trying to use a small amount of energy to secure a large amount of energy, you created a gatekeeper. And so you need to flip the pyramid, and that's why there's no shortcuts here: There's no such thing as a free lunch.

And it doesn't terribly bother me that Bitcoin channels a lot of energy. By the way — through encryption facilities. A miner is an encryption facility, it's generating SHA-256 hashes and you're gonna have to get through that hash wall, and the only way you're gonna get through it is you're gonna have to buy yourself 51% of all of that equipment on Earth, and then you're gonna have to harness 51% of all the energy, and then you're gonna have to attack and it still may not work.

Robert Breedlove: And you're still incentivized to honesty, even if you're executing a 51% attack.

Michael Saylor: And the joke, of course, is, if you really could buy 51% of the encryption equipment and harness 51% of the energy, you could have 51% of the block rewards and 51% of the transaction fees, and the value of owning half of the transaction fees is higher. When the thing's worth $10 trillion and the transaction fees are 1%, there's gonna be $100 billion a year worth of transaction fees, so it's $51 billion a year. If you capitalize that at a 20x multiple, it's $1 Trillion just to go along with the program.

So someone's gonna have to have $100 trillion and be really mad in order to want to spend $1 trillion to destroy other people — it just doesn't make any sense. Plus, you're not gonna get to attack the thing without a counterattack, right? Because at the point that you have 100 million people that all have all of their life's energy — all of their hopes, their aspirations and all of their economic security and physical security invested in the network — they're probably not gonna roll over and let you do it.

It's pretty hard for one evil genius by himself to plot a trillion-dollar war on a cyber network. How are you gonna gather the software and the hardware and the monetary energy and the political support by yourself? And if you're not by yourself, I'm reminded of the phrase, "Three can keep a secret if two of them are dead."

So when I think about Bitcoin I think, it's harder, but it's harder because it's an evolving herd creature, an evolving swarm creature. And it's evolving as fast as it can, and it feels pain. As Nicholas Taleb says in Antifragile, pain is a good thing. You can tell the difference between people that get it and people that don't get it. There's a group of people in life that run from pain, and they're attempting to anesthetize themselves and always isolate themselves from pain. And there's a group of people that run toward pain, or at least they embrace pain. Pain's a good thing. The more pain you feel, the faster your reflexes.

Put your hand on a hot stove, it moves quickly — you have reflexes. So that system is learning and pain is one of our number one information signals — maybe the most important information signal, in order to provide a living organism with vitality.

Robert Breedlove: That's right. Taleb says that, "Stressors and pain are indistinguishable from information at the organic level." That it's the only way that an organic system learns is by exposure to something that is resistive or conflictive to it. And even at a genetic level, when a virus invades a host organism and destroys its cells, there's always a few cells that survive. And those cells that survive actually take some of the DNA from the virus that wiped out 99% of their cellular comrades and incorporates that DNA into its own such that it is resistant now to that virus. So it's happening at an informational level even deep in biology. This isn't just something that happens kind of at the physical layer here. It happens really — it's quintessential to life, is I guess the point I'm getting at.

Michael Saylor: I agree. And I think that if it's not a living thing, it can't be hard, it can't be hardest. People think gold is the hardest thing because it's a rock, but it really isn't the hardest thing. The hardest thing is like the Mongolian horde if you're in its way. When the horde with 100,000 comes at you and they're terrorizing, blood-curdling with their missiles — and if you read the history of Genghis Khan, they would ride across the plain, they'd find a city, the city is surrounded by a moat, they would go upriver and divert the river, they would divert the river, choke the city, starve them of water for three days, and they'd ride underneath the gates in the riverbed. It's just horrifying, blood-curdling, living horde creature. And you want to attack it? It's moving. You could go toward it, they feign retreat, then they stop, they turn around, they throw you off balance, and they murder you. That's what happens when you're dealing with living creatures. Go into a hornet's nest, attack the hornets with your revolver — see what happens. The most horrifying thing, right? You have 1,000 hornets — you see these videos of 20 hornets taking apart 20,000 bees?

Robert Breedlove: Yeah I saw that on Twitter recently.

Michael Saylor: Frightening. Frightening. And they're all examples of that. There's bacterial examples, there are cancer examples, there are single-celled organism examples, there's invertebrate examples. When something is a decentralized, organic creature that is rapidly evolving and adapting, it becomes excessively antifragile, because every time you kill it or kill an element of it, the elements you don't kill get that much stronger.

Robert Breedlove: I'm reminded of a quote too when you're talking about the possibility of Bitcoin being disrupted or attacked from the outside, that you are attacking the life energy of the HODLers, if money is energy. And there's a Sun Tzu quote from the Art of War and it said that — I'll paraphrase — but basically you never want to attack an enemy whose back is against the wall. They have nothing to lose. Therefore, they will come at you with everything they've got. It's almost just an unstoppable force. And it seems to me that's what you're facing when you face Bitcoin. You're facing this swarm intelligence of individually self-interested but collectively organized life force that's defending their own life energy. So it seems just really hard to attack that. Almost as if you're attacking the honey in the wasp nest — they're gonna come at you with everything they've got.

Michael Saylor: Yeah, gatekeepers are like centralized banks or centralized regulators. Those are gatekeepers and political financial systems and gatekeepers on the wall of China on a wall are literally gatekeepers. And the result is: one person has the power to destroy the lives of a million by opening the gate. Because Bitcoin doesn't have a gate there's no gatekeeper, therefore there's no asymmetric payoff. It's not economically feasible to break through.

Now another big advantage in the antifragility of Bitcoin is this idea that I can run my own node, I can take my own keys off the network, I can choose to trust them with a custodian, or I can choose to take ownership of them myself, and I can choose for example to put a million dollars with a custodian that's going to run a bank or maybe lend it out or give me yield on it or protect it.

But then at the point that I lose confidence in them, I can shift all those assets to another trustee, or I can take physical possession of them myself, and because you have lots of different entities — individuals and corporations — choosing to do different things at all times, even though someone might entrust their keys to a bank, the security of the entire system is being protected by the someone that doesn't.

So the diversity with which people choose to engage and support the network actually makes it more antifragile. And the potential—for example: if I know that you are gonna take possession of your keys, and then if you set up an organization that's custodians that will hold my keys and charge me 10 basis points, then if another organization charges me 50 basis points, I would just say to them, I'm gonna shift to the 10 basis points. The competition drives their rate down or I move to 10 basis points.

If I hear that your exchange or your custodian is not secure or it might not be secure, or if I heard that there's a new multisig protocol that's more secure and I ask you and you don't support, I move my keys or I move my assets to the next wallet, the next technique. That means that you cannot afford to ignore me because you don't have a monopoly, right?

There's no regulator saying I have to leave my keys with the Breedlove Trust in order to get a tax deduction, or in order to qualify for a 501c3, 413, 509, etc. IRA, this or that. As you start to have pure competition, you get this ferocious evolution in all aspects of the ecosystem. And we saw it with miners for example: they started with CPUs, and they flipped to GPUs, and then they went to ASICs. And like every 3 years there's another generation.

The thing that really destroys the quantum computing argument is, if anything, Bitcoin has proven that people that believe in the Bitcoin network are going to pursue the highest performance special-purpose client and server hardware before everybody else in the world. If you look at the state of the art with Bitcoin wallets, they're more secure than any other mobile client device. And if you look at the state of the art with Bitcoin ASIC miners, they're higher performance than AWS or any device you'll find on a public cloud. And that's because Bitcoiners have skin in the game.

Back to one of Nicholas Taleb's books, it's like everybody that writes the software, runs the software, uses the software — everybody has skin in the game. If you go into this business you're gonna be obliterated if you fall behind. As a miner you can't make any money. As a wallet you can't sell anything. As a crypto bank exchange — look how fast the money moves out of one exchange to another this week. Everybody has skin in the game, nobody can ignore the pain, agency bias is very hard to come by, and you can expect every year forever, it's gonna get harder.

Now back to this issue of Apple and Amazon: Apple's the most valuable company on Earth because they can ship a software update to a billion people for a nickel. That's pretty powerful. Well with Bitcoin, you can rewrite the software that runs the nodes to make it smarter, right? You can rewrite the software or the firmware that works on the mining devices. And so the hardware cycle — what's the upgrade cycle on hardware—2 years maybe? 1, 2, 3 years? In that range. What's the upgrade cycle on software? Well it depends on whether it's the node or the client software — 1, 2, 3, years. In the greater scheme of things, it doesn't have to be super fast, you just have to compare it to the number of hardware and software upgrade cycles on gold.

Robert Breedlove: Zero.

Michael Saylor: In 5,000 years. In 5 billion years. Gold's pretty much 5 billion years — God made it, and left it that way. And there's a lesson to be learned from that, but the point is: it's not upgrading, it's not going to move. And if it doesn't move, invariably when human beings come in conflict with mountains, we move the mountains. We move the rivers, we chop up the granite. Go look at a citadel, it's literally moved the mountain. You might have to move it 2 tons at a time. Look at the pyramids. 5,000 years ago someone figured out how to move a mountain of rock. And that's 5,000—4,000 years ago. Are you gonna bet against 100,000 humans with their channeling gravity, channeling water, channeling whatever? Or are you gonna bet on the mountain? And the answer is: you're gonna bet on the human beings.

When you roll forward and you look at human beings versus animals, it's the same thing: us versus elephants, us versus lions, us versus nature — anything living. Generally, we figure out how to beat it because we channel energy better. So now we're talking about, what are we competing with here? Well, if we're competing with gold, it just seems like no contest whatsoever. Let's talk about smarter, stronger, and faster. Because harder is relevant to money, but smarter, stronger, and faster is the source of all tech value in the past 20 years — maybe in the past 2,000 years.

Let's talk about smarter. If Bitcoin is programmable money — it's a pure energy token and I can move it around — that means that I can write software on the clients and I can write software on the server that will actually move the money. And that means you can hold keys to 100 Bitcoin and then you can write software that will prove that you're credit-worthy, that you own the keys to 100 Bitcoin. Then you can actually use it to register and certify a transaction or information or a title channeled into that.

I can write a piece of software that will do what I want. What if a million people wanted to borrow money pledging Bitcoin? And what if another million people wanted to loan out money and generate interest? And what if I write a program where everybody on one side says, I'll pay 1, 2, 3, 4, 5, 8% interest, and everybody on the other side says, I'll make a loan assuming a collateral coverage of 4:1 in Bitcoin and I can mark it to market every hour. Okay?

You're offering me terms, you're offering the loan, I'm offering to borrow, we create a piece of software, it runs every day, every minute, every second. Robert, what if you had 10 Bitcoin and I scanned everybody — 100 million people — what if I scanned them today and I found someone willing to pay you 8% interest with a loan to value of 10% and you could mark it to market every minute? Good deal? Maybe. Maybe a good deal. Maybe you wouldn't be willing to loan it, but if you did — what if I went through all 100 million of 'em and I found someone that needed the money real bad and would pay you 27% interest? Smart. Smart.

What if I actually ran the algorithm every hour? What if I ran the algorithm every second? What if someone wants to borrow the money for a year and they'll pay you 10%? And what if they want to borrow for 30 years and they'll pay you 15%? What if they want to borrow for 1 year and they'll pay you 5%? What if they want to pay you overnight and they'll pay you x%? You could generate the yield curve with a piece of software, and then that software will connect all of the lenders with all of the borrowers on a server. It could be a centralized server. The logic doesn't have to be decentralized. In fact a centralized server runs a billion times faster than a proof of work network, so it's likely that the logic will be smart, and that's an idea.

Another idea is: put the intelligence on your iPhone and be able to talk to your iPhone and when I talk to my iPhone I can tell it to chop my money into 37 pieces and send it off — Send my money to my daughter and let her spend it on ice cream but not on Uber rides because I don't want her running off with her boyfriend somewhere. So I can condition the money using all manner of software, and I can turn on servers that will move my money around while I'm sleeping, and I can turn on clients that will handle my money.

And of course, if I walked into your bedroom and I picked up a gold bar and I walked out with it, the bar's not gonna complain — it's a dumb rock. On the other hand, if I took your Bitcoin and I wrapped your multi-factor authentication around it and your retina scan and your voice scan, I could geofence it and I could say, "Nobody can move a mobile device with this Bitcoin key outside 100 meters of where Robert lives." You could do all sorts of magical crazy things with software that you can't do in mechanics with machines.

So gold is in essence mechanical, and Bitcoin is virtual. And of course, it's tokenized, and that means it can get as smart as the computer can get smart, and eventually the computers can talk to other computers and that opens up all sorts of applications like credit ratings, authenticity, insurance.

Maybe I want to buy insurance for you but I want you to pledge the full value of my house and I want to know that you have the proof of the reserves to pay off the insurance, right? So you can create very interesting pieces of software with less risk or more transparency and more speed. We know it doesn't work with gold. It doesn't work with tokenized gold, because you've got the counterparty risk and you don't know whether the tokens actually are backed by real gold. It doesn't really work with other tokenized assets because stocks, bonds—they move too slow — and there's no API between the bonds and the tokens and if there was an API you end up with regulatory compliance issues and you're not delivering the physical instrument, ever. So that's just not likely to happen.

Robert Breedlove: It seems to me the general theme is we're betting on dynamism over the static state. Like the reason the human beings can overcome the mountain is because we have time and dynamism on our side and the mountain is static. It's not responding —

Michael Saylor: Dynamite, nitroglycerin — literally we can blow through the mountain by unchaining chemical energy.

Robert Breedlove: And the mountain is not mounting a defense of any kind. It's not adapting, it's not changing. I wonder if — so today, to kind of change the lanes a little bit, the US Treasury is considered the risk-free rate in the world, being that it bears low to no interest, but it presumably has no risk of default, or the lowest *risk of default. As you see us moving into more of a Bitcoin-denominated world, there have been talks of a risk-free rate for Bitcoin developing on the Lightning Network, such that you would fund these Lightning channels in time-locked contracts that the market would determine — anywhere from a minute to 30 years — and it would bear some interest rate. Do you think that's the direction the risk-free rate goes? Is that how we develop a Bitcoin yield curve? Or how do you see that evolving?*

Michael Saylor: Yeah. I think on a chain like Lightning above the base chain — by the way you could do it with a centralized or a decentralized solution, right? So Lightning is one solution, but you could do it on any exchange too if they chose to do it. There's a lot of ways to solve the problem.

The key idea is to create a free market yield curve. Rational people of their own volition would never willingly loan you for 30 years for 1.4% interest. There is no person that would think that's a good idea to do it. In a free market you would expect that you would see — 30-year rates would definitely be north of 10-year rates and north of 1-year rates — you would see things like 3, 4% short term, 5, 6% 10-year, 8% long term risk-free rates. And that's what they used to be.

When Volcker finished his work and he (kind of like) reset everything, for a while — the conventional wisdom back in the 80's was the long term risk-free rate was about 7–8% and then the risk premium was 4% and so the cost of capital was about 12% on making an investment. And that's how they came at it. And the interest rates typically mirrored that — you could get a savings account for 4–5% — anywhere from 4–8% on the yield curve. It's only state intervention that will deflect it below that. We could talk about that when we get to interest, but that's my thinking on that.

With regard to Bitcoin as an asset, we talked about why it's harder and why it's smarter — because you can program it — but it's also stronger. And the stronger is an important thing. Strong is about channeling energy with force and acceleration. I walk into the ring— we're gonna fight. If I punch slowly, I could be strong. But if I'm slow I don't win. The fighters that win have explosive force. You could be stronger than me but I can strike you before you strike me and I can knock you out. And so it's all about how explosively you can channel that energy.

And that's what an arrow is, or a sling, or a bullet. It's not the guy that's the biggest, strongest, fastest, that wins — it's the guy that puts a bullet in your head first that wins. And so we know that in military combat. If we think about stronger in the context of money, it's all about channeling monetary energy with force and acceleration.

So that means on a Saturday afternoon from a standing start, if you have $10 million, can you figure out where to put the $10 million to generate the most yield? Like, I need software. If I had $10 million in gold, I can't do anything with it on a Saturday afternoon. On the other hand, if I have $10 million of Bitcoin and I wrap it with a server, and if the server scans through 1,000 institutional counterparties and then I finance the bid, then I could chop my $10 million of Bitcoin into 10 pieces. I could send $1 million to Tokyo, $1 million to Beijing, Berlin, London. I could leave some of it outstanding for a 1-week loan, some for a 1-month loan, some for a 1-year loan. If their circumstances change — if you're pledging some other kind of collateral and I'm marking it to market and it changes, I could snatch back my assets.

And so the ability to channel the energy and move it where I need to move it intelligently, fast — what if I make you a loan of $1 million at 12% interest on Saturday, and then you come back to me 4 minutes later and you want another million and you'll pay me 12% more? And what if I like your credit? In the virtual world, in the Bitcoin world, I can just send another million.

What if you come back and say, okay, I'll take $10 million and give you 14% — I just need it for 3 days? 3 hours? That's fast, right? You can't do that with gold, ever. And if I had a bond or Apple stock or something like that on a Saturday, it's not moving. But realistically—I'm giving you an extreme example because it's kind of magical that your computer server is making you money by scanning the networks on a Saturday afternoon—well I don't have to be that extreme.

For example, if you have $100 million of Apple stock sitting with the wire house, a big bank — pick up the phone and tell them you found someone that's willing to give you 3% interest on the Apple stock — they would laugh at you. Ha ha ha, ha ha ha.

"Well we don't give interest on that."

"Well, could you just like loan it out to this other party that's willing to give me —"

"Ha ha ha, no."

When's the last time you got interest on your stock assets from the bank or the trustee that's holding the stock? Never. They don't do it. Okay so let's go through the thought experiment: You have $100 million in cash, you put it in a savings account back in 1988 that gives you 5% interest. You have $100 million and you put it into Apple stock and they give you a dividend. Okay good. Now the bank loans your Apple stock to someone that's gonna short the Apple stock, they short it, they sell it for $100 million, they take their cash, they put it into something yielding 5% interest, they keep the 5% — what you get? Nothing. Your Apple stock got hypothecated, you didn't get paid, and yet if you had $100 million in a market basket of stocks, why shouldn't you get a yield on it? The reason you don't get a yield on it is because the conventional financial industry doesn't have an incentive to do it nor a need to do it, they just have the power and you don't.

And what causes people to do that is competition. Sometimes, by the way, you don't see competition across regulatory domain because if I can get all the regulators in my country to agree that we won't do this then it's collusion and regulatory capture. And so when you have a cross-domain asset, it's less likely you'd have regulatory capture.

Ultimately, the strength of Bitcoin comes from the fact that an individual can self-custody and so if the individual can custody and they can do it in any of 200 different regulatory jurisdictions, they can find a jurisdiction where they will have the strongest money where they can generate the most yield and they can move to that jurisdiction. Whereas you're not gonna be able to convince that megabank to move to that jurisdiction and give you the most favorable terms, because they have no interest in that. So that's what stronger is all about.

And then faster comes back to the idea that we've dematerialized the gold, and because we've dematerialized the gold there's no mass. And if $E=MC^2$ and the M goes to zero, then we get the C^2 with very little E. Well, at least get the C. We can actually get to the speed of light because we have no mass, and when we think about $100 million of gold, back to my 3,000 lb metaphor, that's $250,000 every time you go around the Earth — so what if I want to send it around the Earth every week? Every day? Every hour?

Robert Breedlove: Can't even do it, yeah.

Michael Saylor: The amount of energy that you consume is just prohibitive. And so faster means that I can send it at the speed of light, and if I can send it at the speed of light, then there's all sorts of high-frequency transactions and microtransactions that make sense now that never made sense before.

How about a world where one rich person makes loans to 10,000 middle class people every week? That's inconceivable with gold. That's not possible with stocks or bonds. Because sometimes it's regulatory compliance issues, or it might be systems issues, right? But it is possible to imagine with Bitcoin — someone will just create the eBay of social finance.

What about one rich person that wants to lend money to 10 institutions? What about 10 institutions that want to work with each other? Ultimately, the things that make something slow, it might be physical mass like the weight of gold that makes it slow, it might be impedance, and the impedance might be a systems impedance. Like it's Friday afternoon and I cannot move money between 4pm on Friday and 9:30am on Monday. That's a systems impedance — it's not a law, it's just a custom. And if it's on Thanksgiving or Christmas, well then that doesn't work either.

And then there’s compliance: there is a law. There might be a regulation in a city, a state, or a country, that keeps the energy from moving. There might be a system that keeps the energy from moving. Or there just might be laws of conservation of mass that keeps the energy from moving. And so by moving out of that domain into the cyber domain, you get speed, which is somewhere in the order of 1,000 to 1 billion times faster. You put all those together: something that’s just continually getting — and by the way, it’s getting smarter every year, right? We’ve all seen on Google the computers beat humans on chess playing. The computers are beating us all on everything. The algorithms just keep getting smarter. Wouldn’t you like an algorithm that figured out how to make you money while you’re sleeping? Because I would. Programmed trading, right? The manifestation is programmed trading right now. And the issue is: consumers don’t have the power of programmed trading algorithms to protect their financial interest when their financial wealth (their monetary energy) is denominated in traditional assets because they can’t plug their assets into that new system.

Key Insights from Chapter 6

- Bitcoin is the hardest money that’s ever existed because it’s the most adaptive to entropy. It’s the most antifragile money we’ve ever had.

- A good defense doesn’t necessarily matter if there is a gate or a gatekeeper because you can bribe the gatekeeper. Bitcoin outcompetes the legacy system because the latter is built on gatekeepers. That’s what central banks and governments are: gatekeepers for the flows of economic vitality or monetary energy.

- The more monetary energy is stored on the Bitcoin network, the more incentivized all network participants are to defend it. Even outside of Bitcoin there are massive incentives to develop quantum resistant encryption in response to a quantum computing breakthrough.

- For the first time there's the possibility of facilitating a yield curve for Bitcoin.

- Gold will always suffer from the oracle problem. You'll always need to ultimately trust the custodian of the gold at the end of that chain of security features, whereas Bitcoin can be directly integrated with software security features.

- In the Bitcoin ecosystem, trade and settlement are now effectively the same thing — you can demand final settlement 24/7 to and from anywhere in the world.

- Bitcoin has collapsed money into Constitutionally protected speech.
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Chapter 7 | The Virtues of Strong Money

"Bitcoin as religion — the question comes naturally: What are its values? Truth, natural law, thermodynamics, self-reliance, honesty, fairness, technology, self-sovereignty. All of these things are embedded in the value system of true Bitcoiners, and they carry that into the world — in their word, in their deed, in their investments."

Robert Breedlove: Bitcoin is a monetary technology that we're able to deliver with much more force. Force in the physics definition being Mass x Acceleration, or Mass also being equal to Energy. So we're able to channel this energy in a very targeted and specified format at a very high speed and recalibrate almost instantaneously to always optimize our yield.

Michael Saylor: Doesn't it sound like a cruise missile?

Robert Breedlove: Yeah.

Michael Saylor: Or a projectile weapon? And we're back to the issue of what happened to the guys without the guns when the guys with the guns showed up? What happened to the guys without the airplanes when the guys with the airplanes showed up?

Robert Breedlove: So you're holding the proverbial high ground behind the wall of encrypted energy, but you can also send these financial — I guess you can call them financial weapons — out in a very targeted fashion based on what the market's signaling. Right? What's the demand for loans or what have you.

Michael Saylor: Yeah, in theory isn't a crypto bank the smartest, fastest, strongest financial entity in the world? Because it's going to be working while everybody's sleeping. Just like YouTube and Facebook and Apple networks — they're working while you're sleeping, 24/7/365. And to a certain extent you see that metaphorically if you just look at a crypto exchange and you look at the trading of Bitcoin and it's working while you're sleeping. And if you try to watch it you get exhausted. All traditional assets are constructed to trade from 9:30 in the morning till 4 in the evening because human beings need to watch over them and that's about the maximum endurance of a human being. And Bitcoin — when you go beyond that, you're going from 35 hours a week to 168 hours a week — it's 5x as much. People think, oh, it's just a little bit more. It's not a little bit more — it's 5x the bandwidth, just the trading. And then when you consider that it's trading every hour in every currency pair —

Robert Breedlove: Everywhere, yeah.

Michael Saylor: Everywhere, on a host of exchanges. This is extremely high bandwidth price discovery and transparency, right? The highest bandwidth price discovery, the highest bandwidth market of any security, of any asset ever. I mean, Bitcoin really is the perfected asset, or at least the apex asset in the financial jungle, because it's the creature that never sleeps, that's an octopus that's working everywhere all the time. The metaphor — Kraken — it's a pretty good metaphor because it's always going and it's never stopping.

And remember back to our previous discussion, I was saying in Jurassic Park the guy picks a fight with the little dinosaur and then he realizes that there's a hundred of those little dinosaurs and they don't sleep. And when you pick a fight with a swarm and the swarm doesn't sleep and you sleep, you realize you're gonna lose. You're doomed. There's that point you're gonna realize you're doomed because you can't keep up with a software creature with a million heads that is continuously working everywhere all the time.

And here's the thing you know: the thing that's working, that's dominating the market in Bitcoin all the time — it's the strongest version of the creature in that domain, not the weakest. The weak parts of the herd get culled out, they're being deprived of their capital, they're being squeezed out. You see it with the Mt. Gox disaster — anything that doesn't quite work. And so this asset class is a living asset class. And its strength comes from the fact that it's being developed — it's not constrained by the lowest common denominator, it's strengthened by the highest common denominator.

If there's somebody in the world, if there's an exchange in the world that can actually do this better, faster, stronger and it comes onto the Bitcoin network, they will set the price. Even if you're just HODLing, if you're just sitting there holding an asset — let's say I'm HODLing $100 million of Bitcoin and I'm wishing it would go up in value. And then let's say there's a million kazillion laws in the United States that prevent people from trading index futures on Bitcoin. But somebody in Malta is able to do it, or somebody in Singapore is able to launch that exchange.

So Singapore launches an exchange that matches billions of dollars of forward index buyers or option buyers to billions of dollars of call buyers, and that attracts tens of billions of dollars of capital in the market because it solves the yield curve problem.

Let's say hypothetically somebody on a centralized or decentralized or Lightning Network exchange solves the yield curve problem. If I could give you 8% risk-free yield on 10-year money, presumably you can take $100 billion and you can take that money and you can short fiat. Short the dollar, go long the Bitcoin, squeeze the 8% yield on $100 billion, make yourself $8 billion a year. If you can figure out how to do that in Singapore, you set up in Singapore, you do it in Singapore — what happens to the price of Bitcoin? Goes up. What happens to the HODLer sitting in Schenectady, New York — that never did anything, that didn't touch it, that doesn't know about it, that doesn't understand it — with their little 1 Bitcoin?

Robert Breedlove: It's been strengthened.

Michael Saylor: The network is getting stronger based on the highest common denominator. Anybody in the world with a better idea that plugs into the network is lifting everybody. And that's totally different than a conventional centralized traditional structure where one regulator might create one set of rules that constrain and hobble.

Robert Breedlove: Right. The asymmetry has been inverted from the Great Wall of China. If there's one failure at any point the whole thing is compromised, but now you've inverted that asymmetry such that if there's one benefit it benefits everyone.

Michael Saylor: And that's what it's like when you've got this swarm. If the herd is attacked by a predator and one of the creatures figures out how to kill the predator then the ones that couldn't figure it out die, the ones that do figure it out live, they procreate, pretty soon they can all kill the predator. They all have that immunity. That's herd immunity, whatever that might be. And that's the beauty of Bitcoin. I had this tweet—I put it out there, I said, "You know, lions get tired of chasing the antelope. Lions complain to the ranger. The ranger hobbles the antelope. Lions get fat, dumb, and happy. Antelope all die. Lions all die. Ranger blames it on the weather."

And that is a metaphor for a lot of stuff. Generally, it's a metaphor for how you destroy a crypto network by trying to make transaction fees lower. It's like, someone complained about transaction fees so we tried to change everything to drive the transaction fees down because somehow it's an abomination in the eyes of God that people get charged for transactions. The antelope ran too fast. Not fair. Slow them down. The transaction fees were too high. Not fair. Make them lower. We'll hobble them for you. Okay?

It's also a metaphor for interest rates. Interest rates are too high. I can't afford a loan. Make them go down. It's also a metaphor for competition. My competitor goes too fast. They're making it too difficult. Slow them down. Make the foreigners stop that. Make somebody do something. Right? It's somebody trying to regulate or manage something because they think it's a problem.

But at the end of the day, you're messing with Mother Nature. You're in a war with Nature—it's not gonna end well. At the end of the day lazy lions should die, and slow antelope get eaten, and fast, strong, healthy antelope procreate, and fast, strong, healthy lions feed, and the lions get better, and the antelope get better, and they live in an ecosystem. And if you remove the predator, all sorts of crazy, bad things happen.

Robert Breedlove: Right. You throw off the balance. This goes all the way back to your original point that there's no fair fight in the world. And that when human intervention tries to make that fight more fair, the intervention into a complex system throws off all these unintended consequences. That is one of the things that's gotten us into the situation that we're in today with low and negative interest rates. We've constantly tried to introduce an economic analgesic to paper over the business cycle or paper over losses. And we've distorted the natural price discovery function of the market, and so now we're in these totally asinine times with stock market at an all-time high and 40 million people unemployed. And this is one of Taleb's main theses is: Human beings have to strive to not intervene with Mother Nature. Mother Nature has been executing a certain strategy for a long time — we have to assume that it's being done for a reason, no matter what Science tells us.

Michael Saylor: You know, interest rate is the time-value of money, but if interest rate's the time-value of money, it's the time-value of energy. And if it's the time-value of energy, interest rate is the value of time. And maybe if we come back to thermodynamics — the rule of thermodynamics is: time cannot go backwards. Entropy is at work— time must move forward.

Trying to drive interest rates to zero or negative is a war on time. You're trying to make time go backwards. You're trying to make water flow uphill. You're trying to reverse gravity. You're trying to reverse the laws of thermodynamics. It's me saying to you, Robert, will you give me everything that you own — give it to me — you go without, I'm going to keep it, and when you die I'm going to give a third of it back to your heirs. And so the only way you would do that is if you thought the future of your life had negative value.

If you knew you were gonna be serial axe-murderer killer and you knew it in advance and you thought, "You know, I should be deprived of the future of my life because I'm really a liability to humanity." Maybe. But if you actually thought that the future had any value, you couldn't make that trade. No rational person would say, I'm gonna give you everything I have in return for half of it back when I'm dead. It's so moronic it's just insane.

And so when I'm trying to drive the interest rate to zero or negative, I'm actually trying to reverse time and make it run backwards. I'm first trying to stop it, and then trying to make it run backwards. I'm trying to reverse entropy. I'm trying to put the genie back in the bottle. Thermodynamics, the laws of physics, the laws of humanity, they all say: It can't be done, and only something catastrophic will ensue.

Robert Breedlove: Let me ask you. So — this is interesting — we have a legacy financial system that's trying to grind against or move countervailing to the laws of thermodynamics or the thermodynamic arrow of time, and here we have this new system introduced in the form of Bitcoin that nearly perfectly mirrors the laws of thermodynamics. It aligns itself with the thermodynamic arrow of time. It seems almost serendipitous that Bitcoin is released at this time when the system is starting to come apart in this decade. Do you think that this is almost like an autoimmune response by the global human hive mind?

Michael Saylor: Yeah. I do. Because look at what Satoshi put in the Genesis Block. It's pretty clear that Satoshi was troubled, was inspired, antagonized, irritated enough to do this. Humans solve problems, and so if I introduce a pathogen into your body, your body reacts. Living creatures react to protect themselves.

So if someone is sensitive to a given issue — and Satoshi was sensitive to financial integrity, and obviously had some decent sensitivity and awareness of Austrian economics and the perils of inflation and the moral hazard of bank bailouts — so that was a sensitive individual that tapped into a bunch of other sensitive nodes

— individuals — who shared that. And it's almost like: I pricked you with a needle and I introduced this little pathogen and then you swelled up with some inflammation. And the inflammation grew and festered and some antibodies build and the organism builds and the organism got bigger, and it got bigger, and it got bigger. And then it got fed. If we had zero inflation — if the US dollar monetary supply expanded by 0% for the past decade — how much passion would there be in the Bitcoin community compared to the amount of passion there is today?

Robert Breedlove: It'd be much less.

Michael Saylor: So we're back to this issue: It's a great technology, and it's inevitable. We're back to this issue of: Zoom was inevitable. YouTube was inevitable. Virtual business models were inevitable. Bitcoin was inevitable. But it sure did get accelerated by certain things that happened.

Bitcoin had 10 years of I would say a 7% forcing function with the dollar, and it got goosed a bit harder with Argentina and other developing world countries and, you know, the anxieties of Syria and the anxieties of Iraq and the anxiety in Africa and the anxiety in South America and the like, so those stressors — that goosed it. And the currency wars pushed it a bit harder. And then I think the pandemic crisis lit it on fire.

I could say Robert, like, there's no way I'm talking to you if there's no pandemic. But I don't think for a second that Bitcoin wouldn't be successful without me, and I don't think anybody's gonna stop it, but I do think there is an avalanche of energy — individuals and corporations — that got inspired and driven into this ecosystem because of the pandemic.

Again, back to war: Wars cause paradigm shifts. And this year there's a currency war, and that's a war on money as a store of value, right? And that war on money as a store of value creates massive dislocations in the bond market and the equity market, and there are consequences to everything. Ultimately, Bitcoin is an antifragile but scalable platform serving as a store of value. And so the best possible circumstance would be if the entire world plunged into a war where value was dissipating in every currency everywhere at a rapid rate. And I think it describes what we have today.

Robert Breedlove: I think that's exactly correct, and it's almost as if the Mongols breached the Great Wall of China — the Mongols being central banks that are basically robbing value from currency, and it's giving people more of an impetus to evaluate alternatives and — for those that see it — to retreat behind the wall of encrypted energy. You have to defend your life force, your energy, your money, from confiscation. And the most prevalent form of confiscation in the world today is inflation.

Michael Saylor: I guess that takes us to our next subject: it's that Bitcoin is a store of value. Bitcoin is an incredible store of value. I think that's its primary use case, or its killer value proposition now and probably for the next decade — maybe forever. The entire world is looking for a store of value right now, and you have to look across $250 trillion of assets. If we think that asset inflation is running north of 10% — and I think it is, I think it's pretty clear it is — and if you have capital, you have to choose between bonds, $80 trillion in sovereign debt or corporate debt or municipal debt, or mortgage-backed securities, or you gotta choose equities, tech equity, conventional equity, or you gotta go to precious metals, or you go to real property, real estate property.

And when you look at all of those things, the problem is half of all real estate is impaired because of the political response to COVID, and it's not likely that's gonna change in the next 10 years. We've probably got 10 years of uncertainty about real estate assets, especially commercial real estate assets. The other challenge with real estate is the taxes on it. It's illiquid, immobile, and highly taxed. Generally real estate is taxed annually everywhere, it's just a question of whether it's 20 basis points or 200 basis points. So that makes real estate a challenging store of value.

And that takes you to bonds: Bonds have worked as a store of value when the interest rates keep going lower. You can see that everybody that's been on the bond train and benefiting from it, they're screaming as loud as they can: They want negative interest rates. Because they're like — The secret to success is lower the interest rates 50 basis points a year or 100 basis points a year and it's a no-lose proposition — for them. But at this point it's getting kind of ridiculous and silly because when interest rates go negative everybody takes their money out of the bank, it creates bank runs, all the bank systems break. It's kind of morally bankrupt — I mean a lot of people, they don't really understand that they're being abused at 2% instead of 5%. But pretty much everybody can figure out that when you're being billed 1% of the money you have at a bank, you're being abused.

Robert Breedlove: Yeah. And again, if you look at money as an insurance policy on uncertainty and all of a sudden that policy has negative value — it's just asinine. It doesn't make any economic sense.

Michael Saylor: Yeah, so that doesn't work, and so where does that take us? It takes us to equity. We've got an equity bubble that's a very crowded trade, but the real issue with equity as a store of value is that the revenue gets taxed as sales tax, the cash flows get taxed as income tax, the expenses — the cost structure gets taxed as employee and payroll tax. Then the trade gets taxed — there's a tariff.

Then you have the existential threat or regulatory risk of onerous regulations. Pick up the paper and see maybe Australia's gonna bill Google every time they link to a newspaper article, and then if Google doesn't link the newspaper articles they're gonna get fined for not linking the newspaper articles. And as these things become more powerful, they become regulated utilities and politicians start to think that they can and should be regulated — and maybe with good cause.

If there's only one provider of information anywhere in the country then it definitely becomes a political issue. So that's the challenge with equities: they might work for 2, 3, 4 years, but they're valued as a multiple of cash flows. So if all of the liquidity of the civilization gets squeezed out of debt — when interest gets to zero — the government and the Fed owns all the debt. Okay, they bought me out — now I gotta put it in something else so I jump on equities so now equities double. But now the P/Es double or triple, and the revenue multiples triple.

What's the value of an equity? The value of equity should in theory be tangible assets on the balance sheet plus the sum of the discounted cash flows. And so people are using equities as store of value today. In fact, you could make the argument that equities are the most popular store of value for the majority with their Robinhood trading — they're all buying Apple, Amazon, Facebook, NASDAQ, SPDRs — everybody. Even though nobody thinks revenues are going up this year, nobody thinks earnings are going up this year, but equity values have doubled this year. Well that means that they're getting riskier.

If you contrast Bitcoin to equity, the problem is if the price of an equity goes up by a factor of 10, you've got more risk because it's delaminating from its underlying cash flows and its fundamentals. Because it is a centralized, regulated entity, and the only way that the cash flows are going to grow into that value is they keep raising the price. And if they're a monopoly and the raise the price, the regulators will react. If they don't raise the price, they can't grow into the multiple. And so you kind of have a chicken and the egg thing, right? And if there's any competition and their cash flows deteriorate, eventually you're trading at 200:1 or 200 P/E, and then any degree of disappointment causes massive volatility.

How are they different than Bitcoin? Well, Bitcoin's value proposition is the liquidity, it is the store of value — that is it. If you're going to function as money, you want to be a single-celled organism — like the algae of the ecosystem. The base layer of the ecosystem that's plankton or bacteria or single cells. You don't want to be a vertebrate. And these companies are vertebrates with a brain and a backbone, and that means — for example, WeChat or TikTok has a headquarters — and if it's in the wrong country, it gets its head chopped off. And Apple and Facebook, they're subject to a certain court, a certain country's jurisdiction. So that means they're foreigners somewhere else. They're vertebrate.

If you're a vertebrate, I can kill you with a needle. A human being I can figure out how to kill: there's a heart, I take a needle, I poke there — you're gone. You're very fragile. Hard to do that with a swarm of hornets. Hard to do that with all the plankton in the ocean. It's very difficult to do it with an amorphous decentralized invertebrate of some sort. So if you want a store of value, you don't want a company that's valued based upon the ability to engineer hyper-complicated products that have to keep getting upgraded. You want something which is simple, that can just keep things simple.

This is again where some of the crypto-enthusiasts, they keep wanting to tinker with a better blockchain, a better crypto. You know, they never saw an upgrade they didn't like so they just wanna keep revving it every year. Like it's the iPhone version 37. There's a fundamental difference, which is: if there's a bug in the iPhone version 37, everybody in the world's heart doesn't stop.

Robert Breedlove: Right. I think this analogy you're using with the simplicity of plankton being the base layer for the ecosystem is apt. I want to say plankton makes up the majority of the biomass on the planet by a pretty substantial margin because it's so simple and it's so efficient at converting solar energy into biological energy. And then it is the base layer for this multiplicity of layered ecosystem that we have in the world. And another thing I think is interesting is that you make a great point that equities are becoming more risky as they increase in price, because they're delaminating from their valuation fundamentals as you said. And Bitcoin's the opposite. Actually the more valuable Bitcoin is, the more secure it's network, the greater the liquidity, the more resistant it is to attack. So it's a very interesting counter-trade to equities as a store of value.

Michael Saylor: And of course, if you have an individual entity with an individual headquarters and a CEO, as it gets bigger it becomes a bigger target. But maybe everybody in your home country loves you. But what about everybody in every other country? So you don't want a head, and you don't want to be a target, and you don't want to be valued based on cash flows if you're going to be money. They're the right creature to be building a device or maybe creating an exchange or creating an application, because — there — I want the software to run a billion times faster, and so it's okay to have one company write software.

The question is: I want it to run a billion times faster, but do I need it to last for a thousand years? And the answer is: I could throw my phone away and Western Civilization will not end if I lose my phone or if you screw up my phone, right? It will not end. If I put all the energy of civilization into a crypto network, I can't afford for someone to, like, ship a buggy release. And again, what people forget is: if I put $1 billion into Bitcoin on January 1st, 2021 and I don't touch it for 100 years, the thing is working.

Truly insanely great technology is — okay we're back to Nicholas Taleb haunting our thoughts — *via negativa*, right? Add by taking away. Insanely great technology is when it does a thing without you doing a thing. You know, a junior technologist, they create gadgets. I have a mobile application that has 150 features and 150 buttons and then you click and there's a billion different things it can do depending upon the combination of the buttons and the features you click on—okay that's one thing.

How about another mobile app? You download a mobile app and everywhere you walk on Earth it kills all your enemies and gives you infinite food and water and protection and plays whatever music you wanted to hear around you without you touching it — hands-free. If I walk around and someone walks behind me and they do everything I want before I ask them to do it — without me opening my mouth — isn't that a heck of a lot better than a gadgety thing with features?

Robert Breedlove: Yeah. It's higher utility.

Michael Saylor: Well, it's Saint-Exupéry — the design's not done, it's not a perfect design until there's nothing left to remove. And so if I told you: Take all your money, put it into Bitcoin, and then you'll be rich and happy and prosperous for all of eternity without doing a single transaction — that's a lot better idea.

Robert Breedlove: Yeah, that's a great point.

Michael Saylor: I don't need more features and more gadgety things. I just need it to always work. And all that work is vapor around me.

Robert Breedlove: You're right, absolutely. Just to give the listeners a little bit of context for via negativa *— for those who haven't read Taleb: His canonical example in his book* The Black Swan *where you can see as many white swans as you want, but you can never prove by virtue of that evidence that* all *swans are white, but with a single sighting of a black swan you have* disproven *that all swans are white. So the moral of the story is that disconfirmation is more rigorous than confirmation. And I think that's getting to your point here is that it's getting stronger by taking away. Your factual base on which you're building your premise and strategy in the world is strengthened by disconfirmation more so than it is by confirmation.*

Michael Saylor: And I'll give you another example that we see all the time with Google: You go into Google, you ask the wrong question and misspell it and it gives you the right answer. That's a truly great piece of software: You ask the wrong question and it gives you the right answer because it knows which question you should've asked and it knows how you should've spelled it and said, "We're gonna answer *this* question for you instead because the odds that you really wanted to know *this* question answered is 99,000,000:1.

The odds that you really were asking a unique question that seems foolish and misspelling a popular name while you're asking it —that's 1 in a billion. So if you really wanted to ask the wrong question the wrong way you try it twice, but 99.999% of the time they give you the right answer to the wrong question and they do it because they built this very fault-tolerant, common-sense, rational interface.

So back to store of value, right? Bitcoin is an ideal store of value because it's got the ability to convey your energy not across 10,000 miles but across 10,000 days. 30 years into the future. 100 years into the future. Most people when they're investing in assets, they're taking this very short term view of like the next month, the next year, the next two years, and I find that if you're looking at a 3-year time frame, everything gets very noisy and complicated and there's all these debates.

But if you really want to end the debates, go out to 100 years and just take $100 million and go through the exercise of giving it to someone — your heir's heir's heir in 100 years. And then all of a sudden, all this noise drops away. Can I put it into real estate? No, it'll be taxed out of existence in 100 years. Can I put it in gold? No, 98% of it will be gone if 100% of it isn't gone in 100 years because we mine it to death. Can I put it into fiat? No it's gonna be inflated to death. Can I invest it in a company? No, name one company that's around today that was around 100 years ago that hasn't been diluted, recapitalized, etc. Can I put it in a stock index? Well, you're trusting a human being to rebalance the index over and over again. What stock index do you trust for the next 100 years? And by the way it's got counterparty risk at the nation state level. The nation might not be there, and 90% didn't make it. So what are you left with?

And the truth is, when you just do that thought experiment it's pretty obvious: You would put it in a crypto network, in a decentralized proof of work network if the adherents — the maximalists — were fanatic zealots about protecting the integrity of the network against meddlers who would screw it up. If the network is supported by those with a religious conviction to the network such that you could imagine 100 years from now there will still be people protecting the network — the phrase is, Keepers of the flame. Every great religion, every great institution has keepers of the flame, and there must be passion.

Do you believe in your religion? Do you believe strong enough that you'll flee persecution to continue to practice it? The United States was built on the foundation of separation of church and state, and it's a pretty important and interesting metaphor: People came here because they could practice their religion. That's why they came here. And they came here because their religion was above their government. They would not sacrifice whatever it is they believed in. If you look at every institution that lasts more than 100 years — name them: Harvard University, Cambridge, Oxford, the Catholic Church, Islamic sects, the Jewish faith — there's not that many.

There's churches or religious sects and then there's some educational institutions — and I see the educational institutions being shaken at their core this year. I mean literally you could've said for 500 years — Ivy Leagues, the elite universities — they're stalwart institutions. They have lost a huge amount of credibility this year. When you send all of your students home and you close the campus and people are studying virtually, people's affiliation to the bricks and mortar of the institution has been dramatically weakened unless they morph into a virtual institution.

The virtual institutions, by the way, have dramatically strengthened, right? Your affiliation with YouTube and Facebook and Apple TV and Bitcoin and Square Cash has dramatically strengthened. If it's virtual. And your affiliation with the bricks and mortar physical institution has been weakened. And those that will survive have dematerialized and virtualized and they've learned how to project their ethos in cyberspace.

So yeah, back to the store of value: You need people — human beings, flesh and blood people — that are going to keep the flame. And the flame of Bitcoin is the node and the mining rig. We're getting into Bitcoin as religion or as faith now, but imagine 1,000 years ago, I want to keep a religion alive — I have an altar in my home. Every wealthy person had a chapel in their home. If you look at religions, go to the Far East, there are altars in Buddhist, Shinto, other faiths.

The idea of an altar or a shrine or a cathedral or a church — these are structures where people go to worship, and the worshiping is the feeding of the flame. And oftentimes during the worshiping they're tithing and they're channeling 10% of their money as energy into these religions in order to keep them relevant. If you look at fantasy fiction, in fantasy where they have gods, the strength of the god is a function of the devoutness of the worshipers, of the followers. Such and such was worshiped as the god of the forest, and they are worshiped as the god of the forest and all of their adherents — all their acolytes — are feeding energy to the god of the forest. And when they're no longer worshiped, their energy goes away. If you lose your faithful, if they won't feed you with the fire of truth, with the energy that you need, your efficacy falls and you die.

And so how do you feed a fire, right? If it's a physical fire you have to throw wood on the fire. How do you feed a religion? You have to tithe — the Catholic Church, or any church — you have to give it money, and maybe you have to give it your life service. Onward Christian soldiers. I will fight for the cause. I will donate to the cause. That's how you feed a religious institution. How do you feed a crypto network? You gotta spin up facilities of encrypted energy that adhere to the protocol. Every time a miner comes onboard it's feeding the fire. Every time a node comes onboard it's gonna certify and validate. It's protecting. It's creating one more chain in the fault-tolerance structure.

And that's why a smaller crypto — a crypto that's a technical experiment, that's an application — isn't the good store of value. Because if the people are willing to fork it, if they're willing to fork it and abandon it in order to a new technology, then the flame dies. And so it's pretty clear that Bitcoin maximalists, they have a lot of the faith and the conviction of true believers in any religious faith for the past 2,000 years. And why wouldn't they? Because they share the same values. Their values are: Truth, Nature, Natural Laws, Laws of Physics, Laws of Thermodynamics, Math, Austrian Economics, No such thing as a free lunch, Self-Reliance, Honesty, Fairness, Technology Advances. They share those values.

They're taking their monetary energy and tokenizing it on a Bitcoin network with Bitcoin. Bitcoin is that single shared store of value. If money is energy and energy begets life and a crypto decentralized network gives you sovereignty, that's a path to immortality, and that means that everybody in the Bitcoin community that believes in this for the long term is engaged in the pursuit of immortal life. And pursuit of immortal life sounds like a religious mantra, I think. And I grew up in the Southern Baptist faith so I'm very familiar with the ideology of Christianity, and pursuit of immortal life is pretty paramount there.

Robert Breedlove: I did as well. That's interesting.

Michael Saylor: And if Bitcoin is a store of value for 100 years, then it is a technique through which you can project your values through time. So if you're not actually providing for a better life for yourself, you're providing for a better life for your family or your loved ones of your friends. Or perhaps your values are you want to support a dog park and you want to endow the dog park for 100 years. Or what if you want to endow an environment cause or Save the Seals or Save the Whales.

It doesn't really matter what the cause is — cure cancer, do this, go to the stars — when I die I want all of my wealth to be used to make education free for everyone forever. That actually is one of my values and I have a foundation, the Saylor Foundation, which gives away free education to hundreds of thousands of people. That's a value. Other people want to go to outer space. If I can channel my energy and put it into a network and that network can be used to fund and power an endowment that will do that thing, then that Bitcoin network or that crypto network is going to be my mechanism for achieving all of my hopes and aspirations from now to eternity.

Robert Breedlove: That's an amazing point. And I would even conceive of that as a mechanism almost of the afterlife. It's a way to carry your will beyond your own life. And the American mythologist Joseph Campbell, he described religion as a story that points toward *the transcendental mystery that we all experience but cannot articulate. And so all religious traditions, mythological traditions, are stories pointing toward a higher truth. And I find it interesting that Bitcoin has higher truth embedded in code. This 21 million number is quite literally transcendental: We can't touch it, we can't change it, we can't do anything about it. Every 10 minutes it's promulgating the most indisputable truth that we've ever had. It's true global consensus. So it's not just a metaphor I think to call Bitcoin religious, it quite actually is religious.*

Michael Saylor: If you worshiped Science, if you worshiped the Laws of Physics and the Laws of Thermodynamics and Mathematical clarity, then Bitcoin's your religion. And that's what takes us out of the range of simple asset debates. If your time horizon is 10 years we can debate Bitcoin this versus Apple stock that versus Amazon versus bonds versus whatever. And when your time horizon is 3 years, by the way, it all is just in the domain of macro traders and cute arbitragers and everyone wants to tell you about the Fibonacci this, triangle thing, and that's over-bought and this is over-sold and my head kind of explodes trying to figure that out, but the truth is I just don't care.

Being right in the next 2 years strikes me as being a bad idea because in order to be right as a trader in the near-term, I have to turn off the part of my brain that thinks about what's true and honest and morally hazardous or rational. You literally have to be like, I know it's stupid to go this way but since everybody else is gonna go this way I'm gonna do something stupid now because I think I'm less stupid but more stupid than they are stupid — it's just not a way to live.

Robert Breedlove: The market can stay irrational longer than you can stay solvent, right?

Michael Saylor: Yeah and the problem is if I lose trying to act stupid then I really was stupid. And if I win acting kind of irrational, crazy, then I don't respect myself. And meanwhile a much simpler idea is just figure out what's gonna go up by a factor of 100 or 1,000 and just go stand there and wait for entropy to take its course, and wait for gravity to take its effect, and let the water flow downhill, and let the fire burn, and don't dash around while the fire's burning everything — stand up and watch it.

So the solution there is: You move from 3 years to 10 years, and then you move from 10 years to 100 years. And at the 100-year time frame it's pretty clear what's an asset and what's a store of value. And it's just very obvious: (A) Bitcoin is a store of value if its believers have religious conviction in it. It becomes very simple. Now, if you step out to a 1,000-year time frame and you say, what's this thing gotta be in order to last 1,000 years? It better be the worship of Math and the Laws of Thermodynamics and the Laws of Physics and Einstein and Newton. That might make it. And if we just focus upon that and don't let all this other stuff get mired — I mean, that's reasonable.

People have been studying and honoring Math and Algebra for 2,000 years, they've been honoring Calculus for 400 years. You could say this is truly the adoption of economics as a science. It's that critical inflection point where economics went from being a political preference to being a science, and if you adopt it as a science then you actually get massive advantages technically. And if you reject it, I guess it's kind of like the guys in the Dark Ages and they could accept Calculus or they could reject Calculus.

Robert Breedlove: Right.

Michael Saylor: No bridges. If you reject Calculus — make a bridge, see what happens, have at it. There's a lot of stuff in life — if you look back at the guy that did more than anything — probably Isaac Newton is responsible for 90% of just about everything we have around us, and if you rejected *Principia Mathematica* because you just thought it was inconvenient, you probably rejected 90% of everything that we have today that we hold near and dear.

Robert Breedlove: And natural selection takes care of the rest. If you reject these truths that are uncovered, it's the end of your legacy in the long run.

Key Insights from Chapter 7

- Bitcoin has the highest bandwidth price discovery, transparency, and security of any asset in the world.

- Bitcoin's paradox: By being totally open to inspection, Bitcoin actually resists manipulation—or even emulation—in that everything about it is out in the open.

- Satoshi was clearly an individual sensitive to the negative socioeconomic consequences of central banking, and his own sensitivity was a response to the sensitivity of others: People who had been marginalized or victimized by the system throughout the ages. You could call that a form of inflammation in the socioeconomic super-organism. Thus, Satoshi was the cell of the organism that figured it out. He figured out the correct defensive response, launching it into the world in the right way, at the right time.

- Real estate is the easiest thing in the world to tax because you can't hide it, and that is the dominion on which the government projects its power.

- Bitcoin is kind of like a single-celled organism. You don't want it to be a vertebrate with all these complex features, because they open you up to more risk and attack surface. What you want is something very unidimensional, single-cell, that just holds value in a trust-minimized way across time and space. And that's what Bitcoin is. That is the big breakthrough.

- Satoshi took away all the indivisibility, all of the decomposability, all of the unrecognizability, all of the immovability, and all of the unpredictability from money. He removed all these negative aspects of money to deliver us the quasi-perfection that is Bitcoin. And that speaks again to this minimization of attack surface. You've reduced all of the features of money to strip it down to its bare bones. Bitcoin achieved attack surface minimization via monetary property optimization. This is *via negativa*: The principle that we know *what is wrong* with more clarity than *what is right*, and that knowledge grows by subtraction.

- The lesser feature set of Bitcoin is a feature itself, not a bug, because its whole value proposition is survivability. It cannot be stopped.

- Bitcoin as religion — the question comes naturally: What are its values? Truth, natural law, thermodynamics, self-reliance, honesty, fairness, technology, self-sovereignty — all of these things are embedded in the value system of true Bitcoiners, and they carry that into the world in their word, in their deed, in their investments.

Chapter 8 | Bitcoin and Immortal Sovereignty

"Bitcoin doesn't make sense as a medium of exchange right now. And maybe never. I tend to think never."

Robert Breedlove: I'm gonna bring up one more point on the religious aspect of Bitcoin that I've thought about. My religious tradition too was Southern Baptist — I was raised Southern Baptist — so I have Judeo-Christian underpinning in my thought. But the ancient idea in Genesis was that God was the force that confronts chaos with courage, truth, love, and converts that chaos into good, inhabitable order. So when we look at that through a thermodynamic lens, we know that the whole universe is everywhere and always trending towards increasing chaos and entropy, and that the only thing that works against that thermodynamic arrow of time is life, actually. Life converts entropy into order.

So in a way we could say that life or even the God-aspect of life is the anti-entropic principle that propagates through all of life. And I can't help but relate that and see Bitcoin in that lens, too, where we do have this system that's actually reaching deeper into chaos than anything else. The SHA-256 mining algorithm is quite literally finding a single atom in an entire universe — that is the mining race, if you will, is identifying the single atom in the universe — and with that reach into chaos it is forging the most profound order we've ever had. You can't argue with the Bitcoin blockchain. Whoever has those UTXOs has them. So it seems to me like there is a profound connection between truth and this principle of converting chaos into order, and the quote that took that home for me with G. K. Chesterton was, "A dead thing can go with the stream, only a living thing can swim against it." So Bitcoin in that sense is quite literally alive.

Michael Saylor: Bitcoin and beavers. The beaver is engineering the stream to create a dam, human beings are engineering systems.

Bitcoin is an engineered monetary energy system. It's the first well-engineered monetary system that we've built. Human beings through their intellect channel that energy, and within the Bitcoin system we extract order and we have a very efficient structure, and the price we pay is some disorder outside of our system. Some energy dissipation. But I'm not terribly concerned with that. The universe has got more energy than it needs and we're just dipping our fingers into 0.000 — with a million kazillion 0s after it — 0001%, right? So it can spare a little bit of energy. And when we create something beautiful — be it a railroad or Bitcoin network or social network or the like — we find a way to do something millions of times faster, smarter, stronger than before, and that drives forward the human race. So this is not the only great achievement of the human race, it just happens to be the most interesting achievement of the human race.

Robert Breedlove: Would you say that's the struggle then? We're sort of pushing back entropy? That's what civilization is, right? We're creating a larger bubble of order within the universal bubble of entropy.

Michael Saylor: I would say the highest calling of Man is to extract order from the chaos. And you could also say humans are divine when they engineer — through their intellect — a better habitat for themselves and those that they love. The human that bridges the chasm. The human that constructs the city. The human that creates the machine. And that's been going on for human history. That's the most honorable and admirable thing that human beings can do. And if we didn't do it, we'd be running around naked on the plains being chased by wolves and probably eaten by wolves. And that's that. So to engineer is divine. Look at a problem, engineer a solution. That's why we respect the engineers. You wanna talk about Bitcoin as a medium of exchange?

Robert Breedlove: Let's do it.

Michael Saylor: You know, a lot of people, they focus upon the whitepaper and we debate it ad infinitum, and I think that everybody that we agree with notes that peer-to-peer cash means a settlement network for a bearer instrument. And it's important that, in that regard, you be able to exchange assets.

It's, in my opinion, utterly unimportant that you be able to engage in high frequency small transactions on the Bitcoin network. And I think that it's dysfunctional to pursue that vision. I think that Roger Ver is just wrong. I mean it was a nice idea. Roger Ver thinks the magic in Bitcoin is to be able to pay for coffee with your phone. And it's a tragic error of insight — that was not the magic of Bitcoin then and it isn't now. The magic of Bitcoin is being able to catapult energy 100 years into the future and not lose the energy. And then the other magics are the magics that come from wrapping the Bitcoin with software to make it smarter, faster, and the stronger — that's the magic. The ability to do a minor transaction, it's trivially inconsequential.

Number one, only 1–2% of your wealth needs to be in these high frequency transactions. So a logical thing for anybody is just to keep 1, 2, 3% of their wealth in fiat. And then operate it on the Paypal network or the Visa network or the Apple Pay network or some centralized network. I think Saifedean's already made the point: A proof of work network is a billion times or a trillion times less efficient than the centralized version. So you're gonna get a billion times more speed and a billion times more economy by running the 1% of your life on a centralized network.

And if you didn't need to move it to off-chain — I don't even think it's a crypto, I just think off-chain's just a centralized network. You might as well just put it on Square or Apple, because nobody cares whether they go out of business in a year, 2 years or 5 years. And if you're in a country, the fact that those transactions are being monitored just means that, well, 1% of the value of your transactions are being monitored. If you want to do something illicit, you would do it 0.01% of the time on the Bitcoin network. So the one time in your life when you need to flee Iraq or Syria as a refugee, and that's when you need to do your on-chain transaction, well then you use Bitcoin and you can wait 30 minutes — you'll probably be okay.

By definition: If you move $1 billion once, it's a billion times more valuable than moving $1 once. And so the use case of moving $1 billion back and forth is valuable, and you can pay high transaction fees. And the use case of moving $1, $2, $3, $5, $10, or $50 or $100 is not valuable, because it falls exponentially or at least linearly with the amount of money being moved.

Now there's some other reasons why Bitcoin doesn't make sense as a medium of exchange right now. And maybe never. I tend to think never. I think what's gonna happen is — well we can debate semantics about what medium of exchange means — but I think Bitcoin is a high frequency store of value, low frequency settlement network. Or low frequency medium of exchange. Once a month is more than enough. Once a year? Maybe enough. Once in a lifetime? Might be enough.

If the rest of the ecosystem develops, there will be a point at which there's a crypto bank and Robert Breedlove has 200 Bitcoin in the account and he points the bank toward the Bitcoin and the Bitcoin's worth $1 million a Bitcoin and the bank will forward you fiat on credit at good rates to do whatever you want your entire life — the Bitcoin will never move. And the bank will do whatever, and if the bank fails, what do you care? The bank failed, not you. It doesn't matter, you've got 1%, you've got nothing at risk, they actually took all the risk. So you don't need to do things with a high frequency to create value other than protect the money.

Now there are some accounting and systems and tax reasons why it doesn't make sense as a medium of exchange. The obvious thing is: Every time you actually take revenue in Bitcoin, it would be a taxable event. You would mark the Bitcoin. So if you paid me in Bitcoin while Bitcoin is trading at $10,500, I have to mark it. You pay me $27 in Bitcoin, I have a ledger entry: $27 of Bitcoin with $10,400 basis. If your friend bought another $32 book, I have another ledger entry. If I sold a million things in Bitcoin, I'd have a million ledger entries with a million different basis prices. And then if I paid you a salary in Bitcoin, I would then have to figure out capital gain on it, and then I'd have to pay the tax.

So if I have $100 million in Bitcoin and then next year I pay $100 million in expenses in Bitcoin and Bitcoin went up by a factor of 2, I would actually generate $100 million in capital gains and I would owe $20-$30 million to the federal government within 12 months for the privilege of running Bitcoin, which runs a billion times slower than Apple Pay.

How many ways is this awful? First of all, it takes a decade to get your accounting systems to work. It's worthwhile to note: How does a global multinational work? I have 27 subsidiaries in 27 countries. I do business in about 10–15 different currencies: US, Euro, Yuan, Yen, Pesos, Real, Mexican Peso, Argentine Peso — you name it. We've got lots of currencies. By law we do that. By law we pay all the medical bills in those currencies, all the insurance, all the vendor relationships — took 10 years. It takes 10 years to set up a global multinational from start to finish.

It probably takes 5 years to get all your accounting systems to work right, $20, $30, $40 million to get it all working right. If I buy $500 million in Bitcoin, I put it in the treasury, I stare at it, I never do another transaction hopefully for the next decade. Until we're in an insolvency event, we're not moving it, right? Why would we? The truth is, we want to put it in the vault and throw away the key. Like, I would say to anybody, I don't want to be able to move it. I just want it to sit there. I'm gonna wait for it to appreciate. Then we sell stuff in Yuan and Yen and Euro and Dollar, and we put that into our accounts.

We sweep all the fiat into maybe one currency, into one thing: Dollar. We balance, and if we need to shuffle some stuff around in working capital accounts we put it back. If we don't need it for working capital we sweep it into the treasury. If it's in the treasury, we convert it to the treasury reserve asset — Bitcoin. It goes in, never comes out. You know, a vendor wants — they send me a bill, I pay them in a local currency. No vendor's gonna bill me in Bitcoin. People say — I've heard this — Well, don't your employees want to get paid in Bitcoin? Well the answer is, No, they don't.

But let's say the answer was yes. Let's say all of them were Bitcoin maximalists. Exactly how much Bitcoin do they want to get paid in? They have mortgages. They have bills. So the answer is, we pay them in the local currency, and if they want to buy Bitcoin, or if they want Bitcoin, they buy it.

Why would it be an awful mistake to pay them in Bitcoin? Because we would accelerate the tax bill by 100 years. And as Bitcoin is volatile, we will generate all sorts of tax obligations. The second thing is, we would increase our accounting complexity by an order of magnitude. And the third is, our employees will then have to sell the Bitcoin and generate their own taxable accounting complexity in order to get cash to pay their mortgage and pay for their kids' school or whatever they're doing, right? So you're creating lots of complexity here.

If I've got 2,000 employees and 200 of 'em want to own Bitcoin then I pay all 2,000 in Euros and Dollars and Pesos and Yen and Yuan and they go on their local exchange and they buy the Bitcoin and they HODL it. Because like, Do I want to HODL it? So the point is, when you're running a company with Bitcoin as an operating system, the logical thing to do is you just sweep your fiat into your treasury, convert your treasury into Bitcoin, let it sit there, and then let every — if a vendor wants to get paid in Bitcoin, I pay them in Dollars, they buy Bitcoin. If an employee wants to get paid in Bitcoin, I pay them in Dollars, they buy Bitcoin.

The whole ethos of Bitcoin is: Take responsibility for yourself. Don't treat adults like children. So it's patronizing in a way for me to say, well, Robert, you work for me, but I'm gonna help you by paying you in Bitcoin. It's like, you're a smart enough person — if you want to buy Bitcoin with it, here's the money. Buy Bitcoin with it — have at it. It's what you do. The money has a constant value at the time I've transferred it from me to you, and the thing that really cripples a cryptocurrency and a thing like Bitcoin as a medium of exchange is the tax treatment that the local regulatory domain makes. At the point that the United States IRS decided that this was going to be taxed as a capital gain on the moment of sale, it was utterly impossible for it to be a currency. It's just that simple. It's done.

For anybody to even maintain it — the only way you could think that you could use it as currency is if you're gonna run a black market, grey market operation with no accounting, no accountants, no lawyers on staff. And it's like, good luck with that. I mean, how many discos and bars got shut down for sales tax or income tax evasion, right? What was it, like — most of them.

Small businesses, they're eventually going to all run afoul of the regulators and get shut down. So it's just not practical from a systems point of view. It would take me a decade to rewire the systems. It's not practical from a complexity point of view. You've just made your accounting a hundred times as complex. And it's not practical from a tax point of view. You've just tripled the tax rate. And it's not practical from a whatever employee point of view because it's making everybody else's life more complicated too.

So that's why I think that Bitcoin really isn't a medium of exchange as much as it is just a low frequency settlement network. I'll make one more point, this is back to *via negativa*: The fact that you could settle peer-to-peer means that you might not have to for 100 years. The fact that I say to you, Robert you do as I say, or else. — if you have the ability to leave the country and depart with all of your wealth and I know you have the ability — right?

Robert Heinlein has a phrase, "An armed society is a polite society." I live in Miami Beach. Sometimes I run around, and people have guns in Miami. You can carry concealed carry permit, right? A young debutante that looks harmless could be walking around with a gun in her purse, and you think twice when it's 2 in the morning and you're in a bar and everybody's drunk, before you're pushy or rude. And then you think, Maybe I shouldn't be in a bar at 2 in the morning when people are drunk. And then you think, Maybe I shouldn't mouth off to someone that just cut me off on the highway. Somebody pulls over and they shoot you dead. And you think, I just oughta be very polite to everybody walking down the street, because every single person could have a gun concealed on their person. And maybe I'll be nice to people today. It's a deterrent, right?

So the ability to cash settle, or the ability to settle and exchange, that's the deterrent. Because the fact that I know you could means that I'm gonna treat you with respect. If I knew you couldn't, if I thought that you were trapped — if you were stuck in my monopoly and I had total power over you—then I start to take you for granted. And then I start to abuse that power. Because as Lord Acton said, "Absolute power corrupts absolutely."

So the power to do a thing is more important than the doing of the thing. And that's what the Bitcoin Cash guys, they miss. They miss the fact that: If I can move all the money when I need to, then I won't need to. And so yeah I need to be able to move it so it will hold $100 trillion in value, but no I don't need to move it now. And maybe I don't need to move it more than once, ever, just to make an example. It's like nuclear deterrence. I use it once, maybe I used it twice — I haven't used it since. But do you think that doesn't — you know, it's not effective? You think it doesn't matter to have it?

Robert Breedlove: Yeah, I think that's a great point. And this gets back to the difference between a free market and a monopoly. In a free market, you know that your customers have options. They can go to other providers, they could settle elsewhere, they could move their capital out. And because they have that optionality, you as the producer or provider are accountable to their preferences. You have to listen to the customers. You have to listen to their wishes and desires and make sure you're satisfying their wants. Whereas on the other end of that spectrum is the monopolist that, to your point, he just doesn't care about the preferences of his customers because his customers have no choice. They're stuck in his little fiefdom. It's the difference between an economic democracy and an economic dictatorship. So the fact that Bitcoin is just totally movable anywhere, anytime and you can't put the gate around it and monopolize it is what makes it such a game-changer.

Michael Saylor: Just because you can do a thing doesn't mean you should do a thing. That's basic Stoic philosophy. And there's the line that comes from the movie, the guy says, "The loudest man in the room is the weakest man in the room."

Robert Breedlove: American Gangster, I think.

Michael Saylor: That's it. American Gangster. The loudest man in the room is the weakest man in the room. If you know I can do it, if Robert Breedlove can teleport anywhere on Earth anytime with all of his energy and I can't stop it, that gives you a lot of power. I'm gonna give you a lot of respect, but that doesn't mean you want to flit around back and forth like a firefly. You know you can do it — now do something constructive with the knowledge that you can do it. It's quiet confidence.

And don't get distracted by shiny objects. Faster transaction rates, low transaction costs — they're shiny little lures at the end of the hook that the stupid fish grabs. It doesn't matter. It's the Cracker Jack box prize or something like that. If you look at the value on Earth, there's $250 trillion of value in assets. What's the value of being able to move money back and forth to buy coffee? And what's the likelihood you're gonna beat Apple in that race?

Robert Breedlove: Right. This is the picking up pennies in front of a steamroller type thing, right?

Michael Saylor: Basically. It's just not worth doing. And given the fact that it's a million times more expensive from a tax point of view, and given the fact that it's a billion times more expensive from a computing power point of view, the only reason to want to do it is regulatory arbitrage, i.e. grey market, black market operation. Not a very profitable pursuit of human ingenuity.

You know, you ask Meyer Lansky, what'd you learn in all your years as a gangster? He said, "It was so hard, I realized if I'd have to do it again I just would've gone straight." It's a lot easier to make money just being legitimate, so inventing some kind of technology to evade the regulators explicitly? It's like, they might let you invent something that will carry energy 100 years forward, because it seems like something that legitimate law-abiding citizens might want to do. They're not gonna let you invent something that allows you to evade taxes, and so: Why try so hard? It's just not worth the trouble. I'm reminded of the stories of Pablo Escobar, and he made the billions of dollars selling all the drugs and cocaine and they said at the end of his life he's sitting on the run and they're burning $100 bills in order to stay warm. It's like, yeah. Good luck with all that.

Robert Breedlove: Talk about money as energy.

Michael Saylor: I mean the vignette, just sitting burning stacks of $100 bills because no one will let you spend it. It's an example of just trying too hard—not worth the trouble. So let's talk about unit of measure. Bitcoin as a unit of measure: I think Bitcoin is the universal language of economic truth. I think it will evolve as a unit of monetary energy. It makes sense in that way. Before Bitcoin, if you go back 50 years, people talked about ounces of gold. So I think a Bitcoin will be like an ounce of gold — an immutable, self-sovereign unit of economic energy — it makes sense to think of it like that.

On the other hand, it will generally always be converted into local fiat in the political domicile in question for buying and selling things. And that's okay. There will be countries that allow stronger, better currencies. There will be countries that'll have weaker currencies, and they'll crash their currencies and they'll replace their currencies. And as long as you HODL Bitcoin in your account you'll be able to go into a new country.

You know, I've been to a country after they crashed their currency. We've all been to Germany, they crashed their currency, we bought the new currency. Then they got rid of the Mark and it was the Euro, we bought the Euro. So the right way to think of it is as just a standard unit of economic energy or monetary energy and not overthink it, anything more than that.

It can be interesting to say divide stock price by gold ounces — you can divide the cost of living by ounces of gold or by Bitcoin —

it’s interesting. It’s okay. It’ll give you a pretty good read on inflation. If 1 Bitcoin buys you a car in the US but 1 Bitcoin buys you a skyscraper somewhere else then you’ll have relative cost of living. People did that with the Dollar, right? They used to say you could live in Mexico on the beach for $187 a month and live like a king. There are relative strengths. I think Bitcoin will be like that. But I don’t think we need to stretch it any further than that. I think that’s enough.

I think ultimately, people focus too hard on medium of exchange, unit of measure, and they underestimate the value of Bitcoin as a store of value and as an asset. I think they undersell it and they don’t do it justice. Because it’s so much better than every other asset. And then I think the entire community of crypto, it’s too much engaged in debates between the various crypto networks, with each other.

I’ll post something — “Bitcoin is a great way to transmit energy into the future” — and then the Ripple people will start fighting, and the Bitcoin Cash people, someone gets triggered and takes offense. The way I see it is: Crypto is a $300 billion pond sitting on the beach next to a $300 trillion ocean of capital, of energy. That’s 99.9% of all energy of humanity, is sitting in alt-assets, not altcoins. Not cryptocurrency. So I see all the crypto enthusiasts working on all the altcoins and they’re just trying to drag down Bitcoin.

It reminds me of a bunch of crabs in a bucket and the bucket’s kind of hot or overheating and the one crab wants to crawl out and the other crabs just want to drag it back in instead of crawling on its back and getting out. It’s like, all the crabs just keep pulling each other back into the bucket. It’s totally self-sabotaging and it doesn’t matter, because what they ought to be thinking is, How do we get $30 trillion of the $300 trillion to flow into our pond? In fact, they ought to be thinking, we want to go from $300 billion to $3 trillion, and from $3 trillion to $30 trillion, and from $30 trillion to $300 trillion.

And the way you’re gonna get there is not by attacking each other. The way you’re gonna get there is by dividing the market and focusing. So if you want to be a cryptocurrency, well, currency means you have to have a stable asset value because the IRS is gonna tax you to death if it changes. So Tether can be a currency, and it cannot change. Do we need one? I don't know. It's a centralized stablecoin, and maybe we need a decentralized stablecoin. The market will decide that. But that's a thing in its own.

If you want to build a crypto application, there are applications of things that people might want that Bitcoin doesn't do like: total privacy, or maybe you want to issue title or provenance, or maybe you want to do a decentralized exchange or wrap something or implement insurance or lending. That's fine — DeFi. There are applications — have at it. You'd be better off to, say, Ethereum is 60% of the application market, Tether is 52% of the currency market — cryptocurrency and crypto app — and then Bitcoin is 95% of the crypto asset market. And if you believe in another type of asset, I don't think that it's impossible to ever come up with one, I just think that right now the only clearly successful crypto asset is Bitcoin.

Robert Breedlove: Right, agreed.

Michael Saylor: And all the crypto enthusiasts would be better just to accept the success of Bitcoin, because it's the gateway drug for all the alt-asset holders to move into crypto, and they really ought to be encouraging the next $3 trillion to come and the number one way to do that is not to relentlessly, ruthlessly attack and sabotage and undermine Bitcoin, but rather to celebrate its success, because as it gets bigger, the opportunities for other crypto networks to interact with it will also get bigger.

Robert Breedlove: Absolutely. And to people that don't understand this space, I like to articulate it that Bitcoin is digital gold. It has essentially already run and won that race. And that for the same economic and game-theoretic reasons there's only one analog of gold, there's only ever likely to be one digital gold. So it's almost like you just have to accept that. And then I describe the rest of the crypto-asset universe as liquid venture capital subjected to *little, if any, due diligence. Some of them may succeed meaningful in markets orthogonal to money, but it's very high risk. And the fact that anyone can spin up one of these assets in 15 minutes on the Ethereum blockchain, there's a very low barrier to entry, very low diligence, so invest accordingly. I'd suggest no one — if you're gonna have a crypto-asset portion of your portfolio, I think the highest risk position you could ever be in is 80:20 Bitcoin to everything else. And the conservative position would be 100% Bitcoin, 0% everything else.*

Michael Saylor: I would agree with you. I think we see it the same way. And I guess it's a good segue just to general crypto theory: the emergence of magic in cyberspace. What's a crypto network, what's a digital network? How do I view it? And you tell me if I'm right or wrong. I'm a newcomer — you've been around for a while.

My view is that a crypto network is a software application running on a decentralized network with a consensus mechanism, ideally proof of work, arguably maybe proof of stake or some other consensus mechanism that theorists would agree promotes decentralization and antifragility. And a digital network is a centralized network. It's a Facebook or Twitter or Google or Apple or whatever—a more conventional client-server network.

So as we look in the future of cyberspace we're gonna see crypto networks and digital networks. Why would you want a crypto network? You want a crypto network for immortal sovereignty and antifragility. We know Bitcoin is doing that right now. If you want to convey humanity across time and space — especially across time — maybe across time and domain, domain might be even better than space.

You know, crypto networks aren't necessarily faster than digital networks for moving shit through space, but they're better at moving it across domains, and the reason for that is that sometimes you have lowest common denominator regulatory or compliance constraints that cripple the innovation in a digital network. Apple Computer will never do certain things because they're subject to the

US law, even though they could do them in 98 countries. And so a crypto network may in fact develop faster across regulatory domains if digital networks are crippled.

There are obvious trade-offs between digital and crypto, and I think about it as: Trust, Security, and Duration are attributes of crypto. Economy, Performance, Functionality, and Compliance are attributes of digital. If I want to do it cheap, it's a billion times cheaper on a digital network. If I want performance, it's a million times faster on a digital network. If I want functionality, it's likely much more functional. There's nobody that would dispute that Apple Pay or Apple or Google, they're beautiful, functional products. Apple or Google Maps. Trying to do Google Maps on a decentralized crypto network would probably be a disaster.

And then compliance — digital networks are better at compliance. Sometimes we regret their compliance. They shut down stuff and they censor things. But they're better at compliance. So the very things that make crypto attractive — Trust, Security, Duration — are purchased at the price of giving up Economy, Performance, Functionality and Compliance. The conclusion is: Some stuff makes better sense to be doing digitally, and other stuff maybe with crypto. It's early days, and the only obvious conclusion we both agree on is Bitcoin is a successful crypto network as a store of value. We know that. Everything else is an experiment and when it hits $100 billion it's not an experiment anymore.

We can debate whether it's an experiment at $50 billion, but I would say at $100 billion it's a thing, and it's pretty clear. And it might be a thing we all know and recognize below that, but you'd need a lot more education to determine that.

Robert Breedlove: I think — just to throw in my 2 cents here — it is distinguishing between a decentralized network and centralized. And this reminds me of an old quote, "If you want to go fast, go alone. If you want to go far, go together." So the centralized solution is kind of like going alone, because it's just according to one single plan, it can innovate very quickly and move in a certain direction very quickly. Decision-making is concentrated. Whereas the *decentralized solution has more of a dynamic equilibrium that lets it persist further across time.*

So the centralized solution is going to be really compliant, really fast, really innovative, but the decentralized solution is really only useful when you need censorship resistance. You need to know that it will persist across the maximum duration of time, and that no individual group or individual for that matter can shut it down or manipulate it. So Bitcoin is at the extreme end of that decentralized end of the spectrum, whereas something like Apple would be at the other end.

Michael Saylor: Yeah you could use a metaphor like an Aspen forest or some massive biomass versus an animal. A vertebrate versus invertebrates. Different life — plants versus animals. They're different creatures. And they have different life spans. You want to live for 2,000 years? Or do you want to live for 80 years? One of them moves faster than the other, but one of them is more vulnerable than the other.

I think from a human point of view, the real question for humans is: Do you want to put your trust in Man, trust in the organization, trust in a company, trust in a government? And/or: Do you want to put your trust in a crypto network? And we're gonna solve our problems with a segmentation, mixture of these trusts. We're probably not gonna put the functionality into Bitcoin—in a total decentralized proof of work network — that will do what Amazon does or Uber does or Pandora or YouTube does. Probably not. But do we really need to? We probably don't need to.

So when we think about the crypto industry segmentation, I think you really want to divide and conquer. A crypto network is a life form. And I think that's different. I mean there's a genetic encoding in a crypto network, and it's very different than solving the problem with a human being or solving the problem even with a digital network. When you design the protocol, Bitcoin, and you let it go, it becomes a life form and it multiplies and all these nodes spread and all these miners spread. And you can't easily pull the genie back in the bottle and re-engineer the genetics of it when you've released it into the wild.

In fact, you would almost say that once plankton is spread out to dominate all the ecosystem — if God had a better idea — he created that and made those minnows. Or he created bacteria — he created a new species. You have a better idea? Create something different, and if that something different is in the same energy ecosystem as the plankton it's not gonna make it. If you take a bird, the bird can coexist with a deer, but it's hard to create two apex predator birds and put them in the same space at the same time—one of them eats the other one. One apex predator creature. And that's the same thing with crypto, I think.

You gotta think of it as an evolution of a new life force in cyberspace, and the genetic code is of paramount importance. So if you really hope to be successful — and this is my opinion — when you release the privacy coin, you need to say, This is for this thing, and this is what it is, and we're going to get it just right and we're gonna let it breathe and let it live and we're not gonna mess with it or monkey with it. And we're not gonna decide that we wanted the monkey to have wings and also have gills and swim. And if I'm watching TV and I see these cool antler things and I want to put the antlers on it.

You can't be a technology-tinkering enthusiast with a crypto. You have to have more of the mindset of, it's the crazy guy in the lab coat in the laboratory creating a new kind of life, virus, creature. It's like you get one shot, and it's at a lab — and you're done. You can't get it back. And so you should be aware of just how important it is. It's a protocol. Life is a strand of DNA and that DNA is going to go and it's going to do whatever it's going to do. You're creating life. And so how devastating is it for God to say, I created human beings. I made a mistake. I'm gonna recall them and then tinker and then do a human being 2.0. It's a problem, right?

In the Bible, the biblical analogy is the flood where everybody just gets killed. Or some crazy stuff in the Garden of Eden. But with things like Ethereum, right? You've got Ethereum 1.0 and now you try to do Ethereum 2.0 — you'd almost be better off just to do something different as opposed to like retrofit something, because it's going back and trying to force-fit the DNA. You could do it with a digital network. It's okay if you're running on a single server and you can throw a switch and everybody switches with it. But the whole point of a decentralized life form is that they're all different and they all get to make their own decisions. And if you can order them to do something, it doesn't feel that decentralized anymore, does it?

Robert Breedlove: It's not decentralized, exactly.

Michael Saylor: So with regard to this, I look at the crypto industry and I think, The logical segmentation is: Assets versus Applications versus Currencies. It is the purpose of the crypto network to be an asset, and Bitcoin clearly is. And then Bitcoin Cash and Bitcoin Satoshi's Vision and the rest are trying to be. It seems pretty clear that Bitcoin is 93–94% of all the assets and the other ones are probably going to zero. And the only way you don't go to zero is: You have to be differentiated in a way that the marketplace appreciates you.

It's a species, right? I can't be almost like a bunny rabbit, I need to be a wolf. I need to be different than a bunny rabbit. And it's an important caveat: Differentiated in a way that the marketplace appreciates. In other words: Differentiated in a way such that Nature gives you an advantage in natural selection. So taking Bitcoin, forking it, and making it do transactions 10 times faster isn't good enough because you're a million times slower than the other way to do transactions. So that's just not a genetic mutation that works. It's an example of a faulty genetic mutation that kills the carrier — it's an extinction-level mutation.

Genetic mutations happen all the time in Nature. Most of them are mistakes. But some of them might work. If I can stand up on two feet and if my eyes start to work 10 times better than your eyes and I can fashion missiles then maybe that's gonna work for me. And maybe I will procreate. So it's not impossible to have a genetic mutation on Bitcoin that would be naturally advantageous. It's not impossible. It's just, the ones you look at right now, like, what do you have? I mean, transaction speed? That's really no good. Privacy? It's a regulatory threat and so therefore it's probably a nice thing but it's a bullseye in its forehead which means that when the regulators go to shut down crypto exchanges the first thing they'll do is pass a law against that, right? And so, give me an example of a mutation on Bitcoin that would make it a distinct asset as a store of value that would deserve to have its own life?

Robert Breedlove: I've thought about this a lot and written about it, and I have yet to identify one. Like Bitcoin sort of perfects the main properties of money that you need, and then the added caveat to that is that even if a mutation or innovation occurred such that the market identified it as useful and you saw this reflected in its market cap, Bitcoin also has this capacity to absorb feature set from competitors. So if something becomes market proven, Bitcoin can assimilate that into its existing UTXO set.

Michael Saylor: Just as living organisms do. Just as companies do. You get a tough competitor: That which does not kill me makes me stronger.

Robert Breedlove: Right. Instagram absorbing Snapchat's Story feature, right? Totally destroyed Snapchat.

Michael Saylor: So we don't see another one right now. It's not impossible to conceptualize that maybe there would be one, but right now it looks like Bitcoin is 94% of the crypto-asset market. You can't include Ethereum because although it's proof of work now it's going to proof of stake, and they don't even have an intellectual commitment to be proof of work. And proof of stake is—it's not clear to me that proof of stake is intellectually defensible over the long term. It's not proven. It seems like it may be defective as an idea, right?

It's like an example — we're back to this issue: It's like I want to hobble the antelopes. It's like somebody complained we're using too much energy. It's almost like the environmentalists took over and they've decided that they're just mad that someone's using too much energy, so they want to pass a law to stop it. But it's a law against Mother Nature, and it's brooding to hobble the antelope.

So the 4–5% of crypto-assets that are left — the Satoshi's Vision, Bitcoin Cash — none of them are differentiated enough that they would seem to have any future in a Darwinian world that I can see. So they're destined to go to zero.

Then we go to cryptocurrencies. That's a simple one — stablecoins. People go ballistic if you say they're cryptocurrencies because everybody wants Bitcoin to be a cryptocurrency. But if the definition of currency is money that you can legally spend in a country without incurring taxes, Bitcoin is not a currency, but Tether is. Because—it's kind of simple — read the IRS tax code. You'll find, most places on Earth, they're gonna tax it as an asset. And there's nothing wrong with that. In fact, that's better.

It's actually a mistake: Why would you want to create a short-term currency that's a billion times more expensive to implement computationally than the other way? It's like, the only reason you'd want to do something a billion times more expensive is for immortality — immortal sovereignty. That's a good reason. But you've got that. If you had that on 99% of your assets 99.9% of the time, then you're good. Then you could afford to have a little bit of risk in order to have a billion times faster performance on a digital network. So that's currencies.

The only reason for a cryptocurrency is for regulatory arbitrage or maybe innovation due to decentralization. If you can make the argument that we're gonna innovate faster because we're governed by the laws of Singapore and people in Malta can use a Singaporean cryptocurrency and it's not technically illegal, it's just a hundred times better because we're able to tap into that regulatory arbitrage opportunity, you could call it a free market, then I think, Okay. Fine. Then I think there's a place for a cryptocurrency.

Robert Breedlove: I guess you could add the speed factor too. That Tether can be moved 24/7, whereas US Dollars would face *certain restrictions in terms of time or jurisdiction — capital controls as well.*

Michael Saylor: On the other hand you can move Apple Pay 24/7, you can move Square Cash 24/7. So there's no reason why you couldn't have run a digital network 24/7. By the way, Binance is a digital network, and you can trade 24/7 on Binance. So Binance could simply move it — if I'm moving money on an exchange or between exchanges, I could do that on a digital network. I don't need a crypto network to do it. So whether or not people use a digitally-based stablecoin or a crypto stablecoin, it's a function of regulatory arbitrage. I mean there are privacy issues that people have. And then there are trust issues.

Robert Breedlove: I guess you could say Tether is a digital network in this framework. It's not decentralized. You trust a counterparty.

Michael Saylor: Because there's a central authority that can actually redeem, right? And so you could almost argue that if Tether is the king of stablecoins, it's running on a digital network, not a crypto network. Like, DAI — the irony is that DAI is a truly decentralized one — but people don't like it because it's inefficient and it doesn't quite peg the value. But you could create a cryptocurrency as a stablecoin using an oracle, using some information feed, and using Bitcoin as the underlying collateral and create a synthetic Dollar and a synthetic Euro. You could. And will we need it? It depends.

I mean that takes us to crypto applications and digital applications. I mean Binance and Coinbase and Huobi and the like, they're digital exchanges. They're digital networks for exchange. And then you've got the decentralized exchanges, right? Uniswap, SushiSwap. There's a very romantic notion, like, decentralized exchanges will beat digital exchanges. I'm not so sure. We'll see.

The point that I made in one of my tweets is: So I buy $400 million worth of Bitcoin. I do it in 7 days. For 7 days, I have some exposure on a digital exchange. I move it to cold storage for the next 10 years or 20 years. So for the next 7,500 days it's off the digital exchange. So 99.9% or more of the time I'm not exposed to the risk of a digital exchange.

To everyone that says, "digital exchanges are risky," well, they're risky in the same way that walking across the street is risky, Robert. But you do it. In theory when you cross the crosswalk, if the red light failed, couldn't you get slammed into with a truck and killed? Okay, so are you gonna take the position that we're gonna have to build flyovers on every crosswalk in the world? Eh, pretty expensive. You know?

So what will happen there? I don't know, but it seems to me that Binance is doing just fine as a centralized exchange. And it seems that Square works fine as a digital client application. And if Apple Computer announced that you could send Bitcoin or cash via Apple Pay or iMessage tomorrow, I think a lot of people would be pretty happy to do that. No one would be complaining about it.

It really comes down to this issue of Economy — it's a million times cheaper, Performance — it's a million times faster, Functionality — it's a million times more functional, and Compliance — they're probably more compliant in the country you're in. So if you can do it on a digital network, you probably will. When you think about crypto, you really have to ask this question: What is so important that I want immortal sovereignty? What is so important? And certainly my life force. If you said to me, Michael, I'm gonna take all of your energy and I'm going to protect it and I'm going to fulfill your wishes from now to 1,000 years from now — I think that's worth it. That's a religion. That's a crypto.

You know, I have a park. I'd like for the park to be trimmed and the trees to be kept nice and I'd like for the park to be beautiful for the next 1,000 years and it'll cost $1 million a year. And so if I put $50 million into endowment and it generates 2% yield, then I could reasonably say to Robert: Robert, why don't you take my $50 million endowment and put it into Bitcoin. And then it becomes the question of: How am I gonna see my wishes through for 1,000 years?

Well, the current way you would do it is you would create an institution like the Rockefeller Institution or the Hughes Institute. And you trust human beings. What we're really talking about is: We're talking about the crypto-asset network of Bitcoin taking the base layer and holding all the monetary energy for 1,000 years. Or 100 years, if you think 1,000 years is crazy — we'll go 100 years. And then we're talking about a digital network. Maybe it's gardners.com or Amazon will probably do it because they'll probably do everything by the time we get through the decade. There will be nothing that they're not doing.

Let me make it easier: Let's just say I want to send flowers to everyone in my family on their birthday for the next 100 years with a little note from me telling them how much I love them — that's a simpler thing. I endow an account with Bitcoin, I create an Amazon account, I wire it up with flowers, I create some kind of mechanism so that when children are born they get added to the family unit, and the thing drips out a little bit of monetary energy in Bitcoin, converts to fiat, buys the flowers for $187,000 a dozen roses in the year 2047, and it ships them for the person, and I die happy, and for the next 100 or 1,000 years, flowers get delivered to people in the Saylor family line because that was my last will and testament.

How would I do it if I was John D. Rockefeller 100 years ago? I would have a crapload of assets, probably a lot of gold and stock — and he owned every oil company and they're still around — and then you would put some of that into a foundation. A non-profit foundation, because you need tax-free, tax-advantaged. That's why you need to go non-profit. By the way, all of your capital gains become tax free, tax-advantaged, and all your income becomes tax-advantaged, right? So that's a good idea. And then you have a board of governance and you have 5 advisors and they appoint a money manager and they appoint a business manager and when one of them retires they have a selection committee process that puts someone else. We're probably on the 7th generation of advisors to the Rockefeller Institute.

By the way, John D. Rockefeller created the University of Chicago. It's still running. Rockefeller created a lot of stuff that's still running. Rockefeller University. So I can do it the old-fashioned way, but to do it well you're probably talking about $100 million of assets and a $5 million a year operating budget. You get down to $1–2 million it's not really — $5 million a year is the minimum. Your best bet is you have to become a university, a religion, a church, or a non-profit, a Hughes Institute. So that's the way that you get immortal sovereignty, or like some long-range sovereignty.

Another standard way people do it is like, Rockefeller bought up all the land on the Hudson River and he gives it to the state of New York, turn it into a state park or national park like Grand Teton National Park, and you get the federal government, the state government to agree to fund it in perpetuity per the Smithsonian Institution. The problem with all those things is that sometimes you just can't even trust the governments, right? But your best bet is you lay it off on a state or a federal government, and that's okay until the government fails or it reneges on something or other.

The bottom line is that it's not really within the grasp of a small or mid-size business or the individual. What is YouTube? YouTube allows you to become your own TV station. So what are we talking about here? The cliché, Be your own bank, if we look at it a different way it's like, I'm a normal person, I endow my own monetary energy into Bitcoin, I plug it into to some digital network, I give the keys to someone I trust — family member, heir — maybe I do multisig, right? Instead of having 5 people on a board — if you have 5 people on a board of an endowment with $100 million or $1 billion in the endowment, they're kept honest by filing annual tax forms with the IRS. And in theory they could be personally liable if they defraud the foundation or abuse their fiduciary trust. But you know it's kind of like a 20th Century idea.

A better idea would be: I put the money into Bitcoin and I have 3 or 5 digital keys. And it takes 3 out of the 5 to vote on anything. And every 3 months they have to register that they're online and of healthy mind and spirit, and if someone stops registering then that remaining 4 automatically appoint a 5th, and whenever anybody retires they automatically transfer their key, and then what I've done is I've dematerialized an endowment, board of directors, corporation, institution. And it's just become maybe a simple application in cyberspace plugged into a base layer of monetary energy. And it's doing something. Whatever the something is. And so that's a combination of (1) people and (2) digital networks and (3) crypto networks in order to allow people to achieve their hopes and desires over long periods of time.

Robert Breedlove: That's really interesting that you can lock up the base layer money and then appoint this multisig schema as almost a decentralized organization, in a way, among these 5 members. And then that gives the ability for your will and testament to be carried out in a more dynamic way across time. As governments fail, organizational landscape changes, or other macroeconomic conditions change, these 5 people can adapt. So they're sitting on this money that you've put behind the wall of encrypted energy, but it also gives it some dynamism in a way, that they can adapt to your will and testament.

Michael Saylor: Until computers get so smart that they do it all for us, the most likely outcome is: I appoint 3 or 5 people, and then they plug into digital networks that do all the stuff that we want done. Maybe you want to host all of your writings and all of your videos for the next 1,000 years in cyberspace. Okay so you appoint 3 trustees of the Breedlove Institute, you endow it with some amount of Bitcoin. You live a fine life, you depart, and their rules are: Every 10 years they have to find someone to replace them. You put your own term limits in there.

By the way, you could put all sorts of rules in. You could say that everyone that actually reads my stuff and passes the test can vote on who the 5 trustees are gonna be. Or, you can't even be a trustee unless you've watched and read everything and understood everything and passed my quiz that I left for you.

What if Ludwig von Mises left all of his stuff in cyberspace and we'd need to read all his stuff and pass it and pass an oral exam by him in order to become a fellow. And if we become a fellow we're invited into the academy of Austrian Economists, and that goes on to all eternity. And then the academy adds to the body of knowledge and they can upgrade the protocols and they can upgrade the tests and they can pass the flame down from generation to generation and the entire thing becomes antifragile and it's the religion of Austrian Economics. I'm thinking the guild system.

If you want something to be an immortal sovereign crypto network, then you think like the guilds. It's gotta be something that people are so passionate about they'll pass it down from parent to child, father to son, mother to daughter, master to student. There are things like that: schools of martial arts, Brazilian jiu-jitsu, religions, etc. And the medieval guilds were like that. Maybe there would be a — for example, I like environmental causes. What if I had a park and I said, well, people have to go in and they have to work in the park and support the park, but the park is this private/public entity where you can be a member and you get to use it but only if you can contribute to it and protect it. And then I endow it and it becomes a self-sovereign entity for 1,000 years. It's like a park-owner's association, right?

Robert Breedlove: Yeah that's really interesting. You've called domain names akin to digital real estate. And so what's coming to mind is that you could also do this with a domain. You could fund a domain in perpetuity and then populate it with these institutional factors that you've described.

Michael Saylor: Bitcoin.org.

Robert Breedlove: Yeah.

Michael Saylor: Or any domain. It could be wisdom.com and people put wisdom in and there are curators and there's a governance mechanism and there's a support mechanism and you just want it to go on forever like a monument in cyberspace. Think of it as a monument in cyberspace. It's something that has some functionality. But you know the real question here is just: What do you love? And what is it you want to achieve in your lifetime? Are you a family man? Are you a political man? Are you religious? Do you love art? Maybe you want to create a website that allows people to upload their art. You can upload digital art, and then other people can check it out or use it subject to some licensing and you just want that to become a Wikipedia-type thing, right? It's a commons service that you just want to go on forever, not subject to fragile or human failure.

Robert Breedlove: It's so interesting that another way to think about money is a medium for expressing preferences. When you buy a car or sell a house the economy responds dynamically by making more cars and less houses. So now with Bitcoin you can actually express these preferences transcendentally of space and time. You can express them, in theory, for all eternity.

Michael Saylor: I'll give you two examples: All the time you have examples of rich donors if they endow a professorial chair at a university and the money gets hijacked to build a new football stadium. Or, I want to endow a professorship of Austrian Economics — but then someone takes over the university and so they divert the money to buy a new hot dog truck. So—I'm dead — how do I make sure that someone doesn't, that they really fulfill my last dying wish? How do I make sure?

Well, I can do it if I have a Rockefeller Institute and they hold the purse strings and they dole it out annually and then they watch. But then I have to actually have a bunch of people that I trust that 100 years from now will remember what my wishes were. And maybe nobody cares about that. It's a challenge for people.

I went to Naples this year. And if you go to Naples, you go into the port, you have all these superyachts in the port, and very wealthy people on the superyachts. But if you walk out of the port there's puddles on the floor and there's this park in the middle of Naples right next to the port and it runs about 10 blocks. And 50 years ago it was beautiful and the jewel of Naples and it was like Central Park and everyone could come there and enjoy the day.

Now if you walk through it, all the buildings are spray-painted, they're all closed, there's pooling stinking water, no grass grows, it's just mud. It looks like something 40 years after the apocalypse. It's in the middle of like the third largest city in Italy. There's money everywhere, and yet somehow neither the municipal government nor the state nor the federal government of Italy, nor any wealthy person, can manage to come up with $100,000 Euros a year to water the grass and mow the lawn.

In the middle of a city of a million people — the public jewel, the single most important piece of real estate in the entire city—is suffering from a tragedy of the commons. Now I ask myself: If I was that wealthy person and I wanted to bring that park to life, if you wanted to do it top-notch you need $1 million Euros a year. You want to do spectacular? $10 million Euros a year. If you want to just mow the lawn and water it and keep it from being a stinking, mud-infesting war zone? $100,000 Euros a year.

How do I actually endow that park without having the money hijacked by whoever hijacked the money? Because obviously none of the economic energy is making it to that park in the middle of Naples. It might be a more worthy cause just giving a third of my money to the government as tax on my death, or two-thirds when I die for them to — whatever they're doing, like, what could you say about a civilization that can't keep its central park in its most important city in the province from being a stinking, post-apocalyptic zone?

Robert Breedlove: Right, it's terrible.

Michael Saylor: It's one of my values. So people have wishes. They have a hard time realizing their wishes. There's a lot of ways to do it, and the theoretical magic way to do it is: I create a program on a crypto network in cyberspace, it runs on a blockchain network — nobody can stop it, ever, it's unstoppable code — it's literally a magic spell. A less intense way is: I power it with crypto network and I use a combination of people and digital networks to do it.

I think ultimately people can achieve what they want in a variety of ways. If I want to be fed, I can feed myself by hunting a deer, hunting some chickens, growing some corn — these are all different ways to achieve what you want. The real key is: I need to have mastered the technology of hunting, cultivating animals, wildlife, and farming. And if I have all three of those technologies on my fingertips that I can mix and match, then I can actually do something interesting.

I think that's the potential here. We talked about: Money is power. Any sufficiently advanced technology is indistinguishable from magic. So if you apply advanced technology to money, then you can give someone magic power. The source of magic power is providing an instruction into the digital and crypto network space and powering it with crypto network space such that you cast a spell in the future, or cast a curse into the future when you're long gone. That's real magic, right? Like, casting a curse 30 years after I die — that's impressive magic.

Less interesting but more practical is: I just want to cast a spell while I'm alive. And the magic spell I would cast, were I a magician, is I would go to that park in Naples and I'd wave my hand and it would become a beautiful paradise with gorgeous trees and nice grass and water flowing and cleanliness and safety and free ice cream cones for the kids and flowers and happy music and concerts on the green. And it would be like that forever, because that's my magic.

By the way, in fantasy, the great magicians, the demi-gods, the powerful wizards, they have magic power — they can do that. And the more powerful they are, the longer the spell lasts. So if I have $1 million, I can do it for a year. If I have $100 million, I can do it forever. If I have $1 billion, I can allocate 10% of my power to do it there forever. Magic power, right?

Money is power. You got enough money, you can snap your fingers, materialize a jet, it'll fly you to Tokyo on 10 hours' notice. It'll cost you about $250,000, if that's how you want to expend your energy. If you don't have enough energy it doesn't matter. If you don't have $250,000 — right? The other part in those magic stories is the mere mortal that wants that same trip, their life force is sucked out of them and they're aged 82 years and they die. If I take a middle-class person making $72,000 a year and they need to fly from New York to Tokyo for $250,000, that's all the savings they're ever gonna have. I just sucked their life force out of them to cast that spell.

And so how much power do you have? And how much magic can you bring to bear? Look, John D. Rockefeller had magic. He just did it the old fashioned way. And his son did, John D. Rockefeller, Jr. They went to all the beautiful places on the Earth, they bought them all up, and then they ended up turning them into national parks. The US Virgin Islands is a Rockefeller Park. St. John, they bought. Acadia, they bought a lot of Mount Desert Island, made it a national park. Grand Tetons, they bought all that land, made it a national park. Most of the land up and down the Hudson River — between they and J.P. Morgan — they bought it and made it a state park. They were casting spells.

Robert Breedlove: And they turned it over to the governmental organization which was the best method he had at the time based on the technology, right?

Michael Saylor: They gave it to the longest-lived sovereign power. They did the best they could do. And they played the cards that were dealt them: they had gold, they had state governments—by the way, most of the state and federal trusts, that law was written by lawyers paid for by Rockefeller — so they created a political system. You think 501c3 law just came out of nowhere? I mean, all of these laws, they were created. John D. Rockefeller's son married Abby Aldrich. And Abby Aldrich, who became Abby Rockefeller — the richest woman in the world at the time — was the daughter of I think Nelson Aldrich. And Nelson Aldrich was the head of the Senate Banking Committee — happened to be key in the creation of the Federal Reserve — but he was one of the most powerful men in the country for quite a long time. And so there was an example where they were working to influence the political system, and critics say they influenced it to the detriment of people, but you can also see many ways they influenced it to the benefit of people. If you've ever enjoyed a national park, half of them are endowed by guys like Rockefeller.

Robert Breedlove: Right. Interesting angle, yeah. Aldrich, of the Aldrich Plan, that failed right before the Federal Reserve Act.

Michael Saylor: Yeah he was — from all the way through 1920 or something — for 30 years he was dominant. The senator from Rhode Island. So I think that in a nutshell is my views on crypto. I think ultimately we're moving into an exciting world where people are gonna be able to achieve great things with a variety of digital networks and crypto networks. What we can see right now for sure is: There is at least one successful crypto network — Bitcoin — there are many successful dominant digital networks — YouTube, Google, Facebook, Amazon, Apple, etc., plenty of them —and the future is gonna be defined by these networks, and you just gotta decide what's the best tool for the problem?

You can solve problems with people. I can pick up the phone, I can call you and I can say, Robert, buy me $100 million worth of Bitcoin. And I solve the problem with a person. Or I can solve the problem with a digital network. I can go online on Binance and I can start like typing and trading, and I can buy $100 million worth of Bitcoin. Or I could solve the problem with Uniswap — with a decentralized total crypto network.

What will happen? The market will determine—what we know is that over time, what was done by people will start to be done with digital networks, and what was done by digital networks may in fact float into crypto networks to the extent that sovereign immortality is worth the trouble. And if you're gonna make a crypto network — if they're just doing it for regulatory arbitrage it's probably got a 10-year life span and it's shaky — because who's gonna lay down in front of a tank for something like the difference between Binance and Uniswap? Would you sacrifice your life to try to trade on Uniswap instead of Binance? Because I wouldn't.

Let me tell you what they'll lay down in front of a tank for: They'll lay down in front of a tank for Freedom, for their Religious Values, for the Future of their Family, for the Future of their Ideology. If they have a passionate belief, they will lay down in front of a tank in order to defend it. The principles of Science and Justice and Humanity and Truth. Not everybody. But you don't need everybody. You need the keepers of the flame. You need the maximalists. The 1% who are willing to fight and die for the principles.

"We hold these truths to be self-evident." We didn't come up with it here today in this call. It was a basic truth that founded a lot of things including the United States of America. And if you go back to the Roman Republic, there were a lot of Romans when the Roman Republic was healthy, and they were willing to fight and die for Roman principles. To a certain extent, a lot of good Romans assassinated Julius Caesar because they thought they were fighting for Roman principles. And a lot of wars were fought over it.

If you really want a crypto to be successful over 100 years, the technology is only a part of it, right? It's the ideology paired with the technology. And you're gonna have to have an ideology that is so pure and so straightforward that people will fight to the death to defend the ideology. And that's why I'm probably not gonna sacrifice my life for the 13th iteration on smart contracts. It's not that important. On the other hand, if you tell me that we're about to suck all of the economic energy out of the civilization and plunge ourselves into the Dark Ages, then I think I'll fight for it. That's worth fighting for.

Key Insights from Chapter 8

- The only market-proven crypto network in the world today is non-state digital store of value: Bitcoin.

- The ability to do a minor transaction, it's trivially inconsequential. Only 1–2% of your wealth needs to be in these high frequency transactions. So a logical thing for anybody is just to keep 1, 2, 3% of their wealth in fiat.

- Bitcoin doesn't make sense as a medium of exchange right now. And maybe never. I tend to think never. It really isn't a medium of exchange as much as it is just a low frequency settlement network.

- All the crypto enthusiasts would be better just to accept the success of Bitcoin because it's the gateway drug for all the alt-asset holders to move into crypto. They really ought to be encouraging the next $3 trillion to come, and the number one way to do that is not to relentlessly, ruthlessly attack and sabotage and undermine Bitcoin, but rather to celebrate its success, because as it gets bigger, the opportunities for other crypto networks to interact with it will also get bigger.

Chapter 9 | Economics, Inflation, Interest Rates, and Natural Competition

"The creation of the first effective crypto network is an invention akin to the discovery of fire or atomic energy."

Robert Breedlove: The purpose of humanity has been to channel energy through our intellect. It's how we've developed everything, essentially. And I'm reminded of a quote by Alfred North Whitehead that I'll paraphrase. He said that it's common wisdom for people to say you should think before you act, but that in fact civilization advances by us being able to execute more important operations without having to think about them. So we're actually — when we can embed these certain important actions in a protocol that we don't need to think about as much, it frees us up to do other things — those are the kind of the layers upon which we've built civilization.

It just seems like this Digital Age we're going into is something radically new. It seems to be as profound as the Renaissance, or as profound as the Enlightenment. Do you see it that way? Are we co-evolving with the tools that we're creating and that the next 500 years are gonna be something so fundamentally different than what history has been that it will be difficult to recognize it in a few hundred years?

Michael Saylor: I do think that the creation of Bitcoin and the creation of the first effective crypto network is an elemental force that is a true invention akin to the discovery of fire or the discovery of atomic energy or the discovery of — we can make a list of a lot of fundamental things.

Maybe one interesting thing is just the science of sterilization. Germs — modern medicine and the awareness and the importance of sterilizing instruments and the way that the disease spreads.

Immunology. Once we figured that out, we were able to go from living 50 years to living 70 years because we realized that every time we entered into a medical procedure including the birth of a child, we were entering into a non-sterile environment that was life-threatening, soul-sucking, life-stealing.

The death rate from childbirth was huge, right? The average life expectancy was short, and we needed that breakthrough to realize that we were swimming in germs, and the very simple solution is: Wash your hands. Sterilize your instruments. And you put that together with antibiotics and human life expectancy jumps by 50%.

What if we're actually doing economics with dirty money? And so we've been using monetary energy which is bleeding, right? It's the same way as operating with non-sterile instruments, and the patient keeps dying and we don't know why. The significance of Bitcoin is: We're going from defective money, which is somewhere between toxic — it may just been ineffective, bleeding 2, 3, 4% a year—or it's toxic when it gets to -10% or -15% real yield. So using toxic money, how is that different than using toxic instruments when I commit surgery on you? How is it different than feeding you toxic food?

I think we're breaking through this new world, we sterilize our instruments, we encrypt our money, we're moving to a science of non-toxic economic energy. Madame Curie died of radiation poisoning. She died of cancer from the radiation. They didn't realize that radiation killed you, that it caused cancer. There's a lot of basics in life that we don't understand. A big breakthrough in health was when we realized that sugar was toxic. My mother didn't know that sugar was toxic. Conventional wisdom and governmental advice was you need your four favorite food groups and pursue a low-fat diet but starch and sugar was fine. And of course now we know that too much starch and too much sugar makes you insulin-resistant, makes you Type-2 diabetic, gives you cancer.

My mother became diabetic, became overweight, got cancer. We thought it was just unfortunate. The doctors said, We don't know why these things are happening, it's just unfortunate. If I could go back in time I'd be like, I know exactly why it happens. I know exactly how to solve it now. Like, don't eat sugar, don't eat starch —

stop eating. Fast. I never eat before 1 in the afternoon, I only eat in an 8-hour window and I'll go 2–3 days and I won't eat. I adopt fasting and I won't drink anything with sugar in it. You want to live a long time, don't dose yourself with sugar, it's toxic. The instruments are toxic. The germs are toxic. We killed George Washington — we bled him to death. Toxic. The money is toxic.

That's fundamentally the issue. The money is toxic. I mean, that's the fundamental issue with inflation, and if we segue into the discussion of inflation, everybody keeps thinking there is no inflation because everybody focuses upon a market basket of consumer goods. And if you look in the US and you look in Europe, they leave food and energy out of the basket of consumer goods and they say, "We left out the highly volatile food and energy from the index." Well, highly volatile means it went up. And if you put them in, the number would change. Volatility, in other words, is the signal. We've left out everything that actually changes in price from our price index.

It literally is like a Jedi Mind Trick, and it's like a triple mind trick. It's like: We have a consumer price index and we've left the prices that vary out of the index. We have a consumer price index — well first of all it's not a scalar, it's a vector. It's an in-dimensional vector dynamically changing in time — you've just created a scalar. We've created a market basket of things that we think you want. Yes, your basket is what I want. The market basket of things that you want does not include assets. No. I would never want to buy assets, only rich people buy assets. Poor people do not buy assets.

How did poor people get rich? You have to buy assets to go from being poor to being rich. I didn't go from being poor to being wealthy by not buying assets or not creating assets — you create them or you buy them. So the entire field of inflation is defective, and the irony is that 99% of the economists that talk about it, they've already accepted the notion that a market basket of consumer products and services is acceptable, and it's acceptable to throw out energy and good.

I've never seen an economist say, "Why don't we actually define the things that a working 22-year old is gonna want to buy by the time they're 32?" And here's one thing: Early retirement. I want to buy early retirement by the time I'm 32. How do I do that? I need to buy a bond that pays me $75,000 a year in interest risk-free, and I need for the $75,000 a year to pay my living expenses. And if the interest rate was 7% then I'd need $1 million for that. But if the interest rate goes to 0.7% I'd need $10 million for that.

So from 2010 to 2020 the interest rate went down to 60 basis points on a 10-year government bond which meant that the bond went from $1 million to $10 million, which meant that the 22-year old was suffering from 22% inflation on early retirement. But because that's not in the basket —because that's not something that you would ever want to give them — there's no inflation. You can actually track it and you can see that the inflation rate changes across a thousand different — if you just started with a simple principle: Inflation is a basket of products, services, or assets. If you just did that, that's kind of the equivalent of saying, Oh it's possible the Sun revolves around the Earth, it's possible the Earth revolves around the Sun — let's find out which.

Has anybody ever asked the question whether or not the basket should include assets? Or products or services? No one's even questioning the most basic premise. It's a pernicious rule of propaganda, and it's attributed to Joseph Goebbels in the Nazi regime who said — and it's also attributed to Ogilvy, so maybe it's apocryphal — but he said, "All of our focus groups show us and tell us we can't tell people what to think. But we can tell them what to think about."

I cannot change your mind once you've made it up—and if you have an opinion — but I can get you to focus upon something. So if I just keep saying inflation, CPI, and it didn't go up and this is what it is—When's the last time 100 million people said, What we really wanted to buy was early retirement? I didn't even know that was a product I could buy. Because I couldn't conceive of it. Well it is a product you can buy: It's a government 30-year bond that yields 6% interest in a non-inflationary environment — that's a risk-free retirement. You can buy that. Right now, the problem is at 140 basis points that would cost you $30 million for $60,000 a year. How do I make that at $75,000 a year salary saving $15,000? If I'm making $500,000 a year and I save $100,000—you know, I pay $200,000 in tax, I make $300,000, I save $100,000 — if I save $100,000 for 20 years I've got $2 million, and investing that in a government bond at 60 basis points or 100 basis points, you get nothing.

So the problem starts with the fact that inflation is mis-defined. The right way to think about it is: Every single product, service, or asset has an inflation co-efficient. And the inflationary co-efficient, that's the rate at which the price will change as I pump fiat into the money system. And so the coefficients vary, and they're a function of the scarcity of the asset, the demand of the asset, the information content of the asset, the material cost or the variable cost of the asset, and then the modularity of the asset.

If I can stamp out the asset a million times out of a factory, it's gonna be less inflationary because the fixed cost is higher and the variable cost is lower. Mobile phones will not be inflationary because everything's in the fab. Software will not be inflationary because there's no variable cost. Streaming music on Amazon Music or Apple Music will not be inflationary because I can stream it a billion times. A Picasso will be inflationary because there's only 20, 50, 100. The best 5 acres of beachfront property in the middle of Miami Beach will be inflationary to the extent people want Miami Beach. 5 acres in Ohio will not be inflationary because there's a lot of land in the world. The only land people want is in the middle of New York, the middle of London, the Hamptons, Miami Beach, the middle of LA, the middle of San Francisco, the middle of Tokyo. You fly across this country and look down, there's enough land to park 10 billion people on 5 acres each. It's just gotta be in demand and scarce.

Robert Breedlove: May I ask you a question about this? So the coefficient itself in my mind would be a product of the scarcity of the good or service relative to the scarcity of the money it's denominated in, such that if the money supply is outpacing the production of the good or service, that good or service would inflate in price, right? So to your point it's not a single variable — it's not CPI is inflation — everything has its own inflation rate. The second *part of that question would be: Why is the narrative surrounding inflation so distorted? Do you think it's intentional by governments that are clearly heavily indebted? I don't see any equitable benefit to inflation whatsoever, it's purely a mechanism for reallocating wealth, and I don't understand why the narrative's so distorted.*

Michael Saylor: 200 years ago people thought they had to bleed George Washington to death to save his life, and everybody agreed on it. The whole point of paradigm shifts is: Everybody agreed that the Sun revolved around the Earth. Everybody agreed the world was flat. Everybody agreed that humans would never fly. Everybody agrees on stuff until they realize that they're just utterly and totally and horrifically wrong.

So in this particular case, I blow a bunch of liquidity into the system. Let's say there's $50 trillion worth of energy and I blow $10 trillion worth of money in the system. So there's still $50 trillion worth of energy, but now the money is diluted by 20%, right? So if I have a product and I can measure the pure energy content of the product, then if it's 100% pure energy, the price has gotta go up by 20% if I expand the money supply by 20% assuming it's completely liquid and in demand. What's an example of that? Maybe a bond. A pure financial instrument, something that —

Robert Breedlove: A ribeye.

Michael Saylor: Something that is tangible and you cannot produce it with any less energy. This is why proof of work in Bitcoin is critical. If it takes me a tangible amount of energy to produce that thing then it's inflation co-efficient is going to be like 100%. And on the other hand, the cost to produce a streaming YouTube video, the energy content is 1% of the value added or the value of use. And 99% of the value-in-use is information and non-scarce information. So it's got a variable cost of 1%. An iPhone's got a variable cost of 30, 35, 40%.

Everything's got a different variable cost. Gold's got a much higher variable cost, right? Because it's holding its energy. So when you look at all these things you'll be able to calculate different inflation coefficients and therefore different inflation rates across an array of thousands of things. It'd be different inflation rates in New York City, Manhattan versus farmland in Kansas. It'd be different. So you can't really say, Oh this asset class. There will be different inflation rates on different stocks. You notice that if the cash flows are likely to continue from the stock, or less affected by inflation, then it's gonna go up.

I think the pernicious mistake everybody makes is: They don't really think about energy density and information density of their products, services, and assets. They're not applying conservation of energy. If the law of conservation of energy applies, when I increase the money supply by 20% and if the energy is constant, then all of the numbers have to change. And if they didn't change on the deflationary products, they must have changed more on something else. So you can't very well be printing 10% more money and not have the inflation. It's just: We're choosing to pick just 1% of the things that we track in our inflation bucket, and it tends to be the deflationary items that we put in that bucket. And it's almost too easy. If I get to throw out all assets, all real estate, all stocks, all bonds, and then I get to throw out energy, and then I get to throw away food, well how could you possibly generate inflation? Because you could print a hundred-kazillion-trillion-billion dollars and the cost of free, streaming Twitter and YouTube is not going up.

Robert Breedlove: Yeah. You're throwing out anything that changes. So it's self-defeating.

Michael Saylor: So bottom line is: There's no such thing as a free lunch, but the inflation that's being reported is an irrelevant metric. I call it a metaphysical metric that's been artificially defined in order to provide some comfort. And it's working. The great majority of people, not only do they not think there's inflation, you literally have politicians lamenting that they can't create inflation and how important it is to create inflation, even as they're inflating every scarce asset on Earth to the point where no one can afford — look, Robert, like, I'm a rich man. I'm a very wealthy man. I can't afford to buy a house in the Hamptons. Like I'll go look at these things and I'm like, Who's paying $47 million? They're selling houses for $25 million on 2 acres. I'm like, Are you guys out of your minds? Or you go to New York City and someone's paying $25 million for a 5,000 square foot apartment. $5,000 a square foot. At the end of the day, it's obscene. And what you can see is: We're running 10–20% inflation for the past decade, we're just running it on all of the scarce luxury assets that have high energy value. I mean, what's the definition of scarce, right? Maybe the definition of scarce is that it has high energy value, because if I could stamp out a billion of 'em for the same amount of energy I must be diluting the quality of something, right?

Robert Breedlove: Right, right. Just take things that are hard to produce. So gold and Bitcoin are all inflating. And I would say that it is a lie, right? I'm not sure necessarily about the intentionality, you could argue about that. But it's definitely a lie, that CPI is inflation. And it seems like it's being used to cover up this widespread system of theft that is monetary inflation.

Michael Saylor: I'm not even sure that they realize that it is theft or that they're doing it. I'm half-convinced that 80% of the people in government don't even realize that the inflation metric is a wrong metric and irrelevant. It's like I'm burning myself to death and I'm calculating the temperature on the counter 6 feet away and I'm burning but I need to keep turning the thermostat down. And I guess it's like they're just not feeling the pain.

And because of that it takes us to the issue of interest, right? If you think inflation is not coming so you keep printing money and you keep driving the interest rate down, the problem we have is really just a war on currency, a war on time, we render the money toxic if you hold the money. Once you understand that the real inflation rate is 10–15% because that's how the assets were clocking, then you realize that any currency you're holding is draining energy from your life at 10% a year.

It’s almost like I put in a battery that drains 1–2% a month. I can’t store energy. You know another metaphor for what happens in the human body when you can’t store energy? It’s like, Robert if I took you and I dropped you in the middle of the Arctic Circle and it was 20 below, your body would start losing energy at a rapid rate, you’d freeze to death. It’s literally like I come into your office and I crank the temperature down 20 degrees and I’m freezing you to death because I’m pulling the energy off your skin. And so what do you do? Insulate, cover up, right? But what if you can’t?

If you’re a wealthy person you put on a fur coat, right? Or maybe you’re smart enough to realize: What do wealthy people do? You drop them in the Arctic and they get on their jet and they fly to the Caribbean where it’s warm because they can. And what do you do if you’re poor? I drop you in the Arctic, or even worse I go to your hometown and I just turn the temperature down to 20 below and you can’t leave. But you know it’s like I slowly freeze you to death. I don’t do it fast. You don’t even realize it’s happening if it happened at a gradual enough rate.

It’s like, I know I’m working hard, I’m just not getting ahead. I’m working hard but I’m not getting ahead, because every time I put money in the bank the price of everything keeps going up. The price of a house in Miami Beach it was $1 million on the street where I live, and then it’s 2, 3, 4, 5, 8, $10 million. I’m not talking about every decade, I’m talking about every year. I’m talking 2000–2010 during that administration we were printing money so fast that we had this housing boom and everybody that owned houses were happy — they’re refinancing their houses — but you look at it and you’re like, how is it possible that people bought this house in 1998 for $1 million and I’m being asked to pay $10 million for the same house? If you happen to be working for cash — it’s what Pomp would say — if you’re working for cash and paying taxes and then you’re putting the cash in the bank, then you’re suffering from inflation —

Robert Breedlove: Shadow tax.

Michael Saylor: Your life energy is being robbed from you. And so that actually takes us to this issue of real yield, right? If the actual nominal yield is 1.5% on the 30-year T-Bill or it's 0% on short-term money, and if the asset inflation rate blended across all liquid assets — stocks and bonds and the like — it's probably 15% right now, maybe 12, 13, 14, 15%. But let's just say it's only 10% just to be nice. Well, then you're looking at a real yield of -10%. You've never seen that number printed in any kind of public media. No one would dare say we have a negative real yield of -10%, it would create a panic. But if you did think negative real yield of -10%, what happens next? You cycle through and you say, well, if my cash flows of the stock aren't going up by more than 10%, that's diluted.

The only equity you can buy where you're gonna make out on is where the company's able to grow its cash flows more than 10% a year, right? And then you gotta buy it at a decent price. So if your cash flows are growing 20–30% a year maybe it's good deal. That's why people like tech like Facebook or Google or Amazon because they did for a while. I don't know if they will going forward. It's a lot harder, over the next 36 months it seems much less likely that you'll see 20% cash flows. Will you see 10%? I don't — what percentage of the S&P 500 will grow cash flows more than 10% this year? Any? 5%? 10%? Probably not more than 10%, right? We could figure it out but if you're not doing that then you're diluted. Of course that means that any fixed bonds that aren't generating 10%, they're long-term diluted. So where does that leave Bitcoin? Well Bitcoin's got a positive real yield because you're not getting hit with that -10% currency debasement.

Robert Breedlove: Let me ask you: Just to jump back a little bit to Bitcoin as a unit of account or a financial frame of reference. Do you suggest here that it is actually useful—I guess you could do this with either Bitcoin or gold — to look at the historic price charts denominated in Bitcoin or gold to strip out a lot of this central bank induced market manipulation via inflation?

Michael Saylor: I think that will be a lot more useful in the next 10 years with Bitcoin. The first 10 years with Bitcoin, it was so developmental going from zero. You have this asymptotic zero number, so I think that if you look over the next 10 years that'll become a valuable thing. People have done it in gold and I think it's a more stable application because gold is a bit more stable through this time period. But again it's manipulated to a certain extent and its got its own problems.

Robert Breedlove: This would help eliminate some confusion I think for people that think the S&P is just going up forever. If you actually denominate it in gold the chart doesn't look that great, right? It had a boom in 2001 but it's not been good ever since.

Michael Saylor: If you simply divide it by the monetary supply, if the monetary supply is going up by 7% and the S&P is going up by 8%, then the overall market's flat. And that makes sense because why do people think that stocks should always go up 8%? I'm in business, Robert, it's hideously competitive. Do you think that we don't have a competitive market for everything in this country? It's obscenely competitive.

And so what you've got if you look at the NASDAQ is you have like 5 companies — Apple, Amazon, Facebook, Google, Microsoft — those 5. And aren't they responsible for like 80% of the game? Everybody else is competing and it's a competitive market. And what does that mean? It means it's hard to grow 20% a year because whenever you do anything someone else is copying you and they're pushing on you. So unless you get a dominant digital network with a near-monopoly with these massive, exploding economies of scale on a zero variable cost, low variable cost, it's very very difficult to perform. And most of the S&P isn't.

To the extent that the S&P isn't Apple, Amazon, Facebook, Google, Microsoft, they're just a bunch of companies competing with each other, so you would think that they would grow at the productivity growth rate of the overall company which is 2–3%. If you had hard money — by the way, coming back to that theory of

Bitcoin network value — Bitcoin network value goes through the roof, skyrockets in the early days when there's massive adoption and massive technology explosion, but in the later stages of the S-curve when it's fully diffused and when it's mature, it just grows with the GDP. It grows with the productivity of the people in the network.

If they grow 2% a year, it grows 2% a year. So in a mature equity market you would expect equity indexes, equity prices to grow with the GDP. If they're growing faster, it's the monetary expansion, right? Expand the monetary supply 7%, expand the GDP 1%, S&P should go up 8%. It's gonna be disproportionate. The big tech, the leading-edge, innovative tech, is gonna be double, triple, quadruple that, and then the trailing-edge laggards are gonna be tanking, and then everyone that's working their asses off as hard as they can is gonna be barely keeping up. Because you have to do 100,000 things right just to stay in business in a real Darwinian capitalist economy. It's like: Being flat means defeating 99% of the rest of the market — being flat. To be up, you have to, you know — Amazon's up because they beat 15,000 companies. The next two are just slightly okay, and there's some that are flat, and everybody else is destroyed because of the natural effect there.

Robert Breedlove: This reminds me of the Red Queen from Alice in Wonderland who said, "In my kingdom, everyone has to run as fast as they can just to stand still."

Michael Saylor: I'll give you another example. There are 3,500 publicly traded companies. To be the best you have to better than 99.99% of humanity. That makes you the number 1 out of 10,000 people. If you were smarter than 99.99% of humanity, there are 750,000 people on the planet smarter than you. And 99% of them want what you have, if you have a billion dollars. If you have a publicly traded company, 99% of the people that are smarter than 10,000 other people don't have what you have, and they can probably raise $1 billion and chase you, right? They're harder, they're smarter, they're faster, they're stronger than you are.

Like I'm sitting at a company — one of 3,500 — and the world is full of people that are smarter than me that can raise $1 billion that want what we have, want what I have. That's Darwinian competition. There's the view from one side of the table which is, oh yeah, well you made it, you're successful. There's the deal on the other side of the table, which is: There's a guy that's gonna work 80 hours a week that's gonna be surrounded with 100 other people that are gonna work 70 hours a week that are gonna raise infinite money that are going to target you and do everything they can to take their market from you. Now that's a very humbling observation, that's why you can't rest on your laurels. There's something beautiful in that terrifying concept. That's what drives humanity forward.

Robert Breedlove: It keeps you honest, right?

Michael Saylor: It is the core of Stoicism, and it reminds you your best chance is to focus all of your energy, all of your assets on just this one thing that you're gonna be the best in the world at, and you better stay humble. If I take my own business — I became public in 1998 — there's a 99% mortality rate. 99 out of the 100 companies I competed with are gone. Out of my peers, I'm the only person — I'm talking about 100 publicly traded companies — there's probably 500 CEOs that launched a company with 20, 30, 50, $100 million of capital, and they're all gone. That's what the open market, the free market, will do. And it is what it is.

I mean, that's why the human race is what they are. There's always someone. And when they attack, if you're distracted, if you're arrogant, if you're fat, dumb, happy, comfortable, they're going to eat you. And if you're half-focused or de-focused, they're gonna take your arm off. And if you're completely focused, you can react. If there's something they're doing that's good, you channel it, you inherit it, you evolve, you live and you grow stronger. And otherwise you shrink and they squeeze you out of the entire market.

There's something I observed and again it's very Stoic: Everybody thinks when you're young you wanna acquire as much as you can acquire. So young men are acquisitive. Young business people are acquisitive. Can you acquire the thing? That's generally the easiest hurdle to jump. The next question is: Can you maintain the thing? Can you stay competitive? That's 10 times harder. And the biggest hurdle is: Are you going to be able to commercialize the thing or profit from it? Can you buy something or build something and continuously improve it forever so that you're competitive and then do it in a manner that is cheaper such that you can charge more for it than it costs you to do that thing. That's obscenely hard.

So typically everybody thinks that they can acquire something, then when they realize the maintenance requirements they fail, and then very few people ever get to the point where they could commercialize something. By the way, you can apply it to a boat: Everybody wants to buy a boat, and then they're like, oh my God. This is really expensive to maintain a boat and I can't afford to maintain the boat. But if they buy the boat, they gotta spend 10% a year to maintain it. And then at some point the question is, can you enjoy the boat? They're like, Oh. I'm spending all this money on the boat but I never have time to go and use the boat — this is just crazy. This is weighing around my neck. I gotta get rid of this.

It's an example of being too ambitious in your acquisitiveness, and it illustrates the law of decimation. And the law of decimation is: In the ancient Roman republic if the legion screwed up, they killed 1 out of every 10 men in the legion. Actually they made the 9 kill the 10th in order to remind them that they should stay disciplined.

Robert Breedlove: At random, right?

Michael Saylor: Yeah, random. They didn't kill them all because then there would be no legion left. But 1 out of 10 is gonna die if you break ranks and retreat so it was their ultimate punishment, the law of decimation. But you can apply it to anything in life, Robert, but it goes like this: The universe tends towards entropy and disorder. If something will go wrong it does go wrong. That's Murphy's law. The law of decimation is: 1/10th of all the moving parts in anything will break in any given year. If you have 10 employees, 1 will quit or become unhappy. If you have 10 moving parts, 1 will break. If you have 10 plans, 1 will blow up in your face. If you have 10 features, 1 of them will stop working. If you have 100? 10 of them will stop working. If you spend $100 million on something, you have to spend $10 million to maintain it. You're gonna have to divert 10% of the cost of anything to maintain something.

I talked about how steel will last forever — if you maintain it. Most people don't maintain it. It costs a lot of money to paint a steel ship. Most people, they budget for the acquisition, and then they underestimate the maintenance because they don't have the humility or the life experience. This is the problem with building Rube Goldberg devices in the crypto network. That's the problem with all the complexity with Ethereum and all the complexity with some of these things: It sounds good on paper, but when you put 187 moving parts into something and when 1 of them breaks and the entire thing crashes and burns and you die, it wasn't worth it.

When you're young, you overestimate the value of functionality and acquisition, and you underestimate how expensive it's going to be to maintain things, and then you really underestimate this last issue: Can you enjoy it? This is a basic rule of life: Can I buy the thing? Can I maintain the thing? Can I enjoy the thing? Men are always reaching beyond their fingertips. Sometimes women too. They want too much. They're empire-builders. This is why Napoleon should not have gone to Moscow. This is why Hitler should have not have gone to Moscow. This is why you don't fight a war on two fronts. And this is the essence of Stoicism, but Stoicism is really a philosophy that is very consistent with thermodynamics and entropy and complexity theory. And if you've ever run anything complicated or built anything complicated or have been responsible for anything complicated, you know stuff breaks.

Robert Breedlove: Do you think this — we'll call it a law of nature, this 10% of the components in a complex system break down yearly, annually, and require maintenance — is this connected to the religious tithing, do you think? Where you're actually *supposed to feed the flame with 10% of your profit to maintain the institution?*

Michael Saylor: I think it's interesting the extent to which you see this 10% number pop up on an annual basis. 10% is the maintenance obligation on a boat. 10% is the tithing obligation for thousands of years. 10% is a reasonable estimate for a house with 187 light bulbs, 18 of 'em will burn out, right? It just pops up over and over again. And my only real explanation is just: friction, randomness, chaos, life, corrosion, weather, termites, bugs, bacteria.

The same would be true with your body, right? If you're talking about maintaining yourself. You've got to actually allocate time and energy to maintain yourself and a lot of times people under-invest in their own health. And then when they under-invest in those things, they blame it on genetics or they blame it on some unfortunate occurrence. We don't know why these this happened it's very unfortunately — these things just happen sometimes.

I'll end with one thought on Stoicism. And Nicholas Taleb would appreciate this one too. It's like, the words don't matter — the action matters. Okay? Words are just words. And that applies to Stoics: you and me and Marcus Aurelius. I think one of the great paradoxes of history is Marcus Aurelius was the last emperor in the line of the Antonines during the golden age of Rome and there was Trajan, and there was Hadrian, Marcus Antoninus, etc., and for about 100 years that was the Pax Romana, and each of those emperors was elected based on virtue as an adult and he adopted his heir, and they typically adopted a 40-year old emperor who had had a career in the military of virtue and he was tough and responsible and grounded in reality.

And if you're a general in the army campaigning and you get drunk and screw around your soldiers put a knife in you. You're not gonna make it. In order to keep the respect and stay alive in war-time around a bunch of guys with weapons, you better be a good leader and they better respect you because you're leading them to their death, if they don't respect you. So if you actually rose through that meritocracy, maybe you had a chance.

So Marcus Aurelius writes the Meditations and it's the quintessential text on Stoicism. And he says, just because you can do a thing doesn't mean you should do a thing. He said, "Soon you will have forgotten all and all will have forgotten you," and you should know your place in the universe, and you should submit yourself and do the right thing for everyone else, and that's all good.

But at the end of the day, the single most important decision Marcus Aurelius made in his entire career, in his life, was the decision on an heir, and when it came time to make that decision he failed miserably and he appointed his son Commodus to be the emperor. And Commodus was a minor and of weak moral and intellectual constitution and in no way shape or form qualified to be emperor of all the known world. And by so doing that Marcus Aurelius plunged the Roman Empire into chaos and turmoil for hundreds of years, resulting in the deaths of millions if not tens or hundreds of millions of people — awful. Awful. And there's your philosopher-king. He's remembered today as somebody that's written a good book and had been a great Stoic. But if you look at his actions, the actual action he took was the least Stoic, most foolish, most painful action of any of his forbears and it just makes the blood curdle.

Robert Breedlove: Yeah. I've been anxiously waiting to talk about this actually because I'm a huge fan of Marcus Aurelius. That particular episode is documented somewhat well in the movie Gladiator for people that want to go out and watch it. But he is also known to have been one of the greatest emperors of all time, right? Up until that point where he made that fateful decision.

Michael Saylor: He was great until the succession. He had all of the power of the Western World in his hands. He had the keys, right? The crypto keys to all the wealth and power in the Roman Empire. In Gladiator it was implied that he was murdered by his son, but in the history books they're pretty candid that he handed those keys to Commodus and Commodus was a disaster.

Robert Breedlove: And he was that Platonic ideal of the philosopher king. Arguably the only successful philosopher king throughout history. And I think one of his other quotes that I really liked is, No man can lose any other life than he now lives, nor can he live any other life than he now loses. Stoicism's been big in my life personally and I think it's necessary for everything we've talked about today, for this eternal contention we have with reality. If you don't adopt a Stoic philosophy, how do you keep yourself together?

Michael Saylor: I think Stoicism is critical, and I think he was a good writer. I would even probably admit he was a good emperor until that final decision, which I just lay out as a paradox. And maybe it's a warning and the warning is: You could live a great life and you can be a great writer and you can be a great thinker, but at any given point it's always that last decision. You still have time to snatch defeat from the jaws of victory.

Robert Breedlove: Did he choose love over his principles? Was that what it was? Love for his son over principles of succession?

Michael Saylor: Presumably. You can read up on it and come to your own conclusions. It's a short chapter.

Our last point just on vitality: A synonym for antifragility could be genetic vitality, Darwinian vitality. If I'm evolving in response to threats as a life force, then I'm antifragile.

Robert Breedlove: And this was Darwin's famous quote, "It's not the strongest, fastest, or smartest species that survives, but the one that's most adaptive to change." Which makes it antifragile.

Michael Saylor: Which makes it over time the strongest. It's just not in the near-term. Yeah, there's a certain terrifying beauty in nature. There are no ugly animals. You look at a bird that is beautiful. You look at a lion in the wild, it's beautiful. No one's got a mangy coat, there's no unhealthy anything, and that's our ideal of beauty, right? We think nature is beautiful, all the trees are beautiful, the plants are beautiful, the birds are beautiful, they chirp their beautiful sounds, the flowers are beautiful.

What they don't really think about is what's going on behind the scenes, because they've got this simple zoo backyard view of nature. The truth is, everything's at its finest when tomorrow is uncertain. When the life of the creature is uncertain. I've actually got this beautiful banyan vine tree in the back of my house in Florida. One day on a beautiful sunny day I walked by and I stood and stared at the tree and I saw a bunch of ants running up and down. And when I traced the ants I saw that there were thousands of ants and I saw that there was a centipede. Some kind of millipede about a hundred times bigger than a normal ant. And those ants had decided that they're going to eat that. They were actually gonna haul that millipede back to their queen as dinner. And they attacked it relentlessly, relentlessly.

And I watched hundreds and hundreds and then thousands and then thousands of ants going into this one millipede. And it's fighting for its life and I swear I watched for 45 minutes like a war non-stop on a beautiful sunny day. If you looked around you would've seen grass and blue sky and pretty water and birds chirping, but there was knock-down, drag-out war to devour this millipede and it's fighting for its life. And I watched it crawl up the tree and the ants dragged it down and it did everything it could and they kept coming and it just had this horrific, terrifying, sad conclusion. It's gonna die. Unless a massive rainstorm spins up to blow water down and create some disruption it's got no chance. It's going to get eaten alive. And it's horrific and it's terrifying, and that's life. That's nature.

And then you start to realize, on all those pretty National Geographic TV shows, you see the lions attack the antelope or the gazelle and they try and this miss like, well, no dinner for you tonight. No, the antelope trot off happy with their babies and the lions trot off happy with a little smirk. And everybody's like, that's about as much nature as they want. Nobody wants any more nature.

When you think about it a bit more you realize, well, they're gonna miss 3–4 days, they're gonna hit 1 of those a week, and the 99% of them are gonna live but 1 of them is gonna die, and over 3 years they're all gonna die. And over 3 years they were all gonna be eaten by the lions. And that's nature. Every week that goes by, it's like they're clicking on a carousel. And the oldest one is getting slower, and a little bit more tired, and a little bit less flexible, and if they don't get the old one, they're gonna get the unlucky one. And that's why every one of them is beautiful. Because they're all in the prime of their life.

The same is true with all the predators: they're all in the prime of their life. When they get a little bit old, a little bit hurt, they get driven out of the tribe or out of the pride and that's the end of it. So in nature the life expectancy of those wolves or those predators or those lions is 5 years, and in the zoo it's 15 years. Now if you want to see a fat, mangy, lame one, you'll find one in the zoo. You won't find one in the wild. And the same is true with the rest. And when those two herds, when they go at each other like that viciously, they're both strengthening. You take away the wolves from the deer, the deer overpopulate, they eat all of the trees, the trees all die, the trees die, they destabilize the embankment of the river, the river erodes, the riverbank gets screwed up, all the greenery dies, all the deer are gone, all the wolves are gone.

You wanna fix the river? You put the wolves back in. The wolves scare away the deer, the trees grow, the roots stabilize the bank, the river flows, all the wildlife returns. This beautiful thing we call nature is in continual dynamic equilibrium, and everything about it is getting stronger and harder and faster and getting culled all the time. And mother nature is supreme, and men with delusions that they will defeat her are gonna be disappointed, right? I guess the great challenge is the paradox of the engineer versus the zookeeper. We see nature, we want to engineer a better world for ourselves, and it can be done. But we can also reach too far and try too hard and try to make water flow uphill and try to make time flow backwards.

We can try to shake our fist at mother nature, we can defeat all of those natural forces, and if we try to do that, the energy consumption goes up exponentially. And eventually it goes up to such a level that we deplete ourselves of energy, and we end up like those natives on Easter Island where you chop down every tree to build your monuments to your gods and pretty soon there's no canoes and pretty soon there's no fish and pretty soon there's no food and pretty soon there's no you. All you've got is your monuments to your god and you're all dead because you depleted the energy in the ecosystem in pursuit of over-engineering your reality.

Robert Breedlove: Let me ask you about that point, which I think is fantastic: it seems to me like the free market is the economic expression of that Darwinian equilibrium, and that possibly with the implementation of central banking, which is antithetical to a free market institution, that is it's a monopoly. I guess in our attempt to over-engineer the economy we have disturbed that Darwinian equilibrium in the economy and that's why we have all these haywire consequences like inflation and negative rates and so on and so forth?

Michael Saylor: Yeah we've stopped it, right? We've attempted to stop time and interfere with nature. We're trying to freeze that dynamic equilibrium that's being continually calculated all the time. We're trying to turn nature into a zoo.

Key Insights from Chapter 9

- Fiat currency is toxic money. It's just not sound. It's infected with entropy, and this has all these second-order effects with everything that it touches.

- The destiny of all money is to be encrypted. We can think of encryption itself as a sterilization function or process for money. We've actually disinfected the money by encrypting its rules and its supply.

- The creation of Bitcoin and the creation of the first effective crypto network is an elemental force that is a true invention akin to the discovery of fire or the discovery of atomic energy.

- Most people, they budget for the acquisition [of wealth or assets], and then they underestimate the maintenance because they don't have the humility or the life experience. This is the problem with building Rube Goldberg devices in the crypto network.

- The inflation of fiat currency has only one purpose: To reallocate wealth away from some and to others. I don't know what else you could call it, really, besides theft. And it seems to me like Keynesian economics is an academic cover story.

- Unless I'm growing my business or my own personal cash flows by more than 10% year over year, then I'm being diluted. I'm losing money. So this is the hurdle rate, basically, to use another investment term.

- Stoicism is really a philosophy that is very consistent with thermodynamics and entropy and complexity theory.

- You could live a great life and you can be a great writer and you can be a great thinker, but at any given point it's always that last decision that has the potential to define your legacy [as the story of Marcus Aurelius illustrates]. In other words, everyone always has time to snatch defeat from the jaws of victory.

- Words don't matter, actions do. It's more about what actions you take with skin in the game versus your cognitive beliefs.

Chapter 10 | The Death of Gold

> *"We're moving from politically engineered money to scientifically engineered money."*

Robert Breedlove: So, you know, I've told you about the feedback we've gotten on the Saylor Series, it's been absolutely tremendous. People are reporting that it completely reshaped their worldview. Sent them down the rabbit hole. It's been very valuable. And so I commend you for that. And I think the right place to jump in here today — since we're taking the big picture approach — is to just answer the question, what is money?

Michael Saylor: You know, I never in my life had anybody asked me that question before you asked it to me the first time for our first series. And when you asked me that question, that catalyzed the whole set of thinking. And it got me thinking: money is energy. And of course we did the nine-part series and we spent a lot of time talking about energy systems and engineering as a discipline of manipulating energy and why it was critical to the human race and the human condition. And so we won't re-cover any of that, because that's well-trod territory.

But coming back to the question of what is money and just focusing on it, I would say, for our round two or part two of this series, I'd say: money is tokenized energy within a socio-political framework. So we know money is energy. But tokenized energy — that is: a dollar is a measure of energy, a gold coin is a measure of energy, a bale of tobacco that you're trading is a token of energy. And how much energy? Well, that's where the socio-political framework comes in.

You can measure actual physical or engineering energy and its objective, but money ends up being to a certain degree subjective because if I show up on a deserted island and I've got a stack of gold coins and everybody has guns, and they're shooting each other, they might not accept my gold coin. Whereas if I actually offer the bullets for their gun, they might think the bullets are actually better money than the gold coin, just like cigarettes are good money and in a POW camp, but gold coins wouldn't be.

So this question of what is money gets you really going. If money is tokenized energy within a social-political framework, then we can look over the history of mankind and we realize that we've chosen many different ways to tokenize that energy, and we'll skip through most of them because they came and they went, but there are three that popped to mind that are interesting.

One is the gold standard. The second is the fiat standard. And the third is the Bitcoin standard. And when you start asking what is money, you start asking, well, what's the best monetary system? And we've got these three systems to talk about. I think one thing that's pretty clear is that most people can't get their heads around Bitcoin because it's an utter paradigm shift. And why is it a paradigm shift? Well, because I think Bitcoin is the first digital money. It's the first digital money. Gold wasn't digital money and fiat is not digital money, either. And so economists and politicians and investors, they all lack the right mental model to understand Bitcoin because science and engineering were so intrinsic to gold that we took it for granted, because gold is a 10-pound lump of something and if you if you get slugged in the head with it, nobody had to explain that science and physics mattered.

And then science and engineering kind of became like quasi-irrelevant in the fiat world. Everybody just kind of ignored science and engineering and there were no immediate clear consequences. There were just the hyperinflations and the collapses of those fiat systems. But because they abandoned gold and there was no digital money, the only alternative to one fiat, you know, the Weimar Republic fiat currency, was the [British] Pound. And that alternative was an alternative to the next thing and the next thing.

So we're comparing fiat currencies or fiat monies to each other. And so there was no real need to embrace science and engineering. And Bitcoin is this paradigm shift where we have digital money crashing into, I guess what we'll call analog money, or maybe a political money is a better way to describe it. And now we start to ask the question again, what is money? And I think tokenized energy isn't good enough to explain what is money.

Another way to describe money is by coming back to the ideal model. What is the ideal model of money? Ideal money is a shared, immutable, correct ledger. You know, you've heard the phrase, Bitcoin is a shared immutable ledger and people debate about whether it's truly immutable or not, but a crypto-asset network, fully decentralized and mature, is the closest thing we can get to an immutable shared ledger in the history of the world. I think a lot of times when people describe it as a shared immutable ledger, they leave off the correct or the mathematically correct, because it's almost implied.

But I think that if you were to focus upon the three critical dimensions of ideal money, you would say: (1) it's a ledger that's shared, everybody in the political system has to have the same access to the ledger. (2) It has to be immutable, no one can doctor it. But it has to be (3) correct, mathematically complete or mathematically proper, because if it's incomplete or incorrect, then it's not ideal money. So if I take that as a model, money is a shared immutable, correct ledger, then I can imagine the perfect money would be: some godlike being comes down on earth and they create this perfect uncorruptible system and they telepathically drop that shared immutable correct ledger into the heads of every human being. And every time you incur a debt, it updates the ledger and when you incur a credit, it updates the ledger. And no one can corrupt the ledger.

Robert Breedlove: This is getting right to the heart of property, right? Where it's essentially a list of who owns what, and that ownership is premised on what favors have you rendered to the market, then you've earned some right to redeem favors from the market. And money is just kind of the ultimate form of property and one that can be redeemed for any other form of property.

Michael Saylor: Yeah, I mean a shared immutable correct ledger could be used to allocate ownership of all intellectual property. You kind of need it to keep track of your music rights or your video rights or your movie rights on your iTunes Store. What do you own? Or the right registrations and licenses? Can I enter a building? Can I exit? Can I cross a border? What are my rights? And what are my obligations? And what do I own? And what do I share, and it all comes down to just a shared database, but that database needs to be immutable and correct. And that is property, the essence of property.

And I suppose, when you own a piece of property, and you buy it — a piece of land five acres somewhere, then there's a deed and you go and you file the deed with the courthouse. And that's kind of critical, right. And so much of property law revolves around making sure that there are no liens on the property before I buy it, making sure that you have the proper title to it. And that you can transfer that title. And I think that money, to your point, it's kind of like the apex property, it's the sum of all property or at least it's the most important property, I suppose, because it would be the property that I could use to trade for any other property. Any other property or any other product or service. So that is my tokenized energy.

Robert Breedlove: Now, what you're describing here is this movement of money out of the sphere of politics into an engineered standard. And one definition of politics I really like is the discussion of how to apply coercion. So Bitcoin seems to be this type of money that's actually moving economics out of the sphere of politics into a more hard science. That seems to be part of this paradigm shift. We've always thought money was in the domain of who could apply force. But now it's moving into the domain of what's scientifically proper.

Michael Saylor: Yeah we're moving from politically engineered money to scientifically engineered money.

Robert Breedlove: Yeah. From social engineering to scientific engineering.

Michael Saylor: Yeah. Is that money because the most important person in the village said it's money? Or is that money because the best engineer offered it as money? Satoshi engineered money, whereas every government has created money. So if we start with that model, right: shared, immutable, correct. And now we go back to all the monetary systems in history.

Start with ancient coins, right? Coin networks, well there have been so many coin networks starting with the Lydians. Each coin network is: well I have a gold coin, I have a silver coin, I have a copper coin, oftentimes, there's three different coins, you know, one for 1,000, one for 10, one for 1, they have to be at three different scales. And the shortcoming is you create a system of coinage or custom tokens, and then you start shuffling them around in a network, you can only trade to the extent that you have those tokens. And if people lose the tokens the money is gone. But then people try to counterfeit the tokens, they try to cut the coins different ways. Eventually the coins might wear down, the coins have to be carried.

And so why do all these coin systems fail? I guess on one hand, there's never enough of them. There's always a limit. On the other hand, they're discrete. So you can go from 1000 to 100 to a 10, or from 1000 to a 10 to a 1 but how do you get to a 0.1 or 0.01 when you want to get to orders of magnitude more or you want to make change in the middle. You end up with all these money changers, and the result of the money changing is a huge amount of friction or impedance. And for thousands of years, people couldn't figure out how to manufacture coins that couldn't be cut or counterfeited or that were durable and so ultimately, every single mercantile system has its own internal coin system. And when you swap from one system to the second, there's this money changer in the middle and they take you for 10%.

I remember traveling from London to Rome back before the EU was formed, Robert, and I showed up with dollars, and I converted my dollars into pounds. And then I think I took a plane to the Netherlands and I converted my pounds to whatever the Dutch currency was. And I went to Belgium and I converted the Dutch currency to the Belgian currency. And then we went through France, I converted it again. And then we went to Italy and I converted again, I ended up with like 50% of what I started with after about 72 hours.

Robert Breedlove: Which makes sense to that energy analogy, because every time you have a transformation of energy from thermal to kinetic to chemical, whatever direction it's going, you have a loss. So there's this problem we've always had, transforming one form of energy into another and back.

Michael Saylor: Crossing a coin network is: every single time you cross the network, there's energy loss. And then there's loss over time as people counterfeit the coins, and you can't tell. And then there's losses as the coins get debased. And so the result is what, like 5,000 different systems of coinage over time. And every one of them struggled. Ultimately, they were the basis of the gold standard.

We have this myth that there was a gold standard, the good old days, but it seems like the gold standard never worked as far as we can find anywhere. I mean, we have the stories of the Romans debasing their coinage and the collapse of the Roman Empire. And we know the Lydians eventually lost theirs. And we know there must have been hundreds of systems of coinage in Greece, and one city would sack another city and melt down their coins and create the next set of coins. So I mean there's a problem with the whole idea of coinage.

And the problem with coinage is it's based on metal — metallic money. And of course, the highest form of money is the gold. So let's say it's called the gold standard. But what's the problem with the gold standard? I mean it never worked that well, because it's got some crippling flaws, and just some nagging flaws. There's inflation. Inflation is a nagging defect, it doesn't inflate fast, it inflates slow, but it inflates at 2% or 3% a year.

And because it's inflating, because you're mining more gold, and you're striking more coins or whatever it is you're moving around, it's not conservative. And by that I mean it's not mathematically conservative, or it's not conservative from a scientific point of view. You start with 100 units, and then you've got 102 units, and then 105 units, and 108 units. So whatever percentage of the gold supply you have is leaking at 2% to 3% a year. And it could be worse if it's not a closed system.

So if anybody introduces new gold into the system like when the Spaniards found the Inca gold and they brought back the gold from the New World, you get a massive inflation. I think I remember reading a history of Caesar where when he came back after the Gallic Wars, he had seized so much gold that that created a hyperinflation in Rome and the interest rates leaped. And there was just too much money. And so, either there is inflation because you seized gold or another nation flooded your market with gold, or there's inflation because the gold mining process continues. And so against our ideal model of money, that means that gold's not mathematically correct. It's not conservative, nor is it correct. It's an approximation. Under the best of circumstances, it's got a 2% error a year and so over the course of a decade, the best of circumstances is like a 20%, 25% error rate.

But the worst of circumstances is: I double or triple the amount of gold when I come back from sacking the Aztecs or the Incas, or I sacked Gaul and I just brought back all this gold and so then you've just got a massive error. So gold is mathematically erroneous because you're just shuffling metallic tokens and it's not a closed system.

The second problem with gold is confiscation because it's physical. The custodial rights of gold aren't really — they're not great. I guess they're a little bit better than fiat currency. But in some ways they're not as good. If I have a lot of gold, if I have a million dollars worth of gold or $10 million worth of gold, it's heavy — really heavy. I can't get it through an airport. I don't even know like — you know I tried to — I went online on Amazon once a few years ago and I wanted a little pocket knife. And I ordered this pocket knife — you know how pretty much you can only carry a blade, which is like, I don't know, half an inch long or what's the number? It's like some infinitesimally small sized blade, like a quarter inch or a half an inch blade. So I went on Amazon and I found someone advertising this really super small, dinky little knife that you could use to cut a string with. And it was advertised as being a TSA friendly, compliant little pocket knife. And I was so proud of myself, because I must have paid like $39 for it.

And then I took it to the airport to go through the metal detector. And I swear that the TSA agent stopped — you know, I put all this stuff through, they stopped the luggage and said like, Show me that. And they pulled out this little half inch blade to slice a string with, like, Sir, you can't take this with you. And of course the defense of: Amazon said it's okay didn't really fly. Right? And the point of the story is, you can't even slip a needle-like metal blade through an airport. I wonder if you could carry five gold coins and get away with it? Or ten gold coins, if you can't get my little pen knife thing through? And so the answer is: it's easy to confiscate [gold coins], custodial rights are really weak. The fundamental problem with the gold standard is, the custody is difficult and the security is hyper-expensive. And by hyper-expensive, it's — how much does it cost to secure a billion dollars worth of this stuff? Like you hire 24 armed guards and you have a vault in Fort Knox?

Robert Breedlove: Cost of custody, it scales with the amount stored. Which is something that's very problematic.

Michael Saylor: Yeah, the cost — you would say if there's a small amount of it, it's impossible to secure and if there's a large amount of it, it's exponentially expensive to secure. And the security cost scales in some ways exponentially with the amount you have. And also with the number of nodes. If I have 100,000 nodes, I have 100,000 points of failure. And so it scales with the number of nodes, the value of the nodes and the velocity of the nodes, or the velocity of the money, right?

Say I wanted to move a billion dollars of gold every day of the week, for 365 days in a row, 1,000 miles. Figure out the security cost of that, right? The conclusion is, this money is so heavy it has no velocity. It goes into Fort Knox, it sits there for 30 years, and nobody audits it. And under that circumstance you can almost delude yourself into saying it's secure. But it's only secure because there's no money velocity and there's no distribution. If I wanted to give gold to 8 billion people and wanted to move it every day, then the cost of the security would go up to consume all of the energy that humanity produces, right? Probably the cost of security for 8 billion people moving gold every day is more than the sum of the entire gross national product of the world. So the security cost doesn't work — doesn't scale.

Robert Breedlove: This gets into then the decomposition of money into an asset and a currency. So gold is functioning as an asset but it's not useful as a currency, something that circulates. Therefore, we introduce a currency. And the other thing on gold, I guess it's the best approximation of that immutable ledger that you described as ideal money. But it has still has this error rate for both inflation, which is relatively small, but the bigger error rate has to do with the violence and confiscation risk, that it's unpredictable when the market is going to be flooded, or if you're going to be confiscated, or if your custodian is trustworthy, etc.

Michael Saylor: It's not correct. It's tokenized energy. And it was the best idea for tokenizing energy in the Bronze Age, I suppose. But it's not a [mathematically] correct token. And because it's so easy to confiscate, it's not really a shared ledger anymore. It's not shared tokens, right? I can't share it because I can't carry it with me.

And then that leads us to the third problem with it, which is the hypothecation. It's too easy to counterfeit and manipulate. And it's easy to counterfeit because I can either debase the coins themselves, or I can lie about the asset in the vault while I give you a gold note. I either create the gold note and I lie about it or I just debase the coin, and that creates a problem of authenticity.

And that leads us to the fourth problem, which is authentication. It's too expensive to audit and authenticate. In fact, out of everybody that I know, I don't know a single person that's ever authenticated a gold coin. You wouldn't know. Coinage was an attempt to make authentication self-evident. Right? And Isaac Newton worked on how do you create a good coin, and most coins were crappy, but we still struggle with this issue. And so if you can't authenticate it, then that really undermines the "shared" part of the shared ledger as well, and maybe the correct part.

And so transportation is also a defect. It's too expensive and slow and difficult and dangerous to move. And distribution's a defect because it's difficult to move, difficult to authenticate and easy to confiscate. Then that means: how do you distribute it? It's too difficult to distribute, and too expensive to distribute. And when I talk about distribute I mean, how do I give it to a billion people? Right? I mean, practically speaking, if you distribute up to $1,000 of gold coins to a billion people today, there would be a 35% markup/markdown every time it trades, right? You pay $90 for $60 worth of gold. If you have $60 worth of gold, you'd be lucky to sell it for $40. And so that transfer cost is obscene.

And then finally, you've got this division issue. How do you divide a one ounce gold coin? You can't. And so I can't divide it. So I have to create silver coin and copper coin. And now I never have the right combination of change. So prices don't work quite right. So gold has — fundamentally, it's the best tokenized energy in the Bronze Age. But because of all of these defects, by the Middle Ages it was clear it was going to be replaced with some some type of fiat or some kind of checking system or other paper ledger system. It's not clear to me that they didn't replace it — I talked about the gold myth — it's not clear to me they didn't replace the 2,000 years ago. Like they talk about the Sumerian tablets, right, and the Sumerian tablets of clay — they have ledgers on them. So, isn't it quite possible, if not likely, that we had checking systems or ledger systems that were privately enforced by banks thousands and thousands of years ago.

So the gold myth is: there never was a gold standard. There was never a time when all money was gold. What you had was a time when the principal asset for a store of value over the long duration was gold. And it's quite likely that you always had — you know, the merchant had their ledger and you would have credit with the merchant. And maybe that was a credit with the ancient Roman merchant. And maybe that was a credit with the town. Or maybe that was a credit — if you were on a ship, anybody that's ever been on a ship sailing for months at a time, the quartermaster has all the supplies.

And if you wanted some of those cigarettes or some of that alcohol, they might let you sign it out, but then they charge your account. And when the voyage is over, they debit it against your wages. And so this has been going on for as long as people have been sailing ships around. What we have is the myth of the gold standard, but we've always had an asset currency system, and the currency was a checking system with a central counter-party — we can call them banks today. But we had goldsmiths that had gold notes back in the Middle Ages. And I think you've always had merchants, and you've always had quartermasters. And you've always had the guy in the army that said, "You lose it, we're going to dock your wages for that," whatever it is you lost or you spent or you consumed.

Robert Breedlove: I think this is a really important point to zero in on: so what gold gave us was this reliable medium for final settlement. But it can only be used for large transactions, essentially, because the economics don't make sense to use it for small transactions. So it doesn't circulate well but you can settle large transactions with it. So due to that technological limitation of gold — effectively it had such a high value to weight [ratio], I guess you might say — that we needed these cheaper systems, these credit systems or derivative systems — systems of deferred settlement built up around it. And that has been kind of the problem throughout history, because we have this system for final settlement, but we build systems of deferred settlement around it, which are *economically more efficient. But they introduce all of this need to trust counterparties, which comes with counterparty risk which blows up time and time again.*

Michael Saylor: And one of the implications of that is: the socio-political systems we can create are small and local. The implication is city-state, because I have to get my credit from the local merchant. So if the local merchant is 1,000 miles away, the credit system doesn't work. It breaks down. So I think you can create a trust network that goes about 10 miles or 20 miles — a trust network out to the suburbs. And then once you get beyond the suburbs, the trust breaks down. And if the trust breaks down, that means you've got Renaissance Italy, with 100 different city-states and they've all got their own little system of trust and ledgers and coinage. And there is no universal money. And so you can't have an easy rise of the nation-state under a gold standard or a coinage standard, or at least it's kind of challenging.

Robert Breedlove: It's fascinating to me that the actual shape and configuration of our institutions is derived from the nature of our money in a way. The reason we have a central bank is because of gold's technological limitation. So what it gets me really thinking about is when you swap out gold for Bitcoin, how transformational that potentially is to all the institutions we take for granted today?

Michael Saylor: Now I think that what we see is the progression of money as technology through the ages. And if you have a better money, you have a better economy. And as the money gets — if we come back to this issue of: Is it shared? Is it immutable? Is it correct? The greater the sharing, the greater the economy. The more immutable, the higher integrity. The higher the efficiency of the economy, the more correct. The more effective the economy, the faster the network updates, the faster the economy.

So just finishing up on gold — why does gold fail? In theory — why does gold fail in theory? It fails because the base layer protocol is okay. It's kind of God-given by nature. It's the creation and the smelting of gold. It’s okay, but it’s not conservative. The base layer protocol is not conservative.

The application level protocol — that one layer above the base layer, the layer 2 — the application protocol is difficult and dangerous. But what does that mean? Like so I give you 10 blocks of gold. That’s the base layer. Okay, well, so what are all the applications of gold? What can you do with it? How many people do you know that can actually refine or melt down molten gold and create something with it? That’s the dangerous part and the difficult part.

So gold applications, they’re difficult and they’re limited by the laws of physics. And that just means that — and there is no application protocol. So if I want to do anything with gold above the layer 1 it’s either a very difficult, dangerous application like gold goblets or gold coins or something like that, or I have to create the equivalent of a gold [paper] check, which is a manually implemented protocol.

And now we’ve got the same problems we have with a fiat currency. At the point you implement a gold check, you’ve in essence moved on to the Fiat standard with a gold reserve of some sort. So gold fails primarily because there’s no good application protocol. It’s not conservative, it’s too difficult, too slow, too dangerous. And the physicality of it invites violence. I can literally firebomb the city, kill everybody in it, and the gold won’t be damaged. And so when in doubt, shoot first, and then sift through your clothing and take your gold. I don’t have to worry about any collateral damage.

The last thing in the world you want to do is be carrying around immutable money on your person. When someone has an incentive just to kill the people and take the money. So that’s the fundamental problem with gold. That’s why the gold standard ultimately has always just been an invitation to war. And never-ending. I think thousands and thousands and thousands of wars — many wars, right? And an invitation to criminality and an invitation to violence. If the criminals don’t take your gold then the counter-parties take your gold and the counter-parties don’t take your gold then your own government takes your gold and your own government doesn’t take your gold the hostile government takes your gold. But ultimately, gold — because of its physicality — is imperfect property. It is imperfect property. And it's imperfect money, because the token itself is so cumbersome and unwieldy to utilize.

Robert Breedlove: And so this is Bronze Age money as you said, but this contention over gold, it extends right up into the 20th century in World War I and World War II. There's still massive gold flows taking place geopolitically while those nations are at war. So it's almost — I think it's an ill-understood aspect of human history that a lot of the violence between countries has been over the gold or about the gold. But it's not often discussed in the history books.

Michael Saylor: Yeah I mean when you think about it, you never read a historical account where someone says, we invaded their country and we killed everybody so we could take their paper currency. I could give you 5,000 accounts of: we sacked the city, we burned it and we took their gold. We might have sacked the city and hauled the people off in slavery. We might have taken their livestock, we prefer to take — normally the livestock is all dead by the time we take the city though. So the people are dead, the cattle are dead, the food is gone, the water is all gone, but the gold is still there.

So generally, if you look at the first 5,000 years, it's: we sacked the city, we took the gold. Then even in the modern era from 1700, 1800, 1900: We sacked the city. Okay, what do we make off with? There's not that many diamonds — normally they're not easy to find. There's not a lot of that. Sometimes, you know, the Nazis took the art, right? And the Nazi took the gold. And so there's a lot of that: we took the gold from the Treasury, Stalin seized the gold during the Spanish Civil War, and the Poles had to smuggle their gold away to keep it from the Nazis. And there's all sorts of examples of somebody sacking some city or rolling over someone's border to take their gold. That's because it's take-able, right? It's like, why do you want to take it? Because it's take-able. But not a single example [for paper currencies]. Like the Nazis didn't want Norwegian paper currency. They didn't want Swiss paper currency, they didn't want the Dutch or the French paper currency.

And not many people ask the question, Why? We didn't take their checks. We didn't take their securities. Because at the end of the day, securities and derivatives and all these things have no value. Maybe the factory has value. If it's not destroyed. Maybe the people have value if you get them to work for you. The gold has value but the paper doesn't.

Robert Breedlove: There's some examples of the opposite, actually. I think in Japan they had the Noborito laboratory where they were running experiments where they would bomb their enemies' territory with counterfeit currency. The idea was to go and bomb, say, England with a bunch of counterfeit pounds so they could hyperinflate their currency and disable their economy.

Michael Saylor: Psyops.

Robert Breedlove: So it's like not only do you not want the paper [currency], but you try to actually inflate the enemies' paper.

Michael Saylor: Yeah I think there's a lot of examples where hostile governments would attempt to just counterfeit the currency of their enemy to destabilize the regime. But it doesn't get reported a lot in political history or even military history. I mean some nations successfully did it to the US, but we don't want to talk about it, right? So you won't read that much but it's an effective thing you could do.

Well anyway, that ends my thoughts on gold. It's basically a metallic token, and it was our best idea, but it seems to have stopped working thousands of years ago. You could say 2,000 years ago almost certainly they had shared ledgers. And they just used gold as a method of final settlement. And we know that it failed at different points. We know that it was debased and resulted in the collapse of the Roman Empire. And so if Rome was the greatest empire on Earth, when their final settlement network failed, then the empire collapsed and what you have is: history is the endless succession of successive empires rising up with the new gold standard that was not debased, and then generation after generation the coinage of that next empire — the successor empire — would be debased, and then that empire would fail and collapse. And then another empire would come along and they would start the cycle over and over again.

And yet the myth of gold as immutable money or the sovereign store of value — it stayed with us for thousands of years into the 20th century. I would say that the gold standard has been dying a slow death — a death by a thousand cuts. 1914 comes along — you know we've got the golden age from 1870–1914 but then World War I comes along and then in 1914 every country abandons the gold standard. And maybe that's the final cut.

And after World War I, the Treaty of Genoa, we came back to a gold reserve standard and in essence we had gold backing the pound and the dollar. And that degree of backing successively slid, so there was a debasement of the pound and the dollar, consistently and gradually. And then of course the pound and the dollar became layer 2 applications if you will, and then every other currency became a layer 3 derivative. And then everything else in the economy was built on those derivatives, so you had basically layers of derivatives of gold that got progressively less backed by tangible energy — or de-nurtured. And that resulted in who knows how many collapses: the Great Depression.

Eventually we get World War II which you could say came out of just a bunch of economic collapses like the Weimar Republic. And all the gold ended up getting centralized and seized by the Americans in Fort Knox, and then we wrapped Bretton Woods around it, and Bretton Woods was the second gold reserve standard of the century, except this time just the dollar was the reserve currency, backed some percentage by gold — say 40% or 30% by gold. And then every year thereafter it slid from 40% to 30% to 20% — I shudder to say it must have been less than 10% by 1971.

And in 1971 we defaulted on the gold standard. In essence, at that point the gold reserve standard was effectively dead. Gold still has a fiction of being a store of value and an asset, and if you're looking for a non-sovereign store of value between 1971 and the invention of Bitcoin, you could've gone to gold, I guess you could've used property like land or commodities, timber rights, oil rights, something like that. And you could've used art. There's probably no one king, right? People dabble with, Is silver a store of value? I think the free market went back and forth but it's pretty clear that gold kind of died — it started dying if not had died about 10 years ago as far as I can see, when Bitcoin was formed.

And if we look at performance in the last 12 months, just for kicks let's just go and look at 12 months of performance. So it's quite a day, right? In 12 months Bitcoin is up 240%. Gold is down 9.82%. Over the course of 10 years Bitcoin is up 132% compounded annual growth rate (CAGR), gold is 0.94%. The S&P Index is 13.9% over 10 years. 33% over one year. Summary is: gold is not a store of value over the last decade, it's something opposite of a store of value. If the S&P is up 33% in 12 months, then a reasonable surrogate for the collapse of purchasing power of the currency in 12 months is the inverse of that, right? But you could say you need 33.7% more money to buy the same share of the S&P. So a store of value has to clock at 33% or better, and gold is -40%, and Bitcoin is +200%. So that's the marketplace screaming at you that no-one really sees gold as a monetary asset anymore except for the gold bugs.

Robert Breedlove: Yeah. One of the key points that jumps out at me here — getting back to your framework of ideal money — is that: immutable quality necessitates a proof of work, and gold again was just the best approximation of that. That's really the only thing that ever made gold money was the proof of work necessary to produce it. It was just really difficult to produce, therefore it minimized counter-party risk.

Michael Saylor: It was the best token that you could work to produce that you could possibly mold into a coin, right?

Robert Breedlove: But it then lost relevance in a globalizing society because it wasn't fast enough.

Michael Saylor: Yeah. Not fast enough, not smart enough, not strong enough. Everything that lives in a Darwinian world has to be faster, smarter, stronger.

Robert Breedlove: Yes, adaptive.

Michael Saylor: The cicadas came out after 17 years and I saw them a month or two ago, and they all come out of the ground at the same time and I watched the cicadas fly around my home and I thought, those are the stupidest, slowest, weakest creatures I've ever seen. Sometimes they would like fly to a branch and they couldn't land on a branch. They were literally so stupid they couldn't land on a branch, they could barely fly fast, a lot of them would just accidentally fly into the dirt and slam into the dirt. So when you looked at them and you compared them to other insects, you saw that those other insects are so much better at flying. You take it for granted, but you never really think, Oh those birds are good at flying, they're fast and strong and smart. Until you see something that isn't fast and isn't strong and isn't smart. And you might say, well, this thing is pretty much history. How could it possibly survive? And the answer is: they all have to birth once every 17 years, and there's a million of them in the air, and they're all so stupid and slow and inept that they're definitely going to die, but all the other creatures aren't going to be able to kill them fast enough to keep some of them from procreating.

Robert Breedlove: Right. The quantity over quality strategy.

Michael Saylor: Yeah and I think gold eventually just fails because it's not strong enough, it's not fast enough, it's not smart enough. And the world goes to the next best thing, and fiat arises not because it's better than Bitcoin but because there's no such thing as cryptography and computing. So in a world without computers and without cryptography and without networks, you ask yourself: What are you going to do next? The answer is: I'm going to come up with a shared ledger — I do have math. We've got algebra. We got calculus. I do have writing. I can create a shared ledger. And the immutable part is going to come from the institutional credibility. The credibility of the proprietor and the reputation of the proprietor, whether that's a king or a mayor or an owner or a religion.

But the immutability and the credibility comes from an institutional human source. And the greater the institution, the greater the monetary system. And you have the greatest monetary systems of history associated with the greatest institutions. And then you have the weakest ones: all the way down to somebody on an 80-foot sailing ship in the middle of the Atlantic and there's a ledger and there's a quartermaster and there's 42 people and that's their money system and they're trading. It's a money system which is good for 6 weeks or 8 weeks, but it is life or death for the 8 weeks. And that is the fiat standard and that's backed by the force of the captain. You have a good captain you might make it, and if you have a bad captain there's going to be a mutiny and everybody's going to die.

Robert Breedlove: Yeah it's a great framework. I love the framework of property and energy because there's also this deeper notion that most species are territorial. Most social species especially, and that includes humans. And I think we actually express and manage our territoriality through property. And so every little enclave, whether it's a boat on the ocean or a large piece of land tries to control its most important property because it's an expression of managing its territory. And that's why we have these regional monopolies on money we call central banks.

Key Insights from Chapter 10

- We’re moving from politically engineered money to scientifically engineered money.

- The main problems with gold: (1) It’s mathematically erroneous because the supply isn’t a closed system; it’s subject to disruption. (2) It’s subject to confiscation and the security cost doesn’t scale. (3) It’s too easy to counterfeit and manipulate. (4) It’s too expensive to audit and authenticate. (5) It’s difficult to distribute, or to divide. (6) There is no good application protocol.

- The gold standard ultimately has always just been an invitation to war. And an invitation to criminality and violence. If the criminals don’t take your gold, then the counterparties take your gold, or your own government takes it, or the hostile government does.

Chapter 11 | The Failures of Fiat

"If you want to drive people insane, change the rules."

Robert Breedlove: So gold fails [in a] globalizing, digitizing society. And then we have fiat introduced—and where does that take us?

Michael Saylor: If I can't use gold, I need money that's going to be stronger and faster and smarter. So what is an example of stronger money? Well, I'm in London. I have $100 million worth of money. I need to get it to Rome on a telegram, or I need to move it to Rome at the speed of a single horse, and I can't haul that much gold. So I use a letter of credit. If I can project money via some credit network — whether it's the Rothschilds network or it's an imperial network or something else — then I've got stronger money, and I've got faster money.

And if I can offer you that money in return for 3% interest, it becomes smarter money — smarter meaning I created an application that does something complicated. Like when I give you money for 30 days and at the end of 30 days you have to pay it back to me plus 4%, I created a derivative of the base-layer money, or a security of sorts. So the ability to create securities and the ability to project them distances with speed enables a more sophisticated economy and it's good for economic productivity. Hence the idea of a fiat.

And I think the earliest fiats were like the trading stations in the French territories of Canada. They had credits: credit systems on ships, merchant credit systems. There have always been private credit systems. And then eventually there became municipal credit systems, like the mayor of the city-state or something, or mayor of a hamlet, and then you get to statewide systems and federal systems and agency systems and who knows whether or not churches generated their own credit systems.

And we know in the military, the military generates credit systems. They'll have credit vouchers or travel vouchers in the military. I remember the airlines used to give you these tickets where you could fly "space available", and they would give them to their employees like flight attendants. So the flight attendants are able to use that and give one to their wife or husband. And they sell those things. There actually became a secondary market in travel vouchers because they had monetary value.

So you have private monies and you have public monies and then you have quasi-public monies from any kind of food stamp or institutional voucher or military voucher that gives you a right to something of value — it becomes money in and of its own, in time.

So the real question is, why does fiat fail? We know why it works. The reason it works better than gold is because you put it on a piece of paper, and ultimately the rise of databases along with the ability to shuffle paper around meant that it's a lot easier to create an application of fiat base-layer money — all I've got to do is take a check. Every time I write a check, I created an application, right? "Hey, pay Robert Breedlove $792.27."

Doing that in a fiat standard takes about 20 seconds. Creating $792.27 of gold is impossible for all but one in 100,000 people and would take a day. So it's pretty obvious why it works — because you can create applications faster, you can create more beautiful, complicated applications, types of credit, you can move them around faster. And you can travel with them further distances. And also, they don't invite violence to the same degree.

A very famous example of a private money that was created in my youth was American Express traveler's checks. And I don't know if you remember American Express traveler's checks, they actually became popular before the credit card. If people were going to travel internationally, and you were going to go to some place, they would say, well, before you go, you should convert your money into American Express traveler's checks. I mean, literally, this is a big business. And the number one use case was: that way, if you lose them or they're stolen, you can get them replaced. The idea was you were less likely to be robbed if you were carrying the traveler's check.

Robert Breedlove: It's like an analog multi-sig, maybe something like that?

Michael Saylor: Yeah, like a multi-signature version of money or wrapped money. And American Express made all their money because people would buy $100 million to traveler's checks and only redeem $95 million. And there was the float. And then there was the unredeemed amount. And that was the difference. People used to be very excited about these sorts of things. I mean a traveler's check is almost like a derivative of a derivative. The paper money is on top of the gold or itself, although now I guess paper money is the base-layer money, and the traveler's check was on top of that. People liked that idea, and then I suppose the credit card is just another example of a fiat application.

The idea is: you would send your kid on vacation with a credit card feeling comfortable that you could reload the credit card if they ran out of money. But if your kid said, Dad, I need $25,000 in cash just in case I have an emergency and I need an emergency appendectomy or something like that — there's no way you want them traveling with that. And of course, they couldn't even get across the border with $10,000 of cash.

So 99% of the world has more than $10,000 that they could put on a credit card. But nobody can take $10,000 in cash. So the fiat standard is appealing because of the portability of the purchasing power and because it doesn't invite violence. If someone steals your credit card, you can cancel the credit card. If they stole your traveler's checks, you could cancel the traveler's checks — things like that. It's like, make sure you write down the serial numbers of your traveler's checks — all of these things.

But why doesn't it work in the in the modern era? I mean the fundamental problem is: we start with inflation again — the government can create more of it. But here the inflation problem is not a minor problem like gold. It's a major problem. Instead of getting a guaranteed 2% inflation with an occasional 20%, you get a guaranteed — I think Saifedean suggested the number was more like 10-11% guaranteed. 7% guaranteed in the US, more in other countries. So you're getting a guaranteed inflation of somewhere between 7-11% a year, and you're getting occasional inflation of 30% or 40% a year. So the base-layer protocol doesn't work so well. It's not conservative.

So if you have a shared ledger that's not conservative, it's collapsing. And there's just nothing — I say to people, there's no engineered machine or structure that works with that degree of error or inflation. If you had a 2% leak in your gas tank — over what period, right? If you're leaking 2% a day, your car won't work. So leakage of your swimming pool or a balloon or any kind of electrical system or any thermodynamic system at all, just doesn't work with that much leakage. So the inflation — ironically, right?— inflation implies you're getting more but really, it's dilution, in a way. It's guaranteed dilution or depletion.

So the first problem with fiat is it's just depleting energy at too rapid a rate. If we just said 10% a year on average, then a system which is losing 10% of its energy every single year or almost 1% a month — it's kind of a crippling first-order problem on the base-layer. It's almost like if I'm building on sand, and the sand was sinking 10 feet a year. What structure can you build? When you're building on something, you want to build on granite and you want the minimum deflection. Granite doesn't deflect, steel doesn't deflect. Sand and clay deflects, swampland deflects. So the problem that we start with is the base-layer is collapsing.

The second problem of the fiat standard is confiscation. The government can seize your currency or they can seize the asset. To a lesser extent other people can seize it. I mean if you carry your cash around there's that famous image of Pablo Escobar sitting and burning $100 bills to stay warm. If you can carry it around someone with a gun can seize it, so that's a problem. If you don't carry it around then the counterparty that you trusted it with — generally the bank — can seize it. And in all cases, generally the government can seize it.

And so these are two fundamental defects with the fiat standard: (1) your property rights are weak under the confiscation risk, and (2) your property is defective because of the constant inflation. You literally bought swampland in Florida that's sinking. It's worse than that. Because swampland doesn't sink forever — it's pretty much as bad as it's gonna get. This is worse than that, because it just keeps getting worse and worse and worse.

Robert Breedlove: Yeah. There's like a conundrum of money here, where this is the tool or the economic medium that's intended to be trust-minimized. So you can put your wealth there and you don't need to trust anyone else. It's like a safe place to park wealth. But to get that you come with all these limitations of gold, historically. So to pick up these stronger, faster, smarter qualities of money comes with a cost of counter-party risk. So it's like, the conundrum is: gold, you can trust nobody — you don't need to trust anyone — but you can't really trade with anyone because of all of its limitations. Or you can use fiat, in which you need to trust a lot of counterparties, but you can trade with a lot of people. So there's this kind of conundrum we've been stuck between, historically. And Bitcoin is the answer: it's the collapse of the counter-party risk.

Michael Saylor: Yeah, we go from heavyweight, indestructible money that's a brick too heavy to pick up off the floor, to lightweight cotton candy, which is really easy to move around but ultimately not satisfying and blows with the wind, doomed to waste time. Yeah. I mean, and the counterparty risk, you know, is the explanation for inflation, right, the government's inflating it, and the counterparty risk is the root problem with confiscation.

But, you know, again, even if you had your cash in your pocket, you don't have counterparty risk in the near term. But what you do have is an invitation to violence, and, and it's too difficult to move it, then then you've got counterparty risk in the form of hypothecation, which is the third problem, which is, you put your money in the bank, you know, maybe they won't steal it. But they're making more of it, right, because they're loaning it out with a reserve ratio of 10 to one one to 10, or one to 20.

And so you put a million in the bank, and they create 10 million more, which dilutes your million. So the hypothecation is, is the third failing of a fiat standard. The fourth is the authentication again, because remember, we want it to actually be able to move it distances, but you can't authenticate it over a distance in its bearer instrument form, because you can't pay for something (large/remote) with cash. So the way to authenticate it is through a counterparty.

So I have to put my money in a bank, the bank has to issue a credit card, then I have to input the credit card into the website, then maybe the credit gets approved, then I have to do the transaction, and maybe the transaction gets approved, and maybe it doesn't get approved and it doesn't cross all borders, it would be impossible to do a transaction via credit card, you know, with a vendor in Cuba, or Nigeria, or China or I mean, all sorts of places where if you cross central banks, the credit cards don't cross those jurisdictions. So that is another way of saying it's an authentication and a transportation problem. How do I transfer/transport fee on currency? Well, how do I transform/transport a million dollars? Do you know any private individual that's ever moved a million in cash over a commercial airline? Like it or not, there's almost a presumption that if you had a million dollars of cash in your luggage, that you're a criminal, right? Isn't that interesting?

There's a presumption that if you try to do it yourself as a bearer instrument, that you're a criminal. And so what we have is a monopoly on transportation (of money) through the banks. But the banks only can really move within the central bank network, otherwise, they have to cross central banks. And so there is a monopoly at the bank level, then the central bank level, and then the central banks answer to the government state departments. So the central bank of the US won't do business with the Central Bank of Cuba, right? Period. Won't happen. So transportation across jurisdiction is difficult and dangerous, where it's possible. You're either in the zone where it's working, like you're in the US moving money between Bank of America and Citigroup at 3pm on a Tuesday, and then maybe it works.

Although the truth is even I get denied. I tried to do little credit card transactions between my credit card and my phone. I get denied like a third of the time.

Robert Breedlove: What type of transaction?

Michael Saylor: Try to move $2,500 to Cash App from your your bank, and the bank might actually stop it. Yeah, or anything like that. You know, if it looks like it's a transfer, you get all sorts of all sorts of limits on the rate at which you can move money. And if you want to go to higher volumes, if you wanted to move $50,000 or $100,000 invariably you have to have a person, a trusted banker, a human being that's calling you on the phone to make those wires.

And so the way that we move large sums of money is between 9 and 5 pm. With a trusted banker, with a manual authentication, and I mean, it feels to me like that hasn't changed in 30 years. So you can't, you can't move outside of the 9 to 5 (banker's window), and you can't do it. Without the trusted banker you can't cross any kind of technical jurisdiction or any kind of any kind of political jurisdiction that's not in their free flies zone. And the result of that is that the impedance to move that money is extreme.

So on one side, yeah, My take is 72 hours. And the real cost is hundreds of dollars. On the other side, if we go back to the credit card network, though, you have a monopoly in the credit card network. And the result is a two and a half percent credit card fee, right. So where the money does move, there's a two and a half percent fee. And, and that's under the best of cases, right? But the remittance network gets up to 10% sometimes. So you've got, you've got fees that range from two and a half to 10%. If you're lucky, you get one of those cashback cards. And so the effective cashback is one and a half percent or something. So the effective cost is only 1%. But it's 100 basis points, which is a huge amount of money, insanely large amount of money, and the network really isn't competitive. And it hasn't changed much in 30 years, and there's massive fraud chargeback issues.

Robert Breedlove: And all of these impedances you're describing, I would argue are essentially an impairment of property rights. Where in a pure property right, you should be able to send an express, receive, move, move the asset, however you choose. But when you have to then answer to a counterparty or go jump through these hoops, or wait, all of these things impair your property rights and money.

Michael Saylor: Yeah, I would agree with you on that. I think that your property is impaired, it is taxed. It's either explicitly taxed by a government or municipality, a state or federal government, or it's privately taxed by the money transfer network charging you a fee to move it around. And that's a challenge. So transportation is either difficult or dangerous, or it's impossible, or it's expensive. Those are the problems with that.

And then that leads to two other real deficiencies in the fiat standard. One, you got a lot of credit failures, because lots of complex debt is layered on top of the base layer money. And that's complex debt and credit instruments with counterparty risk. Now check kiting, right, check fraud, or any anything like that, where someone's stood in to pay lots of fraud and lots of credit failure.

And then you've got securities fraud. And I guess with security fraud, what I mean by that is, there's too many complex applications that are constructed based upon a counterparty's representation to you, and there's no way for you to transparently authenticate or assure yourself that the application is properly constructed and properly backed, right? And you're destined to have that. Someone says, oh, we have $2 billion in reserves, but they don't, but you don't know they don't. And then there's a collapse of the securities they issued or the credit or the vouchers or whatever they issued. And so that those things are inevitable.

And the question is why? Why do you have credit bubbles? And then why do you have securities fraud on top of the fiat standard?

And I think the answer, if we come back to our base first principles, is the base layer protocol of fiat is defective, the base layer money is defective, right? Is it a shared immutable, correct ledger? And the answer is no. It's not shared. It's not immutable. It's not correct. Right? It's a quasi, it's an imperfectly heterogeneously shared, mutable, incorrect ledger. So I guess what you could say is it's a ledger. Right?

Robert Breedlove: Yeah.

Michael Saylor: It's a ledger. But it's not shared, it's not immutable, it's not correct. Yet, you know, you don't have final settlement. So you have massive fraud on the credit card network, you have transfers that didn't take place that are represented to have taken place, you have balance sheets that don't exist that are represented to exist, though. You have an issue there with the base layer.

The second problem you have is the application layer protocol. It's random and manual. So the protocol by which I would create application's a base layer money, like a security is an application, a credit card is an application. All of these things are, there's no technical protocol, or programmable. There's no API. It's a manual protocol. So every single bank creates their own application, like a bank loan is an application of fiat. And every bank executes their own set of loans. And they're all expected to maintain a certain degree of reserves, right? But they're audited. And the fact that they have to be audited is proof that they're all just manually executed and heterogeneous.

So there's no transparent perfected application protocol. And the result of that, or the implication of that is that the applications and the fiat standard have no integrity. They have no integrity in a mathematical sense, or a technical sense, or an engineering sense. You know, a physical metaphor is, in the real world I'm standing on solid ground. And in the fiat world, my eyes are closed, and someone told me, I could take two steps to the left, and I would be standing on solid ground, right. But I don't know that I'm standing on solid ground. I don't know that I'm 100 feet from the cliff, I might be one foot from the cliff. Someone told me, and I don't have a protocol to ascertain whether it is true or not true. And so, you know, you could generously say, people just make mistakes all the time, I thought you were 10 feet from the cliff, and you only were nine. And so that's why you're dead. Maybe it's just an honest mistake, right? But then again, maybe it was in my vested interest to tell you that you were 10 feet from the cliff. And you were really five feet from the cliff.

Robert Breedlove: And this gets straight to the heart of Bitcoin, right? Where it's removing the need to trust, you know, "don't trust, verify." And this securities fraud you're describing, I mean, it culminates in the weapons of financial mass destruction with this huge gigantic derivatives market, which is really premised, I would argue largely on this, this gap we have between trade and settlement in the fiat system. There's no final settlement occurring, which is the verification mechanism. So in that gap, we have an accumulation of hidden risk that ultimately becomes systemic and leads to giant systemic blow ups. And, you know, to your point, there being no integrity, it's like, of course, there's no integrity. I would say the lack of integrity is synonymous with corruptibility. And so not only does it not have mathematical integrity or, or financial Integrity, but you could ultimately say there's no moral integrity, because these systems that just don't hold up to corruption.

Michael Saylor: I can't take personal custody of a billion dollars of gold. So therefore, I trust it to the bank. And I take their gold derivative. And I assume that that is base layer money. And I can't take delivery of a billion dollars of currency or fiat. So therefore I entrust it to the bank, and I take their credit instrument as my base layer money.

And you could say that what you've got is two fundamental shortcomings here that cause these two systems to break. One is the base layer protocol is defective. You need a base layer protocol where you can take final settlement. At any scale. Can you take final settlement of $387? Can you take final settlement of $38 million? Can you take final settlement of $38 billion? Can you take final settlement every day, if you can take final settlement at any scale at any frequency, then the underlying gold standard, or fiat standard would have some integrity, at least a modicum.

The second problem is there's no application protocol. So there's no way to create a security or derivative or some Layer 2 application on top of the Layer 1 monetary token. We need the applications, right? The world needs more than just base layer money, like you need credit or you want yield. Or you want to build an insurance contract. Or maybe you need to be able to post a security deposit. And you need to be able to get the security deposit back or maybe you want to put something in trust with a multi signature arrangement for 37 years. There are lots of applications that make the world work. But there's no application protocol. So in both cases, with gold and fiat, the base layer protocol is defective. And the application protocol is either non-existent or defective. And what makes Bitcoin special? And what makes a digital money is Bitcoin gives you a base layer protocol to take final settlement. And it gives you an application-level protocol, too. We could debate back and forth, right? Is the application protocol on the base chain the blockchain? If so, it's a Bitcoin transfer. Or is the application protocol on lightning, you know, using a layer two? Either of those, you know, could be viewed as an application protocol.

Robert Breedlove: Yep. And you're still maintaining the option for final settlement even on Lightning. And this is so critical for people can understand this, the antifragility of the world economy, even, is dependent on this simple fact. If we can have higher frequency final settlement, we can have more verifiability in the economic structure, which means less corruption and less blow-ups. Whereas the fiat system right now is the complete reverse of that: very low frequency final settlement. Very low verifiability, tons of corruption. Lots of blow-ups.

Michael Saylor: Yeah, final settlement with high frequency creates reality.

Robert Breedlove: Yes, that's what the blockchain is, right?

Michael Saylor: It's reality. And every single second, gravity is testing your structure and testing your orientation. You know, there's that phrase, can you defy gravity? And the answer is, maybe if you're a great athlete for a second or two, right? For a very short period of time, can you defy gravity, it requires extraordinary strength, extraordinary agility. And it still comes at great peril. I mean, you could try to do a backflip and you can land on your head. And you can break your back and you'll be dead or paralyzed for life within a couple of seconds. So there is there is a final settlement.

You know, if you're considering our crypto asset network like Bitcoin, and Bitcoin is the greatest, maybe the only successful example but certainly the greatest example of a crypto asset network, the frequency parameter, and the block size parameter for a crypto asset network is the equivalent to the space time constants in a universe. So the frequency, the 10 minute frequency, that's how fast the universe evolves. And the block size is the gravitational constant: the relationship between mass and energy or how tightly packed is matter.

And once you've established those two constants, you've defined the constants that run the universe. Everything else evolves around those two constants, and this is why you would never want to change them, because changing them is playing God. And you only get to play God once when you start the universe. If you change the frequency and the block size of the coin, you invalidate all work that's come before you, you imperil all structures that have formed in the universe since the beginning of time. You impair every mechanism that's been created within that set of structures, you impair them, and maybe break them and you throw the future into chaos.

All of those things, it couldn't be clearer to me, but if you want the physical metaphors, like what if I triple gravity on the surface of the Earth right now if I could snap my fingers and triple gravity? How many pieces of furniture do you have that crumple? How many chairs collapse? How many people's hearts stop? And how many ships sink? How many planes fly? How many factories keep working? Stuff breaks—not just factories start breaking, buildings start collapsing, bridges collapse, tires deflate. Right? And the person that worked for 100 years to create that great company, or the Rockefeller Center and just crumbles. Okay, so you worked for 100 years – you did something and it was beautiful until everybody's dead. Yeah, what happened? Well, you changed the gravitational constant, you changed the space time constant. And therefore the structure which had been carefully constructed since all of eternity forward, the structure is now rendered null and void and it collapses.

Ships are designed based on the Reynolds number, there's something called a hull speed. And the hull speed is a function of the shape of the hull and that's a function of the way that fluid flows and the end, these are basic constants. And no matter how fast you push that hull, it won't go any faster because the water pushes back right as hard as you push it. That's why a 150 foot vessel that's 30 foot wide will cruise 12 and a half or 13 knots and it doesn't matter the horsepower of the engine in it. It's just the way that the world works.

Another example of that is shockwaves and the speed of sound. There's a reason for the speed of sound, I mean, it's a fundamental constant. You go faster than the speed of sound and then you're moving faster than the air can move out of the way you get a shock wave, lots of devastation. What happens if you change the speed of sound? What if you change the speed of light?

Robert Breedlove: Oh yeah, so I love this because maybe we're creating another meme here that Bitcoin is the E = MC2 of money. And that, to your point, you only get to play God once. Satoshi did that. Satoshi set the rules and this you know, the strategies we build based on those constants, they are dependent on the invariance of those constants. If you go and change them then all the strategies that have been built up around them collapse.

Michael Saylor: Yeah, like the entire Bitcoin mining industry: it's all predicated upon Bitcoin coming out at a certain speed and the protocol tapering off, and that the value of all transaction fees are predicated upon the block size. If you increase the block size, the transaction revenues collapse. If the transaction revenues collapse, the Bitcoin mining profitability collapses. If the Bitcoin mining collapses, the energy usage collapses. If the security collapses, the network collapses.

You know what's going on with some of the proof of stakers is: if you flip from proof of work to proof of stake, you destroy your entire mining industry. If you destroy your entire mining industry, all of the security — all the social, political, economic, thermodynamic security — collapses. If it collapses, the integrity and the durability of the money collapses. There are all sorts of second-order, third-order, fourth-order consequences to this.

The fundamental basis and integrity is: one has to know that God is not going to change the rules on you if you spend your entire life working on something. If you want to drive people insane, change the rules. Like if I just change the speed of light and the speed of sound and I quintuple gravity before you compete in the Olympics? And then when your competitors start competing, I dial down gravity to the moon gravity and they jump 180 feet in the air and they beat you. You're like, well, that guy had moon gravity and I had Saturn gravity and my legs are broken and my hips are broken and I'm lying on a gurney with an IV in my arm and my heart is about to stop. And the other dude is bounding over the stadium. It doesn't seem quite fair! Okay, at some point when the Gods toy with you like that, you say, "Well I'm not going to play this game anymore." And people withdraw from the ecosystem. It's like, what would you do other than run as fast as you could away from a random universe that wanted to kill you?

Robert Breedlove: We're back to the original introduction of immutability — how important that is for money. And I've said this before, but I think the way you're articulating it here really unpacks it, is that this is really what Bitcoin's doing: it's radically changing the *world by virtue of being unchangeable. It's the first unchangeable set of rules we've ever had. So it's causing us all to reorient our strategies. Whereas fiat is a constant changing of the rules. So you're talking about driving everyone insane — that's why we're going mad in the world.*

Michael Saylor: With every election cycle, you could argue, with every political appointee. Because I could even appoint a different head of the Fed tomorrow. The immutability of Bitcoin, though, is an emergent property — people would say, there's nothing that's absolutely immutable short of being God — it's an emergent property, but as Bitcoin becomes more decentralized, the thermodynamic inertia of the thing increases and it becomes more immutable. When there were a hundred HODLers it was something, when there was a million it was more immutable, when there was a hundred-million it was more immutable. When the Bitcoin mining spreads further, when the HODLers spread further, when there are more institutions and organizations and jurisdictions HODLing — the more it spreads, the more immutable it gets, and therefore the higher degree of confidence you can have in the integrity of the entire thing.

That is what's interesting about this entire phenomenon. I've described it as a fire in cyberspace, but a better metaphor might very well be a monetary virus in cyberspace. It's a living creature — and we can go back and forth over what kind of living creature it is —but it's a living creature that's massively decentralized, massively fault-tolerant, that's got a genetic code. And it's a swarm creature: it's genetically reproducing DNA and the more it reproduces, the more it spreads, the more decentralized it gets, the more immutable it gets, the more vital it gets, and the stronger it gets.

Robert Breedlove: It's paying us to live in symbiosis with it. That's how it's growing, right? It's incentivizing everyone to interact with its network favorably, whether you're a miner, a HODLer, a developer — everyone's aligned with the success of Bitcoin by virtue of its incentive design.

Michael Saylor: Yeah, it's some kind of swarm cloud of truth. A living truth nebula.

Robert Breedlove: I tweeted this out the other day, I thought you might like it: because fiat is the opposite — it's like a cloud of self-deception, we think we can just print more money and things are okay, paper over past mistakes. And the remedy to self-deception — I've been talking to this guy John Vervaeke — he says: is wisdom. So you might say it's this cloud-swarm-organism of wisdom, too. That we're getting back to the principles of money.

Michael Saylor: Yeah. An extraordinary thing. Extraordinary.

Robert Breedlove: And to outsiders we sound crazy.

Michael Saylor: So, the human race tried with metal money, and then it went on to fiat money, and then we invented computers, and then we put them on a network, and then we perfected cryptography. So once we checked off the computer box, the network box, and the cryptography box, then you had all the components you needed in order to create a crypto-asset network. And that's what Bitcoin is, right? Maybe not the first, but the first successful crypto-asset network that emerged as the ideal technology for money.

Key Insights from Chapter 11

- The fiat standard is appealing because the purchasing power is portable and because it doesn't invite violence.

- The first problem with fiat is it's just depleting energy at too rapid a rate. If we just said 10% a year on average, then a system which is losing 10% of its energy every single year or almost 1% a month — it's kind of a crippling first-order problem on the base-layer. If you're building on sand, and the sand was sinking 10 feet a year, what structure can you build?

- If you flip from proof of work to proof of stake, you destroy your entire mining industry. If you destroy your entire mining industry, all of the security — all the social, political, economic, thermodynamic security — collapses. If it collapses, the integrity and the durability of the money collapses. There are all sorts of second-order, third-order, fourth-order consequences to this.

- If you want to drive people insane, change the rules.

- The human race tried with metal money, and then it went on to fiat money, and then we invented computers, and then we put them on a network, and then we perfected cryptography. So once we checked off the computer box, the network box, and the cryptography box, then you had all the components you needed in order to create a crypto-asset network. And that's what Bitcoin is.

Chapter 12 | Why Bitcoin Succeeds

"The private keys of cryptography are the strongest property rights in the history of the human race."

Robert Breedlove: We've talked about how gold emerged, its limitations, why it ultimately fails. We talked about why currency was introduced to augment some of its shortcomings. But also suffers from its own problems and it ultimately fails as we have seen and are seeing. So how does Bitcoin fit into the picture?

Michael Saylor: So Bitcoin as money. Simply put, Bitcoin's the best money the human race has ever had the ability to embrace. We know gold is defective because of its mechanical nature, and we know fiat's defective because of its political nature. Why does Bitcoin work? Bitcoin has the things that you need in order to construct a shared, immutable, correct ledger. That's our target. Shared, immutable, correct ledger. And Bitcoin's got a base layer protocol. The base layer protocol was genetically embedded in the system from day one — the 21 million coins is one of the critical parts of it. The 2.1 quadrillion satoshis and no more than that.

The second part of the protocol is not just the mathematical coin parameter, it's also the proof of work security model. So when we created it, [Satoshi] created it with the 21 million coin limit, he created it with the proof of work security model, and that is the part that makes it more likely to be immutable and shared and correct.

The third part is the balance of power between the miners, the nodes, and the wallets—or the HODLers. It wasn't inevitable that you had to build it this way. Bitcoin is an engineered monetary system, so it's a feat of engineering. They built a protocol, they chose components — they chose cryptography, they chose proof of work — all of these are engineered components.

And the balance of power between the miners, the nodes, and the HODLers with the wallets — that became clear in the block size wars I think. And following the block size wars — I mean, first miners were the nodes and then the miners and the nodes split, and eventually we saw that the nodes themselves had a huge amount of power to check the power of the miners, but now we're seeing the emergence of the power of the HODLers, and of course — all three of these dimensions are decentralized.

So I would say that we need the protocol, the base layer, that base layer protocol that I just described that replaces the gold that was here at the beginning of the universe, and it replaces the base layer protocol that's the political artifact of any kind of fiat currency.

The second thing that Bitcoin has that makes it a much better money is it has an application layer protocol. The base layer protocol ensures that we'll never have more than 21 million Bitcoin. I mean that's the most important thing. And then it ensures — what do I need? I need durability, integrity, and security. So I need to know that there will never be more than 21 million Bitcoin even 1,000 years from now, I need to know that people aren't going to monkey with material things like the frequency or the block size — that's the integrity part of it — and I guess durability and integrity they go arm-in-arm in a way because if you change the frequency and the block size you may inadvertently end up changing the limit [of Bitcoin] at some point because you break it.

And then you need security: you need to know that no malefactor can hack it or break into the network. So we're getting all of those things — the security, the integrity, the durability, via the base layer protocol. But the application layer protocol is also important, and that's the ability to do final settlement transactions at the base layer with a Bitcoin wallet address,

We could debate, interestingly, Robert, like: do you need Lightning or not? And that's an interesting debate, because I think that if we just had the ability to do base layer transactions on the Bitcoin network, the system is still perfected enough to serve as a monetary system. You would end up with the layer 2 application being built via counterparties with some counterparty risk instead of what you can do with Lightning. But the combination of base layer transactions via Bitcoin wallet and then layer 2 transactions via

Lightning wallet, I think those two things together are clearly superior as an application-level protocol. And if you group them together and say that is the non-custodial decentralized open source protocol for moving money around in the Bitcoin standard, then you've got the structure of something really compelling.

Another way to describe this is: Bitcoin gives you a base layer protocol to move large sums of money around at small cost in an hour, and it gives you a layer 2 protocol to move mid-level and small sums of money around instantly for almost nothing. On my Lightning wallet I can move 1,000 satoshis at the cost of 1 satoshi in a second. Which is more efficient. I can move 100,000 satoshis, which is like $30 — I can move $30 for 1 satoshi in a second. That makes that layer 2 protocol faster and more efficient than any other currency transfer protocol that I'm aware of in the history of the world.

Robert Breedlove: Right. And my understanding of it is: as the network proliferates and the network density grows, that the transactions per second actually increase. And I don't think there's an upper bound on that, so the Lightning Network could actually be at full scale this global system of instant final settlement, effectively, which is mind-blowing.

Michael Saylor: Yeah, I mean I think the Lightning Network is really strategic to the proliferation of Bitcoin, but as a theorist I think that what's critical to Bitcoin's success is not the Lightning Network but the underlying final settlement. As long as someone can take final settlement of $100,000 in an hour for $5 or 50 cents — I think right now if you set a lower fee and you were willing to wait longer you can take final settlement in a day for a nickel or some very small amount. And if you want it done fast then you pay more.

Robert Breedlove: I recall that optionality is what keeps counterparties honest — as you said. I recall you describing people being polite in Miami because everyone's potentially armed — that kind of dynamic.

Michael Saylor: Yeah it means you can pull your Bitcoin off the exchange or out of the bank or away from your counterparty in an hour. And so I think that's critical. I think Lightning is probably something that's going to accelerate the spread of Bitcoin, and it's the logical next step. If we can engineer a decentralized non-custodial crypto-asset network based on private keys and cryptography and the like as Bitcoin, then there's no reason why we couldn't engineer a layer 2 of Bitcoin that's also non-custodial and decentralized that makes a different set of trade-offs. In essence, I think the Lightning Network is a decentralized, open, proof of stake network. But the critical issue is: it's a proof of stake network not staked with its own token — it's staked with the underlying token of Bitcoin, which makes it a much higher integrity, much longer-living, more secure open decentralized network, than one that was created with its own token, which would be self-referential.

Robert Breedlove: One other slight difference there is the actual revenues generated on Lightning are for routing transactions versus just staking the asset — you get some percentage yield off the asset — you're actually getting paid for services rendered.

Michael Saylor: Yeah. Which makes sense. So we've got two things: we've got a base layer protocol, and we've got an application layer protocol. And so why does the Bitcoin standard work? Or why is Bitcoin the best monetary system? How does it succeed in reality?

Well, let's start with inflation. The scourge of inflation, which destroys fiat and destroys gold, doesn't exist. In Bitcoin you've got a conservative system and a mathematically complete system. So it's mathematically correct and it is conservative in a thermodynamic sense that we don't add additional energy in the system nor does it deplete energy in the system. And the basis of that conservation is that base layer protocol. The base layer protocol plus the proof of work together. You couldn't design the system which wasn't conservative. Just because you get everything else right — if Satoshi had gotten everything else right but had left some randomness or some inflation in the token count, it wouldn't have been ideal.

The second thing that is a big attribute for it is the difficulty of confiscation. So the private keys of cryptography are the strongest property rights in the history of the human race. So if we think about cryptography as the basis of property rights, then truthfully the only property you can really own is digital property on a secure crypto-asset network, of which the only one we know of is Bitcoin. So right now, the only property you can truly own is Bitcoin.

Look, if we ever went to Alpha Centauri and we went to a new planet and we cut off communication with the Earth, it's possible that the dictator of Alpha Centauri could spin up another Bitcoin network — Centauri network. And if it was a closed system and it had the same exact mathematics as this, as it spread as a virus and infected the billions of people in Alpha Centauri it would be its own crypto-asset network. So you can imagine something like that, but right now what you've got is one successful life form and that is Bitcoin.

And so you've got one successful digital property and now we actually have an engineered property and so you have a scientific basis for property rights. We've never had it before. It's illustrated pretty effectively. It's like, if you have a million dollars and you convert the million into a building, well the building can be seized by the mayor by eminent domain. You can lose your building and you can't move the building. So the building is impaired property and the building can be taxed. It is immobile, it can be seized, it could be taxed, and it also has physical nexus which means a meteorite can fall on it or an earthquake can wipe it out, so there are risks to the building.

The useful life of the building is 100 years, maybe less if it's poorly constructed. And it's only appealing to people in that geographic-political nexus. If the building was in North Korea, how valuable is a building in North Korea to an investment company in New York City? So a building as property leaves a lot to be desired.

Land as property has the same issue. A block in Manhattan is a lot more valuable than a block in the middle of the desert, because the land is geographically located next to a lot of investors with a lot of money that can actually develop the land for some economic purpose. And the land can be impaired and taxed. Property tax — there's a phrase: in the real estate business they always talk about triple net, and triple net is like, after your taxes, after your expenses, etc. So these are high-maintenance properties. They're very expensive and not very portable.

So you can rifle through the properties. Well, there's art, there's gold, there's silver, there's commodities, there's securities, there are bonds, there are currency baskets. They all have some liability, many of them are currency derivatives and so as currency derivatives they have dilution coming from the inflation of the underlying currency. But the ones that have physicality are subject to search and seizure and impairment because you can't move them. Others are unique, maybe, but even though you might own art people are going to keep making more art.

So if you're looking for a property that no one can seize, that is hardest to impair, Bitcoin is the hardest to impair property. Technically I would say it's probably the hardest to tax, the hardest to destroy, the hardest to seize, and the hardest to impair, of any property available to you to purchase at this point.

Robert Breedlove: One thing I think is really interesting here is that if we define property as an exclusive relationship between an owner and an asset, but that relationship requires an enforcement. Typically it's enforced via the government historically, but Bitcoin is different. Bitcoin is this property right that —

Michael Saylor: You can enforce yourself.

Robert Breedlove: Yeah, enforce via the network, and it's actually the first property right completely independent of that monopoly on violence. Like, not only does it not require enforcement by the government, but it's completely independent of the government. And that's just a lot to get your head around, I think – that we've actually invented a property right fully independent of government.

Michael Saylor: Yeah, and we'll talk about it more when we get to talking about proof of work and mining and the like, but you can go really deep into the basis of security for this property. But for the purposes of the Bitcoin standard right now, I think the most important thing to be said is it's the hardest thing to confiscate on Earth. And I joke, you can take it with you. If I shoot you in the head you can't take your gold with you. The Egyptian pharaohs built the Pyramids to take the gold with them—they were all robbed. If I said to you, take the million dollars and buy some property and then you're going to be murdered by somebody in your family next week — buy some property that they won't get after you're dead — what is that property?

Robert Breedlove: There's only one.

Michael Saylor: Securities and bonds and currencies won't last. Gold — all these things are seizable. You can't take them with you. But if you hold Bitcoin and you hold the private keys in your head, when you die, the Bitcoin goes with you and no one else gets their hands on it. And the significance of that is: because it's impossible to confiscate with force and because it disappears with your death, then any hostile actor always has an incentive to negotiate with you because it's better for me to hold a gun to your head and take half of it than pull the trigger and get none of it. That's not true with any other property. In all other cases I can pull the trigger and get all of it. And so you have an unstable situation with that property.

And the interesting thing is: that confiscation element, it works at the individual level, the institutional level, and the nation-state level. So if one nation is going to declare war on another nation and murder everybody, they get their gold. They also get their land. They also get their factories. I mean there's a joke, neutron bomb — I'll just nuke you and we'll leave the cities. I might get the gold, the silver, the farmland, the buildings, the factories, the ships — but I wouldn't get the Bitcoin. And so Bitcoin is not spoils of war. Other things may be, but Bitcoin is not spoils of war.

With regard to a bank or a financier, if you own the Bitcoin and someone puts a claim against you, the way it normally works is: I've got a million dollars with the bank and I borrow against it and then they decide to mark my securities down and they force-liquidate me and they sell all of my securities and take my property. That happened with Archegos. I mean it happens all the time. It happens all the time with crypto exchanges. If you have the crypto on the exchange, or your Bitcoin, and you borrow against it and you can't meet a margin call, they can force-liquidate your assets and they seize them. But if you pull your assets off the exchange and if you hold your private keys, then your bank can't seize your assets in a hostile fashion — they would be more likely to negotiate with you. And because you can take custody of it, you don't necessarily need to rely upon this all-or-nothing: either I give you all my stocks and I borrow against them or I have none of them. We could split the difference with: I give you proof that I have it — proof of reserves — without transfer of the asset, in return for a different type of loan.

Robert Breedlove: Let me ask you a question real quick: you touched on something really important, that Bitcoin alters the economic logic of violence, which is a very important element of how humans organize themselves. And you mentioned previously at the beginning of the series how war has this tendency to accelerate learning for people. Under the duress of war, the desperation of warfare, we figure out new ways of doing things — do you think armed conflict in the 21st century could accelerate nation-state adoption of Bitcoin as this advantage dawns on them that they can be plundered of their gold but not necessarily their Bitcoin?

Michael Saylor: Yeah. I think all stress will accelerate the adoption of new technology. For example, when we were fighting wars we were hauling over C130s or air freighters full of pallets of cash, right? So normally the way you fight wars is with gold or with cash or with something. And if you're cut off and you need to buy or sell materials of some sort, you need hard currency. So I think that yeah it will accelerate the adoption of technology like Bitcoin.

And coming back to this issue: because it's not confiscatable, it doesn't invite violence on your person from other family members. Like for example, do you ever worry about being murdered by a family member because of whatever your will says? If you had your Bitcoin in a multisignature relationship or a multisig something-or-other, then maybe it changes that dynamic. So I think all sorts of violence against the individual, against the company, against the institution, or against the municipality or the nation-state — I think those are all less likely. Less likely to happen and more likely to result in negotiation.

So why else is Bitcoin going to succeed? Why does it succeed? Well, because hypothecation is so much more difficult. I'm not saying impossible, but the layer 1 and the layer 2 protocols — Bitcoin and Lightning — those mitigate the need for hypothecation. It's less likely you would need to trust all your asset custodians. If I can move 1% of my assets a day for a satoshi, then I wouldn't put all my assets in a bank. Right now all of your assets, all your stocks, are probably in a bank. You don't hold any of them. In theory you could actually take possession of your stocks as certificates and put them in a safe deposit box — take them out of street name, they say.

Robert Breedlove: Yeah, I don't really know if you can do that anymore. I may be wrong about this but I think the DTCC now officially owns all stocks even if you own the certificate.

Michael Saylor: Yeah, so it's harder and harder. So the protocols mitigate hypothecation, and also they punish hypothecators. If someone did go naked short Bitcoin in order to make a quick dollar and you pulled all your Bitcoin off the exchange and they have to then settle, it creates a massive short squeeze. So whenever you pull those securities away from the hypothecator then they get squeezed. So if I knew that it was impossible, like if you had a billion in gold and I knew you're never going to take delivery of it, I can short it 100:1. And if I knew you'd never take delivery of the Bitcoin I might get a little bit cute, but in a world where people can — there are exchanges that fail just like there are bad banks that fail, it's just that they're failing continually in Bitcoin because that's the nature of the free market and that's how the free market squeezes out defective actors and malicious actors. And the ones that have perfected security and the like, they succeed.

So the fourth dimension of virtue for Bitcoin is superior authentication. And again the layer 1, the layer 2 protocols and cryptography in general what you can do — proof of keys and the like, proof of reserves — they solve the authentication issue. And maybe it solved for the first time, effectively, when you want to settle and you want to buy a piece of real estate sometimes it takes a week or two weeks to do a title search and it's a manual process so you're talking about a hyper-expensive manual process to even do a title search on a small amount of property. And Bitcoin gives you a way — I mean presumably it's somewhere between 1,000%-1,000,000% cheaper and faster. Like it's not an order of magnitude, it's many orders of magnitude better on the authentication. And maybe even more important than that, you can automate the authentication. So that means that a computer program could authenticate 100,000,000 different HODLers every hour for a nickel.

And when you get to that level that opens up the possibility to create new types of businesses and new types of applications that otherwise would have been cost-prohibitive because of the friction involved. Bitcoin succeeds over fiat based upon transportation dynamics too — the fifth element of virtue.

It's fast on layer 1, and it's instant on layer 2. Moving $100,000,000 of property on layer 1 in an hour is fast, but moving $100,000 on layer 2 in a second for nearly free is instant. And so it's just much faster. And it comes with no artificial geo-political constraints. Even when you have fast on fiat, you have fast but with the geo-political constraints that are continually shifting. Whereas fast on Bitcoin has no geo-political jurisdiction, it's cyberspace. It's orthogonal to geo-political space. And what you want is a situation where a billion people in China could all authenticate themselves and trade on a value network through the great Chinese firewall to a website sitting in Cuba or sitting in the US. You'll never do that with any kind of fiat system.

In fact, the entire Chinese system is constructed to create a capital firewall so that capital can't flow out of China or in China. The central bank chokes it in order to maintain the RMB peg versus the dollar. So that idea that I can cross that layer instantly, that means that I can build a value exchange into a billion transactions a minute, and I can build a value exchange that flows in a billion transactions a minute for a billion people across 100,000,000 different applications.

And they can all move fluidly, right? So I guess it's the difference between, if you want a metaphor — liquid fluid flow into a container — that's Bitcoin and Lightning. Or, you take 1 kg gold blocks or even 400 oz London good delivery bars, you can build whatever you want with those. But one of them is like LEGO that are 400 oz each, and the other is water — liquid, or air. There's just no comparison between the two.

Robert Breedlove: Static versus dynamic

Michael Saylor: What I want you to do is construct a tornado with London good delivery bars of gold. You can imagine a tornado with air and water, or a whirlpool, or a cyclone. You just can't imagine that with big bars of gold, because one's discrete and the other's continuous. It's the difference between discrete fluid dynamics and maybe arithmetic with LEGO or something. You're like, Well LEGO is fine — yeah it's fine for 3rd graders. But LEGO — you can't design a rocket nozzle with LEGO. You're not going to design the wing of a supersonic airfoil with LEGO. You need continuous fluid dynamics.

Robert Breedlove: You need more optionality. We're back to this — there's some core point where again earlier the customer optionality to withdraw their Bitcoin out of the bank at any time keeps the service provider honest, so then the service providers actually have to compete for the business which gives us lower prices, better services, better innovation, more wealth, move *freedom. And it's all rooted in this precision of customer choice. Similar with the LEGO, you don't have as much option with the big block, but if you're at the molecular level of the fluid you have a lot of optionality.*

Michael Saylor: There is a dance between counterparties. And are you dancing using 2-ton granite blocks? Or are you dancing using water? With air? And what structures can you create? If we take all those things together and you consider the consequences for distribution, Bitcoin is something that can be distributed to everybody on Earth for effectively free, so it's truly liquid. What I need is if I wanted to distribute something to 8 billion people and I wanted 8 billion transactions an hour forever — like I want that degree of fluidity — I need to do it at the price of a satoshi a transaction. I need to do it with very low friction. So this low-transport cost, this low friction, results in massive distribution.

And we can see with gold, for example, there isn't 1 oz of gold for everybody in the world to give to them, and so there's no way to distribute money with gold. And if we take fiat we know there's 1.7 billion people that are unbanked — maybe more — and we know the ones that are banked are imperfectly banked and the banks don't cross-connect, so we can't find a good way to distribute money via fiat or via gold, but via Bitcoin we do have a clear way.

The last element is this division, the ability to break it down. I can't break down a gold bar or gold coin. I can sort of break down fiat, but even 1 penny is like too much sometimes. Being able to go down to 1 satoshi is better than 1 penny, especially because of this one thing: one of the big problems in cybersecurity is spamming and scamming. So I can't use the DMs of my Instagram because 99% of the communication is spam or scam. I can't really use the DMs of non-verified or non-followed individuals on Twitter because of the same thing. I can't really use e-mail without very expensive e-mail filters that are continually filtering out spam and scam and the like, so there's massive amounts of cyber insecurity.

Every single day — if you have a website — it gets attacked by a denial-of-service attacks all the time. The reason for DDoS attacks and scams and spam is there's no cost to attack it. There's no skin in the game. And so what we need is to introduce skin in the game. The way to do it is to require a value exchange whenever you move through a router. If you had to pay a price to send a message, at some point you would attenuate large amounts of that scamming, spamming activity. I mean that's why proof of work was created in the first place: Adam Back was trying to come up with a solution to stop spam.

But the other thing you can do is you can actually tie the value, say 100,000 satoshis, as a security deposit into your persona online. If you posted 100,000 satoshis to your Twitter account and if they gave you an orange check and that was your security deposit, then the understanding would be, if you misbehave you get fined, and if you're a malicious actor maybe you forfeit the security deposit. This is the way it works when you're renting a home. This is the way it works in all the real-world — you're a bonded laborer? You can't drive a car without proof of insurance, right?

So we have these kind of insurance policies or security deposits or performance bonds that pop up all through the real-world economy. And in the absence of all of those, because you're a real-world actor, you have a risk. And the risk is: if you misbehave, you have culpability, you're responsible. You may get pulled over by a cop. If you say something rude to somebody's wife you may get punched in the face.

You have either economic skin in the game or you have actual skin in the game. If you walk out onto a bridge that rickety and you screw around and you do back flips you fall off the bridge, you may die. There are consequences in the real world and the economic world but there are no consequences in cyberspace for many of these activities. And so the result is: people actually even create programs to create bots, and I might spit out 100 million fake Michael Saylors to do irrational malicious things, and they're not even me — they can't die.

By the way, when you do that that's a denial-of-service attack. The result of that is cyber insecurity. So the fluidity, the speed, and the efficiency of the Bitcoin and Lightning Network has the potential to give us a solution to the cybersecurity problem if we simply start to tie a small Lightning wallet or an amount into some of these systems that YouTube, Google, Facebook, Amazon, Apple, Twitter, Medium, Reddit — all these other places where people — there's not a day that goes by that someone doesn't post a Michael Saylor video on YouTube offering to get you free Bitcoin. I'm like, that's not me. But there's no verification on YouTube or registration so if you had to actually post some deposit and if you forfeited the deposit — by the way, it's not that they don't take down the accounts — they do. They take them down within a day of when I report them. Every day there's ten Michael Saylor accounts that get taken down on Twitter. It's just there's no economic cost. And I understated that, really. The truth is that on my news feed there's a thousand Michael Saylors that have stolen my image and my name. If you charged them all a quarter each it would become a massive revenue source for Twitter, the scammers would immediately realize that there's a cost to attack the system. The amount of malicious behavior would drop by 99%.

Robert Breedlove: This is a tragedy of the commons, effectively. The freeloader problem where there's no cost to these imitators, so they just keep repeating. And apparently that's a profitable enterprise because there's new ones every day.

Michael Saylor: And the reason they exist is because the fiat standard system makes it too difficult to actually post something of monetary value. The only way you can post something of monetary value in the fiat standard is with a credit card and then you dox yourself and then you're not anonymous anymore. And that's offensive to one group of people that don't want to be identified — they want privacy. But it's also prohibitive to another group of people that don't have a credit card. And the third group of people that have a credit card but the credit card won't clear across nation-state boundaries. Bitcoin solves that problem. And it fixes it.

So on one hand, one extreme benefit is, yeah, Bitcoin's a way to store a billion dollars for 100 years. On the other extreme, Bitcoin's a way for everybody on Twitter to post 30 cents in order to eliminate

99% of the malicious behavior and improve the quality of the experience.

Now back to your dance: if Twitter can ask for 100,000 satoshi deposit and if people give it and if Twitter's quality of service goes up by an order of magnitude and their cybersecurity enhances, that might put pressure on Instagram. That might put pressure on Google. And ultimately now what you have is every big tech company competing with every other big tech company to improve their service by building Bitcoin into their service, because Bitcoin can eliminate fraud on YouTube. And Bitcoin can eliminate fraud and cyber in-security on Instagram.

And so you've got something — it's like, why do I adopt this new technology? Because I can, or because I need to? The early adopter does it because they can, and the later adopters do it because they need to. If PayPal doesn't roll out Bitcoin and Square does, then Square starts to eat PayPal's lunch. And if PayPal and Square both roll out Bitcoin and Apple and Google don't, then they're at a loss, right? So you have competition that's embracing new technology to progress the civilization to a better space. Bitcoin's the future for cyberspace. It's the promise of cybersecurity. In fact, that's the great irony: people criticize Bitcoin as maybe being a tool of cyber-hackers, but the ultimate destiny of Bitcoin is to make cyberspace safe for 8 billion people by building it into every cyber-offering and creating skin in the game for every malefactor, and consequences that are not based upon counterparty risk and don't have to be threaded through nation-states or central banks or credit card companies.

Robert Breedlove: Yeah, it's truly transformational. Just simply by introducing that risk of loss for agents in the system, you're incentivizing honesty which is again this grand theme of Bitcoin. And one other benefit that — and this is like a promise of Bitcoin now, it's still experimental — but the idea of building social media applications on top of the Lightning Network in a way that they're actually censorship-resistant. I think that experimentation is very important especially as we see more people being de-platformed, cancelled, *censored, etc. So it just seems like there's this potential for Bitcoin to restore honest, unstoppable discourse in the digital age.*

Michael Saylor: Yeah, Bitcoin is the crypto-asset network and Lightning is this crypto-transaction network, and it's possible for us to have other crypto-application networks. Decentralized, open, non-custodial, and they all have promise based upon the core idea.

So if we wrap up this entire Bitcoin standard discussion: why does Bitcoin succeed in theory? Well because the base layer protocol is mathematically sound and secure, and the application-layer protocol is sound and it's open to upgrade. And therefore the property rights that it delivers are perfected, and they're open to upgrade. The open network means that you can upgrade—and when I say upgrade, maybe upgrade means keep making the Lightning Network better, maybe upgrade means roll out Square cash app with Square cash tags, maybe the upgrade means building it into Apple Pay — what does it mean? It means 100,000 different things.

There's a free market, a Darwinian competition to build applications and other types of platforms that are upgrading the Bitcoin protocol, and they're drawing on the promise of digital property in order to meet some other use case. And there will be some that will succeed, there's going to be some exchanges that fail, some succeed. There's 100,000 different ways to think about a wallet, right? Some are better than others. The market will sort it out. We've got a very competitive Darwinian system, but the result is: every intelligent person that encounters Bitcoin, they're incentivized to find a way to make Bitcoin more valuable.

They might do it by a creative financial application, they might do it with security, they might do it with a device, they might do it with a software program, they might do it by plugging into a different network a different way, and some ideas will be great ideas and other ideas will be good ideas, and some ideas will be bad ideas. And Bitcoin will benefit from the good ideas, it will accelerate from the great ideas, and it will ignore and slough off the bad ideas.

And that's something that gold and fiat, they don't have working for them, right? You could imagine some gold applications but you're crippled — gold jewelry, gold goblets, gold coins. And then that's the end of the road with gold. With fiat applications we took that much further — heck we created sovereign debt, we created all sorts of interesting money markets and all sorts of other securities with fiat, we created traveler's checks, we created credit cards. You can't say there aren't interesting applications — the entire Visa and Mastercard network is kind of a thing of awesome beauty if you compare it to the system of Roman gold coinage, clearly. It got the civilization to where it is.

Bitcoin is really just still early days. We're just into the second decade, and these applications and these networks that will be built on it are in some ways a blink of an eye. But you see in the conventional world companies like NYDIG, [who] has built a CeFi platform and that platform is being used to deploy Bitcoin-type services to banks downstream. They've created a platform so that you could create a credit card that integrates with Bitcoin in some way. So that's layer 2, layer 3. It's a custodial layer 2, not a non-custodial layer 2. You have counterparty risk in NYDIG, but there's nothing that says that it couldn't become a non-custodial Lightning-based platform at some point. Who knows?

But that's one approach, and then Lightning is another approach to a layer 2, a non-custodial decentralized layer 2. But there's no monopoly on this, you could create a competitor to it if you wanted to, and then you can create applications like Square Cash App which is another custodial application, but it is an open custodial application so to the extent that you want to you can move your Bitcoin out and move it to another layer 2 platform or to another application or just take self-custody. So there's all these different ways that the network is evolving.

Robert Breedlove: Yeah, even the custodial risk is different with Bitcoin because it's trivial to take delivery of it. It's relatively cheap to secure. So even if you are taking some counterparty risk with some of these layer 2, layer 3 technologies, a lot of them can be set up to either you withdraw manually or even you automate the withdrawal into self-custody.

Michael Saylor: Yeah, sometimes the hardcore Bitcoiners are a little bit hard on everybody and it's like the perfect being the enemy of the good. For example, if I buy the Bitcoin and I trust a counterparty for a week but I have the option to take it off the exchange or out of the mobile app when I want, that's a totally different situation than if I bought a million dollars worth of land in California. You will never in a million years be able to move the million dollars of land in California out of California. So the custodial risk of a building or land is orders of magnitude greater. You're trusting the mayor of the city and the governor and the nation-state and every bureaucrat, and you're never going to change that.

Whereas — look, even if you bought Bitcoin on PayPal when they didn't have the wire out option, people had a big fit on Twitter. But the truth is, so you had some Bitcoin on PayPal and for 6 months you couldn't take custody of it and then after 6 months and the big protest about it eventually they upgraded their wallet and now you can. So it is possible if you buy Bitcoin in certain places that you'll be able to custody it or move it somewhere in the next year or two or three years, but it's absolutely impossible that you will ever convert a building in New York City into something which moves at the speed of light on the Lightning Network. Not without selling it, liquidating it, paying the capital gain tax on it and converting it.

So I think that generally just about all these flavors of Bitcoin are higher-quality property. And sometimes people want to own the Bitcoin outright and I get that because our company owns the Bitcoin outright because we want to be able to transfer and take custody and do whatever we want with it — we want sovereignty over it. But if you were to own Bitcoin in a fund or in an ETF, it's still the second-highest quality property you could own. It's not the best property in the universe because the best property in the universe is own the Bitcoin. But the second-best property in the universe is own a Bitcoin derivative that 1-for-1 backed by Bitcoin from a counterparty that you can reasonably trust.

There are plenty of examples where people have buckets of money and they can buy the security but they can't buy the Bitcoin. It's like in some retirement plan and they can buy the security but they can't buy the Bitcoin. And there are lots of investors — trillions of trillions of dollars of money managers — and they've raised money, I have 10 billion dollars in my portfolio and because of tax code or because of law or because of contracts that I have with my limited partners or my charter, I can't buy that Bitcoin buy I can buy a security based on Bitcoin. And so the real world is a nuanced one where there are lots of different types of capital.

Not all capital is fungible. If I have a billion dollars it doesn't mean I can just convert a billion into Bitcoin overnight. I might have 100 million and all I can do is buy land. And I might have another 100 million and all I can do is buy a security. And I might have another 100 million and all I can buy is debt. If that was the case, I would prefer to have debt backed by Bitcoin, land underneath a Bitcoin miner, and a security that is a Bitcoin-backed security, because those three things would be sitting on a stronger foundation of collateral, than to buy three things that have no relationship to Bitcoin.

So that's the value of Bitcoin. Ultimately you compare gold standard to fiat standard to Bitcoin standard and your conclusion is: Bitcoin is simply digital money and digital money is an engineered system to create a shared, immutable, correct ledger. If you were Satoshi and you were doing it again, your checklist would be: Okay how do I make it shared and open? How do I keep it from changing? How do I make it mathematically correct? And then what kind of things can I put in it that will cause the entire world to want to protect it and secure it and improve it over time? Instead of attack it and destroy it? And I think that's what Satoshi created, is this engineered digital asset network that happens to make the best monetary system that the human race has come up with. If there's ever a better one, you'll start from those principles of digital money and you'll say, how do I make it better? And you could imagine maybe something better, but what's most likely to happen is the Bitcoin blockchain just migrates into the better thing and it goes on, right?

Robert Breedlove: Yeah the adaptivity.

Michael Saylor: It checks itself into the better thing and carries on. I don't know why it wouldn't last thousands and thousands of years.

Robert Breedlove: You would assume that the UTXO set would just migrate wherever it needed to go. And again, it's that open source adaptivity of Bitcoin makes it seemingly disruption-proof. Because what can you do? How do you disrupt it? It's really interesting, and I think this is a brilliant way to look at it.

So one last question here: clearly it is superior property for all the reasons we've laid out. What is the impediment to people understanding this? Is it just the proof of work necessary to study Bitcoin to understand all of this? Or is it a misunderstanding of property? Is it a combination?

Michael Saylor: I think it's just the rate at which the human race can embrace a profound idea. There's the speed of light, and that's a limit Einstein spent a lot of time talking about, and the speed of light is the rate at which the universe can communicate with itself. And the speed of sound is the mechanical rate at which air can communicate. If you're moving faster than the speed of sound, you're arriving before the molecules can get out of the way. And that's why there's a shock wave, because you're moving too fast. The speed of light is like, that's the speed limit.

Now there's a speed of which an idea can move through a social-political economic fluid that we call a culture. How fast can an idea move through a culture? And it would be a function I think of cognitive bandwidth of human beings and distraction and life expectancy of human beings and how fast they speak with each other and how much you sleep.

If you ever look at animals, you ever see some of the animals, the smaller ones seem to move faster than either you or me? They're at a much different frequency. They can agree, disagree, careen into a brick wall, come back, regroup, careen into another one, agree again and move forward, all in the amount of time that we're deciding whether we should make a decision. It's a frequency.

So there's a socio-political frequency, and in the history of science, the structure of scientific revolutions, they just speculate generally new paradigms they take place when the entire old generation dies, so in 30 years an entirely new group of decision-makers take over the government. The person running the army is 50, and the 20-year old will be running the army in 50 years because that's the life expectancy or the career expectancy.

Normally it's life expectancy to take a paradigm shift, or a generation expectancy, so about 30 years. But sometimes if there's a way certain things get jammed faster and they get jammed faster because your life expectancy on Normandy Beach wasn't 30 years, right? If you do something egregiously — like assuming you cling to horses and you reject machine guns, that was an irrational behavior that could last from 1905 to 1914. But it didn't last from 1914 to 1915 because there was a war and it was like an instant consequence. So I think that war and disruption accelerates consequence. In the absence of that it's a generational thing, and then you'd have to crank in the rate at which information moves now. For example I feel like information moves on YouTube much faster now than 5 years ago or 10 years ago. Information moves on Twitter faster.

Robert Breedlove: Yeah, there's almost this liquidity coefficient for ideas that has just gone through the roof with digital tech.

Michael Saylor: I mean it used to be that someone would retire and then they'd go back to Harvard University and they're going to teach. Okay so you retire and you're going to teach from age 50 to age 80, so you teach for 30 years and you have 100 students a year, 200 students a year. And so you spend 30 years of your life and after 30 years of your life you've taught 10,000 people. And now you get 10,000 views in a podcast and that's like a boutique thing, right?

So information is moving at a broader scale, there's a much more competitive battle over ideas. Good ideas travel faster, mediocre ideas don't travel at all they get stamped out, the race is evolving a bit faster. But having said all that, like Warren Buffett has still got a lot of money, right? Has he listened to 10 hours of content on Bitcoin yet? Probably not, right? So somebody controls $100 billion and they haven't spent 10 hours thinking about it. So what happens? Well that $100 billion will be locked in the old system until some cataclysmic change takes place. So I think that it's reasonable to expect it takes a decade. And a decade is fast.

Robert Breedlove: So you think full maturity? Well clearly not full maturity, but —

Michael Saylor: It took one decade for the thing to become—look at the life cycle of Google: weren't they founded in like 1998? The first decade they were growing fast but people weren't convinced that Google was going to rule the Earth, right? The second decade — now we're at the end of the second decade, and now what do you think of Google? So I think normally 10 years to get going and then the next 10 years is the big 10 years. I think we just finished year 1 of the second decade. Year 1 started like Black Thursday of March last year, and think about everything that happened between that March and the next March.

Robert Breedlove: Yeah. Such an inflection point for digital technology, the digital age even, and for distrust in analog institutions. It just seems like they both diverged almost in an instant.

Key Insights from Chapter 12

- Gold is defective because of its mechanical nature, and fiat is defective because of its political nature.

- The Lightning Network is a decentralized, open, proof of stake network. But the critical issue is that it's a proof of stake network not staked with its own token — it's staked with the underlying token of Bitcoin, which makes it a much higher integrity, much longer-living, more secure open decentralized network, than one that was created with its own token, which would be self-referential.

- The private keys of cryptography are the strongest property rights in the history of the human race. If we think about cryptography as the basis of property rights, then truthfully the only property you can really own is digital property on a secure crypto-asset network, of which the only one we know of is Bitcoin. So right now, the only property you can truly own is Bitcoin.

- Bitcoin is the hardest to tax, the hardest to destroy, the hardest to seize, and the hardest to impair, of any property available to you to purchase at this point.

- Because Bitcoin is impossible to confiscate with force and because it disappears with your death, then any hostile actor always has an incentive to negotiate with you, because it's better for me to hold a gun to your head and take half of it than pull the trigger and get none of it. That's not true with any other property……And that confiscation element, it works at the individual level, the institutional level, and the nation-state level.

- The great irony: people criticize Bitcoin as being a tool of cyber-hackers, but the ultimate destiny of Bitcoin is to make cyberspace safe for 8 billion people by building it into every cyber-offering and creating skin in the game for every malefactor, with consequences that are not based upon counterparty risk and don't have to be threaded through nation-states or central banks or credit card companies.
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Chapter 13 | Bitcoin is Cybernetic Life

"Capitalism is a proof-of-work system with a difficulty adjustment."

Robert Breedlove: We're going to jump into the Bitcoin mining network and proof of work, which is this unique security model that secures Bitcoin and distinguishes it from many other forms of money, although it draws parallels to what gold represented historically but in a new form. Where should we start, Michael?

Michael Saylor: You know, Bitcoin is its own self-contained monetary system and the Bitcoin system is replacing the fiat system. So I think a good place to start is exploring the way that we did everything in the 20th century with the fiat system and then comparing the implications when you switch to the Bitcoin system. In essence, what's the difference between an analog monetary system and a digital monetary system? And I find it helpful to try to model the two.

If someone said to me, what is Bitcoin doing? Well, Bitcoin is dematerializing the accountants, the auditors, the compliance function, the information technology function, the human resources function, the security function, the facilities function, the police force, the military, the function of war, and the function of politics — all of those things are dematerializing and they're collapsing into the Bitcoin network.

On one side, the 20th century — if you want to operate a business like I'm a publicly traded business, I need accountants. And every three months we close our books. Every three months. The gold standard of business operations is you close your books quarterly and you have finance people and accountants that do the book close. And I've been a public company officer for like 22 years, so I've done 88 quarterly closes — actually more than that.

Well, Bitcoin closes the books every 10 minutes, right? So you've got an accounting function but instead of a quarterly close you have a 10-minute close. So then you have this other issue which is that in my world — the public company world — after you close the books you have auditors. So the auditors look at the books to verify that they've been properly closed and they sign off on your quarterly results and then there's a 10-K where they sign off on the books at the end of the year.

The audit process is manual and expensive in the fiat world. In the Bitcoin world you've kind of got triple-entry bookkeeping — not double-entry but triple-entry — because the books get burned cryptographically and they're truly closed. They're closed with an encryption whereas the books of a public company they're kind of open and then they're sort of closed. With a private company they never quite close — you can never be quite sure. And you've got manual accountants and manual auditors.

So those two functions matter a lot, but then the third function, if you look at the conventional bank or any kind of fiat bank, you've got a compliance function. And the compliance function is to make sure that you didn't enter in any kind of transactions which are illegal or non-compliant or not permissible. You can't enter into any inappropriate transaction.

That compliance sometimes comes after the transactions are entered into. You see with like Archegos where you had UBS write off billions and billions of dollars. Well, they entered into some non-compliant transactions where they took risks they shouldn't take, and they found out after the fact and the cost was billions of dollars to their shareholders. Well, Bitcoin solves that compliance issue with the nodes that are verifying transactions are compliant and you've got like six verifications before you're sure it's compliant.

On the first verification it's sort of compliant. But the assurance of the compliance comes from the mining plus the nodes themselves, and it's 100% — what is it 99.999999% certain after like the sixth confirmation — so you've got an automated process which is distributed which is almost incorruptible. Whereas compliance in the fiat banking world is a manual process—and you’re not even checking compliance of every transaction, right? In Bitcoin you’ve got 100% compliance by the time you’ve got six verifications down, which is 60-minutes.

You could be 6 months after closing a quarter in a publicly traded company or at a bank — think about your credit card company. You’re Mastercard or you’re Visa and you’re doing all of these transactions — what percentage of them are 100% compliant, and when? In fact you could be six months after the quarter closes and you would still have lots and lots of non-compliant transactions they passed through the system. And that’s fraud, and the economy pays the price on that.

You go to the next level — IT — a bank or a money transfer agent in the fiat world has a big IT department and a bunch of data centers. Whereas in the Bitcoin world the miners and the nodes — that is the IT infrastructure. So it’s completely decentralized and self-maintaining and self-healing. And it’s continually upgrading itself through the process of mining. Whereas the banks, they have to have an IT department and an IT budget and they have to continually invest in this or they become obsolete.

The fiat bank also has a human resources department. You have to hire people to order the equipment, to configure the equipment, to upgrade the equipment, to do the compliance, to do the accounting, to do the transactions. Well Bitcoin doesn’t have a human resources department, because there are no employees. You’ve kind of dematerialized it. Why is that interesting? If there’s no human resources department there’s nobody to quit, right? There’s no one to fire, there’s no one to complain, there’s no one to harass, because there isn’t a human resources department at all. So you can see we have taken all of these human factors and we have dematerialized them into the protocol and the decentralized hardware which is continually being upgraded — unless it isn’t.

If it isn’t upgraded — if you’re running a node and you don’t upgrade, you fall out of consensus and you just get sloughed off the network. And if you’re a miner and you’re running on four-generation old mining equipment then you become 100x more inefficient and then you can’t mine any more Bitcoin and you get pushed off the network. And if you're running a miner and you don't pay your electricity bill you get pushed off the network. So the network is self-healing, self-sealing, self-upgrading in the Bitcoin world. Whereas in the fiat banking world it requires human intervention to do all these things.

So — complicated, we're doing it — what's the next step? Well, security. In a real bank I've got to have security professionals. I've got all sorts of IT security, physical security, people with guns, people hiring and training the people with guns, etc., and I've got to worry about someone hacking into that security system. As long as there's someone responsible then you have to worry about that person. But in Bitcoin the security is intrinsic to the network, it's basically built into the protocol. It's making itself more secure.

And then we get to facilities. The fiat bank has to go and have a facility in the middle of New York City and they've got a landlord and they've got a lease and they're paying money and they're paying taxes on the facility. The facility in Bitcoin in essence becomes the mining rigs wherever they might be, or the nodes. And so you've got a much simpler facilities model.

In order to make fiat banking work you have to have a police force, because the next layer is physical or civil security and that's provided by a police force. If you have a billion dollars in New York City or a bank with a billion dollars, then you need a police force in New York City to keep you from getting mugged. But if you have a billion dollars in Bitcoin and you're in New York City, you don't rely upon the police force, nor does anybody know you have it. But if they did have it, the security for New York City comes from a Bitcoin miner in Iceland.

So security is not co-located with the asset in the Bitcoin system. In fact, security doesn't scale or it doesn't increase with the number of nodes of asset or the velocity of the asset in the Bitcoin system. In the fiat system security becomes a big problem, because maybe you have a billion dollars and you feel safe in New York, but when you go to Brazil, you don't. Or you go to whichever place it is that you don't feel secure in, at some point you're required to rely upon the police protection as you cross jurisdictions. And that makes security heterogeneous and manual and expensive.

With Bitcoin you have the same security on a cruise ship in the mid-Atlantic as you have in a vault in the basement of J.P. Morgan in Manhattan as you have on an airplane. And even if you're sitting on a sailing boat in the Caribbean and three people have guns to your head, you've still got the Bitcoin miner in Iceland securing your asset and it's a negotiation. The Bitcoin is quite secure. It's more secure than you are. You have a physical security issue, but your financial security is the highest degree that anybody in the human race has ever had before

Robert Breedlove: It is a great point that the asset and security as disentangled or separate, because that's a huge benefit to Bitcoin is that it actually has this non-locality property. Where especially if you've properly custodied it, even if you've got the three guns to your head in the middle of the Atlantic, if it's in a multisig done properly you're not going to lose custody of that asset. And even if they take you out in theory you have an inheritance or a succession model behind that such that the asset rolls on to your heirs or wherever you determine — you can't really do that with anything else, at least in a bearer asset form. You always end up trusting some intermediary to carry out your will, so that's something that makes Bitcoin very unique.

Michael Saylor: Bitcoin is the apex property, and it's the highest form of property rights. It really is the only thing that you can take with you to the grave. And I joked about it. I was online going back and forth with Peter Schiff the other day. He was saying, "Why would anybody ever want something for 100 years, you'll all be dead by then?" And I said, "Well you can take it with you to the grave," and he's like, "Well you can take gold to the grave." And I posted an article on ancient Egyptian burial tombs, and if you go back and you do even just a few seconds of history and do a few minutes of research, you know that for the last 5,000 years people have been trying to figure out how to bury gold with them — they've all failed.

2,500 years ago the Egyptians created pyramids and hidden tombs. They still had the tomb robbers — people would break into the pyramids. The only reason that King Tut had a treasure that lasted into the 20th century was because it was buried by a sandstorm and people didn't know his tomb existed. It was lost. But every other protected tomb with a treasure in it was raided and looted. And if you pause and think about it a bit, and if you're really serious and thoughtful, what you realize is: there is no form of property that you can have custody of while you're alive and after you're dead, and there's no form of property that respects your wishes after you're dead, other than Bitcoin. And the significance of that is not you need the money after you're dead — that's not the point. The point is much deeper than that. The point is: should you wish for something to carry on, if you want a park watered for 100 years after your death, you have a method to ensure that takes place.

And if I hold a gun to your head and I say, give me all your property or I will shoot you, with all other forms of property, if you say no, I shoot you can I get it anyway. So the game theory says I should just go ahead and kill you and take your property. But with Bitcoin the game theory says if I shoot you I get none of it, because you go to the grave with your keys.

So with Bitcoin the game theory says I should ask for half, right? And it turns out that throughout human history, people have always struggled against governments and onerous groups be-it a monopoly or the like, but in nearly all of those cases, if they got half, they would've turned out okay. Like, normally if they kept half — you don't like taxes, but if you kept half of your stuff, you would still be okay. The problem with taxes is they take all of it over the course of some period of time and you've got none. You've got no recourse, so you can't even negotiate.

And it turns out that Bitcoin is the first thing where you can negotiate it. And because you can negotiate it, it's property that encourages peaceful resolution. And all other forms of property encourage violent resolution, because the phrase is, "Winner takes all," right? All the other properties are winner takes all. I can either negotiate with you and take half — if the bullet costs a dollar and if the gun costs $387 and if you have a million dollars, I can either spend the dollar and take it all, or I can negotiate peacefully with you and I can take half of it.

But, you know, the ROI on that last $1 bullet is $500,000. It's 500,000:1, and so the problem really is: other forms of physical property incentivize violence, and the game theory dictates a violent resolution. And it's even worse than that, because the person that lines up ten rich people against the wall and shoots all ten of them ends up getting all their money and they're able to raise an army and they're Mussolini or Hitler, and then they get to go line up the next 100, and then you get to line up the next — and then pretty soon you're Stalin, right? There's a benefit to the most violent, and they get powerful.

But with Bitcoin, it works the opposite way, because first of all you can't get all of it with violence. You get none of it with violence. And the second thing is: as soon as I said, Robert, I'm going to seize all your Bitcoin, your answer is, No, then when I say, I'm sending guys to your house with guns to seize all your Bitcoin, the most likely result is you're going to wire your Bitcoin somewhere else in the universe that I don't know about, or you're going to put a multisignature on it and you're going to say, look, you can put your guns at me, you can shoot me, you can pull out my fingernails, but I can't give it to you.

It's like, okay, well tell us who else has got the signature? Oh, somebody on a satellite has the signature. How about: a computer program on a satellite has it and they've locked it up and I can't get it out until you release me and I've lived happily for 5 years and then I can get it out. How do you negotiate with a computer program on a satellite? But if you don't want that you're just like, okay, well I just send it to somebody else. And now when that happens, the first person that attacks the Bitcoiner, he realizes, that didn't work out so well. And all the Bitcoin leaves your jurisdiction. And so the attack causes a response.

What I'm trying to say is: if you have 100 people with a billion dollars of gold each in your country you can shoot all 100 and take

100 billion dollars and the game theory says you should. If you have 100 people with a billion dollars of Bitcoin in your country, you could hold a gun, you could incarcerate them all and get half, but you won't, because as soon as you incarcerate one or two of the people, the other 98 will leave the country — the Bitcoin. You will end up with a country where you managed to get half of two people's Bitcoin, and 98% of the Bitcoin will exit your jurisdiction. So the real game theory says: Bitcoin has an immune system and 98%-99% departs from a hostile political jurisdiction when you start trying to seize the asset. And that's what makes it the apex property. And in the same way, that's what makes it anti-violence. And that's what makes it a cleansing technology for the human race, because Bitcoin promotes civility.

Robert Breedlove: Yes. So the physicality of the property itself is what makes it vulnerable to coercion, effectively. The ideal property would be something that's non-physical like Bitcoin. And this really gets to the crux of the value proposition of government and banking — frankly, it was the provision of the security for property. The bank was going to custody your gold, government was going to secure the territory so you could have peaceful commerce. And Bitcoin, by being the asset that has the security innate to it, it enables this radical new level of social scalability, so we're eliminating this duplication of efforts that every bank has their own, like you said, accountant, auditor, compliance, IT, HR — you've now just eliminated the need for that. So it's such an economizing force as well. It's like, not only is it creating more civility, but it's creating more economic efficiency just by virtue of it being non-physical and having in-built security.

Michael Saylor: Bitcoin is a bank in cyberspace. And 100,000 fiat banks collapse in the Bitcoin network, and all of the energy and all the mass that was allocated to running those 100,000 fiat banks, that gets released back into the economy to be recycled.

Robert Breedlove: And this gets back to your book where you're making the point that to get rich in the digital age is like find a critical social function and dematerialize it such that it is orders of magnitude more efficient, and Bitcoin is essentially dematerializing these critical functions of banks and governments — the largest institutions in the world. So maybe that's one lens to view the significance of its value prop.

Michael Saylor: Yeah I think so. And we even got to the best part, which is: Bitcoin has dematerialized the accountants, the auditors, the compliance function, the operations function, the HR function, the facilities function, the security function, of every single financial business — money mover or bank — it's also dematerialized the civil police and municipal security function everywhere where that function was there to protect property. All property policing.

Now the next thing is what happens when two countries disagree? In order to protect your money you need a military, and if you don't have a military, fill in the blank, right? Ask the Spaniards during the Spanish Civil War or ask the French or the Polish people during World War II. If you don't have a military then you lose all your property rights. So the fiat system provides the geo-political security via military, via war, and via politics. And this is where I'll make this interesting point: Bitcoin's the first multi-national money. All the fiat currencies are national monies. Gold was a physical money. We've replaced gold with fiat and with the banking system because we needed to move the money faster, at a higher velocity, and we needed to build new applications on it. So the price we paid was a central bank plus fractional banks.

But even so, every single money is a national money, and if you want it to work in the world over time, you got the military, you got the state department in politics, and you got war, and that's the real expense of the fiat system. The fact that every 20, 30, 40 years you have to fight a war—be it World War I or World War II or [whatever] in order to defend your property.

So how does Bitcoin avoid that? Well Bitcoin is a multi-national network and because it's challenging to locate the network — like you might be able to locate some of the miners but some miners you can't locate, they're off the grid. And half of the nodes are off the grid coming through the TOR network — you don't even know where they are. So the network is running orthogonal to geography or political jurisdiction. It's in cyberspace.

And that means that geo-political disputes are settled in Bitcoin, but they're settled without war. So when one nation wants to tax a certain property, they can tax it. Maybe they can put a tax on a Bitcoin miner. But they will either put a tax that's so high that the miners shut down and then all the mining relocates, or the network just reconfigures, or they will tax the Bitcoin miners at a level where the miners continue and the network just adjusts. So you've got a peaceable resolution via the political process and there really is no incentive [for violence].

If I actually raised an army of a million people and I have atomic bombs, I can't use that to get your Bitcoin, and no one else can use it to get mine, so again the trans-national violence is not really encouraged with Bitcoin. And what we've done is we have taken a fiat system that was constructed to make gold money faster, smarter, stronger, and done with the best technology of the 20th century, we have taken that 20th century model and we've replaced it with a 21st century model — Satoshi's model — that makes money faster, smarter, stronger again.

But in the doing of it, it's hard to understate how much inefficiency we eliminate. Like if you replaced all the banks, all the money exchanges, and then half the police department and half the militaries, take half of that entire fiat system that's meant just to protect fiat money flows, take it away, and then what you realize is that Bitcoin is providing that monetary system that does something better a million times cheaper. To say it's a thousand times cheaper might be an understatement, but it's at least a thousand times cheaper.

It's a massive crystallization or a collapsing to a lower energy state where you're able to provide a million times more security and you can move money at the speed of light and you can make it a million times smarter and you can do it a million times cheaper. All of those things at the same time. That in and of itself is like inventing cold fusion. I could take a sugar cube and I could split the atom or I could take this bottle of water and I could run a fusion reactor on it that could power up the entire civilization for a year off the bottle of water — that's what it will do to the human condition. That kind of breakthrough technology.

Robert Breedlove: Yeah, it's such a massive reduction to scarcity. And that's really the pretext for war, is people fighting over scarce resources. And if there's one universal, I think it's: human beings could agree that war is a bad thing. It's the most uneconomic activity we can possibly engage in. You're mobilizing capital to go and destroy capital and clearly destroy a lot of human life as well. And I think a good way to look at it is that war—in the 20th century, at least —this was the violent mode of reaching consensus on property. To decide who owns what, countries would raise armies and go to war and draw the lines on the map and say this is mine, that's yours. And Bitcoin is this alternative mode, it's a peaceful mode for reaching consensus. It's like a peaceful alternative to war in a lot of ways.

Michael Saylor: Certainly the Franco-Prussian War was fought over land, and World War I was fought over land, and World War II was fought over land, the Mongols fought over land, the Crusades were over land, the French-Indian Wars were over land, you could say the American Revolution was over land. Mexican-American War over land, Spanish-American War over land. Yeah, a lot of wars over land. If half of the money in the human race was in Bitcoin instead of in land, then the property isn't the spoils of war.

And I think the phrase spoils of war is important. Gold is spoils of war, land is spoils of war. Bitcoin — digital property — is not spoils of war. So if you have a lot of money and you put it in gold, I shoot you and take your gold. Put it in land I shoot you and take your land. If you put it in Bitcoin my gun is worthless. I've got to talk you out of it. I have to negotiate you out of it.

Robert Breedlove: Or you have to serve, right? You have to serve the customer and provide a valuable service to someone that's willing to pay you for it. So it's the spoils of peace instead of war.

Michael Saylor: And once you understand it like that, you think, why would I want to store my time, my energy, or my life's work in something that's a spoil of war? Why wouldn't I want to store all my life's work in something which is a spoil of peace? It's just better every way.

So just a few more thoughts on the Bitcoin model and proof of work before we go deeper down the rabbit hole into the engineering dynamics. The proof of work model is: I do a bunch of work with energy, I generate a bunch of hashes, I try to solve the problem, I solve the problem, and then I claim the reward. I get paid for my work.

The work that the Bitcoin miner is doing is somewhat uncertain — there's no guarantee that you'll actually win the block. But somebody is guaranteed to win, right? What you've got is a bunch of people vying for the energy in the system. Every day, 900 Bitcoin get mined, a bunch of transaction fees get paid. All the Bitcoin miners are competing via honest work to win their share of that energy.

The other part to the Bitcoin system is the difficulty adjustment. As more miners enter and as the hashrate increases, the difficulty goes up. So you could think about difficulty as a protocol for competition and the proof of work algorithm is the protocol for work.

Another way to say this is: the Bitcoin network rests on competitive work. Now that's unique to a financial system, right? We never did it before with the fiat system. You could argue we kind of did something similar with gold. I mean gold is: you do a lot of work, the work is mining the gold, and there's no guarantee that you'll find the gold, but if you do find the gold then you get to sell the gold and you get paid for it. So there's some benefit. And then as more people get into gold mining, it gets more competitive one way or the other. So gold is a little bit like this, but I think that the more interesting idea here is that work and difficulty and Bitcoin feel a lot like the work and the difficulty in capitalism. So Bitcoin is a microcosm of capitalism.

If you look at every single industry, what is the industry? Well, I work in this industry, I work really hard to win customers, and I get paid — there's no guarantee I'll get them — if I launch a restaurant my restaurant may fail but it may succeed. If more people set up a restaurant, then it gets harder for me. Maybe the market will grow, maybe it won't. I don't know. There's uncertainty, there's competition. What's pretty clear is that I'm incompetent if I don't get up and go to work every day. I can't guarantee you you'll mine the next block, but I can guarantee you if you turn off the Bitcoin mine, you won't mine the next block. I can't guarantee you that your movie will be a hit, but I can guarantee you that if you don't make the movie or you don't finish the movie it won't be a hit.

So in capitalism you've got thousands and thousands of industries and markets. You've got groups of competitors, everybody is working. Some people come up with better technique. There's S19 miners — they're better than S9 miners. Invent a new Bitcoin miner, right? You can invent a new technique, you can run it harder, you can find a cheaper raw material. The same thing in Bitcoin mining goes on with every other industry. So capitalism is a proof of work with a difficulty adjustment. The difficulty adjustment is taking place — try launching your own mobile phone company in the year 2021 against Apple Computer. The difficulty has gotten pretty high.

Robert Breedlove: And this is so key because — that's a great way to look at it — the profits that emerge from a successful entrepreneurial venture, that actually draws in more competition. If you're generating 30% profit margin, other entrepreneurs come in to try to feast on that profit margin. That is the difficulty adjustment in capitalism. The industry then becomes more competitive, it becomes more difficult to succeed. And then the opposite is also true: if you start generating losses, you actually expel competition. People go out of business. So it's so interesting to look at it that way, and I've called Bitcoin in the past kind of a fractal of the free market. Like it's *its own little free market that imposes free market principles at large.*

Michael Saylor: Bitcoin is modeled on capitalism. You could say that Satoshi was creating a competitive, honest work system to allocate the responsibility for the network and the rewards of the network. But you could also say it's modeled on a classic engineering control system. It's a first-order negative feedback loop. And if you look at the way that any aerospace airfoil — any airplane, any control system, any ballistic system, any targeting system works — you have these control systems with cybernetic mechanisms and it's like, I try this, I miss, I adjust, I try again, and you've got a negative feedback in order to converge.

The negative feedback of Bitcoin is the difficulty adjustment, and the control parameter is trying to converge on one block every 10 minutes to stay on schedule. It's not unlike an autopilot if you're trying to get from Point A to Point B over the course of a hundred units of time, and you've gone 1 minute and you want to get there in 100 minutes. If you're behind, you check, and you go faster. And if you're ahead of schedule, you check, and you go slower. Because you want to end up doing 100% of the work in 100 minutes, so there's your negative feedback loop. You'll find this is built into thermostats, it's built into autopilots, it's built into all sorts of things in the engineering world. And it's built into every capitalist market.

Now, I'll make one more point: Bitcoin's based on capitalism. Capitalism is based on nature. Where else do you see an example of work and difficulty? It's like hunting. I'm a predator, I get up in the day, I hunt. Is there any guarantee I will get anything? No. What happens if I don't hunt? There's a 100% guarantee I will starve. What happens if I do hunt? Maybe I will be the best hunter that will catch the gazelle. Maybe I will get unlucky and someone else less worthy than me will catch the gazelle. There is uncertainty. Maybe I will catch all the gazelle and there's no more gazelle and I will have to change my parameters.

In nature the prey [and] the predator are in this constant equilibrium — all the plants, all life forms — they're all working.

What are they working to do? To capture energy from the ecosystem. The trees are trying to get the sunlight, right? The prey are trying to eat the leaves of the trees. And the predators are trying to eat the prey. And the difficulty is — if all the giraffes eat all the leaves of the trees then only the giraffe with the longest neck can reach the leaves that are left, and all the giraffes with a short neck starve to death. That's the difficulty adjustment. The difficulty adjustment is the slowest or the unluckiest creature gets eaten, and that's good. But the next week, all the wolves have to catch creatures that are faster than any creature they've ever caught before. And vice versa. So you've got continual feedback and evolution and survival of the fittest.

Robert Breedlove: Could we say that the difficulty adjustment actually makes Bitcoin the first money in history that is adaptive to human action? And then if you're looking at it through this Darwinian lens, I think he said something to the effect that, it's not the smartest, fastest, most intelligent creature that survives, it's the one that's most adaptive to change. So Bitcoin benefiting from this — whatever you want to call it, Bayesian Inference, the OODA loop — it's constantly adapting to its environment. That's why it out-competes all other forms of money.

Michael Saylor: The word that rang through the halls of MIT when I was there was adaptive control system. You want to build complicated things? You want to go at the speed of sound? If you want to go faster than human beings can perceive, you have to build an adaptive control system because human beings can't do it, right? Tesla is trying to put it into their Tesla autopilot — the adaptive control system. So the cornerstone of most human advance is some kind of adaptive control system. You could even say probably the Google people with their AI, they're building adaptive control via AI with machine learning systems that are learning and adapting and evolving. All based on feedback. And the whole principle of training of neural nets is just feedback. Training network. I train it. I tell you this and I tell you that's good. Now I show you the next thing and I tell you that's bad. The more that the training set gets bigger, and the more training I do, the faster I learn, assuming I've got that —

Robert Breedlove: Training to modify environmental fitness constantly and continuously.

Michael Saylor: So Bitcoin's proof of work network — what is it really? It's an adaptive control system for money. And maybe it's the first adaptive monetary system. And what is it adapting? What is the control function? The control function is — if we come back to our last set of discussions — what is the critical definition of ideal money? Shared, immutable, correct ledger. So what is Bitcoin? Well, Bitcoin is an adaptive control system and it's targeting correct. It's self-correcting. It's a self-correcting monetary system.

For example: how does gold go off the rails? Well, I started with 21 million gold coins and then someone sacked the Inca empire and brought back 10 million more. Now I've got 31 million gold coins. Well where's the self-correcting part? I don't want 31 million, I want 21 million. I just diluted the value of the civilization by 30%. It's not self-correcting. Someone made 2% gold — it's not self-correcting. Okay well if you have 2% error every year in a thermostat, well the thermostat you grew up with as a kid if going to fry you by the time you're 47. What happens if you add 2 degrees a year to a thermostat? Well, it's going to be 182 degrees by the time you retire. Your books are going to burn at Fahrenheit 451 — just give it a few years. You can't tolerate 2% drift, much less 7% drift.

All of these adaptive control systems — like the steam engine is based on negative feedback, thermostats, even longitude. John Harrison when he invented longitude — we talked about it — how do you find longitude in the ocean? You need two clocks. How do I actually get two clocks to keep time when you have heat that's compressing and expanding the material in the clock? The answer is: I have to sandwich in or wedge three different layers, and one layer has to contract as the other expands so they off-set so that you eliminate the bias. You can't afford to have any flex. You need perfect — like how do I get a watch to run true? I can't have it lose

2 seconds a day or 5 seconds a day. I need perfect timings. Otherwise you're going to miss by 100 meters, right? And so all of engineering is all about error-correction and adaptive control systems.

So Bitcoin is a self-correcting monetary system, and fiat isn't, and gold isn't, and commodity money isn't. Because there's nobody that can play God. If God could reduce the number of gold coins to 21 million every time someone tried to slide another one in there, it would be self-correcting. When you lost a coin or you shaved it or counterfeited it or whatever, if you had some deity then maybe you'd have a self-correcting system but we don't. We know the fiats are — there's no truth to them. No one even knows how many dollars there are, right? There are two things we don't know in this world: how many ounces of gold there are, and how many dollars there are. And there's one thing we do know: how much Bitcoin has been created and how much will be created. So I think that taken in its entirety, what Satoshi engineered was an adaptive control system for money. A self-correcting system. And ultimately a natural system, a natural, conservative, organic system.

If I look at creatures in the wild and they have to compete in their ecology, in their ecosystem all the time, then they're healthy. And when I make them zoo creatures they stop competing and at the point that you become a zoo creature and you're fed and you're sitting in a chair watching Netflix, you become progressively less healthy, and you become more fragile. So all of these fiat currencies are zoo creature money. They're not natural. When we say, that's not natural, when you think about it you realize the things that are healthy are natural, because nature is conservative. And by conservative I don't mean politically conservative, I mean physically, thermodynamically conservative.

Nature never takes 22 creatures and turns it into 57 million creatures overnight. If you have 22 and 3 die you have 19. Nature is conservative — there are consequences. And if a wolf eats all the deer and there are no more deer, then the wolf starves to death and then the deer come back because there are consequences. Bitcoin is wonderfully engineered because A) the protocol is conservative and mathematically correct, B) the proof of work algorithm is an honest work, it is an open egalitarian work, anyone can do that work — I can do that work with brute force energy, I can do that work with new machinery, I can do that work anywhere in the world, no one can stop me from doing that work, ergo it's an open natural system, it's not a closed system.

Your zoo creature is living in a closed, air-conditioned system, and the wild animals living in an open system, wild animals are going to evolve and adapt to an open environment which is much more challenging. Zoo creatures are going to adapt to a closed environment. What's an example of an adaption to a closed environment? You're 300 lbs overweight because you sit in a chair watching Netflix. There are no fat predators.

Robert Breedlove: Yeah. This is then maybe the commonality between life and Bitcoin, is that each is really just a survival strategy at the end of the day. Your DNA is the collective lessons of your survival strategy across history, and it's kind of like an organic blockchain and Bitcoin is just this survival strategy for money. Life is that strategy propagating through flesh, Bitcoin is propagating through code, and both of them are error-correcting their own behavior. Life is intelligently figuring out what it needs to do to advance itself to extend its DNA into the future, and Bitcoin is adapting to human behavior to extend itself into the future. So its not just an analogy that we say that this thing is an organism, it feels pretty damn close to true.

Michael Saylor: Bitcoin is cybernetic life. And when you play God — Satoshi created life, right? There's a genetic code to it, and then you decide: are you going to release it into your aquarium, or are you going to release it into the ocean? Is it going to be a barnyard animal, is it a domesticated pet? Is it a zoo creature? Or did we release it into the wild? And Bitcoin was cybernetic life released into the wild because proof of work is inherently permissionless, which means that it was meant to live in all environments and to go everywhere and you can't keep it caged. To the extent you try to keep it caged, then you lose control of it.

There are other forms of cybernetic life that have been designed not to go in the wild, like a proof of stake system. A proof of stake is more like a barnyard animal or a zoo creature — you've designed a domesticated pet which is kind of toothless and maybe it's got pretty feathers, but it was never meant to live in the jungle. It would be no fair to let that loose, it was a domesticated creature.

Robert Breedlove: This is such an apt analogy too because that is the only way fiat currency survives is through these layers of legal insulation. If you put fiat currency in the wild so to speak and let it compete on its own merits, it would immediately collapse to anything — to gold, or to Bitcoin.

Michael Saylor: Yeah, fiat is another example of a domesticated money.

Key Insights from Chapter 13

- Bitcoin is dematerializing the accountants, the auditors, the compliance function, the information technology function, the human resources function, the security function, the facilities function, the police force, the military, the function of war, and the function of politics — all of those things are collapsing into the Bitcoin network.

- The Bitcoin network is self-healing, self-sealing, and self-upgrading, whereas in the fiat banking world it requires human intervention to do all these things.

- There's no form of property that respects your wishes after you're dead—other than Bitcoin. And the significance of that is not you need the money after you're dead — that's not the point. The point, which is much deeper than that, is that should you wish for something to carry on—if you want a park watered for 100 years after your death—you have a method to ensure that takes place.

- Geo-political disputes will be settled in Bitcoin, but they're settled without war.
- All the fiat currencies are national monies. Bitcoin is the first multi-national money.
- Capitalism is a proof-of-work system with a difficulty adjustment.
- Physical property incentivizes violence, and the game theory dictates a violent resolution. Bitcoin is anti-violence and promotes civility, and that's what makes it a cleansing technology for the human race.
- Bitcoin is based on capitalism. Capitalism is based on nature.

Chapter 14 | Bitcoin's Seven Layers of Security Part 1

"Bitcoin is the best customer for your volcano."

Robert Breedlove: We've talked about Bitcoin and how it's dematerializing these critical security functions traditionally provided by banks and/or governments, and it's doing this through this proof of work security model. But you've laid out this framework that I think is very brilliant and it's this: proof of work has many layers to it, that these are protecting it from disruption. Maybe you could just introduce that model and we could walk through it.

Michael Saylor: Sure. I've called it the seven layers of security, but I think we should start with a fundamental observation: what we're trying to create with Bitcoin is a digital property network. Something that will last for 1,000 years or 10,000 years. How do I create digital property in cyberspace? And we can call it money if we like — digital monetary network or digital property network. But what I want is I want a decentralized network that will last past the average life expectancy of a human being, or an institution, or a nation-state. So how do I do that?

The beauty of proof of work is that the network derives its integrity from the interplay of the miners and the nodes and the HODLers, or the wallets if you will. With a special emphasis on the miners. What do I define a security? Well, property needs to be optimized for integrity and durability over time, integrity meaning the protocol can't be corrupted. It can't mutate. If the protocol mutates, we go from 21 million to 48 million, you get a monstrosity. So you don't want a mutation. And then I don't want it to die.

So if I said I want to design an immortal creature, something that will live for a million years, and I want it to maintain its integrity, then I've got to protect the protocol, and then I've also got to feed it energy and resources continually, and I have to protect it from a hostile attack in the near-term. Sometimes when we think about security we think just about avoiding an attack, but really security is: you can't [be attacked] in the near-term, we don't want it to die, and most importantly, if you live but you go insane or you get a grotesque mutation, then did you really live? So how am I going to achieve my quest for immortality in a network? And the beauty of proof of work is it is an open protocol that is Darwinian and competitive in a way such that it is not just self-correcting on the protocol level to ensure correct money, but it's also self-healing and it's adaptive to the environment around it.

I like to call these the seven layers of security, so what do you have? You have: an energy layer, a technology layer, a political layer, a financial layer, a network layer, a spatial layer, and a temporal layer of energy. Those seven various dimensions.

So I'll start with the first dimension of security: energy. I have a hashrate — say we've got 100 exahash right now. Well, I've got to feed it energy in order to create the 100 exahash. Bitcoin is a big customer of energy. In fact, it turns out that Bitcoin is the most valuable, most compelling industrial customer of energy in the world right now. You can generate about 40 cents per kilowatt-hour in revenue mining Bitcoin. So that makes it interesting because that co-opts all the energy industry.

One way that Bitcoin spreads and stays organically vital is it forms partnerships with the energy industry. If you have a volcano in the middle of Central America, the value of that energy sold within 500 miles of you is like 1 penny a kilowatt-hour. Unless you sell it to Bitcoin, and it's 40 cents a kilowatt-hour. So Bitcoin is the best customer for your volcano. Bitcoin is a customer of stranded energy, intermittent energy, energy on any frequency, and energy at any scale. And you don't need to be co-located with an industrial customer or a consumer. And you don't need bandwidth.

What's the second-best use of energy? Maybe like running Netflix or Google data centers? But if you want to run a Google data center, you need to be within a fiber-optic distance of the major switching networks because you're streaming video and streaming a lot of high-bandwidth queries. So they're bandwidth-intensive, and they need to be triple-redundant because they can't fail. If you're trying to operate a car with Google Maps, you can't afford to have an outage in the network. So Bitcoin miners are so much better than a typical data center because you could monetize the entire volcano or you could monetize the entire waterfall or the entire nuclear reactor — the full bandwidth of it. You could do it at the North Pole. You could do it with a satellite up-link. And you could monetize intermittent energy coming off of wind or wave or solar. And you can also shut it down in a heartbeat with fairly low economic cost. So that makes Bitcoin the most compelling technology for the energy industry that's come along this decade — maybe in a long time.

The last big energy technology was aluminum, when people would use their excess energy to smelt aluminum. Well, Bitcoin re-built that but in spades. To do Bitcoin you don't need a port to ship aluminum in and out of. You don't need an aluminum factory, which is difficult. You don't need aluminum ore. And you also don't need the industrial demand for aluminum, which eventually saturates. Aluminum is a specialty metal.

So what is Bitcoin? Bitcoin is money. So half of everything that the human race has is Bitcoin, or will be Bitcoin in time. So Bitcoin mining is securing and producing something which is of enormous value on a very thin (communications) line based upon any flavor and frequency of energy, and there's really nothing like that.

Why does that make the network secure? Well, Bitcoin miners are continually looking for new energy sources and they're negotiating for cheap energy, and when they buy it they become a big customer. So as they become customers of energy producers everywhere in the world, the energy producers become political allies of Bitcoin. The energy producers are all regulated entities, but they're all very credible, and they're all wired into the political fabric of every society. So if you are the best customer of the utility in the city or the state or the country, then that utility has the vested interest in protecting your interest.

So Bitcoin co-opts energy companies, and also Bitcoin is hope for a utility. If you are a value stock 100-year old utility company, you're not getting a high multiple in the stock market, there's not a lot of excitement. In fact, you're regulated and there are limits on how much money you can make by selling your energy to consumers or to businesses. If the power company in New York City just jacked the price of energy by a factor of three they would get shut down by the regulators. On the other hand, if you get into the Bitcoin mining business, you don't have those kind of regulations on you. So Bitcoin is a chance for energy companies to become technology companies, and it solves so many of their problems.

I can generate more revenue—check. I can generate revenue at much higher price per kilowatt-hour — it's a very high price customer. The value and use of electricity for Bitcoin miners is higher than any other customer they have, so that's good — check. It's portable. And so that means every place I've got stranded energy production facilities I can move the Bitcoin mine to the energy — check, that's interesting. It's an energy shock-absorber. If I have a renewable energy network say with wind and solar, what I want to do is I want to build it to the level of peak demand, but if I build at the level of peak demand I will over-build it by a factor of two or more, and that means I'll be running at 50% utilization for my consumer and other industrial customers.

So how am I going to find a customer that's willing to actually take up half of my energy production and then turn off for 10, 20 hours a year when there's a brown-out without complaining? Like, the hospital's not going to do that. Homes aren't going to do that. So I need some kind of industrial producer that becomes a shock-absorber — they're going to drive up my utilization to 98% and then when I brown out they're going to turn off and give back energy to the network. So Bitcoin represents a driver for the development of sustainable and renewable energy, it also represents a technology to provide more reliability and fault-tolerance to a power grid. If you think about the role of capacitors or batteries or shock-absorbers in electrical or mechanical machines, that's what Bitcoin is for a power network.

Robert Breedlove: Right. So you've described money as energy, socio-political energy. So in that framework I guess we could consider banks historically where they were actually providing custody of gold, they were kind of like energy stores in a way — we're storing this excess productive economic energy in gold that banks were custodians for us. And now with Bitcoin we have this more direct path to monetizing energy, and so an interesting thing here — you've heard about a power bank, these little devices that will charge your phone that stores energy. So it's like banks and power banks — do you see the energy producers and banks somehow converging? Because it seems like on a Bitcoin standard you eliminate the need for a custodial bank, they become kind of synonymous with energy producers. Do you see those industries coming together in the future?

Michael Saylor: I think once you understand that a bank is storing monetary energy, then you've got a bank of batteries that's storing electrical energy, and then Bitcoin is sitting in the middle. It's encrypting the energy from electrical energy into monetary energy. And so on one hand it's upgrading the raw physical energy through the encryption process, and then on the other hand it's digitally transforming analog fiat energy into encrypted money — digital money. And that makes Bitcoin a bank, an energy bank, a monetary bank. It makes it the world's greatest battery, right?

You could look at it from two angles: banks are bleeding off 1% of your energy a month due to inflation. Batteries are bleeding off 1% of your electrical energy a month just due to battery engineering. You've got two inefficient energy systems: batteries and power grids don't work that well, and fiat banks and monetary grids don't work that well. And Bitcoin plus Lightning gives you a battery to store all the world's energy forever without power loss.

The other day I sent 1,000 sats on the Lightning Network in 1 second for 1 sat. So literally you've got speed of light, nearly — you know, what is the cost of 1 sat, it's such an infinitesimally small number, a fraction of a fraction of a penny — so Bitcoin's a very efficient energy system. I think that the significance here is that

Bitcoin is disruptive technology for the entire energy market, and it's co-opting the energy producers and energy investors into Bitcoin.

You see a lot of traditional power companies that used to be in the energy business, they're getting into Bitcoin mining. People with natural gas, people in the nuclear industry, people in the wind and solar industry — they're getting into Bitcoin mining or they're partnering with Bitcoin mines. So Bitcoin recruited energy capital, and Bitcoin secures the network with energy capital. Hundreds and hundreds of producers of energy are plugged into the Bitcoin network creating that hash wall.

So that's the first layer of security for Bitcoin. And if you're going to attack it, it's no small feat to go and collect that much energy. You almost can't. You're not going to collect twice as much energy, or as much energy as is currently in the Bitcoin mining network. Politically, economically, practically, I just don't know how you'd do it, right? Very, very problematic. And there's no short-cut. In that regard, there's an energy wall and you have to jump over it. And it's a lot, to the point where you're talking about 100 terawatt-hours of energy or something. It's a lot of energy.

The second layer though is technology. And this I don't think people really understand that well, and when I talk about technology I'm referring to Bitcoin ASIC miners. Bitcoin is a wall of encrypted energy, so the energy is the first part, but the encrypting, the encryption, the crypto is the second part — the crypto-wall of energy. It's not raw energy. If it was raw energy you just need 100 terawatt-hours to start to get parity. But when it's encrypted energy, I need 100 exahash of SHA-256 mining equipment, and since it's like 10,000 miners to get to an exahash right now, then that's a million mining rigs of the latest build — how do you come up with a million mining rigs? It's like, no one's got a million mining rigs sitting around. It's not clear how long it would take to manufacture a million mining rigs.

Bitcoin's co-opting technology companies. So Bitmain is a big player, but there are other big players too, and as Bitcoin gets larger, I think you'll see the Intels of the world and you'll see other semi-conductor companies designing Bitcoin mining rigs and getting into the space. And this is another example where this is open and competitive. Who wants it the most? Whoever wants it — and what capital do you need? This is semi-conductor fabrication capital. You have to fabricate these ASICs. The S19 family is five times more efficient than the S9s. So if you wanted to generate an exahash on the S9 family, it took 150 megawatts of power — that's a lot of power. If you wanted to do it with S19s it's 30 megawatts of power. A 5x difference in a generation.

If you get through two generations every 10 years — or three, right? — you're talking about a 20x improvement. I mean we're 100x more technically efficient today than we were 8 years ago. Where will we be in 10 more years? We go back and forth — there's all sorts of people with opinions. There's Moore's Law, some people think that ASIC technology efficiency will slow down, but then again people have been predicting that we would hit a wall in Moore's Law for a while and they're still not quite right.

So when you understand the Bitcoin network, what you can think is that the Bitcoin mining network 10 years ago was very energy-intensive and technology-poor. There was a time when you were just using off-the-shelf chips, and then you were using GPUs, and then you were using ASIC first generation, and then ASIC next generation. So we're like five generations into this now. In the early time-frame, you threw probably 1,000x if not a million times as much energy to generate a hash. And now you're throwing 30 megawatts to generate an exahash. But when we get to the next generation it'll be 5 megawatts.

Well, what if we get to 1 megawatt? So as you go out, energy is still important, but it turns out that the network is rotating from energy-intensive security to technology-intensive security. What if you imagine one day there will be 10 million mining rigs but they'll be seventh-generation ASICs and they will be a hundred times more efficient than anything we have today? What that means is that 99% of the hash power will come from that generation of mining rigs, which means that all this stuff that we have today becomes 1% of the network, right? Which means it gets squeezed out of the ecosystem. All those miners that are obsolete go out of business.

And so that technology intensity has a couple of dynamics: 1) that's what keeps the network from consuming all the energy in the world. I mean, people say silly things like, oh, well, the energy usage is going to increase with the price. Well, no it's not. The energy usage maybe will increase with the log of the price. That is, if the price goes up by a factor of 10, energy usage might go up by a factor of 2. And if price goes up by a factor of 100, energy usage might go up by a factor or 3 or 4. But what you've got is a situation where the halvings are cutting the block reward in half every 4 years, so that's a built-in 5x improvement in efficiency every decade, and then if you crank over that another 10x improvement in hardware, you're talking about a 50x improvement in tech efficiency in a decade, or a 50x50x in 20 years.

So the price has got to go way up, and even so, what you've got is a rotation: the network is rotating from, I burn a lot of energy to generate my hashes to, I burn less energy to generate more hashes. It's not hard to see this, right? It's in your iPhone: your iPhone has more processing power than the space shuttle had 40 years ago. You've got a supercomputer in your pocket, so you've got to assume that the security chips that generate the hashes just keep getting better. And because Bitcoin is unique in the protocol, all of the other processing power is irrelevant, right?

Bitcoin is a Darwinian competition to generate the SHA-256 hashes. And it's 95% dominant in that game right now. There's a few other crypto-networks that use that but they're like 1–2%. So that uniqueness with regard to the proof of work protocol protects it from an attack from all of the AWS network or all of the Google network or if everybody turned their iPhones to Bitcoin mining it wouldn't help them that much, right?

Robert Breedlove: So is the common theme here between the energy layer of security and the technology layer of security, is that — you said Bitcoin is co-opting these industries — so it's a source of demand for energy that's basically unquenchable, and it's a source of demand for semi-conductors.

Michael Saylor: For specialized mining semi-conductors. People think, well, 60% of the miners' costs are going to energy. Well, actually the miners are spending money on energy, and then they're spending money on semi-conductor mining rigs, and then they're spending money on the mining centers themselves. The heat sink, the heat engineer and the generators and the like. So there's a competition. The question is: are you better off having inefficient miners and cheap power? Or are you better off having efficient miners and expensive power?

Robert Breedlove: And do they co-exist? That was my next question. So the new generation miners go to the more expensive energy sources, and then the older generations roll off to cheaper energy sources and they both operate side-by-side?

Michael Saylor: Well, you've got three generations of miners. People are running old, inefficient miners in China on free power, and then in the US and North America publicly-traded companies are buying the latest-generation expensive miners because their power is more expensive but they have more capital. If I have $500 million in capital, spend $60 million to buy 10,000 S19s, mine 1 exahash on 30 megawatts and make $120 million a year. Good deal? Probably a good deal, right? So in that case you would rather buy the better equipment.

I think you find at any given time there's like three generations of mining equipment. When the price goes high, if you look at the hashrate of the network back in the first quarter, there was a point when the hashrate got to like 190 or something, and I thought the peak was 170 but there was a surge up by about 10–15 exahash and I posit that that surge was because people turned on third-generation mining equipment, like the oldest, least-efficient stuff. And there's a relationship: an S19 is generating 40 cents a kilowatt-hour right now. Ergo, it is profitable to buy electricity up to 39 cents a kilowatt-hour. An S9 was 5x more inefficient, so S9s were profitable at 8 cents a kilowatt-hour. But if you go back one generation before that, if you're 5x less efficient than that you're profitable at 1.5 cents a kilowatt-hour.

So the break-even point is a function of the semi-conductor, or the joules per terahash. And that's a technology driver, it's not an energy driver. And most of the people that model Bitcoin mining, they don't really focus upon the technical efficiency of the miners. And by miner I don't just mean the one mining rig, I just mean the mining operator that has a mixture of first generation, second generation, third generation equipment. And you may turn off — like if the price of Bitcoin went to $5,000 a coin right now, people would probably turn off the old mining equipment unless the energy was free, right? You can bootleg Bitcoin on free energy if you're stealing the energy. But if the energy costs 3 cents a kilowatt-hour you're going to turn off your old equipment, you're going to keep the new equipment running. And that's an adaptive correcting system itself.

When the price goes up you're going to turn on the old equipment. And over time though, these are transient phenomena because the more important phenomena to focus on is that over a decade, the technical efficiency of the mining network is going to increase by a factor of 20x–100x depending upon the rate at which people upgrade the semi-conductors.

There are a lot of ways that you can enter into the network and you can disrupt the equilibrium. One way as a nation-state is you can start mining off of a volcano. The other way is you can design a totally new family of ASICs. Maybe Intel or Apple wants to get in the space — they could disrupt it. A big energy producer could disrupt it. Right now, the limiting factor is the number of mining rigs you can get your hands on — that's good. That creates some predictability to it. And that keeps us from going chaotic. So if you look at these two layers of security, the energy layer is co-opting energy providers all around the world, and the technology dimension is co-opting semi-conducting manufacturers. And if one of them like Bitmain gets too much power, you know Jeff Bezos's famous line, "your margin is my opportunity." So if one company gets too much power then, well, this is not that terribly complicated engineering. It's well-understood how to design SHA-256 ASICs.

Robert Breedlove: There's a capitalism difficulty adjustment.

Michael Saylor: Yeah. And there are a lot of companies like Qualcomm, Intel, when they get squeezed, they need a new market — this is the new market for them as the price goes up. And the interesting thing of course is: if there's only one provider of SHA-256 miners — Bitmain — doesn't that create an artificial scarcity to hashrate? And if there is a scarcity to hashrate, the profitability of Bitcoin miners goes up, right? And so if the profitability of Bitcoin miners goes up, then don't more people want to get into the business? And if more people want to get into the business, don't they need to buy Bitcoin mining rigs? And if they need to buy it and there's infinite money, don't they go and find someone else to go break that monopoly?

So energy and technology are a key part of the equation. But they're not the only part of the equation, because if I want to generate hashes I have to set up a Bitcoin mining facility. There's physical nexus — I have to do it in Iceland or El Salvador or Canada or Texas. And you could probably run your stake validator on a secret machine in your basement, but you can't run a 150 megawatt facility — you'd need the support of the mayor, the county, the governor.

You could look at this as a negative — it isn't, it's a positive. What it means is that Bitcoin miners are knocking on doors everywhere in the universe and they're looking for supportive political jurisdictions. If New York City doesn't want to mine Bitcoin, maybe Texas does. Maybe Florida does. Maybe Wyoming does, right? And so if China doesn't, well then maybe Kazakhstan does. In fact, the more irrational one country is, the more appealing Bitcoin mining is to the others. When China actually cracked down on Bitcoin mining, the revenues of Bitcoin miners doubled. The profits quadrupled. And the incentive to get in the mining business jumped.

So what you have is an interesting dynamic where Bitcoin miners partner with energy companies, and they partner with technology companies, but they also partner with political jurisdictions and politicians. And this is an interesting situation too because this is so

Darwinian: what happens if you're a Bitcoin miner and you set up a billion dollar Bitcoin mining rig in New York City and there's a new mayor and he outlaws Bitcoin mining? What happens to your capital? You can't move it, right? So Bitcoin miners have skin in the game. And because they have skin in the game, that means that they have to do their due diligence and they've got a long time-horizon.

A person that's mining Bitcoin is looking out 5 years, looking out 10 years. You kind of want to be in a jurisdiction where you think that a decade from now you'll still be able to mine Bitcoin, and so it forces you to be more thoughtful. And the next step is, well, I just invested hundreds of millions of dollars in this county, I guess I probably better contribute to the local election and get the mayor elected, right? You put down roots, because you have a vested interest. Because you have something to lose.

So the Bitcoin mining network actually becomes a politically lobbying network for Bitcoin. And then every politician that has a Bitcoin mine has tax revenue. You know, when New York wanted to ban Bitcoin mining, that was blocked by a politician because the politician got a letter from the union, and the union said, we're going to lose jobs. So all the money that flows into creating mining centers creates jobs, creates tax revenue, supports politicians, creates massive revenues for engineering companies, for real estate companies, for landlords, etc. It's good for the local economy. Bitcoin brings prosperity anywhere on Earth. You might be a limousine liberal in the upper-east side and not like the use of electricity. On the other hand, but if you're living in the middle of central Africa and if you have a volcano or if you have a waterfall and 90% of your population is living below the poverty level and there is no industry and someone shows up and offers to give you billions of dollars if they can attach a turbine to your waterfall — how do you feel about the situation?

Robert Breedlove: Yeah, it's incentivizing everyone to interact with it in a way that contributes to, and I would argue ultimately ensures, its success. That's the thing that's so hard to get your head *around here is like it's not even something that's being imposed. It's really just a game theory that's turned on people, so people in a way become a core component of the Bitcoin system. It's like we at every level are incentivized to help it proliferate.*

Michael Saylor: 100 megawatts is worth $300 million a year. Netflix is not going to show up in El Salvador and offer them $300 million a year in order to run a data center plugged into a volcano. Google is not going to show up in Zambia, central Africa, and offer them $300 million a year to run a Google data center. It's not going to happen. So what you have is Bitcoin is hope for energy, it is hope for your semi-conductor company, it is hope for your political jurisdiction, it solves a problem, and because it can go anywhere on Earth with a satellite uplink, it's probably a more compelling solution to someone that has natural resources in a poor country even than in a wealthy country.

And what you end up with is a self-distributing dynamically distributing Bitcoin mining network, and it has solved for the problem, "where do I find the cheapest energy in a politically supportive jurisdiction?" That's the problem you're solving for them. That's not the same as the cheapest energy. It has to be cheap energy from an energy provider you trust, in a political jurisdiction you trust. Okay well what's the answer? The answer is that it's a dynamic equilibrium. It's changing every single week. It was different 3 years ago. 3 years ago there was free electricity in provinces that had empty excess coal electricity capacity in China, and so you could get effectively free energy. Okay, it was free but it wasn't politically stable.

Now you can go to a politically-stable place like Texas and get it for 3 cents. Or maybe you can get it for 1 penny in Kazakhstan, or maybe you can go some other random place. You could go to Cuba or you can get a bootleg thing in the jungle from some warlord. For how long? Well, the Bitcoin mining network is continually distributing itself, and it's organic. It's just like asking the question, when we settle around the Earth, where should we settle where we'll be able to make a living and we won't suffer from famine or a tornado? And the answer is: you don't know.

Until you've lived 100 years you don't know that place floods. What's the likelihood of an earthquake in your favorite place? Well, you'll find out. So what happens? Well, the network distributes itself, some people make good choices, some people make unwise choices. This goes back to antifragile or the like. You make really awful choices, you'll probably be put out of business within 36 months. If you make okay choices, you might last 3–10 years.

Then maybe there are other choices where you'll be good for 100 years. But maybe you're better off to go to the place where you're good for 6 years where the energy is 1 penny instead of the place where you're good for 100 years where the energy is 5 pennies. And maybe we could debate it. And maybe you have an opinion, and we could argue until the cows come home, but what does it matter? It's better to just release the entire creature, let the entire world work in a Darwinian, Adam Smith evolutionary fashion, a competitive way, and the lesson of the markets is that the market is smarter than anybody. Every time I think about what I knew, half the time I thought I knew it and I was wrong and sometimes I was right, but the market generally outsmarts the smartest person. So Bitcoin mining is a dynamic market in security and it's balancing energy versus technology versus political support. And that takes us to the fourth layer of security —

Robert Breedlove: Let me just ask you one question about that. So, it is shopping all of these dimensions, but it's also stabilizing them. And I wonder does that apply to politics as well?

Michael Saylor: Yeah. It's a shock-absorber. It is stabilizing. It's a political shock-absorber and it's an energy shock absorber. If you have a political regime and you've got natural resources, but your economy is tanked, Bitcoin comes in — on one hand the Bitcoin miner can generate $500 million a year in revenue—okay. On the other hand, the Bitcoin miner can say we're not coming unless we actually feel this is a stable regime to operate in.

You know how there's that phrase, I got married and my wife made me want to be a better man. It's like, okay, well, if you get married you have to give up some of your bachelor ways, right? And if you have a child, you have to be even more responsible. So the commitment to a family is stabilizing, because you're thinking, I can't do the crazy things I did when I had no other responsibilities. If you don't actually have any industry you have nothing to lose. And so a country that has a Bitcoin mine has something to gain and something to lose, and if you were to actually seize the Bitcoin mine, no other Bitcoin miners are coming to you, right? It's the same as seizing the Bitcoin.

If you make Bitcoin illegal in your country and you make mining illegal then you lose all that. But if you make Bitcoin legal or welcome and you make mining welcome you gain something. And then if you're rational you might remember that your prosperity came from the new job you had, and then that way you don't want to lose the job. If you're jobless and homeless, well, first of all there's nothing to lose, and then if someone puts a $37 revolver in front of you maybe that's more appealing to you to use that.

If I had nothing, I might be more prone to violence. And then why do people resort to violence? Maybe because it's hopeless? If I feel like things are hopeless I either kill myself or I kill somebody else. Bitcoin is hope, and so yeah it is politically stabilizing, because it's a way out for someone that can't find another way out, and once you adopt Bitcoin you see something that worked, and you have something to lose, and you're part of the multi-national Bitcoin network and that is civilizing in its own special way. Because after the mining it draws you into Bitcoin HODLing and then Bitcoin banking and then Bitcoin commerce and Bitcoin philosophy and the like.

Robert Breedlove: So, I guess you could say in a broader scope that digital technologies are already changing the distribution of population thanks to remote work and things like this, but Bitcoin mining itself sounds like it could have a significant impact on where *people live. We're going to migrate to these cheaper energy sources and set up a politically stable environment and get to work.*

Michael Saylor: Yeah. Bitcoin mining lets you create an oasis of prosperity around a stranded energy source. If you have stranded raw materials, whatever the energy is, you can drop the miner on top of it and then you can create prosperity, because it'll monetize any flavor of energy, any frequency of energy, with no bandwidth and minimal infrastructure around it.

Key Insights from Chapter 14

- The 7 layers of security: an energy layer, a technology layer, a political layer, a financial layer, a network layer, a spatial layer, and a temporal layer of energy.

- Bitcoin is the most compelling technology for the energy industry that's come along this decade. Bitcoin mining lets you create an oasis of prosperity around a stranded energy source. If you have stranded raw materials, whatever the energy is, you can drop the miner on top of it and then you can create prosperity, because it'll monetize any flavor of energy, any frequency of energy, with no bandwidth and minimal infrastructure around it.

- Bitcoin is a chance for energy companies to become technology companies, and it solves so many of their problems.

- As miners become customers of energy producers everywhere in the world, the energy producers become political allies of Bitcoin.

- Bitcoin miners are knocking on doors everywhere in the universe and they're looking for supportive political jurisdictions. The mining network actually becomes a political lobbying network for Bitcoin. And then every politician that has a Bitcoin mine has tax revenue.

Chapter 15 | Bitcoin's Seven Layers of Security Part 2

"As it recruits investors, Bitcoin gains a whole network of legal, marketing, and political lobbying capability."

Michael Saylor: We're talking about the seven layers of security of the proof of work network, and the first layer is energy, the second layer is technology, the third layer is political. In all these cases, Bitcoin is becoming more secure, more long-lived, because it's co-opting important actors in each of these three areas. We need a network of politicians to defend the network. You need a network of technology providers and semi-conductor providers in order to secure it. We need a network of energy providers to secure it.

The fourth area that's fascinating to me is the financial dimension of security of Bitcoin. Because miners are capital-intensive, you need capital to build the mining facilities that require the rigs and to do the engineering. So that capital comes from outside the Bitcoin system. So it's US Dollar capital for the most part, or its currency equivalents. Bitcoin miners therefore are recruiting a network of fiat investors. There's a whole set of conventional investors, they can't buy Bitcoin or they wouldn't buy Bitcoin but they would buy an operating company that mines Bitcoin. Kind of like people that don't want to own gold but they want to own gold mining companies.

So this is incredibly stabilizing for the network because, for example, today the news was: Fidelity had bought 7% of Marathon. So a big, credible, traditional investor — I think Fidelity has like $10 trillion dollars, trillions of dollars of capital, 20 million customers, and they've been around forever—that company is now invested in a Bitcoin miner.

How does Bitcoin benefit? Well, now you've got hundreds of millions of dollars to lose if the Bitcoin miner fails, and so the investor has a vested interest in protecting their interest. So Bitcoin recruits a whole network of legal, marketing, and political lobbying capability as it recruits investors. There are trillions and trillions of dollars in the conventional economy that for legal reasons or political reasons or business reasons or tax reasons or practical reasons and technical reasons they can't buy Bitcoin, but they can buy a security based on Bitcoin. So in fact from their point of view, they would prefer to buy the publicly-traded company, or if you're a public company investor, you would prefer to buy a publicly-traded company, and if you're a private investor you'd prefer to buy the stock of a privately-traded company, and that's just what their pooled capital does.

If you want to co-opt that financial network of investors, you need a capital-intensive use of proceeds. So Bitcoin mining is capital-intensive and proof of work is capital-intensive. If I snap my fingers and I go to, say, proof of stake, and you don't actually need to buy equipment and you don't need to engineer the facility and you don't need to buy the energy, then you don't need the capital. Well, then you don't need the investor, then you don't need the investment banker, then you don't need the market they trade in. I guess then you don't need the support of the congressmen, the senators, and the regulators they know, and you probably don't need the media journalists that they talk to. You're losing all of that, right?

And you're also starting to look like you've created your own different system. You've created a virtual world where the politicians and the media and the capital and the bankers in the other world are irrelevant. You created your imaginary world. And Bitcoin mining bridges the Bitcoin universe — the crypto-asset of Bitcoin — with the physical and the political universe. And if money is socio-economic energy, well there's two elements to that. The energy part says that it needs to be thermodynamically sound and conservative and correct. But the socio-economic part says: you need the politicians, and you need the media, and you need the influencers.

So the process by which I construct Bitcoin mining companies is a process of recruiting the support of bankers, investors, politicians, and influencers, as well as technologists and energy producers. And that is a virtuous process. That is actually the hard work. You can think of it as when you're building the Bitcoin mine, that's the foundation upon which you're building the bridge. You ever look at these bridges that cross the Chesapeake Bay, they have to dig down and they have to put in place massive pilings in order to build the bridge on top of it. So if the sub-structure and the foundation is weak, then the bridge itself is prone to collapse. The Bitcoin miners themselves, they do all four of these things.

And then the fifth layer of security is the network itself. And by the network I'm really referring to the Bitcoin mining centers. Every place that Bitcoin miners set up their own mining center, they're plugging into the energy grid, they're bringing in the mining rigs, they're engineering them and suiting them, they're getting the political support, they're raising money, and now the miners themselves have their own companies. And there's a lot of public companies and a lot of private companies. We'll have like a dozen publicly-traded Bitcoin miners by the end of the year, but let's say there's hundreds of private companies.

All of these companies are part of the network of security for Bitcoin. They all have skin in the game. They all have invested human capital, financial capital, technical capital, energy capital, time. As you can imagine, they don't want to lose that, right? They feel pain. And so because they feel pain, they don't want to lose that. They invest to protect that.

And so how do you invest to protect $100 million invested in Bitcoin mining? You're going to contribute to political campaigns, you're going to hire lawyers — you know there are experts in Texas energy law. You're going to hire lobbyists, you're going to hire marketing people that are going to explain why Bitcoin is good for the world, not bad for the world. You're going to hire HR people to manage all your HR issues. You're going to hire financiers to report this to the investors. You're going to spend a lot of time talking to all these constituencies to make the investors happy, keep the politicians happy. You're going to be an early warning system against mishap, like if there's a bug in the software of Bitcoin, or if there's a bug in the hardware of Bitcoin, if you don't trust the mining rig manufacturer you're going to push back.

If you think that the software is not being fast enough or the hardware is not being upgraded fast enough to keep the network secure and to protect your investment, you're going to lobby and agitate. And you're going to do that in a context where there's a balance of power between you and the node operators, the validators —who is a lot, like there's a lot of them—and there's also a balance of power with Bitcoin investors.

If you try to change the number of Bitcoin or you try to change the block size you'll probably have a lot of Bitcoin HODLers that'll push back on you. And then you're going to stay in balance with your fiat investors, the people that hold your securities, or the derivatives. And you're going to be managing the business to try to stay in alignment with regulations in 100 different jurisdictions. And you're going to be communicating all the time.

So the Bitcoin mining network is the fifth layer of security of proof of work, and ultimately when you get to a point where you have 50 well-capitalized Bitcoin miners and they've got billions of dollars each at stake and they're spread across all the countries in the world, then they are in essence the first line of defense for Bitcoin. You can call them different things — I refer to them as a motor of sovereignty. They're running the motor that keeps the sovereign network fresh, and they're providing the transactions and the security.

But you could also look at them as the roundhouses on the wall. They are the first thing you're going to attack, right? When you decide you want to attack Bitcoin, you're going to attack the Bitcoin miner. They're the first thing to get attacked. Since they have the most at risk, if you're a Bitcoin miner in China, what have you suffered? A lot. They have the most to lose. And so the Bitcoin miner in Switzerland is going to be spending a lot of time thinking about whether there's a political attack coming from Switzerland on Bitcoin. They will be the early warning system, the canaries in the coal mine, if you will.

So those five layers, they're implied by the proof of work algorithm, and we get energy, technology, politics, we get finance and we get network support, and they're all sources of inertia and antifragility, and they all draw forms of capital that are not Bitcoin. Bitcoin is capital, but dollar capital, semi-conductor capital, political capital, energy capital, human capital — all of these forms of capital are creating a genetic diversity and a Darwinian vitality for the network.

If I get rid of all that capital, if I just said, well, Bitcoin's going to become a proof of stake network and we're just going to stake Bitcoin to run Bitcoin, you see what you've done is created again a hermetically sealed, closed system, but that becomes an inbred system. And that means the biggest Bitcoin HODLer, when they go insane, the security's not going to get upgraded, right? We're not going to continue to distribute the network, we're not going to upgrade the technology anymore, we're not going to spend much time complying with the local political rules.

Howard Hughes was the richest man in the world, and he ended up retiring into the top floor of his hotel and didn't come out of his room for 7 years. He went bat-shit crazy. You know you can do that when you're the richest man in the world, when you don't need anybody else, but you can't do that when you have responsibilities. You have to get up every day and you have to go and put on a suit or a tie or at least conform with the local mores, and that keeps you from going off the deep-end or off the abyss.

Robert Breedlove: So this vortex you're describing of energy, technology, politics surrounding Bitcoin, that's it's kind of assembling for itself, it's centered on the expenditure of energy. That's what makes it real, that's what gives all these actors skin in the game. And the proof of stake model would be something that severs that bridge. We have this digital reality and physical reality that's anchored by energy expenditure, but if you moved to proof of stake you've now severed that bridge.

On the network mining specifically there's this dynamic equilibrium of the governance between the miners, nodes, and all these various stakeholders. And the last point that I think is really important and we saw exhibited in this Chinese mining exodus was: the network is very amorphous. If one jurisdiction tries to stamp it *down, what happens? A lot of these miners get boxed up, shipped elsewhere, plugged back in.*

Michael Saylor: It's a swarm creature. But it's a swarm creature that lives in the physical world and in the political world. It is not a creature in cyberspace. It's not a disembodied astral projection. If you cut the cord, if you don't need energy, if you don't need the semi-conductors, if you don't need the political support, if you don't need external capital, and if you don't need to engineer a facility, if you don't need a corporate structure, then you've created a virtual world like Second Life.

And now if I create a virtual world and I design in it a virtual money or a virtual token, and then I stake that virtual token to secure a virtual network so that people can trade virtual goods and virtual things, I can create virtual security but it's virtually certain that someone's going to create another virtual world, right? The problem is, you outsmarted yourself. You created something which was free and easy, which means that the guy that copied whatever to create Ethereum, well, then the guy copied Ethereum to create Cardano and then someone copies Cardano to create the Uniswap and then they make SushiSwap and then they make the next thing. Because you could do it fast and easy.

What you've done is you've created a virtual casino or a virtual world and virtual money. And it's not that you can't create a business around it. Like Second Life is a business. Fortnite is a business. Online gambling is a business. You can create a virtual casino or a virtual game or a virtual world — Dungeons and Dragons — and you can live in it and you can buy your virtual Vorpal Blade and you can use it to kill the virtual dragon. But the problem is: you don't have a bridge to the thermodynamic real world, nor do you have a political bridge to the political world, and you have lost the space-time constants that give the property integrity, and durability—and that's a good segue for me to go to layers six and seven.

The sixth layer is the spatial layer and the seventh layer is the temporal layer. By spatial layer I mean that the proof of work Bitcoin mining network forces you to spread out throughout the world, because you're not going to find 120 terawatt-hours of energy in one building. And even if you did in one building, the heat engineering is such that you can't stack the Bitcoin miners up over each-other like that. So you have to spread out, and the natural competition means that, well, we find a volcano here and a nuclear reactor there and we find some stranded gas here. It's an open system and it's naturally diffusing and decentralizing, and that's important.

If you have a proof of stake network and I have 50% of all the money, then I can just stake the entire network in Monaco. Why would you need to spread it to the four corners of the Earth? The problem with staking the network in Monaco is that maybe Monaco has a political attack, or maybe the building gets wiped out. What if I stake the entire network on like 16 computers in the same place? You're centralizing it and there is no thermodynamic incentive to decentralize it, and there is no requirement. But in fact with proof of work and the requirement of energy, it naturally is going to create an incentive for someone to put this on a waterfall in the middle of Africa. And that distributes the security load.

You don't want all the security to come from Manhattan. If all the security comes from Manhattan that's proof of stake, that's centralized. Pretty soon you've got a central bank. Pretty soon three people get together and decide your destiny. Everything just centralizes. You really want to have the security come from a country that you can't get to, because if you can't get to it, your enemy can't get to it. And ideally it decentralizes so it's even hard to distinguish where it's coming from. So that spatial layer is forcibly decentralizing it, and it's not just decentralizing it geographically, it's also decentralizing it politically. And that causes it to decentralize across lots of institutions, lots of political jurisdictions, lots of geographic places. And if you want to attack it, there is no one central place to attack. You can't define the vector. There's no logical nexus and there's no physical nexus. Like how do I find the top 51% — like 51% of the hashrate is in 198 facilities around the world — how do I take control of all 198 of them overnight without anybody noticing? Because that's crossing 100 borders.

Robert Breedlove: Yeah, it's neutralizing all of these central attack vectors. Just by virtue of its own proliferation it's becoming more immune to attack from anyone.

Michael Saylor: It becomes immune to a cyber-attack, it becomes immune to a physical attack, and it becomes immune to a political attack. And it even becomes somewhat immune to a protocol attack where someone tries to get people to change the protocol. It's just much harder to attack the durability and the integrity of the network because it becomes so spatially distributed.

Proof of stake, 10 people might actually stake 51% so now I can actually identify the 10 people. Now I know who to attack. Psychologically, physically — whatever. I don't even know where to start with something which is spatially distributed.

The seventh layer is temporal distribution. And by that it just means that the attack has to come over time, because it takes 12–24 months to create a world-class Bitcoin mining center. It takes another 12 months maybe to order all of the equipment. So how do I go about ordering 7th generation equipment secretly or with malicious intent, and then I have to convince someone to front me the money, right? So I have to hoodwink billions of dollars of investors, then I have to buy billions of dollars of equipment, then I have to wait 12 months, then I have to get someone to engineer the center and I have to get them to agree to do it because they trust me, and they're not going to do it if they think I'm going to attack the network. Then I have to convince the politicians to let me build it — it'd take 3–5 years. Then I have to get all the engineers, then I have to build it, then I have to bring it online so that I can attack the network. And I've got to do all of that without a single person guessing that I'm trying to do it. If one person thinks that I might have malicious intent, they'd leak that on Twitter and 2 million people would know it in 7 minutes. So the space-time functioning of Bitcoin is important. You don't want it to move fast.

This is why if you read the Blocksize Wars, a lot of maneuvering back and forth but at the end of the day, the immediate a priori observation is that they shouldn't have tried to change anything. It was a mistake to try to change the blocksize. It's a mistake to try to change the frequency. Why? Because when you create a crypto-universe or a crypto-creature, the block size and the frequency of the network are akin to the space-time constants of the universe. It's like trying to change the gravitational constant and the speed of light. You don't get to play God except once. You can play God once.

If Satoshi had designed the network with 1.2mb blocks and 8-minute frequency it might have worked. And there's some parameters that wouldn't have worked. But once he started it and it started to spin then you don't change it, because the size of the blocksize is kind of like the packing density of carbon molecules. It's like, if you were to change the speed of light or the gravitational constant, let's say you changed the gravity on the surface of the Earth. Let's say we made it 3x what it is right now? What happens to every creature, every bridge, every building, every plant, every factory? What happens when I crank up the gravitational constant by a factor of 3?

Robert Breedlove: They are failing strategies all of a sudden.

Michael Saylor: If you change gravity — heck, crank it up by a factor of 10. What happens? Every building and everything built up until now collapses, i.e. all work is negated. Not just work by mankind — work by animals, work by mother nature, rivers, mountains collapse, bridges collapse, companies collapse, countries collapse. So you would negate all past work, you would imperil or impair all structures. You would impair or destroy all machinery constructed. Any machine that works would stop working. And you would throw the future to chaos. And it's pretty obvious you're playing God, right?

Change the speed of light — the same thing's going to happen. The Reynolds number determines the hull speed of a ship. Change the Reynolds number, change the way that molecules move around a hydrofoil. Nothing works. Some things sink. At best, everything just gets destroyed, commerce collapses. So when you think about it like that, you come back to the crypto-universe, you say, okay, you created a crypto, you set the space-time parameters. Now the future of mining is based on that blocksize and that frequency. Change the blocksize, change the forecast of the mining revenue, change the expectations of every investor, change every business decision, invalidate all business decisions.

At some point the foolishness of doubling the blocksize — which is what the big blockers wanted to do — if they had doubled the blocksize over and over again they would have driven the transaction fees to zero. When transaction fees go to zero the revenue of the Bitcoin mining business is dramatically changed. Now there's no future revenue model. Now you have miners lobbying to change the block reward. You would basically bankrupt the mining business, screw it all up, and then you've got no security network because I can't afford to actually secure the network.

So they were imagining a world where everything had to be on the base layer, and they're monkeying with the base layer. And instead, what they should have seen is that the world is in layers. The key is: the base layer of Bitcoin is like the granite underlying Manhattan. You never needed it to be lighter. If somebody said, I have an idea: let's turn the granite in Manhattan into cotton candy. It'll taste better. It's like, Are you an idiot? If you turn all of Manhattan into cotton candy all the buildings are going to sink, we're all going to die in a cotton candy swamp. You know it's so silly like we're laughing, but the point is—But the granite's heavy and it's hard for me to move and it's slow and the transaction fees for moving granite around are high — Yes. You're not supposed to move the granite. You're supposed to build on top of the granite.

The same is true with the base layer. You're not supposed to move the base layer. The point of the base layer is to be here 1,000 years from now, right? Yeah, you can wave your arm and convert the granite into sand or into mud and you can move it. You can shovel it out of the way with your little toy shovel from the seashore from when you were a kid, but the point is: you can't build a city that's going to last 1,000 years on mud and on sand.

And if you have an understanding of the universe, the universe is layers and layers, and the consequences of the speed of sound is everything that flies through the air. And the consequences of the speed of light is everything in the universe. And the consequence of the gravitational constant is how the water flows and how the mountains move. And once you understand those things you can build everything. Everything that humanity has done in the history of the human race is predicated on those space-time constants being what they were. And when you decide to routinely change them you're playing God. And when you play God you pretty much destroy everything that came before and throw into chaos everything that comes afterwards after leveling every building and leaving a trail of wreckage in the present day. It's like an earthquake except it's worse than an earthquake. It's like a continual rolling, randomized series of different earthquakes every second. And that's why you don't want to screw with that.

Robert Breedlove: Yeah, the way I'm looking at this is if we go back to life and Bitcoin being the survival strategy propagating forward, those strategies optimize for the environmental invariance. So life is optimizing to overcome gravity in a way, and if you change one of these invariants, all of a sudden you invalidate all of the survival strategies. Like all of the adaptation, all the fitness modifications that's ever been done is just worthless all of a sudden.

Michael Saylor: You double the gravity, you kill all of the birds. Like that. You kill all of the birds. "Why don't we just double the blocksize, like cut it in half?" Because everything will die. Like cut the oxygen content in half. What happens?

Robert Breedlove: And then that very precedent destroys Bitcoin, right? Had there been a successful change — and I've heard it put this way, is that Bitcoin cannot be reproduced because irreproducibility was the invention. To create something that no one can change.

Michael Saylor: Make it immutable.

Robert Breedlove: Yeah, if you somehow change it, all of a sudden you've destroyed that invariance that it was offering. And this is incredible — I've never thought about how all these layers actually preserve it in such a way. And the temporal layer is interesting too because if you try to attack it, it's like there's no way to do it without alerting network participants. So it's like a lag or a buffer in the complex system that stabilizes it. If you try to change it, people adapt.

Michael Saylor: Okay, well that's the speed of sound. Like for example, you're like, wouldn't it be great if sound moved faster or slower? A shock wave is when you attack the air faster than it moves out of the way. Well, do we need a speed of sound? Yeah. You need a speed of sound. What if we made the speed of sound equal to the speed of light? Not a good idea. If you want the universe to work — what if we got rid of friction? The transaction fees are too high, let's take them to zero. Well, that's the same as saying, Let's get rid of friction. And let's get rid of gravity. Well getting rid of friction and gravity is a problem because pretty much everything we've created works with it. How about your tires without friction? How well is your car going to drive without friction? They're not bad things.

In the defense of the big blockers, the Lightning Network was not as manifestly clear today, but all you have to do is look at a Lightning wallet and a Lightning transaction and look at a layer 2 and a layer 3 transaction — look at how Square Cash App works, look at how Muun works — I can send 1,000 sats in 1 second for 1 sat. Once you see that, you realize layer 2 platforms, layer 3 applications are going to move Bitcoin at the speed of light in an almost friction-free way, and what did you really want from Bitcoin? You wanted it to not break. It's foolish to risk the immortality of Bitcoin in order to go from 1mb to 2mb blocks. It's foolish.

How about — you could have had something that would be immortal and you could build a civilization that would last for 10,000 years and you could just do it with layer 2 and layer 3 apps, but instead we're risking putting out the fire of civilization to try to change the space-time coefficients of the one crypto-network which successfully took root and decentralized.

Robert Breedlove: Yeah, and that emulates nature properly, to your point, where nature evolves in layers. So we need an unshakable foundation to support higher layers.

Michael Saylor: Let's talk a little bit about proof of stake, because that's the elephant in the room here. Look, proof of work is an open, competitive, natural system. It emulates Darwinian competition. And it's open. Proof of stake is a closed, controlled system. So the problem with a closed, controlled system is it's not Darwinian and there's no genetic diversity. It becomes inbred.

Let's take a shoreline — a living shoreline. If you're trying to actually maintain such a thing, the way they engineer it is, you would create 200-foot long revetments and you would put a break in it for the seawater to come in, and then you create another revetment to protect the shore from the surf, but you always have to have a break for seawater to come in. If you don't leave that channel for seawater to come in, then the water behind the revetment becomes putrid. You ever seen a putrid standing body of water without circulation? It breeds bacteria, it becomes infested, it becomes dangerous, it's not healthy. And why? It's closed. It's a closed system.

All cities, all civilizations — why do you build a city on a river? Because you need something to carry the waste away. What happens if you recirculate your waste in your own [ecosystem]? How long can you recirculate the waste in your space suit? How long can you recirculate the water and the waste in your apartment? Well proof of stake is recirculating itself and the problem there is, unless you have perfect components and you have perfect conditions, at some point either the toxicity of the closed environment destroys the ecosystem, or it becomes so fragile because of the lack of Darwinian competition and the lack of ecological genetic diversity that eventually it dies.

It’s like a creature endlessly cloning itself. After you endlessly clone yourself, you’re subject to genetic mutations and defects. That’s why every organic creature has sex, right? You actually want to keep mixing the DNA in order to avoid the mutated defects. And you can’t cleanse yourself. You can’t fix the species if you close the system.

Robert Breedlove: This is like the photocopy of a photocopy of a photocopy, eventually it gets very blurry.

Michael Saylor: Yeah. And coming back to the other elephant in the room here: Bitcoin is a successful proof of work network that has created our first digital money, our first digital property. Bitcoin has done that. We can see that. There are no successful examples of a proof of stake network succeeding as digital property. That’s the elephant in the room. There are no successful proof of stake networks. What would be required to be successful? Well, here’s a simple one: if the United States government designated a proof of stake network as property and not a security. It’s never happened. In fact, the head of the SEC just went on television and said, if there’s an ICO and there’s a shared enterprise with intent to profit, then it’s a security.

So as far as I can see every proof of stake network that’s been launched looks to be by legal definition a security, not property. And even Ethereum isn’t a proof of stake network, so there’s a debate about whether it is property or not, and that’ll eventually be resolved. But there’s absolutely no question that at this point the only successful digital property that’s ever been created in the history of the world was created on a proof of work network. There’s nothing else. Everything else is just like a garage-science experiment or a venture experiment.

And on the surface, they all appear to be securities. I mean none of them has successfully decentralized or been embraced as property. And it’s not clear whether they can successfully decentralize. I mean, theoretically new things are possible, but practically speaking we have one example of crypto-property or crypto-money, and that is Bitcoin. And everything else is a speculation.

Robert Breedlove: Let me ask you: it seems to me like central banks in a way are a proof of stake network where essentially the more gold they have or reserves they're staking, the more interest and inflation revenue they can generate, effectively. Could we say that then a proof of stake network could only really exist with — because again there's gold underlying that, a proof of work underlying central banks.

Michael Saylor: You could say currencies are proof of stake. If you own a country, you can designate your currency or your token to be a currency, because you make the law. Venezuela can designate their currency and the US can designate its currency. So you have examples of those systems. They're not crypto proof of stake systems, they're fiat systems.

Robert Breedlove: But they are inherently centralizing. I don't see why a crypto system would be different.

Michael Saylor: They are centralized. They all centralize. It seems logical that other stake systems will centralize. And by the way, centralization is not inherently a vice as long as it's treated as a security. For example, a publicly traded company that has a stock has a centralized token — that is the stock. But they have obligations to disclose the number of those such tokens to their shareholders and it's illegal for them to act in a duplicitous unethical fashion. And it's also illegal for a politician to promote that security, because that gives it unfair advantage to one company over another. So if you treat it as a security, then a staked token is ethical and legal.

Robert Breedlove: Yeah. It can just never work as money.

Michael Saylor: Yeah, it's just not ethical to create private money as a security token and represent that it's property. That's where you're crossing the moral hazard boundary. So I think the summary here is: Bitcoin has created digital money, digital property, if you will, on top of a decentralized proof of work network. There are other proof of work networks that are much smaller and less successful that are less decentralized that are struggling to a degree, not winning against Bitcoin. And there are other theoretical approaches to creating a decentralized network — none of them have succeeded to date, if the definition of success is to be deemed morally and legally as property by a legitimate government.

Key Insights from Chapter 15

- Bitcoin is capital itself, but it also attracts other kinds: dollar capital, semi-conductor capital, political capital, energy capital, human capital — all these forms of capital are creating a genetic diversity and a Darwinian vitality for the network.

- When you have 50 well-capitalized Bitcoin miners and they've got billions of dollars each at stake, and they're spread across all the countries in the world, then they are in essence the first line of defense for Bitcoin. You can call them different things — I refer to them as a motor of sovereignty. They're running the motor that keeps the sovereign network fresh, and they're providing the transactions and the security.

- By staying decentralized geographically and politically, the Bitcoin network becomes immune to a cyber-attack, it becomes immune to a physical attack, and it becomes immune to a political attack. And it even becomes somewhat immune to a protocol attack where someone tries to get people to change the protocol.

- It was a mistake to try to change the block size. It's a mistake to try to change the frequency. Why? Because when you create a crypto-universe or a crypto-creature, the block size and the frequency of the network are akin to the space-time constants of the universe. It's like trying to change the gravitational constant and the speed of light. You don't get to play God except once.

- There are no successful examples of a proof of stake network succeeding as digital property.

Chapter 16 | Bitcoin Economics and Evolution

> *"The simple design idea of the Lightning Network is this: I create a channel with a million times less Bitcoin in it, and I move it a million times faster. And that works fine for small transactions. Once you get that idea, you realize that a layer 2 solution (Lightning) is a lot better idea than a higher performance layer 1 solution."*

Michael Saylor: We just finished discussing the proof of work mining network and the seven layers of security: energy, technology, politics, finance, the network itself, the spatial security, the temporal security.

So what you have is you have Bitcoin as a decentralized crypto-asset network and the thing that pops up a lot is the question of, what is energy usage look like over time? And is energy usage going to keep increasing as the price of Bitcoin increases? And I've seen commentary on this, I think a lot of people get it wrong. They seem to think that as the price of Bitcoin increases the energy usage will increase linearly. If there's a 100x increase in Bitcoin price there's a 100x in energy usage. And I think it's just worthwhile to make the observation that over the past 10 years the mining network has gone from being energy-intensive to being technology-intensive. Another way to say it is that in every single industry you go from being labor-intensive to capital-intensive.

When we started with a million people sowing or farming, farming was a labor-intensive activity. As the capital equipment gets better, first you have horses and carts and oxcarts, and then you have tractors, then you have mega-tractors, then you have factories. All of a sudden, the amount of labor matters a lot less, and the amount of capital equipment matters a lot more. There was a time when 90% of the country was farmers and now just 1–2% of the people in the country produce all the food, because it's become technology-intensive. The Bitcoin network is similar except substitute for labor, energy. And substitute for capital, technology.

If we go back 10 years, it probably took 100x as much energy to generate an exahash as it does right now. An S19 takes 30 megawatts per exahash, but an S9 takes 150 megawatts per exahash, so you've got 5x improvement in energy efficiency over one generation of equipment. We're like on the 7th generation of mining equipment. If you go back 2 or 3 generations then you get that 100x increase in energy intensity. If you look at where we are today and you go forward 10 years, it's reasonable to expect that you will get improvements in efficiency from three dynamics: you've got halvings in the protocol and we have one every 4 years so that means over the course of a decade you have a 5x increase in efficiency from the halvings, then if you have a 4x increase in energy efficiency and you get it twice — or a 5x then a 4x — you get a 20x increase over 2 more generations of hardware, and now you've got a 20x times a 5x or about a 100x increase in efficiency.

So Bitcoin price could go up by 100x and the network efficiency would go up by 100x and the energy consumption could be flat. And that's an important thing to keep in mind because people sometimes think, well, energy is used to secure the network — No it's crypto-energy. It's encrypted energy that's used to secure the network. And in order to get it from raw energy to crypto-energy you have to run it through a SHA-256 miner that's properly engineered in a heat-sink. And so as the heat engineering improves, the miners get more efficient, as the semi-conductor engineering for ASICs improves, the network gets more efficient, and what you have in that dynamic is a never-ending struggle between brute-force and technique.

It kind of reminds you of something — like nature, and competition. There's two ways to do things, right? You either use raw labor, or you use technology. And it's important to have that dynamic, or that yin and yang, because when the technologists get lazy and they stop improving, then the raw material or the brute-force overwhelms and the control of the network shifts back to the energy holders. But to the extent that the technologists upgrade their engineering facilities, their heat technology, their chip technology, then the energy becomes less important. The technology becomes more important.

And the combination of both of these things requires free-flowing capital. In a free market the capital is continually seeking the best use. Should I invest money in creating the next generation of SHA-256 miners? Should I invest money in engineering a more efficient Bitcoin mine with immersion cooling? Should I invest money in commercializing more energy? Or plugging in more energy and just manufacturing the same design over and over again? So what you have is this nice delicate dance or balance of power.

A model you can think of is when we first started Bitcoin mining, it was all energy-intensive. We were using off-the-shelf computer equipment and you were throwing raw and power and raw commodity materials at it. It's rotated through a set of generations of hashing equipment. So now you need energy but the limiting factor is not with the energy, the limiting factor on generating hashes is the mining rigs. And maybe you could say the limiting factors on doing this well are the mining rigs properly installed in the right mining center with the right cooling technology.

If you look forward another decade, you'll see it's going to be much more technical-intensive. You could have all the energy in the world, but you're not going to be able to generate crypto-hashes. The break-even point of an S19 is like 40 cents a kilowatt-hour. You could pay 39 cents a kilowatt-hour for electricity and make money. The break-even point for an S9 is 8 or 9 cents a kilowatt-hour. If you have 20 cent power you can't make money—you've got to turn it off. The break-even point for the generation of equipment before that is like 2 cents per kilowatt-hour. You can't pay 3, you have to turn it off. And the generation before that you're down to half a penny a kilowatt-hour.

So if you're looking to run antiquated equipment to generate hashes you have to be stealing the power. If you have free energy and it's literally stolen or it's given away, you can run 8 year old or 5 year old equipment. But if you're paying your way, you have to run modern equipment. So this is a dynamic model but it's important because it's a dynamic evolution of the sophistication of the Bitcoin security network that has Darwinian overtones. Because the Bitcoin miner that just buys a bunch of equipment 5 years ago and stops upgrading becomes obsolete. You won't last 10 years with that equipment. I mean your break-even point becomes a tenth of a penny once you're four generations behind.

This is like nature healing itself. It's a very natural process. If you're out of touch and you're 1,000 miles away, the free market somewhere else is going to keep moving the state of the art forward, and should you isolate yourself you will find that you can't generate enough hash power to participate in the revenues and the transaction fees. And it's the network's way of simply squeezing you off the network. Now what that means is Bitcoin miners are organic creatures with a negative feedback loop. In essence a market mechanism. The difficulty adjustment is not just tactically every 2 weeks in the protocol. The difficulty adjustment is in the market, because if you don't upgrade your hardware every 4 years you become uncompetitive. And you can't upgrade your hardware every 4 years unless you were a responsible custodian of your capital.

If you spent all your money, if you didn't save any money, if you're not trustworthy, if you don't have credit, you can't buy any equipment. And so to stay competitive on the Bitcoin network you have to be credit-worthy, you have to be competent, and you have to be credible. Someone that has the technology has to be willing to sell to you. And so I would say that the dynamic nature of the network is: you have this competition between miner, operations, and the technology providers, and the energy sources, and the political jurisdictions, and you have to be upgrading, and you probably need to be improving your efficiency by a factor of 1.5–2x every year. You're subject to Moore's Law and this competitive Darwinian pressure.

The result of that is: energy consumption is going to fall per exahash. We'll go from 150 megawatts an exahash to 30 megawatts an exahash, to 5 megawatts an exahash, to 1 megawatt an exahash, to half a megawatt — and on down. We're just going to move down this efficiency curve. You could imagine if I gave you this supercomputer the size of a sugar cube — look at the amount of power on a modern semi-conductor, right? Think about what's in the latest iPhone chip. We just keep compressing computing power and we find a way. And human ingenuity is like that. They will keep finding a way to create more hash power using better technique. So that suggests that energy consumption will probably increase non-linearly, like with the log of the price.

If the price goes up by a factor of 10x, energy consumption might go up by a factor of 2, 2.5, 3x. At some point you start to think you will roll over — you'll peak. We might have already done it. You peak energy consumption and then you taper off or it holds constant as the hashrate increases, and instead of a Bitcoin miner spending half of their budget on energy, you roll over to spending half of your budget on capital. You're buying capital and amortizing it, and pretty soon your variable cost on energy is 5% and 40% (of cost) is capital equipment, and what you've got is not an energy war to see who controls the network, you've got a technology war to see who controls the network.

Why is that good? Well, because energy is a raw material in the universe, but 7th generation SHA-256 miners are specialty equipment. Everybody on Earth can find some energy and throw it at the problem, but throwing the 11th generation of SHA-256 mining equipment at the problem is something that probably no one's going to be able to do unless they spent a decade or two decades engineering that equipment and thinking about it. So you have a specialization in the same way that John Deere tractors are specialized after 100 years.

If you went back to a farmer in 1850 and you describe what a farmer in 2020 can do, it's like night and day. That's what's going on with the network. And I think it's self-healing, self-sealing, self-correcting, and the combination of the Darwinian and Adam Smith capitalist competition is a critical advantage for Bitcoin versus a proof of stake network. Because in the proof of stake network, you pretty much turn off energy competition, you turn off semiconductor competition and innovation, you turn off engineering and heat engineering innovation, you turn off capital financial competition, you turn off the political element, and that means that I can get fat, dumb, and happy. I can just post a billion worth of my tokens and go to sleep for a decade and no one is trying to make a network so it's not being tested, there's no stressing of it, and ultimately the problem with that is you're going to suffer in integrity and durability.

Robert Breedlove: So we're again at this point where the Bitcoin mining network itself is a free market in and unto itself and it reflects many of these properties of capitalism. So there's this yin and yang of efficiency and magnitude going back and forth. And to your point that's reflected in many markets historically where we shift from labor intensity to capital intensity. And this basically means as the capital becomes more plentiful and effective at amplifying labor, you need less labor to accomplish the same results. This is a natural, capitalistic progression. Let me ask you this: the useful life of a miner I'm not sure what that is —

Michael Saylor: 4 to 5 years.

Robert Breedlove: So we have the older generations rolling to cheaper energy to remain in the marketplace.

Michael Saylor: I can steal energy. If I can bootleg energy, I can run older generation miners profitably because my variable cost is zero.

Robert Breedlove: And then so newer generations would then be allocated to more expensive energy resources initially. Is this an aspect of the gradually increasing fully amortized cost to produce each Bitcoin? Because there's more cap-ex — I mean maybe the cap-ex / op-ex mix is changing, but the overall cost to produce each Bitcoin is rising, which is putting upward pressure on its price in the marketplace. Is that another way to look at this?

Michael Saylor: That's not clear to me. What's the price of energy? Is it going up or going down? Maybe the price of energy is going up and then it's going down. What's the price of semi-conductors? Is it going up or is it going down? Well sometimes it's going up. It's going up if there's a monopoly on the semi-conductor chips, and then it's going down when someone else enters the market.

An Intel 386 chip is pretty cheap now, right? So chip prices are coming down but the next generation is going up. But again, it's a competitive thing. What's the price of semi-conductors? If there's only one provider and it's a monopoly then you could say, well the price of your semi-conductors is going to go up every generation forever and the cost of Bitcoin mining is going to go up. But there isn't only one semi-conductor company. There's multiple. If there is one, we're back to Jeff Bezos's statement, "your margin is my opportunity."

Bitcoin is an open market for mining, so anybody can engineer a SHA-256 chip. There is no one type of Bitcoin miner. There are people slapping these things in the back of trucks and driving them around and plugging them into bootlegged power lines, right? It's not very efficient. There are people that slap 500 Bitcoin miners into a container, and they just drop it on top of a pad. Does it work? Sure, it does. Is it properly heat-engineered? No. Does it get hot? Very. What's the consequences of running hot? You have to turn off the mining equipment some of the time so you lose efficiency, and they burn out and their useful life is not 4 years, it's 3 years. Or it's 2 years.

So there's a competition here and just like there's competition for energy, will energy go up forever? No. Energy could go to zero. If I invent cold fusion, will semi-conductors go up forever? No. At some point Bitmain raises their prices and then everybody starts talking to Intel. You go to another semi-conductor company and you say, well, look, Bitmain's tripled their prices and now you can make this much. And now you draw somebody else into the space. So I think there's a dynamic equilibrium between the engineering and the semi-conductor and the energy. And the best capitalized companies, they can buy the new equipment. Like the publicly traded North American companies, they will go and buy all the new shiny equipment, and poorly capitalized companies will run the old equipment. And then you will roll forward like that.

It's not clear to me how much it will cost to create a Bitcoin, because that's a function of — maybe the profitability of Bitcoin is a function of the rate at which we add hash power versus the rate at which we add Bitcoin HODLers. If we increase the number of Bitcoin HODLers by a factor of 10x, and we increase the hash power by a factor of 2x, then the profitability — the revenue per exahash — is going to go up by a factor of 5x, right? And if we increase the exahash by a factor of 10x and the HODLers by a factor of 2x, the profitability is going to deteriorate. So the profitability of the Bitcoin mining is going to have an impact on the rate of development of Bitcoin mining semi-conductor ASICs, right?

Robert Breedlove: So the total op-ex and cap-ex going into the Bitcoin network at a halving — that total allocation of expense goes from producing, call it 900 Bitcoin a day —

Michael Saylor: The revenue gets cut not quite in half.

Robert Breedlove: So the production cost per Bitcoin is effectively doubled. And then my thinking was that this higher cost to produce each Bitcoin is actually incentivizing miners to HODL. They don't want to sell below cost of production. So that's what's bootstrapping Bitcoin's price upward.

Michael Saylor: I can see once every 4 years the energy cost doubles. But if once every 4 years I upgrade the equipment and it's 5x faster, now the question is: how much did you pay for the equipment? And the answer is: if you paid a lot for the equipment — now amortize the cost of the equipment into the cost of the Bitcoin, and you get an answer. But that's a price that a vendor charges you, which could be triple or half depending upon the competition.

In a very competitive market, well let's say it a different way — if you're buying Intel grade 386–46 chips and slapping them into your home appliance, what is the price of the intelligence that you put inside your toaster? As a percentage of the cost of the toaster? It's pretty cheap, right? Like at some point the cost of intelligence starts to drop because it's a commodity market, AMD drove down the prices and so everybody can put intelligence into their appliance because of competition. If there was no competition it might go the opposite way. So I think that protocol guarantees us through halvings that we're going to have to get 5x as efficient every decade, but the other dynamics like the rate of technology advance go even faster. And they're just as much.

I do agree with you on this one idea: we're in a gold rush right now. Or a Bitcoin rush. Between now and like 2035 — in 2035 we'll have mined 99% of the Bitcoin. So you've got 14 years, and during the 14 years the block rewards are pretty high relatively speaking. And so it's very lucrative to be in the business, but the business is going to become less profitable, likely. At the very least we know it's going to rotate into a transaction-fee business. So instead of 90% block reward 10% transaction fee, you could expect 90% transaction fee 10% block reward. And the transaction fees are also going to scale more likely with the log of the price, not linear to the price. If I want to move a million dollars I might pay you $10, and if I want to move $10 million I would pay $12 but I'm not going to pay $100. I'm just going to overbid the million dollar buyer. And if I want to move $50 million I'll bid $14. So the transaction fees will increase with the number of transactions on the base layer, and that'll be a dynamic. But the block rewards go away.

So if you're a Bitcoin miner it's a very lucrative high-growth business right now, but it will move toward an efficient transaction security network later, and you would do well to buy Bitcoin. That's why it makes sense to HODL Bitcoin if you're a Bitcoin miner, because it's your hedge against two things: it's a hedge against the inevitable protocol rewards — which the protocol says Bitcoin mining is going to get 90% less lucrative over the course of the next two decades, cut in half once, cut in half twice. Over the course of 12 years you get cut in half three times. So it's a hedge against that but that's the long term.

It's also a hedge against the near-term hashrate explosion. If you're mining Bitcoin right now and if somebody comes out of the blue and they bring a lot of hashrate on the network faster than you expected then your market share gets cut. In that situation, HODLing Bitcoin may win. If everybody believes that Bitcoin will win and there's a flood of people that enter the mining business, you want to own Bitcoin. Mining's going to get competitive, Bitcoin's going to get more valuable. So you definitely don't want to be in a situation where you've mined Bitcoin, you sell all your Bitcoin for cash — in that case you're attempting to run a cash business in 16 years when the revenue's been cut in half four times.

Presumably if you cut a 100 to 50 to 25 to 12 to 6, when you're generating 6% of the revenue and everybody else has had 16 years to get in the game, it's going to be more of a commodity business. Now that's not to say that a Bitcoin miner can't compete and evolve, it may be a great business then too. It may be that Bitcoin miners evolve to be running Lightning nodes or running layer 2 platforms of other sorts. It might be that as a Bitcoin miner you have a huge Bitcoin treasury and then you can generate yield on it. It might be that there are other applications. We'll talk about some.

It's very logical for Bitcoin miners maybe to get into some of these other application areas that will pop up in time. And there are technology possibilities here. So I would say the illogical thing to do in business is you don't worry about what happened 16–20 years out if you can make a ton of money now. I mean you might very well make enough money now to have 10, 20, 30, 100 billion dollars of capital then, at which point you can go buy something else to get you into another business.

And my other point I guess here is: yes, it's true, Bitcoin miners should keep Bitcoin. But it's also true that non-Bitcoin miners should keep Bitcoin. Like if you believe in Bitcoin, then the logic follows that no matter business you're in, if you have operating income, you should invest in Bitcoin. And if you are able to generate financing, if you can raise debt or equity then you should also raise debt and invest that in Bitcoin. That would be the case for other companies. Microstrategy looks at it that way and we're not even a Bitcoin miner. But that's my thought on the Bitcoin mining network and the dynamic model over time.

I think that some people intentionally misunderstand this, like proof of stakers, they intentionally misunderstand because if they misunderstand it then they can assume that energy consumption is linear, and then they can play the ESG card and pretend that they're virtue-signaling and say that they're doing something good for the environment because they're not using energy. But the truth is that they're not using energy nor are they using technology. They're not using hardware technology, engineering technology, or energy, nor are they submitting their network to competition, and they're not using external capital, political or financial. What they're doing is they're creating a virtual protocol in order to create security. And even there, if you're going to rely upon a virtual protocol to create security you'd be better off to stake the protocol with an asset that is outside of the network, that is external to the network. So for example, the Lightning Network makes a lot more sense because it's staked with Bitcoin, than if it was staked with "Lightning Coin."

Robert Breedlove: Yeah, and it's paying the revenues for services rendered, which is actually the routing transactions, versus just how much Bitcoin you hold — that's not how it works. So I think the key point here is that the proof of work energy expenditure is actually transforming what would just be a video game asset into a macroeconomic asset.

Michael Saylor: It's creating a real-world asset. It's your connection to thermodynamic reality and socio-political reality. You can't be in denial of those two. What people think of you matters. If the person is the police officer on the beat or runs the country, what they think of you matters. And what nature thinks of you matters. If you decide to step off a cliff by looking the wrong way, nature's opinion matters whether or not you respect nature or not. And so it's very important that if you're building something that you want to last for 1,000 years, you respect politics, and you respect thermodynamics, and you respect physics. And proof of work is this very creative invention to continually connect and synchronize the Bitcoin network with political, physical reality. Never-ending evolution. Every 4 years, every 2 years. A new miner, a new chip, a new place. Throw away the old, the old generation has to die so the new generation can form, so that the creature can evolve. Otherwise, the creature stagnates and becomes progressively more fragile until it's no longer capable of competing in the real world.

Robert Breedlove: And that energetic anchor to these realities — energetic which begets socio-political reality — this is what's keeping all network participants honest and accountable. There's this synchronization and rule set imposed upon them that they can't avoid. It's like gravity. You just can't avoid it.

Michael Saylor: Yeah, it's quite a wonderful thing. It's a crypto-universe. Satoshi played God, you either call it creating your own universe and setting the planets in motion by defining the space-time constant, or the other metaphor is you released a creature into cyberspace and set the genetic DNA of the creature. You controlled how it will procreate. And once you release the thing, then it spreads as some kind of swarm life-form continually evolving.

So I think with that we conclude that the Bitcoin mining network and the Bitcoin network in general is quite a wonderful thing. Now the next question is: how does it scale? If we go beyond the network, the primary purpose of the Bitcoin mining network is to provide security and enforce protocol integrity and durability of the system. And it does that very well. But so how do we actually scale out to provide hundreds of billions of transactions a day to billions of people on the planet at the speed of light using the latest computer technology? That's I think where a lot of people fall down, they just want to look at the base chain and say, well, it moves 7 transactions a seconds and so you can't scale. But they lack the imagination to understand the consequences of the layering.

So the Bitcoin monetary network, it scales based upon platforms at the layer 2 level, and then applications above that level, and you could call them layer 3 applications or you could call them applications on top of applications on top of applications. So you could have layers 4, 5, 6, and 7 and you could have interplay between the applications, but for the purposes of our discussion let's just call them layer 2 platforms and layer 3 applications in order to keep from getting too confusing in our semantics. So what's the most important one?

Well Lightning is a very interesting and maybe the most important layer 2 platform. And the idea behind Lightning is: I want something to be exponentially faster and exponentially cheaper and in return I'm willing to secure exponentially less money. So if I have only $100 at risk, then there's no reason why I couldn't do 10 million transactions on that channel. And that totally makes sense because as you scale out, if you look at the transactions that 8 billion people need to make on this planet every single day, the majority of the transactions are actually of value less than one dollar.

In fact, there are plenty of transactions that could be valued in pennies. And then the big transactions might be $10, $100, or $1,000. Probably if you were to look at the transaction stream on the Visa network or the MasterCard network, the average check is like $28 or something. But there's billions and billions and billions of them. So you don't really need the final settlement security of Bitcoin, because that's the highest level of security in the world. And you're getting that to move a billion dollars. You probably only need to move a billion dollars around occasionally.

If I moved seven nuclear-powered aircraft carriers per second anywhere in the universe, I could probably win whatever war I wanted to win if that's what I'm using my teleportation power for. So Bitcoin is good for teleporting large chunks of value, but for all of the routine work, you want something smaller. So the logical thing to do is — the way Lightning gets at it is you're putting liquidity in the channel and the channel is at risk, but the overall blockchain isn't. The beauty is you get to keep your keys and keep custody of that. So that's a very fascinating thing. You can build the universe on Lightning.

Presumably, you could give 8 billion people Lightning wallet or wallets — more than one — you can build Lightning into PayPal and Square Cash and Apple Pay and Google Pay and Facebook and every messaging app, and you could use it at all scale, not just to move money but to move anything. The other day I did a 1,000 satoshi transaction on Lightning for 1 satoshi in 1 second. So Lightning is an obvious layer 2. And the simple design idea is: I create a channel with a million times less Bitcoin in it, and I move it a million times faster. And that works fine.

There are other approaches, of course, but you can give up all of the proof of work because you don't have as much at stake. And you're creating a staked network but you're staking it with an external asset that derives its value from assets external to it — energy capital and political capital and technical capital are flowing into Bitcoin capital and Bitcoin capital is flowing into the Lightning Network in order to secure it. So once you get that idea, you realize that a layer 1, layer 2 solution is a lot better idea than a higher performance layer 1 solution. Because making the layer 1 twice as fast or 3x as fast or 10x-100x as fast doesn't get you to a million times as fast.

It's no different than the common-sense way that human beings solve every problem: if you had a million dollars in the bank and you were going out on a Saturday and you needed to spend money quickly, you would take $100 in cash and you would break it into 20 $5 bills, you put it in your wallet, and you would rest assured that you can't lose more than $100 and you would leave the rest of the money locked up in the bank. And the money in the bank takes 48 hours to get at and takes lots of degrees of authentication that's behind 3-feet of steel, and the money in your wallet is in your front pocket and it takes 1 second to get at and occasionally you drop a $5 bill on the floor when you're drunk and life goes on.

If somebody said you had to take the entire million dollars with you out to a night club with you every Saturday night you would say, that's pretty stupid. And if they said, well, you know the bank vault doors they're too heavy, we're just going to have to re-engineer them to make them 1,000x lighter so that you can open the doors faster so that you can access the million dollars on Saturday night while you're drinking — you might think, maybe it wasn't such a good idea to actually have access to all my money while I was out drinking on a Saturday night? And the same is true with the idea of a base layer. Maybe it's not such a good idea to have that many transactions on the base layer, because every single moving part is just something to break. You've got yourself too many moving parts — you don't want it.

So the layer 2 platform is this idea that what I want to do is I want to just create a cash, if you will, of a much smaller amount of value that is much less at risk that I can afford to move much quicker and I don't need the same degree of security on it. And Lightning is not the only layer 2 platform you can conceive of. Another crypto-network could be a layer 2 platform. In theory you could spin up a proof of stake and you could stake it with Bitcoin and it wouldn't be a theoretically different thing. You can also spin up a crypto-network proof of stake and then you can move Bitcoin through it, but if you're using the native token and moving Bitcoin then your Bitcoin is only as secure as the native token. And of course if the token is ginned out of thin air — Yoyo-coin — then that's a problem.

Ultimately the real issue with layer 2s is you're moving a portion of value into that layer 2, trusting the layer 2 in order to get performance or functionality. So you can do it, you can do it with Lightning, and the logical thing you do is you stake with Bitcoin and then you reduce the risk to the channel liquidity. That's a way to do it and that's logical.

Another way to do it is to build some kind of system like maybe if I was an exchange and I had Bitcoin on the exchange and I created an API to the exchange that was programmable. Then in a way the exchange is holding the Bitcoin, the API is providing you with a very fast speed. Now you've taken a different risk. You've taken exchange risk, which is some counterparty risk of some sort. If you use a Paxos or a NYDIG platform those are more conventional platforms to build applications on top of. Like NYDIG has a platform for you to build a credit card plugged into Bitcoin. And Paxos has a platform plugged into PayPal to their application. So if you want that kind of platform you plug the mobile application into one of those platforms — there is some risk with that counterparty — it is limited to the amount of Bitcoin you're handling on the platform.

You can do all sorts of things to mitigate the risk, but clearly there is a need for layer 2 platforms. And there isn't a need for just one layer 2 platform. Like Lightning is a compelling, decentralized layer 2 platform because it's decentralized. But there are a lot of layer 2 platforms that will be centralized because they need to be regulated in order to meet with regulatory compliance obligations. And there are a lot of counterparties, like a credit card company might prefer to work with a centralized layer 2 than a decentralized layer 2 because they have regulatory constraints.

Robert Breedlove: And this is similarly rooted in something very fundamental, as proof of work, because there's this fundamental trade-off between security and freedom, typically. And so in money we're saying that the security model of Bitcoin is its decentralization, but with that comes a lot of work. It doesn't do many transactions per second. But what you can do is abstract that Bitcoin to a more centralized database whether it's a custodian or a Lightning — a Lightning is kind of a decentralized centralizing force — you pick up all this functionality but you're giving up some of the trust-minimization you get at the base layer.

Michael Saylor: I would say that if your focus is on property, and what you want is digital property, you want the system optimized for durability and integrity over time. And performance and functionality and compliance are not on your list. Your list is durability, integrity, immortality. Very simple.

But as you move towards applications and you move away from property towards applications, you either have to optimize for functionality — like if there is no functionality there is no application — or you have to optimize for performance. If I can't pay for the coffee within 1 second I can't pay for the coffee.

Or you have to optimize for compliance. If I really wanted to issue an insurance policy or I want to issue a security or if I want to issue a yield token, and it's illegal in a certain state, then if I want to do it I have to actually comply. And compliance pops up with stablecoins. Compliance pops up with DeFi exchanges. Compliance pops up with derivatives. Compliance will pop up with anything that looks like a security token. These are all applications, but is there a future for those applications on top of Bitcoin? Yeah. But Bitcoin is not the application. The application will be built on a layer 2 platform. Or it'll just be built naked against the Bitcoin.

Maybe you don't use a platform, like I can implement my Bitcoin credit card using the NYDIG platform, or I could hire an army of programmers and I could build it one-off, right? There's this little battle of: Do I build a custom app or do I build an app using your SDK? So platforms are going to be operating system-type things with an SDK, and it might be that it looks like an AWS where they spin up services. You could imagine an AWS spinning up a whole package — like a Lightning node — if I could spin up my Lightning nodes and run them maybe I would do that. Maybe I wouldn't do that. Probably Lightning is not the best example because people are looking for something totally decentralized there, but a better example would just be an SDK that allows someone to deploy a mobile app that has Lightning and Bitcoin money transfers embedded in the mobile app and I just want to do it quick and easy.

Robert Breedlove: There's a continuum where we have this totally decentralized layer 2 in Lightning, maybe we have SDKs in the middle versus a fully centralized solution via NYDIG or someone else. But it just speaks to the versatility of Bitcoin which again you can't get from a gold. You can't get this versatility of application layer with something like gold.

Michael Saylor: And there's going to be competition, right? Massive competition with regard to: what are the SDKs in the layer 2 toolkits? And even Lightning has Lightning Labs developing a toolkit to help you with Lightning. So that's competition, and it's interesting but it gets theoretical in a way so it's actually probably more instructive to move down to the applications themselves and talk about: what are the applications of Bitcoin that scale the system? Because these are the things that make the different.

Once you start to think about the range of applications, then you figure out what you might build into your platform if you're trying to create a business hosting those applications. Let's talk about that. The obvious application is the individual HODLer that owns Bitcoin or just HODLs it in cold storage for long periods of time. It's like the family or the individual. I think that's pretty well understood and that's been pioneered for a while. The thing that's varying is the way that that individual chooses to HODL the Bitcoin. And that's where it gets really interesting.

For example, one thing that I want to start with is this observation that one interesting application of Bitcoin is a bond. You can actually create a derivative or a security of Bitcoin and scale the network with the bond. So if a government owns Bitcoin, the government is the customer. And if the government then buys the Bitcoin and then issues sovereign debt, the sovereign debt becomes a Bitcoin derivative. So you're creating credit from Bitcoin. Bitcoin is the base layer money, and the debt or the credit or the bond of the government is the layer 2 money. It could be backed in whole by Bitcoin like 100% or it could be backed in part.

If a municipality like a city buys Bitcoin and issues municipal bonds, that becomes another form of an application of Bitcoin. If an agency like Fannie Mae or Freddie Mac or any international agency — United Nations or the like — if they were to buy Bitcoin and then issue any kind of bonds, it's another example of a derivative. When a corporation buys Bitcoin like Microstrategy and then we issue a bond that's backed by the Bitcoin, we created a derivative. And you could even take Bitcoin — there's $700 billion of this Bitcoin out there — and you could issue asset bonds. Call it Bitcoin-backed securities. They used to be called mortgage-backed securities. And with mortgage-backed securities we'd take a bunch of heterogeneous assets and we'd securitize them into a note, and so imagine you had 21 million identical houses called Bitcoins and you decided that you were going to create a bond backed by 11 or 37 of them — an asset-backed bond.

All of those are applications of Bitcoin. They're not applications the way a computer scientist thinks of them. They're financial applications of Bitcoin. You can have technical applications of Bitcoin that are running on mobile devices like mobile apps, you could have web applications, you could have financial applications of Bitcoin. And a lot of times when people think about scaling, they only think about the technical applications and they don't really think about the financial applications.

If we move on the next layer or next type of application, think about mobile payments. A mobile app that does payments is something you can plug into Bitcoin, like Square and PayPal have plugged mobile apps into Bitcoin. What does the app do? Maybe it lets you buy Bitcoin, sell Bitcoin, send Bitcoin, send Bitcoin via the Lightning Network like the Muun wallet — that's an application — but maybe it lets you send Bitcoin on its own proprietary network like Square Cash Tags. You can send Bitcoin between one Cash Tag and another Cash Tag instantly for free. You could send Bitcoin between any two mobile apps within the network using the handle of the network. If you were doing that, then probably the application provider custodies some of the Bitcoin and they're moving it and they become a fractional bank of Bitcoin. Bitcoin becomes the central bank in cyberspace. Our friend Ross Stevens would say the Decentral Bank in cyberspace. All of these other mobile apps or websites become fractional banks plugged into the central bank of cyberspace. And they custody some amount of Bitcoin.

The mobile payments space is particularly promising, because all these mobile applications are one step removed from being mobile banks, and there's no reason why Facebook and Apple and Google and the like don't eventually become mobile banks. It was a matter of time before they let you send photos, and then at some point they decided to let you send videos. And then they decided to give you emojis. And then they decided they'd let you send audio files. And in a way you could think of Bitcoin as just another file type.

Robert Breedlove: Yeah, it's all media.

Michael Saylor: So it's kind of inevitable. Now you start with mobile payments and then when you start to think about moving that around and the services that consumers want, you're now into retail banking. The retail banking applications are in essence savings accounts and credit lines based on Bitcoin for consumers. You've got billions of people that presumably want a wallet with an asset in it, and the asset would be Bitcoin. Then they want to borrow against it.

Say you want to be able to draw down a credit line at an interest rate. So I carry around $10,000 in Bitcoin, I borrow $1,000, I pay 4% interest. Now I've drawn the credit line in the local fiat currency in question — Euros, Dollars, Yen — I pay the interest whatever that is, and then I spend the money. That's pretty popular. I mean, credit cards are quite popular, right? How many credit cards are there in the world? So we're really talking about credit cards drawn against Bitcoin property instead of unsecured credit lines.

The opposite is also the case: maybe the consumers want savings accounts that generate yield. So I have Bitcoin, I either HODL the Bitcoin, or I move the Bitcoin into a yield-generating account, or I move the Bitcoin into a collateral line and I use it and pledge it as credit. Now is that the same everywhere in the world? No not really. You're going to have different regulators in every jurisdiction. Every state they're going to tell you whether you need a money transfer license or a banking license or whether or not you need a securities registration in order to give yield or give loans or move money around.

Of course, there are always nuances. I think this is one of the nuances that caused El Salvador to designate Bitcoin as legal tender so that they could easily move Bitcoin around in mobile apps. In a place where the government has collapsed, then the solution is going to be a mobile wallet with Lightning because there isn't a regulator. And wherever you have the kind of weak governance then it's not going to matter. As the governance gets stronger, then there are going to be questions of: what's your compliance requirements to do a retail banking application? It looks like it's literally different state to state, country to country, jurisdiction to jurisdiction and evolving right now. That's one of the reasons why maybe centralized applications — CeFi — will actually beat the DeFi, because the message of DeFi is: oh it's really expensive going through all this

KYC. Yes, that's true, but it's also illegal not to. So the real question will be, Can you do it and how much validation do you have to go through to do it, and how compliant do you want to be.

Key Insights from Chapter 16

- Over the past 10 years the mining network has gone from being energy-intensive to being technology-intensive. There was a time when 90% of the country was farmers and now just 1–2% of the people in the country produce all the food, because it's become technology-intensive. The Bitcoin network is similar, except substitute energy for labor and technology for capital.

- We're in a gold rush right now. Or a Bitcoin rush. In 2035 we'll have mined 99% of the Bitcoin, so you've got [approximately] 14 years, during which the block rewards are pretty high relatively speaking. So it's very lucrative to be in the business, but it is going to become less profitable, likely. At the very least we know it's going to rotate into a transaction-fee business.

- Yes, miners should keep Bitcoin. But it's also true that non-miners should keep Bitcoin. If you believe in Bitcoin, then the logic follows that no matter business you're in, if you have operating income, you should invest in Bitcoin. And if you are able to generate financing, if you can raise debt or equity then you should also raise debt and invest that in Bitcoin.

- Some proof of stakers intentionally misunderstand proof of work and assume that energy consumption is linear, and then they can play the ESG card and virtue-signal and pretend that they're doing something good for the environment.

- The simple design idea of the Lightning Network is this: I create a channel with a million times less Bitcoin in it, and I move it a million times faster. And that works fine for small transactions. Once you get that idea, you realize that a layer 2 solution (Lightning) is a lot better idea than a higher performance layer 1 solution.

- There are lots of opportunities for financial applications and innovations in the layer 2 and layer 3 space, all built upon the Bitcoin layer.

Chapter 17 | How Bitcoin Changes Everything

"The killer app for Bitcoin with social networks is creditworthiness leading to cyber-security. Safety and civility in cyberspace."

Michael Saylor: So that's mobile payments and then retail banks. If you go to commercial banking, well with commercial banks you've got treasury services for companies, and you've got credit lines. So corporations — like individuals in a way — I've got a bunch of cash, I want to generate yield on it. Call that treasury services. Or I have a lot of assets and I want to generate yield on them. And I also want to borrow against my assets. And so I need a commercial grade bank to do that. I tend to have slightly different requirements and I'm going to move in greater scale and I may have different accounting issues, but the commercial banking application and the retail banking are pretty similar.

Exchanges are another application of Bitcoin. There, people want to trade assets and derivatives against each other. And you can imagine infinite assets and derivatives. They will be compliance-heavy. Do you need to comply with the CFTC or the SEC or who knows what? And that gets very complicated. But it's a big business.

The next big area of applications will be corporations. And here, any company that simply buys Bitcoin becomes an application of Bitcoin. And this is not a technical application. If we take Microstrategy as an example: when Microstrategy buys Bitcoin, then Microstrategy stock becomes a derivative of Bitcoin. And now Microstrategy stock is being traded on a NASDAQ, a regulated exchange. So what you did is you took one asset sitting on the blockchain and you used that to back another asset, an equity, that's trading on the NASDAQ, and then you back a third asset — calls and puts and leaps — that are trading on a futures exchange like the CME, and you may even issue debt that trades privately over the counter through a set of broker dealers. Each of those are applications of Bitcoin.

Well, how big are they? They could be a billion, $10 billion, you could have $100 billion worth of these applications. Why do they exist? Because people want them. Because there are a lot of people that want to trade stocks on NASDAQ, they don't want to trade Bitcoin on Coinbase. Why don't they want to? Because they took a decade to raise $100 billion and they promised a bunch of pension funds that they would only trade equities on NASDAQ and now it would take them another 5 years to reach that expectation with the pension funds.

Maybe they want that because — like there are companies that do convertible arbitrage. They have $10 billion dollars, they raised it from pension funds and endowments, and they do convertible arbitrage. That's their strategy. And that means they have to be able to buy the convertible debt and short the stock and try to generate a yield. And they've got billions of dollars to do that thing, they get up, that's what they want to buy.

There's a phrase in Wall Street: when the ducks are quacking, feed the ducks. If the ducks want to buy convertible debt, give them convertible debt. If they want to buy equity, give them equity. If they wanted to buy Bitcoin, then they would buy Bitcoin. But these applications have different needs. If you're a consumer, you want to download Square Cash App and you want to smash buy $37 worth of Bitcoin and you want to do it in two clicks. You don't really want a super-complicated trading console with like 16 Bloomberg screens with futures and options and 3,700 trading pairs. That's not what you want. You want to smash buy $37. So sometimes simplicity is the right thing.

Sometimes you want simpler than that. Sometimes you've been doing business with the same private wealth or client adviser, you want to call him on the phone and day buy $10,000 worth of that Bitcoin, and you want them to buy the Bitcoin ETF. And the pure Bitcoin maximalist will say you're crazy. You should store $10,000 on your own hardware wallet. Okay, well I'm 82 years old and I'm in a wheelchair and my hands shake and I don't use computers — how am I supposed to store this on a hardware wallet? I'm just buying it as an investment instead of buying gold and I've got 5 seconds.

So you have 5 seconds and you talk to a guy Bill that you've been doing business with from Merrill Lynch. You know Merrill Lynch was a company on its own before it was bought by Bank of America. So I talk to my guy Bill—I've been talking to him for 30 years. I have 5 seconds and I want to buy $10,000 worth of Bitcoin. Okay, does that application matter? Sure it does. What if I want to buy $10 million worth of Bitcoin? Does that matter? Sure it does. What if I want to buy $100 million worth of Bitcoin? It's the same dude with the same 5 seconds, okay? So do applications scale the Bitcoin network? Yeah, absolutely. Would it be helpful to have more? They're all handles.

They're a handle — I'm grabbing onto Bitcoin. People handle it different ways. If I have PayPal it's installed on my machine and it works and I can click on a button in 1 second and buy Bitcoin. I prefer that to downloading a better application that takes 37 clicks and takes 2 days to get authenticated. So at the end of the day, these applications make a big difference, and what you have going on here is the creation of financial applications being wound into other exchanges and other platforms. And the platforms are everything. For example, a corporation on the NYSE takes you to the NYSE platform. A corporation on the NASDAQ takes you to the NASDAQ platform. A corporation in Japan takes you to the Nikkei trading platforms. So as Bitcoin spreads on the balance sheets of different companies and different governments, it spreads onto different platforms and it morphs into different derivatives and different flavors.

Robert Breedlove: This is a very useful framing — I've said that money is the base layer protocol for human interaction. We've had gold functioning for a long time and we could arguably say that a lot of these institutions, technologies, government currencies, these are all applications on top of gold, effectively. So now Bitcoin is like a new base layer protocol replacing gold and enabling lots of other applications to flourish on top of it.

Michael Saylor: Yeah, I would agree. It's the base property protocol or the base monetary protocol if what you want to do is to build something on a stable monetary base, or a stable value base, then you need digital money, digital property. And that's a little bit different than digital currency, right?

If you want a good digital currency, you're best to base it on a good digital money. Currency is an application of money. And if you understand it that way, money is the base layer, the end-all, be-all source of all value and virtue. And everything else is an application on top of it. I mean I don't need 100% of everything I am to be in my currency. I only need 1% of everything I own to be in the currency, and then 99% of what I am I need to be in the money. And in fact, if the money has integrity, the currency doesn't need to. Which is a pretty important point.

If I had 99% of my wealth in the money and if the economy was growing or my assets are growing at 4% a year and if 1% was in the currency and the currency was devaluing at 10% per year, it wouldn't matter because it's on the margin. I'm getting wealthier faster than I'm dissipating energy. And it's the same with a lot of things in life. The $5 that falls out of my pocket on a Saturday night isn't going to impoverish me. And on the margin you could say well you're kind of crazy to carry cash to a bar on a Saturday night. I know but I'll take the risk, for the convenience. Each of these things is putting value at risk.

For example, if you have $1 billion in a company you have the counterparty risk of the billion dollars in the company. If the company fails or the company's defrauded, then the billion disappears but it doesn't imperil the entire blockchain. And publicly traded company that files quarterly results is a lot more trustworthy counterparty. Like Microstrategy is able to issue debt backed by Bitcoin, but we're a public company. If you were a private company and you also wanted to issue debt backed by Bitcoin, it would either be harder, impossible, or you might pay a higher interest rate. And if you were an entity, an individual that nobody trusts, and you went and you tried to sell the same bond maybe you wouldn't get it (done) because people are taking counterparty risk and they're trying to figure out whether they trust you.

But now here you see the Bitcoin which is: everybody can have an application. So the question is: which exchange is going to embrace Bitcoin most effectively? Which mobile app is going to embrace Bitcoin most effectively? Which commercial bank will? Which corporation will? Which government will? Any government or municipality could issue a billion dollars in municipal bonds at 2% interest, they could buy a billion dollars of Bitcoin that has 100% yield, they could scrape the arbitrage and they could generate a billion dollars a year, right? In appreciation.

You play the game right, in theory you could just issue a bunch of debt, buy the Bitcoin, eliminate taxes forever. Now you're like, well, that sounds too good to be true. Well, everybody can't do it. If you're last, it doesn't work that well. For example, Mark Zuckerberg was first to own Facebook stock so he's rich forever. You can't get rich forever by buying Facebook stock now. What's the difference between you and Zuckerberg? The different is he was first and you were last. And the same is true with Jeff Bezos.

So if you're the first city and you move early, then as everybody else enters it's like if you came to America first and you got 1,000 miles of farmland — Lord Baltimore. You get a better deal than if you show up 9 generations late. So with this you've got a market dynamic where the first government, the first city, the first state, the first country, they can all seize an advantage.

And they're competing. I talked to an energy company the other day, I said, well, you're an energy company, but you know what? You can either sell your energy to Bitcoin miners and get paid 2 cents a kilowatt-hour and you could sell your excess energy — small idea, or you could become a Bitcoin miner and you could mine with all your excess energy and make 40 cents a kilowatt-hour — better idea, or you can go big and you can buy a bunch of Bitcoin miners and/or mine it, and then you can go issue billions of dollars of debt at 2% interest and buy the Bitcoin, and then make billions of billions of dollars.

If you want to make $5 billion if you're an energy company, buy $5 billion of Bitcoin and issue a press release and then wait 90 days and you'll probably make $5–$10 billion. Do you want to bend over and do that? Now that's a bigger idea. But the biggest idea is: you do all those things, you reposition your company, and your revenue multiple doubles or triples because now you're not a sleepy regulated entity, you're a high-tech company.

These big energy companies, you could have 20, 30, 40 billion in revenue and trade at $40 billion market cap. Well, it's likely that a $2 billion Bitcoin miner will trade with a $40 billion market cap. What if the Bitcoin miners start buying the energy companies? It happened with UUNet and WorldCom back in the day where you actually saw the Internet companies buying the baby bells, buying the Telcos. And maybe they don't, but the bigger idea is the utility companies are in competition with the tech companies, and they're in competition with the financiers and the hedge funds and the traders and the governments. Bitcoin doesn't care.

Robert Breedlove: At every level there's this game theoretic incentive to move first, benefit disproportionate to later adopters, because to your point, wealth is hierarchical. If you're there first and you're right you're going to benefit the most. This is true in Bitcoin as well if you're there first you're going to benefit more than later adopters. This is also the problem with fiat in a way that the central bank has arrogated itself to always be first. It always gets to allocate itself new shares of currency, so it's kind of hijacked the hierarchy of wealth in fiat.

Michael Saylor: Yeah, and in Bitcoin, the courageous with clarity, a vision, get to make themselves first. It's a private market in money, and with fiat it's those that are politically connected that stay first. So it's public money. It's worthwhile to say: all of those organizations are competing, but we have other types of derivatives that are very interesting, or other applications of Bitcoin. Like, let's take for insurance: if you're an insurance company then every insurance policy you issue or annuity that you sell based on Bitcoin becomes a Bitcoin derivative. So if I want to sell you a life insurance policy that pays off a million dollars when you die and I'm going to charge you x premium — I've got to charge you enough premiums and invest them to pay off the million dollars, so if I charge it as premiums then I invest them at 2% or 3% yield it's kind of hard to pay off the million. I have to charge you a high premium. Whereas if I charge you premiums, invest them in Bitcoin, I can lower the premium. Or I can raise the payout. Or I can make the payout variable.

So now you've got an insurance company that's created a differentiated insurance product, and you can either do it explicitly by offering the policy backed by Bitcoin, or you can do it implicitly on your balance sheet. You could just take $50 billion of your balance sheet that are your collected annuities or premiums and you could buy Bitcoin with it and use it collectively to back all your policies at the corporate level.

And of course insurance companies have long-durations, they have to look out 30–40 years, they're in business forever, they have huge balance sheets, and their traditional funding vehicle is bonds and bonds are broken. And so you need a better type of annuity generating asset or a better appreciating asset and that's where Bitcoin comes in. And that's where investment firms come in.

My next application — you've got an investment firm and you sell mutual funds or you sell investment funds and people invest with you. So if you create the fund based on Bitcoin and then you sell that, that becomes an instrument that's interesting to corporate investors, family offices, individual investors. There's a lot of people that will want to buy that fund, and there's an entire industry that sells those funds.

There's also an interesting hybrid twist on that, like if you look at Fidelity, Fidelity sells all sorts of fixed income funds — municipal bond funds, sovereign debt funds, corporate bond funds, junk bond funds. They all have a low yield and they've got a negative real yield. Well, you can juice them by blending them with Bitcoin. So if I have an instrument or an asset that's been going up 100% a year and I blend it in to an asset that's been paying 3% interest, now I've created a hybrid instrument that's got downside protection of bonds but upside opportunity of something — maybe not as much. It's like sucralose, a powerful sweetener. It's a booster, a sugar. So for an investment fund you can blend various types of funds.

What if you're 50% Bitcoin and the other 50% is Bitcoin derivatives — call and future collar options. Maybe you collar the Bitcoin or maybe you're 50% Bitcoin and then you're selling covered calls out of the money to generate yield. So you're creating a Bitcoin yield fund. It doesn't have the same upside as Bitcoin but it's got a guaranteed — at some point downside protection, upside yield, you're blending the thing and you're selling it to someone that wants to buy that. Are there people that want to buy it? Sure.

In a way, Microstrategy issues a bond, 6 1/8% interest, and we go and buy Bitcoin with it. If you swap the two, you've given up the upside of Bitcoin in return for the guarantee of 6 1/8% interest. And why do you do that? Because the government is paying you 1% interest. Because the spread is 500 basis points over sovereign debt, and because the spread over investment-grade corporate debt is 400 basis points. So you want the yield. Who wants the yield? Well, somebody that actually sold $100 billion of a high-yield fund to endowments — that's who wants the yield, right? So there are structures in the financial universe. You don't just snap your fingers and change those structures in a day, week, month, or a year. In some cases those structures have formed over 10, 30, 40 years. 10 years is a short period of time in the finance world. So what you have here are investment firms that are able to create all their own financial products with any mixture and blend of risk and volatility and they can hybridize them with existing conventional assets. What if you had a fund which was half gold, half Bitcoin? For people that like gold but they can't bear to give up — or they like Bitcoin but they can't bear to give up their gold. It's like an alloy, right?

Robert Breedlove: What's coming to mind here is that you could almost put in your parameters and just back into the right blend customized for you, using Bitcoin to augment your risk profile and volatility and such.

Michael Saylor: Yeah and there's a lot of people that would give up the ability to get 100% return in a year, in return for a downside protection. If you went to every conventional investor and said, are you willing to give up more than 30% upside a year in return for all these downside protections? Yeah, there's a lot. And big banks sell these all the time. They're very lucrative products. Like I charge you a 2% fee to give you a structured product that gives you some mixture of downside protection and upside that we tailored to your needs. So those are just other applications.

Moving on, let's talk about trust and endowments. You've got a family trust that's supposed to last 100 years. Fund it with Bitcoin. The Rockefeller Foundation, right? If you have any kind of endowment, every university, every non-profit has an endowment. They all have the same funding problem, which is: where do you put your money where it will last 100 years? And here this is an interesting — it's a digital transformation of money managers. I can either put $100 million into an endowment and have like 10 money managers and they charge me a 1–2% fee a year to manage the money, or I can just put the money into Bitcoin and I've dematerialized the money managers. We talked about the complete digital transformation or dematerializing of the fiat banking system in our early sessions. In this particular case you're just dematerializing money managers.

But there's an application: the application would be, I need a certain type of reporting and compliance. Like, what if you did run a non-profit foundation and you wanted to be a custodian of $12 million? You want to flip it into Bitcoin and now you want to spend 1 hour a year monitoring your position, filing the appropriate forms with the IRS, and then you need to make disbursements. So if you're an endowment, what you probably want is a credit line so you can borrow some money to pay your disbursements — your fees — on an ongoing basis. And then you want the principle to be protected. Normally you would have to re-invest it. Well, what's the closest thing to Bitcoin? Probably the Vanguard 500 or the like. And if you look at Vanguard and Fidelity and you look at what they do, one big business for them is this endowment management or donor-assisted funds and they just run these funds where you put your money in this stuff and you put it on autopilot and people look at it once every 90 days. And that's a big business.

Right now, $100 trillion of it is broken, because $100 trillion of it is bonds. And so you've got massive amounts of fixed income bonds and other assets that are broken, and those are traditionally the things that are easy to manage. And if you can't do $100 trillion of bonds you have to get into equity. But there's not enough equity. So if the equity markets in the US were $35 trillion and you've got $100 trillion in bonds that are broken, you can't just jam $50 trillion into equity. You can, but what are the consequences? You overvalue the equity, right? I mean I can't imagine any of those examples right? Yeah. Lots of examples. We overvalue the equity as we jam the money into it. And then people are waking up and they're looking for that ultimate asset-class, and the beauty of Bitcoin is Bitcoin can never be overvalued because Bitcoin is a bank in cyberspace. If you jam $50 trillion into it the price would go up by $50 trillion and it would be worth $50 trillion more.

Robert Breedlove: And this is the crazy thing about Bitcoin too, or money more generally, is that it's a Veblen good. So the more expensive Bitcoin gets, the more de-risked it is as money. So therefore, it's almost like a self-catalyzing feedback loop. The more money you jam into it the more likely it is to eat all the other money in the world, the more value you would expect to be placed on it.

Michael Saylor: Yeah, I mean it's the ideal solution for the problem. Now I talked about a bunch of financial applications and also about corporations and governments as platforms, but we should also talk about hardware applications. There's a lot of hardware applications in Bitcoin. There's the miners, nodes, wallets. Any device that exists could have Bitcoin protocol built into it. And the question is: do you want to build the wallet protocols? You can build Lightning and you can build Bitcoin wallet right into any mobile device. You can build it into jewelry, right? You can build multisig into your watches, your jewelry, your phones, your desks, your appliances. And that's the differentiating business. Like people build Bluetooth into stuff. This is more important than Bluetooth. A bit more important.

You can also build nodes and node capabilities right into devices, and you can build mining into devices. You could integrate these capabilities right into firmware, into hardware. And why would you do it? Because it's differentiating. If Apple builds Bitcoin support into Face ID or Touch ID on the iCloud, if the iCloud was like a multisig, multi-factor Bitcoin cloud, then you've got a trillion dollars worth of Bitcoin that can flow into the iCloud. That differentiates iCloud. You can do it with Google Cloud, you can do it in Microsoft. You can do it in Facebook. And so Apple's interesting because it crosses hardware and software and network areas, but if you go into the software area you've got Windows and iOS and Android and these are all operating systems. And to a certain extent things like Chrome are like their own mini operating system. You can build Bitcoin right into those things.

Google built password managers into Chrome, right? And if the operating system has native support for the monetary protocol, then either that's a reason to use the operating system, or you can pass on those primitives and that functionality to the applications running on the operating system. Like, for example, Apple lets banking applications use Face ID for authentication. And that makes the banking application better. So if you want to compete in hardware and devices or in software operating systems, then putting the monetary protocol into it is just like building TCP/IP. I mean I'm old enough to remember when the TCP/IP wasn't built into every computer.

Robert Breedlove: I've seen recently too that the Lightning Network can actually augment authentication, or you can authenticate through the Lightning node so it can be an alternative to password management at some point.

Michael Saylor: Yeah, there are really fascinating decentralized applications like decentralized identifiers. It's probably outside the scope of our conversation here, it's like an entirely other world. But with hardware and software that leads you to Cloud, right, because Cloud is another application. AWS, Azure, Google Cloud and the like are all general purpose, containerized corporate platforms for spinning up massive applications.

To the extent that they have support for Bitcoin or Lightning and certain other monetary protocols, then it makes it easier for their clients like Zoom. You know Zoom went from nothing to 400 million users during Covid because they were sitting on top of Cloud platforms from an Azure or an AWS and if they weren't they couldn't have scaled. So when they're scaling they need to be able to access these underlying services, and monetary services are destined to be built into every application. And they would be a differentiator.

Now if we go to another really interesting application, take social networks. Twitter, Facebook, Instagram, TikTok. Well, one of the killer applications for social networks is verification and credit-worthiness. Another way to say it is to put skin in the game. Right now there's no cost for malicious behavior, so if I open up my Instagram Direct Messages, 99% of them are bots, they're not really people. And when I look in my Twitter comments, a lot of times 90% of them will be bots or malicious actors. So we've got a bot war. People have computers spinning up fake Twitter accounts and then Twitter has computers taking them down as fast as they can, and in the first hour after I post I just see a bunch of garbage. And then after a day after they clean it up I see real comments.

Well, what that means is there's a 2, 3, 4, 5 hour delay between when I do something and when I don't have garbage in the feed. How do you solve it? The logical way to solve it is I put 100,000 satoshis on a Lightning wallet and then I make a security deposit to Twitter. Or a security deposit to Instagram, or YouTube. And if you deposit — that's like $40, pick a number, it doesn't matter what the number is, something less than a lot but more than nothing — I deposit 100,000 satoshis and then people on Twitter can say I only want people with an orange check that are credit-worthy or Lightning-certified to be able to post comments on my Tweets. And I only want to receive Direct Messages from people that are Lightning-certified or verified or I follow — one of those three. And if you're not an orange check we'll call it, then you can't message me and you can't comment.

So what happens? Well first of all if I just turned my entire feed to only orange check and blue check, all the bots disappear. Everything disappears. CZ does this on Binance a different way, he basically blocks everybody but people he follows from being able to post on his Twitter comments. But the problem with that is if you're only following 1,000 people, you're living in this hermetically sealed world of only 1,000 who can comment. What you want is 100 million people to be able to comment but you want them to comment with civility, or at least you want them to be human beings.

So if 100 million people post $30, then you've got $3 billion worth of Bitcoin posted as a security deposit on the Twitter network. Now everybody that comments is a real person — well let's say 98% of them are, and if someone wants to spin up 29 Michael Saylor bots, well they can do that but they're not orange checked and so I'll never see them. And no one will ever see them. Or if they want to get the orange check they'll have to post 100,000 sats 29 times. And if it turns out that they were imitating somebody, maybe they get a fine of 1000 sats or maybe they just lose their orange check and they forfeit the security deposit.

It's like if you trash somebody's house when you're renting it. Whenever you rent a house or whenever you cross public properly, you either have to have insurance like car insurance, or you have to be bonded if you're a worker, or you have to post a security deposit. And if you have that concept that you have to be bonded or have to be credit-worthy, then what that means is that every time a bot attacks you it costs them $30 and you're monetizing maliciousness. If someone's stupid enough to be malicious.

It's not that the platforms don't have rules — they do have rules. They take people down and kick them off and put them into the penalty box all the time. But the rules don't have consequences. If you have a computer that generates 87 million Michael Saylor bots every day, what the heck do you care about the rules. The rules don't hurt robots. There has to be a price or a cost to maliciousness.

And so what you have here is a Bitcoin Lightning network that allows a billion people to post $20 each.

And then you can have micropayments. And when that happens, if I go on Amazon, I look at the opinion surveys, I don't have to worry that someone created fake bots to fill in the survey. Because if you get caught faking it, you get a fine. And if I go on YouTube and someone's running a scam video and they have to have the right green check and then when they get — we report scam videos every 2 hours, Robert. Every 2 hours someone spins up a scam video on YouTube which fakes something and we report them as fast as they spin up. But there's no cost.

Or there's minimal cost to denial of service attacks. There's no cost to spamming. There's no cost to hostility online. There's no cost to being an imposter. There's no cost to spinning up 97,000 fake accounts that actually try to create the appearance of a panic when it doesn't exist — cyber warfare.

So what's the solution? The solution is: in the real world you have skin in the game. If you walk up to women in the street and you insult them, you take the risk that the person next to them or they punch you in the face. And if you drive your car and you get in a wreck you get fined and then you get an insurance claim and there's a consequence. And if you go rent someone's house for the summer and you trash the house and you have a security deposit, they take it. Or if you rent an apartment with a security deposit and you trash it, they take it. So there's skin in the game and that causes people to act with some degree of civility.

There's no civility in cyberspace because there's no skin in the game. Why not? Well because if you use credit cards there's a 2.5% fee, it takes 2 days to clear and 2 billion people don't have credit cards, among other things. You do it through the App Store, well you could try to do it through the App Store, there's a 30% cut. How do you move 1000 satoshis for 1 satoshi in a split second? So having a value network becomes a credit network, and so for all these social networks you can actually inject the quality of credit-worthiness. And you could do it different scales. You could post 100,000 sats to get a small check, you could post 1 BTC to get a green check, you could post 10 BTC to get a purple check, and if I'm going to go ahead and scalp tickets or I'm going to book a flight from someone online, I want them to have a purple check because I want recourse from them if they lie to me or if they've defrauded me.

Robert Breedlove: Yeah. And this makes a ton of sense because the original purpose of proof of work itself was a counter-measure to spam. It would be applied to other levels.

Michael Saylor: That was Adam Back's idea. Proof of work to fight spam. Well, we're living in a world 30 years later and we still have spam. Like we're running two levels of spam filters at my company. 90% of my e-mail is spam. I'm getting spammed on LinkedIn, Facebook, Instagram, Twitter. All of the news feeds are getting spammed. And it's worse than spam, there are hostile, malicious actors that are fake bots that are waging a psychological psyops warfare to destroy people and ideas and demoralize them. And they do it for economic reasons, political reasons, etc. And there is no way to stop that in the absence of some kind of value network. So the killer app for Bitcoin with social networks is credit-worthiness leading to cyber-security. Safety and civility in cyberspace.

Now if you think about where you can apply this idea of just a security deposit — which is the same as the cyber-bond if you will, or a cyber-passport — the beauty is that you don't have to disclose your identity. You could spin it up and spin it down and it's lightning-fast and dirt-cheap so it's open to everybody. But once you spin up this idea, it works across every social network, it cleans up all of the feeds, the news feeds, etc. It also works for all the communication networks. So you've got Whatsapp, Signal, iMessage, Zoom, Gmail, every other mail, Office365 — all of these things have spam imposters, malicious accounts. The phishing attacks are ridiculous.

So if you're a corporation and you want to send something, maybe you ought to post — if someone sent me a mail message and I knew they posted a 1,000 BTC security deposit and if they had it all at risk and if they could lose it instantly on the platform, I would be probably trust them more. Like it's all about cyber-reputation. If you really want me to do something risky, what do you have at risk? What's your value at risk?

Now what's interesting about this is you're picking up all these security deposits; the platforms could use the float to do something if they wanted. And it's generally good for Bitcoin because it locks up the Bitcoin on the platforms. And the sky's the limit.

Once you've done that, you've not that far from implementing fees. You could say, look, you want to send me a message, pay me 1,000 sats. You can have micropayments, you could have streaming payments, you can tie services to credit-worthiness. If you go to a bank and you put a bunch of money in the bank, they'll give you a higher level of service than if you put a small amount. So if you want to have distinguished services, whatever they might be — it might be more bandwidth on Spaces or some other functionality. But everybody eventually can become a bank.

And that takes us to entertainment/news. They have similar opportunities. They either sell access via these micropayments, or they curate the response, they create safer forms and safer spaces by requiring that you be credit-worthy in order to engage. Or they certify their news by posting some credit-worthiness. There's a lot of news circulating so if you're trying to separate fake news from real news, then you could use this to certify.

You could also have service providers like Uber, OpenTable, AirBNB, and for them the idea of credit-worthiness also matters. It matters when you're pairing customers with restaurants. Like, can I trust my customer to show up to the restaurant? Can I trust the driver? Can I trust the passenger? Can I trust the person that's going to come into my AirBNB? They tried to do this with reputation, but there is no more important metric I think than the credit-worthiness. If someone says I'll post $10 million in order to drive your car for the weekend, it's highly unlikely that they're going to trash your car. And one way or the other, it's just an example of an insurance policy or a bond. If you have the recourse to the bond in cyberspace, then it creates accelerated commerce, and that means that millions of people can trade with millions of people without the impedance of "getting to know each other."

Robert Breedlove: Yeah, an accelerated commerce and higher quality. So again the skin in the game we keep throwing around, but it's this balancing of incentives and disincentives that facilitates quality discourse among people.

Michael Saylor: There's not a day that goes by, Robert, that you don't hear about someone getting hit with a denial-of-service attack. If you have an open website, there are hostile actors in Eastern Europe and they'll unleash an array of machines and they'll just start hammering your website to take it down and it happens all the time. It's like spamming except it's worse. That's because there's no cost for denial of service. But a lot of the proof of work and the Bitcoin protocol was to protect the underlying Bitcoin mining network from denial-of-service attacks, right? But this actually solves the problem. If you have a web service and you require someone to have an orange check, you could actually filter on the orange check and you could then prevent the denial-of-service attack. You can protect yourself from it. Again, like having a credit rating in cyberspace. And since it's open to everybody in the world, anybody can have one, and it causes 98% of the malicious behavior to be non-economic anymore. It doesn't make any sense.

So I didn't spend much time on retail but clearly retail is another example of a layer 3 application of Bitcoin. And it's either buying, selling, trading, security deposits or credit checks or the like as you're doing business. If I'm selling something to someone and I trust them and I trust their credit rating, maybe you can do business with retailers friction-free that don't know you and you don't know them, right? And it definitely is the case that if I'm Amazon or I'm setting up a retail operation, people are coming at me that I don't know very well. So what I want to do is find a way to accelerate the transactional speed and I do that with some type of Lightning security deposit and/or Lightning payment.

We didn't really touch on the elephant in the room here which is the Visa and MasterCard network costs: 2.5%. Just the general payments network and the remittance network is another killer application. Just money movement around the planet, because the existing systems either bank wires — and small sums of money is credit cards, and you have knock-off periods. They don't move on the weekends, they don't move from certain jurisdictions to others, you can't program them, they're not automated. So you've got a heterogeneous expensive manual patchwork quilt of payment options. And you can't scale that.

Lightning helps you scale it at the transaction level, and then Bitcoin is the core settlement network. So if you want to plug 100,000 businesses into each other and move blocks of money a million to $100 million or $1 billion at a time you do it on the Bitcoin network, and then when you want to plug the millions and tens and hundreds of millions of people into each other, you've got to do it on these other platforms.

But it won't be one. I think Lightning will be very successful, but I think that Square, PayPal, Apple Pay, Google, Facebook, they will all build their own proprietary networks, and they will have additional functionality or compliance or performance or convenience built into it. Like if I bought an insurance policy a decade ago and if the insurance company just buys Bitcoin for their balance sheet, I now own a Bitcoin derivative insurance policy and I did nothing.

So there are certain subtleties where the corporation or the actor can do something that provides a benefit to millions of people without them doing anything, and that's how you really scale in a big way. Like if you're a country and all of a sudden you start buying Bitcoin, your currency in circulation with 11 million people using has become a Bitcoin derivative, and it happened because someone in the back office started buying Bitcoin.

So the beauty of all this is: there are a lot of ways to scale Bitcoin and there are a lot of applications. There are hundreds of thousands of applications — millions of applications. There will be lots of platforms, tools. If Twitter creates the orange check and they plug it into the Lightning Network, then who's to say that Twitter couldn't open up the API and let Google and Facebook and Apple and Medium and Reddit and Discord inherit the orange check. Twitter becomes a platform for cyber-security, and you carry your Twitter reputation or credit rating to all these other places. I mean that's what Google tries with their Google log-in and Apple has Apple log-in. They're carrying authentication around the network. There's no end to the type of opportunities that there are there, and you're only limited by human creativity.

Key Insights from Chapter 17

- Mobile payments, retail banks, commercial banks, exchanges, corporate finance: all applications of Bitcoin ultimately scale the network, even when the purists dislike the vehicles.

- As Bitcoin spreads on the balance sheets of different companies and different governments, it spreads onto different platforms and it morphs into different derivatives and different flavors.

- Currency is an application of money. Money is the base layer, the end-all, be-all source of all value and virtue. And everything else is an application on top of it. I don't need 100% of everything I am to be in my currency. I only need 1% of everything I own to be in the currency, and then 99% of what I am I need to be in the money. And in fact, if the money has integrity, the currency doesn't need to. Which is a pretty important point. If I had 99% of my wealth in the money and if the economy was growing or my assets are growing at 4% a year and if 1% was in the currency and the currency was devaluing at 10% per year, it wouldn't matter because it's on the margin. I'm getting wealthier faster than I'm dissipating energy.

- There's a first mover advantage for entities that embrace and integrate Bitcoin applications—cities, states, banks, corporations, energy companies, etc.

- New products are coming that will allow investors to put in their parameters and back into the right blend customized for them, using Bitcoin to augment their risk profile.

- Bitcoin can be used to start trusts and endowments.

- There are a lot of hardware applications in Bitcoin. There are the miners, nodes, wallets. Any device that exists could have Bitcoin protocol built into it.

- Social media companies could use Bitcoin for authentication and verification on their platforms. Requiring users to have skin in the game eliminates the vast majority of bots, scams, and trolls—because it becomes too costly for them.

- It's worse than spam. There are hostile, malicious actors that are fake bots that are waging a psychological psyops warfare to destroy people and ideas and demoralize them. And they do it for economic reasons, political reasons, etc. And there is no way to stop that in the absence of some kind of value network. So the killer app for Bitcoin with social networks is credit-worthiness leading to cyber-security. Safety and civility in cyberspace. It's lightning-fast and dirt-cheap so it's open to everybody, and it works across every social network: it cleans up all of the feeds, the news feeds, etc. It also works for all the communication networks: WhatsApp, Signal, iMessage, Zoom, Gmail, every other mail, Office365, etc.

- Bitcoin is a killer app for the general payments and remittance networks (where Visa and Mastercard charge 2.5%). If you want to plug 100,000 businesses into each other and move blocks of money a million to $100 million or $1 billion at a time, you do it on the Bitcoin network.

About the Authors

Michael Saylor is the Chairman & CEO of MicroStrategy (MSTR), a publicly traded business intelligence firm that he founded in 1989. He is also the founder of Alarm.com (ALRM), named inventor on 40+ patents, & author of the book *The Mobile Wave*. He founded and serves as trustee for the Saylor Academy (saylor.org), a non-profit organization that has provided free education to over 1 million students. He is an advocate for the Bitcoin Standard (hope.com). He has dual degrees from MIT in Aerospace Engineering & History of Science.

Robert Breedlove is a freedom maximalist, ex-hedge fund manager, and philosopher in the Bitcoin space. To him, Bitcoin is fundamentally a humanitarian movement exposing the greatest con in human history: central banking. By learning about the connection between honest money, entrepreneurship, and civilization, we are renewing hope for the future of humanity. To this end, Robert's mission is to restore freedom, truth, and virtue in our world by tenaciously asking the question: "What is Money?"

If you enjoyed *The Saylor Series*, Robert has released many more longform conversations with a variety of interesting voices. They are all available at whatismoneypodcast.com.